

Differential and Analytic Manifolds

Nicolas Bourbaki

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WARNING

This new printing of the fascicule of results on manifolds reproduces without change the works published previously (fascicules XXXIII and XXXXI), regrouping them into a single volume.

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N. BOURBAKI

Part I

Paragraphs 1 to 7 (97 pages)

INTRODUCTION

This fascicule brings together the fundamental notions and the principal results of the theory of differentiable manifolds (over the field of real numbers) and of analytic manifolds (over a complete non-discrete valued field). It contains no proofs.

The definitions and results of the first six books are assumed to be known.

NOTATIONS AND CONVENTIONS

(paragraphs 1 to 7)

Base field

Throughout this fascicule, the letter K denotes either the valued field $\mathbb{R}$ of real numbers, or the valued field $\mathbb{C}$ of complex numbers, or a complete commutative non-discrete valued field with ultrametric absolute value (Top. Gen., ch. IX, 3rd ed., § 3).

Topological vector spaces

When K is different from $\mathbb{R}$ or $\mathbb{C}$, we shall call a semi-norm (resp. norm) on a vector space F over K an ultra-semi-norm (resp. an ultra-semi-norm which is a norm) on F (Esp. Vect. Top., ch. II, 2nd ed., §1). If γ is a semi-norm on F , and if $x \in F$, one will sometimes write $\|x\|_\gamma$ instead of $\gamma(x)$.

We shall call a normable space a topological vector space over K whose topology can be defined by a single norm, and a Banach space a complete normable space¹. We shall call a polynormed space a topological vector space over K whose topology can be defined by a family of semi-norms; when $K = \mathbb{R}$ or $\mathbb{C}$, this notion coincides with that of a locally convex space.

If E and F are polynormed spaces, we shall denote by $\mathcal{L}_m(E; F)$ the space of continuous m -linear maps from E^m into F , equipped with the topology of uniform convergence on the bounded subsets of E^m ; it is a polynormed space. If moreover E is normed, and if γ is a continuous semi-norm on F , and if $u \in \mathcal{L}_m(E; F)$, one denotes by $\|u\|_\gamma$ the lower bound of the real numbers $a \geq 0$ such that

$$\|u(x_1, \dots, x_m)\|_\gamma \leq a \|x_1\| \dots \|x_m\|$$

for all $x_1, \dots, x_m$ in E.

Multi-indices

Let I be a set. For $\alpha = (\alpha_i)$ and $\beta = (\beta_i)$ belonging to $\mathbf{N}^{(I)}$, we set (cf. Alg., ch. III, 3^e ed., beginning of the chapter):

$$|\alpha| = \sum_{i \in I} \alpha_i$$

$$\alpha! = \prod_{i \in I} \alpha_i!$$

$$\alpha + \beta = (\alpha_i + \beta_i)_{i \in I}$$

$$((\alpha, \beta)) = \prod_{i \in I} \frac{(\alpha_i + \beta_i)!}{\alpha_i! \beta_i!}.$$

We set $\alpha \leq \beta$ when $\alpha_i \leq \beta_i$ for all $i \in I$, which is equivalent to the existence of $\gamma \in \mathbf{N}^{(I)}$ such that $\beta = \alpha + \gamma$; this element γ is then unique, and is denoted $\beta - \alpha$.

If $\mathbf{x} = (x_i)_{i \in I}$ is a family of elements commuting pairwise in a ring A possessing an identity element 1, we set:

$$\mathbf{x}^\alpha = \prod_{i \in I} x_i^{\alpha_i} \quad (\text{with the convention that } x^0 = 1).$$

One has $\mathbf{x}^{\alpha+\beta} = \mathbf{x}^\alpha \cdot \mathbf{x}^\beta$

For $i \in I$, one denotes by ε_i the element of $\mathbf{N}^{(I)}$ all of whose coordinates are zero, except for that of index i which is equal to 1. We have $\alpha = \sum_{i \in I} \alpha_i \varepsilon_i$ for every element $\alpha = (\alpha_i)$ of $\mathbf{N}^{(I)}$.

Restriction of a map

If f is a map from a set A into a set B and if C is a subset of A , one denotes by $f|_C$ the restriction of f to C .

The elements ∞ and ω

We adjoin to the set of integers two elements denoted ∞ and ω . The order relation between integers is extended by the convention

$$n < \infty < \omega \quad \text{for every integer } n.$$

We set:

$$\infty + n = n + \infty = \infty \quad \text{and} \quad \omega + n = n + \omega = \omega, \quad n \in \mathbf{Z}.$$

The letter r denotes either an integer ≥ 1 , or one of the elements ∞ or ω . When $K \neq R$, the letter r always denotes the element ω . When $r = \infty$ (resp. $r = \omega$), we set $r - k = r$ for every integer $k \geq 1$.

1 Differentiable functions

In this section, the letter E denotes a normable topological vector space over K ; the letter F denotes a separated polynormed space over K .

1.1 Order of contact of two functions at a point

1.1.1. Let X be a topological space and θ a positive numerical function defined in a neighborhood of a point x_0 of X . A function f , defined in a neighborhood of x_0 and taking values in F , is said to be negligible with respect to θ at x_0 if the following condition is satisfied:

For every $\varepsilon > 0$ and for every continuous seminorm γ on F , there exists a neighborhood V of x_0 on which f and θ are defined and such that

$$\|f(x)\|_{\gamma} \leq \varepsilon \theta(x) \quad \text{for every } x \in V.$$

For f to be negligible with respect to θ , it suffices that this condition be satisfied for a family of seminorms γ defining the topology of F . The fact that f is negligible or not with respect to θ at x_0 depends only on the germs of f and of θ at x_0 . One denotes by $o_{x_0}(\theta)$ (or $o(\theta)$ when there is no ambiguity about x_0) the set of germs at x_0 of functions negligible with respect to θ at x_0 : it is a vector subspace of the space of germs at x_0 of mappings with values in F . If f is negligible with respect to θ , we shall write, by abuse of notation, $f \in o_{x_0}(\theta)$ or also $f(x) \in o(\theta(x))$ when x tends to x_0 .

If f and g are two mappings from a neighborhood of x_0 to F , we shall also write $f \equiv g \pmod{o(\theta)}$ if $f - g$ is negligible with respect to θ .

Suppose that K is equal to $\mathbb{R}$ or $\mathbb{C}$ and that x_0 is adherent to the set Y of points of X where θ is defined and nonzero. The relation $f \in o(\theta)$ then means that $f(x)/\theta(x)$ tends to 0 as x tends to x_0 while remaining in Y , and that $\theta(x) = 0$ implies $f(x) = 0$.

1.1.2. Let f and g be two functions with values in F , defined in a neighborhood of a point x_0 of E . If m is a positive integer, we say that f and g have contact of order $\geq m$ at x_0 if we have:

$$f(x) - g(x) \in o(\|x - x_0\|^m) \quad \text{for } x \text{ tending toward } x_0$$

whatever norm is chosen to define the topology of E . For this, it suffices that the preceding relation be verified for one norm defining the topology of E . If this is so, we have $f(x_0) = g(x_0)$.

If f and g take the same value at x_0 , we call the order of contact of f and g at x_0 the upper bound (finite or equal to $+\infty$) of the integers m such that f and g have contact of order $\geq m$ at x_0 .

1.1.3. The order of contact of f and g at x_0 depends only on the germs of these functions at the point x_0 . One can therefore speak of the order of contact of two germs φ and ψ of maps from E to F at the point x_0 . The relation $\langle \varphi \text{ and } \psi \text{ have contact of order } \geq m \rangle$ is an equivalence

relation compatible with the vector structure.

1.2 Differentiable functions at a point

1.2.1. Let f be a function defined in a neighborhood of the point x_0 of E and with values in F . We say that f is differentiable at x_0 if there exists a continuous affine function v from E into F having at x_0 a contact of order ≥ 1 with f . This mapping v is unique; there exists one and only one continuous linear mapping, denoted $Df(x_0)$, from E into F such that:

$$v(x) = v(x_0) + Df(x_0) \cdot (x - x_0).$$

If one chooses a norm on E , this is equivalent to:

$f(x_0 + h) \equiv f(x_0) + Df(x_0).h \pmod{o(\|h\|)}$ for h tending toward 0,
which can also be written in the form:

$$\lim_{h \rightarrow 0, h \neq 0} \frac{\|f(x_0 + h) - f(x_0) - Df(x_0).h\|_\gamma}{\|h\|} = 0$$

for every continuous seminorm γ on F .

The element $Df(x_0)$ of $\mathcal{L}(E, F)$ is called the derivative of f at x_0 . One sometimes writes $D_h f(x_0)$ for $Df(x_0).h$. It is an element of F defined by the relation:

$$D_h f(x_0) = \lim_{t \rightarrow 0, t \neq 0} \frac{f(x_0 + th) - f(x_0)}{t}.$$

1.2.2. We say that a function f is strictly differentiable at x_0 if it is differentiable at x_0 and if, for every norm defining the topology of E , one has the relation:

$$f(y) - f(z) \equiv Df(x_0).(y - z) \pmod{o(\|y - z\|)}$$

for (y, z) tending toward (x_0, x_0) in $E \times E$. For this, it suffices that this condition be satisfied for one norm defining the topology of E . Suppose moreover E and F normed; for every number $c > \|Df(x_0)\|$ there then exists a neighborhood V of x_0 such that $\|f(y) - f(z)\| \leq c.\|y - z\|$ for y, z in V ; this implies that f is uniformly continuous in V .

1.2.3. The property of a function f being differentiable or strictly differentiable at x_0 depends only on the germ of f at x_0 . The germs of differentiable functions at x_0 form a vector subspace $\mathcal{V}$ of the space of all germs and the mapping $f \mapsto Df(x_0)$ of $\mathcal{V}$ to $\mathcal{L}(E; F)$ is linear. The germs of functions strictly differentiable at x_0 form a vector subspace of $\mathcal{V}$.

1.2.4. A function differentiable at x_0 is continuous at x_0 .

1.2.5. When $E = K$, the map $u \mapsto u(1)$ is an isomorphism of $\mathcal{L}(E; F)$ onto F ; if the function f is differentiable at x_0 , the element

$$f'(x_0) = Df(x_0).1$$

is none other than the derivative of f at x_0 in the sense given in Fonct. Var. Réelle, chap. I, § 1, n° 6, remark 2.

1.3 Composition of differentiable functions

1.3.1. Suppose F is normable. Let $x_0 \in E$ and $y_0 \in F$, let U be a neighborhood of x_0 and V a neighborhood of y_0 ; finally, let f be a mapping from U into V , differentiable at x_0 , with $f(x_0) = y_0$. If g is a mapping from V into a separated polynormed vector space G , differentiable at y_0 , the map $g \circ f$ from U into G is differentiable at x_0 , and one has:

$$D(g \circ f)(x_0) = Dg(y_0) \circ Df(x_0). \quad (1)$$

If f and g are strictly differentiable, the same holds for $g \circ f$.

1.3.2. Let f be a map defined in a neighborhood of a point x_0 of E and with values in F , differentiable at x_0 ; if u is a continuous linear map from F into a separated polynormed space G , the function $u \circ f$ is differentiable at x_0 and we have:

$$D(u \circ f)(x_0) = u \circ Df(x_0). \quad (2)$$

1.3.3. Suppose that F is the product of a family $(F_i)_{i \in I}$ of separated polynormed vector spaces; for every i in I , let f_i be a mapping defined in a neighborhood U of a point x_0 of E and taking values in F_i , and let $f = (f_i)_{i \in I}$. For f to be differentiable (resp. strictly differentiable) at

x_0 , it is necessary and sufficient that all the f_i be so; we have:

$$D_h f(x_0) = (D_h f_i(x_0))_{i \in I} \quad \text{for every } h \text{ in } E. \quad (3)$$

1.3.4. If $E = K$, one may replace $Df(x_0)$ by $f'(x_0)$ and $D_h f_i(x_0)$ by $f'_i(x_0)$ in formulas (1) and (3).

1.4 Product of differentiable functions

1.4.1. Let $F_1, \dots, F_m$ be separated polynormed spaces and let u be a continuous m -linear mapping of $F_1 \times \dots \times F_m$ into F . Let U be a neighborhood of a point x_0 of E , and let f_i be a mapping of U into F_i (for $1 \leq i \leq m$). If the f_i are differentiable (resp. strictly differentiable) at x_0 , then so is $u(f_1, \dots, f_m) = g$ and we have:

$$D_h g(x_0) = \sum_{j=1}^m u(f_1(x_0), \dots, D_h f_j(x_0), \dots, f_m(x_0)) \quad \text{for } h \text{ in } E, \quad (4)$$

which will be written more succinctly as:

$$Dg = \sum_{j=1}^m u(f_1, \dots, Df_j, \dots, f_m). \quad (5)$$

In particular, for $m = 2$, we have:

$$Du(f_1, f_2) = u(Df_1, f_2) + u(f_1, Df_2). \quad (6)$$

For $m = 1$, we recover 1.3.2.

1.4.2. When $E = K$, one may replace Dg by g' and Df_j by f'_j in formulas (4) to (6).

1.5 First variants of the implicit function theorem

Suppose that E and F are Banach spaces and let x_0 be a point of E , let U be a neighborhood of x_0 , and let f be a mapping from U into F . Suppose further that f is strictly differentiable at x_0 .

If $Df(x_0)$ is an isomorphism from E onto F , there exists an open neighborhood U_0 of x_0 contained in U and an open neighborhood V_0 of $f(x_0)$ such that $f|U_0$ is a homeomorphism of

U_0 onto V_0 . The map $g : V_0 \rightarrow U_0$ inverse to $f|U_0$ is strictly differentiable at the point $f(x_0)$ and we have:

$$Dg(f(x_0)) = Df(x_0)^{-1}.$$

1.5.2. If $Df(x_0)$ is a surjective map from E onto F , there exists an open neighborhood U_0 of x_0 contained in U , such that $f|U_0$ is an open mapping.

1.5.3. If $Df(x_0)$ is injective and has closed image, there exists a closed neighborhood U_0 of x_0 contained in U , such that $f|U_0$ is a homeomorphism of U_0 onto a closed subset of F .

1.6 Partial derivatives

1.6.1. Let f be a function defined in a neighborhood U of the point x_0 of E and with values in F . Let X be a vector subspace of E and V the set of points x of X such that $x_0 + x \in U$; set $g(x) = f(x_0 + x)$ for $x \in V$. We say that f admits a partial derivative with respect to X at x_0 if g admits a derivative at 0; this derivative is denoted $D_X f(x_0)$; it is a continuous linear map from X to F . If f is differentiable at x_0 it admits a partial derivative with respect to X at x_0 and this partial derivative is the restriction of $Df(x_0)$ to X .

1.6.2. Suppose that E is the product of a finite family of normed vector spaces E_i ($1 \leq i \leq n$) canonically identified with subspaces of E ; let $x_0 = (x_0^1, \dots, x_0^n)$ be in E and let U be a neighborhood of x_0 in E ; finally let f be a mapping from U into F . We denote by $D_i f(x_0)$ the derivative at the point x_0^i , if it exists, of the mapping $z_i \mapsto f(x_0^1, \dots, z_i, \dots, x_0^n)$ defined in a neighborhood of x_0^i in E_i and taking values in F . This is an element of $\mathcal{L}(E_i; F)$ which is called the i -th partial derivative of f at x_0 . If f is differentiable at x_0 , the n partial derivatives exist, and determine $Df(x_0)$ by the formula:

$$Df(x_0).h = \sum_{i=1}^n D_i f(x_0).h_i \quad \text{for } h = (h_1, \dots, h_n) \text{ in } E.$$

1.6.3. More particularly, let $E = K^n$. If the partial derivatives of f at x_0 exist, we denote $\partial_i f(x_0)$ the element $D_i f(x_0).1$ of F . The following notation is often used; suppose that a notation has been chosen for the coordinate functions on K^n , for example u_i denotes the i -th projection

of K^n onto K . One then writes:

$$\frac{\partial f}{\partial u_i}(x_0) \quad \text{or} \quad \frac{\partial f}{\partial u_i} \Big|_{x=x_0}$$

instead of $\partial_i f(x_0)$.

1.6.4. Let $E = K^n$ and $F = K^m$; suppose that the function $f = (f_1, \dots, f_m)$ with values in F is differentiable at the point x_0 of E . The partial derivatives $a_{ji} = \partial_i f_j(x_0)$ then exist (they are elements of K). The matrix with m rows and n columns formed by the a_{ji} (element in the row of index j and the column of index i) is called the Jacobian matrix of f at x_0 ; this is the matrix of the linear map $Df(x_0)$ of K^n to K^m with respect to the canonical bases of these spaces.

1.7 Iterated derivatives

1.7.1. Let f be a function defined in a neighborhood of a point x_0 of E , taking values in F . If f is differentiable in a neighborhood of x_0 , its derivative Df is a mapping from a neighborhood of x_0 in the polynormed space $\mathcal{L}(E; F)$ of continuous linear mappings from E into F . Let p be an integer ≥ 2 :

we say that f is p times differentiable at x_0 if f is differentiable in a neighborhood of x_0 and if its derivative Df is $(p - 1)$ times differentiable at x_0 . We then define the p -th derivative of f at x_0 ; this is the continuous p -linear map $D^p f(x_0)$ from E^p into F , defined by:

$$D^p f(x_0).(h_1, \dots, h_p) = (D(D^{p-1}f)(x_0).h_1).(h_2, \dots, h_p).$$

We also set $D^0 f = f$ and $D^1 f = Df$. If f is p times differentiable at x_0 and if q and s are two integers such that $q + s = p$, with $s > 0$, then f is q times differentiable in a neighborhood of x_0 , the function $D^q f$ (with values in $\mathcal{L}_q(E; F)$) is s times differentiable at x_0 , and we have:

$$D^{q+s} f(x_0).(h_1, \dots, h_{q+s}) = (D^s(D^q f)(x_0).(h_1, \dots, h_s)) \cdot (h_{s+1}, \dots, h_{q+s})$$

a relation which, by abuse of notation, is written in the form:

$$\mathbf{D}^{q+s} f = \mathbf{D}^s \mathbf{D}^q f.$$

1.7.2. Let $(E_i)_{1 \leq i \leq n}$ be closed vector subspaces of E , such that E is the topological direct sum of the E_i . One then defines, when it exists, the iterated partial derivative $D_{i_1} \dots D_{i_m} f$ of a map f from a neighborhood of $x_0 \in E$ into F ; it is a continuous multilinear map from

$$E_{i_1} \times \dots \times E_{i_m}$$

into F , defined by induction on the integer $m \geq 1$ as follows: if $D_{i_2} \dots D_{i_m} f(x)$ exists in a neighborhood of x_0 and has a partial derivative with respect to E_{i_1} , then $D_{i_1} \dots D_{i_m} f(x_0)$ is given by:

$$D_{i_1} \dots D_{i_m} f(x_0).(h_1, \dots, h_m) = (D_{i_1} (D_{i_2} \dots D_{i_m} f)(x_0).h_1).(h_2, \dots, h_m)$$

for $h_k \in E_{i_k}$.

If f is m times differentiable at x_0 , then the partial derivative $\mathbf{D}_{i_1} \dots \mathbf{D}_{i_m} f(x_0)$ exists and is equal to the restriction of $\mathbf{D}^m f(x_0)$ to the subspace $E_{i_1} \times \dots \times E_{i_m}$ of $\mathbf{E}^m$. Consequently, $\mathbf{D}^m f(x_0)$ is completely determined by the iterated partial derivatives of order m at x_0 .

1.7.3. Suppose that E is finite-dimensional and let $(e_1, \dots, e_n)$ be a basis of E . Set $E_i = K e_i$ and let f be a map from a neighborhood of x_0 into F . If the partial derivative $D_{i_1} \dots D_{i_m} f(x_0)$ (with the notations of 1.7.2) exists, one sets:

$$\partial_{i_1} \dots \partial_{i_m} f(x_0) = \mathbf{D}_{i_1} \dots \mathbf{D}_{i_m} f(x_0).(e_{i_1}, \dots, e_{i_m}).$$

2 Real differentiable functions

In this section, one assumes that $K = \mathbb{R}$. The letter E denotes a normed vector space over $\mathbb{R}$; the letter F denotes a separated locally convex topological vector space over $\mathbb{R}$.

2.1 Differentiable functions at a point

2.1.1. Let f be a function defined in a neighborhood of a point x_0 of E and with values in F . Let u be an element of the space $\mathcal{L}(E; F)$ of continuous linear maps from E into F . For f to be

differentiable at x_0 and to admit u as derivative there, it is necessary and sufficient that one have

$$\lim_{h \rightarrow 0, h \neq 0} \frac{f(x_0 + h) - f(x_0) - u(h)}{\|h\|} = 0.$$

2.1.2. For f to be strictly differentiable at x_0 , it is necessary and sufficient that one have

$$\lim_{\substack{(h,k) \rightarrow (0,0) \\ h \neq k}} \frac{f(x_0 + h) - f(x_0 + k) - Df(x_0) \cdot (h - k)}{\|h - k\|} = 0.$$

2.1.3. Let F_1 and F_2 be two separated locally convex spaces and let u be a bilinear map from $F_1 \times F_2$ into F , satisfying the following continuity condition:

(SC) If $((a_n, b_n))$ is a sequence of elements of $F_1 \times F_2$ converging to an element $(a, b) \in F_1 \times F_2$, then the sequence $(u(a_n, b_n))$ converges to $u(a, b)$ in F .

Let f_i (for $i = 1, 2$) be a map from a neighborhood of a point x_0 in E into F_i . If f_1 and f_2 are differentiable at x_0 , then $u(f_1, f_2)$ is differentiable at x_0 and we have:

$$D(u(f_1, f_2))(x_0) \cdot h = u(Df_1(x_0) \cdot h, f_2(x_0)) + u(f_1(x_0), Df_2(x_0) \cdot h)$$

for all $h \in E$.

2.2 The mean value theorem

2.2.1. Let x, y be in E , and let $[x, y]$ be the closed segment joining these two points. Let moreover f be a map from a neighborhood of $[x, y]$ into the space F , differentiable at every point of $[x, y]$. Then $f(x) - f(y)$ belongs to the closed convex hull of the set of points $Df(z) \cdot (x - y)$ for z in $[x, y]$.

2.2.2. Let U be a connected open subset of E , and let f be a map from U into F , admitting a zero derivative at every point of U ; then f is constant on U .

2.2.3. Let U be a convex open subset of E , and f a map from U into F , differentiable at every point of U . Given a continuous seminorm γ on F and a real number $M \geq 0$, the following conditions are equivalent:

(i) For every x in U , one has $\|Df(x)\|_{\gamma} \leq M$.

(ii) For every x and every y in U , one has $\|f(x) - f(y)\|_Y \leq M \cdot \|x - y\|$.

2.2.4. Let U be a neighborhood of a point x_0 of E and let f be a function defined in the complement of x_0 in U , with values in F . Suppose that f has a derivative $Df(x)$ at every point x of U , $x \neq x_0$ and that the function $x \mapsto Df(x)$ has a limit D_0 as x tends to x_0 . Then, if $\dim(E) \geq 2$, f has a limit at x_0 and the function f extended by continuity to all of U is differentiable with derivative D_0 at x_0 ; the same holds if $\dim(E) = 1$ and one supposes that f has a limit at x_0 .

2.3 Functions of class C^r ($r \neq \omega$)

2.3.1. Let U be an open subset of E and f a mapping from U into F . One defines the relation “ f is of class C^r ” (for $r \in \mathbb{N}$) by induction on r in the following manner:

- 1) f is of class C^0 if and only if it is continuous;
- 2) if r is an integer ≥ 1 , the function f is of class C^r if and only if it is differentiable at every point of U and if the derivative Df from U to $\mathcal{L}(E; F)$ is of class C^{r-1} .

The functions of class C^r are also called r times continuously differentiable functions.

One says that f is of class C^∞ (or infinitely differentiable) if it is of class C^r for every integer r .

If f is of class C^r in U , then f is p times differentiable for every integer $p \leq r$ and the function $D^p f$ is of class C^{r-p} .

2.3.2. The mappings of class C^r from an open subset U of E into F form a vector subspace $\mathcal{C}^r(U; F)$ of the space of all mappings from U into F . One has $\mathcal{C}^s(U; F) \subset \mathcal{C}^r(U; F)$ for $s \geq r$.

2.3.3. For a function f to be of class C^1 in an open subset U of E , it is necessary and sufficient that it be strictly differentiable at every point of U .

If E is a product of normed spaces E_i , a map f from an open set V of E into F is of class C^r if and only if f possesses iterated partial derivatives $D_{i_1} \dots D_{i_m} f$ continuous for every integer $m \leq r$.

2.3.4. Let G be a normed space and let U be an open subset of E . Let V be an open subset of G , $g \in \mathcal{C}^r(U; G)$ and $f \in \mathcal{C}^r(V; F)$. If $g(U) \subset V$, the map $f \circ g$ of U into F is of class C^r .

Let F_1 and F_2 be two separated locally convex spaces and let u be a bilinear mapping from $F_1 \times F_2$ to F , hypocontinuous with respect to the set of bounded subsets of F_1 (resp. F_2) (Topological Vector Spaces, ch. III, § 4, no. 2). Let U be an open subset of E and $f_i \in \mathcal{C}^r(U; F_i)$ (for $i = 1, 2$). Then the function $u(f_1, f_2)$ belongs to $\mathcal{C}^r(U; F)$. If E is finite-dimensional, it suffices to assume that u satisfies condition (SC) of no. 2.1.3.

2.3.5. If E is a product of normed spaces E_i ($1 \leq i \leq n$) and if f is a continuous n -linear mapping from E into F , then f is of class C^∞ and one has $D^p f = 0$ for $p \geq n + 1$.

2.3.6. Suppose that E and F are Banach spaces. Let f be a function of class C^r (with $r \geq 1$) defined in a neighborhood of a point x_0 of E and with values in F . Let $y_0 = f(x_0)$ and suppose that $Df(x_0)$ is an isomorphism from E onto F . Then f induces a homeomorphism g from a neighborhood of x_0 onto a neighborhood of y_0 (1.5), and the inverse mapping of g is of class C^r in a neighborhood of y_0 .

2.4 Derivatives of functions of class C^r

2.4.1. Let f be a map of class C^r from an open set U of E into F . For every $x \in U$, and every integer s with $s \leq r$, the multilinear mapping $D^s f(x)$ is symmetric.

2.4.2. Suppose moreover that E is finite-dimensional and let $(e_1, \dots, e_n)$ be a basis of E . The partial derivatives $\partial_{i_1} \dots \partial_{i_s} f$ depend symmetrically on the indices $i_1, \dots, i_s$. Let α_k be the number of times the index k occurs in the sequence $i_1, \dots, i_s$ and let $\alpha = (\alpha_1, \dots, \alpha_n)$. We then set:

$$\partial^\alpha f = \partial_1^{\alpha_1} \dots \partial_n^{\alpha_n} f = \partial_{i_1} \dots \partial_{i_s} f$$

When the coordinates relative to the basis $(e_1, \dots, e_n)$ are denoted $x_1, \dots, x_n$, one also writes $\partial^\alpha f$ in the form:

$$\frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}$$

2.5 Taylor's formula

2.5.1. Let r be an integer ≥ 1 and let f be a map of class C^r from an open set U of E into F . For $x \in U$, $h \in E$ and $p \leq r$, let us agree to write $D^p f(x_0).h^p$ instead of $D^p f(x_0).(h, \dots, h)$. If the segment $[x, x+h]$ is contained in U , one has the formula ("Taylor's formula"):

$$f(x+h) = \sum_{p=0}^{r-1} \frac{1}{p!} D^p f(x).h^p + v_r(x; h)$$

where the "remainder" $v_r(x; h)$ is given by:

$$v_r(x; h) = \int_0^1 \frac{(1-t)^{r-1}}{(r-1)!} D^r f(x+th).h^r dt$$

One has:

$$v_r(x; h) \equiv \frac{1}{r!} D^r f(x).h^r \mod o(\|h\|^r) \quad \text{when } h \text{ tends to } 0$$

and

$$f(x+h) \equiv \sum_{p=0}^r \frac{1}{p!} D^p f(x).h^p \mod o(\|h\|^r)$$

as h tends to zero.

2.5.2. Let moreover γ be a continuous seminorm on F ; if one has $\|D^r f(z)\|_\gamma \leq M$ for every point z of the segment $[x, x+h]$, then one has:

$$\|v_r(x; h)\|_\gamma \leq \frac{M}{r!} \|h\|^r$$

2.5.3. Suppose in addition that $E = \mathbf{R}^n$. One then has:

$$f(x+h) \equiv \sum_{|\alpha| \leq r} \Delta^\alpha f(x) h^\alpha \mod o(\|h\|^r) \text{ when } h \text{ tends to } 0$$

by setting:

$$\Delta^\alpha f(x) = \frac{1}{\alpha!} \partial^\alpha f(x)$$

2.5.4. Let f and g two functions of class C^r on an open set U of E , with values in F . For f and g to have at a point x of U a contact of order $\geq r$, it is necessary and sufficient that one has $D^p f(x) = D^p g(x)$ for every integer p with $0 \leq p \leq r$. When E is of finite dimension, this amounts to saying that the iterated partial derivatives of order $\leq r$ of f and of g (with respect to a basis of E) are equal at the point x .

2.5.5. Let U be an open subset of $E \times \mathbb{R}^n$ of the form $V \times I_1 \times \cdots \times I_n$, where V is an open subset of E and $I_1, \dots, I_n$ open intervals of $\mathbb{R}$ containing 0. Set $U_0 = V$ and $U_j = V \times I_1 \times \cdots \times I_j$ for $1 \leq j \leq n$. Given a function $f \in \mathcal{C}^r(U; F)$ (with $1 \leq r \leq \infty$), there exists one and only one sequence of functions $f_j \in \mathcal{C}^{r-1}(U_j; F)$ (for $0 \leq j \leq n$) such that:

$$f(x, t_1, \dots, t_n) = f_0(x) + \sum_{j=1}^n t_j f_j(x, t_1, \dots, t_j)$$

for $x \in V$ and $t_j \in I_j$. We have:

$$f_0(x) = f(x, 0, \dots, 0)$$

$$f_j(x, t_1, \dots, t_j) = \int_0^1 \partial_j f(x, t_1, \dots, t_{j-1}, t_j u, 0, \dots, 0) du$$

for $1 \leq j \leq n$. In this last formula, $\partial_j f$ denotes the j -th partial derivative of the function $(t_1, \dots, t_n) \mapsto f(x, t_1, \dots, t_n)$.

2.6 Criteria for differentiability

2.6.1. Suppose that F , in addition to its topology T , is equipped with a coarser topology T' , which also makes F a separated locally convex space. Suppose further that T and T' satisfy the following condition:

- (S) For every neighborhood V of 0 for the topology T , there exists a neighborhood W of 0 for T' such that the T' -closed convex hull of every T -compact subset of W is contained in V .

Let f be a map from an open subset U of E into F . Suppose that f is of class C^r ($1 \leq r \leq \infty$) when F is equipped with the topology $\mathcal{T}'$, that $D^m f(x)$ is, for all x in U and every integer $m \leq r$,

a continuous multilinear map from E^m into F equipped with the topology $\mathcal{T}$, and that the map $x \mapsto D^m f(x)$ is continuous from U into $\mathcal{L}_m(E; F)$ (F being equipped with the topology $\mathcal{T}$). Then f is of class C^r when F is equipped with the topology $\mathcal{T}$, and its derivatives $D^m f$ are the same for $\mathcal{T}$ and $\mathcal{T}'$.

Condition (S) is in particular satisfied if there exists a fundamental system of neighborhoods of 0 for the topology T which are closed for the topology T' : this is the case if the dual of F equipped with T is identical to the dual of F equipped with T' . Condition (S) is also satisfied if F equipped with the topology T is quasi-complete (Topological Vector Spaces, chap. III, § 2, no. 5).

2.6.2. Let f be a mapping from an open set U of E to F . If f is of class C^r (with $0 \leq r \leq \infty$), the scalar functions $u \circ f$ are of class C^r for every continuous linear form u on F . Conversely, if F is quasi-complete and if the functions $u \circ f$ are of class C^{r+1} for all $u \in F'$, then f is of class C^r .

3 Real or complex analytic functions

In this paragraph, we assume that $K = \mathbb{R}$ or $\mathbb{C}$. We denote by (E_i) , $1 \leq i \leq n$, a finite family of normed spaces over K , by E the product space of the E_i and by F a separated locally convex topological vector space over K .

3.1 Convergent series

3.1.1. Let $f = \sum f_\alpha$ be a formal series belonging to $\hat{\mathbf{P}}(E_1, \dots, E_n; F)$ (Appendix, no. A.5). If γ is a continuous semi-norm on F and $\mathbf{R} = (R_1, \dots, R_n)$ a sequence of n strictly positive real numbers, we set:

$$\|f\|_{\gamma, \mathbf{R}} = \sum_{\alpha} \mathbf{R}^{\alpha} \|f_{\alpha}\|_{\gamma}.$$

If F is a normed space and if γ is the norm of F , we write $\|f\|_{\mathbb{R}}$ instead of $\|f\|_{\gamma, \mathbb{R}}$.

The set $\mathcal{H}_{\mathbb{R}}(E_1, \dots, E_n; F)$ of $f \in \hat{\mathbf{P}}(E_1, \dots, E_n; F)$ such that $\|f\|_{\gamma, \mathbb{R}}$ is finite for every continuous seminorm γ on F is a vector subspace of $\hat{\mathbf{P}}(E_1, \dots, E_n; F)$. One has

$$\mathcal{H}_{\mathbb{R}}(E_1, \dots, E_n; F) \subset \mathcal{H}_{\mathbb{R}'}(E_1, \dots, E_n; F)$$

whenever $R_i \geq R'_i$ for $1 \leq i \leq n$. The union of the

$$\mathcal{H}_{\mathbf{R}}(E_1, \dots, E_n; F)$$

is a vector subspace, denoted $\mathcal{H}(E_1, \dots, E_n; F)$, of $\hat{\mathbf{P}}(E_1, \dots, E_n; F)$, whose elements are called convergent series on the product of the E_i , with values in F . It depends only on the topology of the E_i and not on their norms. One can therefore speak of the space $\mathcal{H}(E_1, \dots, E_n; F)$ when the E_i are normable spaces, without having to choose a norm on each E_i .

3.1.2. The map $f \mapsto \|f\|_{\gamma, \mathbb{R}}$ is a seminorm on $\mathcal{H}_{\mathbb{R}}(E_1, \dots, E_n; F)$. The topology defined by these seminorms as γ runs over the set of continuous seminorms on F (or simply a set of seminorms defining the topology of F) is Hausdorff. If F is normable (resp. complete), so is $\mathcal{H}_{\mathbb{R}}(E_1, \dots, E_n; F)$. The seminorms on $\mathcal{H}(E_1, \dots, E_n; F)$ whose restriction to each $\mathcal{H}_{\mathbb{R}}$ is continuous define a Hausdorff topology on $\mathcal{H}(E_1, \dots, E_n; F)$, which depends only on the topologies of the E_i . The injection of $\mathcal{H}(E_1, \dots, E_n; F)$ into $\hat{\mathbf{P}}(E_1, \dots, E_n; F)$ is continuous.

3.1.3. The canonical isomorphism of $\hat{\mathbf{P}}(\mathbf{E}; F)$ onto $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; F)$ gives by restriction an isomorphism of topological vector spaces from $\mathcal{H}(\mathbf{E}; F)$ onto $\mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; F)$.

3.1.4. Let $f \in \mathcal{H}(E_1, \dots, E_n; F)$ and let $J(f)$ be the set of $R \in (\mathbf{R}_+^*)^n$ such that $f \in \mathcal{H}_{\mathbb{R}}(E_1, \dots, E_n; F)$. The interior $I(f)$ of $J(f)$ is called the strict convergence indicator of f . It consists of the set of R for which there exists an $R' \in J(f)$ with $0 < R_i < R'_i$ for $1 \leq i \leq n$. We denote by $\Omega(f)$ the set of points $(\log R_1, \dots, \log R_n)$ of $\mathbf{R}^n$ for $R \in I(f)$: it is a convex subset of $\mathbf{R}^n$.

When $n = 1$, the set $\mathbf{I}(f)$ is an interval $[0, \rho(f)]$ of $\mathbf{R}$ and $\rho(f)$ is called the strict radius of convergence of f . It is also the supremum (finite or $+\infty$) of the set of real numbers $R > 0$ such that for every continuous seminorm γ on F , there exists a constant M such that $\|f_m\|_{\gamma} \leq MR^{-m}$ (with $f = \sum f_m, f_m \in P_m(E; F)$) for every integer $m \geq 0$.

The set of points $x = (x_i) \in \mathbf{E}_1 \times \dots \times \mathbf{E}_n$ such that there exists $\mathbf{R} \in \mathbf{I}(f)$ with $\|x_i\| \leq R_i$ for all i , is called the strict domain of convergence of f and is denoted $\mathbf{C}(f)$. It is also the interior

of the set of points x for which there exists $\mathbf{R} \in \mathbf{J}(f)$ with $\|x_i\| < R_i$ for all i .

3.1.5. For $f_\alpha \in \mathbf{P}_\alpha(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ and for every continuous seminorm γ on $\mathbf{F}$, set:

$$\|f_\alpha\|_\gamma = \sup_{\|x_i\| \leq 1} \|f_\alpha(x_1, \dots, x_n)\|_\gamma.$$

We then have the inequalities:

$$\|f_\alpha\|_\gamma \leq \|f_\alpha\|_\gamma \leq \frac{\alpha^\alpha}{\alpha!} \|f_\alpha\|_\gamma.$$

For $f = \sum f_\alpha \in \hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ and $\mathbf{R} \in (\mathbf{R}_+^*)^n$, set:

$$\|f\|_{\gamma, \mathbf{R}} = \sum_\alpha \mathbf{R}^\alpha \|f_\alpha\|_\gamma$$

and let us denote by $\tilde{\mathcal{H}}_\mathbb{R}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ the vector subspace of

$$\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$$

formed by the f such that $\|f\|_{\gamma, \mathbf{R}}$ is finite for every γ , equipped with the topology defined by the seminorms $f \mapsto \|f\|_{\gamma, \mathbf{R}}$. One has $\mathcal{H}_\mathbb{R} \subset \tilde{\mathcal{H}}_\mathbb{R}$ and the injection of $\mathcal{H}_\mathbb{R}$ into $\tilde{\mathcal{H}}_\mathbb{R}$ is continuous. The space $\mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ is the union of the $\tilde{\mathcal{H}}_\mathbb{R}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$, and its topology is the finest locally convex topology making the injections of $\mathcal{H}_\mathbb{R}$ into $\mathcal{H}$ continuous.

If $f \in \mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$, one calls the convergence indicator the interior $\tilde{I}(f)$ of the set $\tilde{J}(f)$ of $\mathbf{R} \in (\mathbf{R}_+^*)^n$ such that $f \in \mathcal{H}_\mathbb{R}$. We have:

$$e^{-1} \mathbf{I}(f) \subset \tilde{\mathbf{I}}(f) \subset \mathbf{I}(f)$$

(where e is the base of natural logarithms). Starting from $\tilde{I}(f)$, one defines as in 3.1.4. the domain of convergence $\tilde{C}(f)$ and, when $n = 1$, the radius of convergence $\tilde{\rho}(f)$. In particular, we have:

$$e^{-1} \tilde{\rho}(f) \leq \rho(f) \leq \tilde{\rho}(f).$$

3.1.6. For $\mathbf{R} \in (\mathbf{R}_+^*)^n$, we call the (closed) polyball centered at 0 and with radius $\mathbf{R}$ in $\mathbf{E}$

the set $\mathbf{B}(\mathbf{R})$ of $x \in \mathbf{E}$ such that $\|x_i\| \leq \mathbf{R}_i$ for all i . If $\dim \mathbf{E}_i = 1$, we also say polydisk. If $f \in \mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$, the domain of convergence (resp. strict) of f is the union of the polyballs $\mathbf{B}(\mathbf{R})$ for $\mathbf{R} \in \tilde{\mathbf{I}}(f)$ (resp. $\mathbf{R} \in \mathbf{I}(f)$).

3.1.7. Let $f \in \mathcal{H}(E_1, \dots, E_n; F)$. Suppose F is quasi-complete. For every $x \in \tilde{\mathbf{C}}(f)$, the family of $f_\alpha(x)$ is summable in F . Its sum, denoted $\hat{f}(x)$ or simply $f(x)$, is a continuous function on $\tilde{\mathbf{C}}(f)$. More precisely, for every R such that $f \in \tilde{\mathcal{H}}_{\mathbb{R}}$, the family of $f_\alpha(x)$ is uniformly summable for $x \in B(R)$. The map $f \mapsto \hat{f}$ is a continuous linear injective map from $\tilde{\mathcal{H}}_{\mathbb{R}}$ into the space of bounded continuous functions on $B(R)$, equipped with the topology of uniform convergence.

3.1.8. Let $F_1, \dots, F_m$ be separated polynormed spaces and let u be a continuous m -linear map from $F_1 \times \dots \times F_m$ to F . Let $f_i \in \mathcal{H}(E_1, \dots, E_n; F_i)$ for $1 \leq i \leq m$. The formal series $u(f_1, \dots, f_m)$ belongs to $\mathcal{H}(E_1, \dots, E_n; F)$. We have:

$$\mathbf{C}(u(f_1, \dots, f_m)) \supset \bigcap_i \mathbf{C}(f_i)$$

$$\tilde{\mathbf{C}}(u(f_1, \dots, f_m)) \supset \bigcap_i \tilde{\mathbf{C}}(f_i)$$

and if $x \in \bigcap_i \tilde{\mathbf{C}}(f_i)$, F and the F_i being quasi-complete, we have:

$$u(f_1, \dots, f_m)(x) = u(f_1(x), \dots, f_m(x)).$$

3.1.9. Let $\mathbf{F}_1, \dots, \mathbf{F}_m$ be complete normed spaces and suppose $\mathbf{F}$ quasi-complete. Let $\mathbf{f} = (f_i)_{1 \leq i \leq m}$ with $f_i \in \mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F}_i)$ and $g \in \mathcal{H}(\mathbf{F}_1, \dots, \mathbf{F}_m; \mathbf{F})$, such that $(f_i(0))_{1 \leq i \leq m}$ belongs to the strict domain of convergence of g . Then, for every $\alpha \in \mathbb{N}^m$, the formal series $g_\alpha \circ \mathbf{f}$ belongs to $\mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ and the family of $g_\alpha \circ \mathbf{f}$ is summable in $\mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ and a fortiori in $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$. Its sum will be denoted $g \circ \mathbf{f}$.

More precisely, there exists $\mathbb{R} \in \bigcap_i \mathbf{I}(f_i)$ and $\mathbb{R}' \in \mathbf{I}(g)$ such that $\|f_i\|_{\mathbb{R}} < R'_i$ for $1 \leq i \leq m$. Under these conditions, the formal series $g_\alpha \circ f$ belongs to $\mathcal{H}_{\mathbb{R}}(E_1, \dots, E_n; F)$ and the family of $g_\alpha \circ f$ is summable in $\mathcal{H}_{\mathbb{R}}(E_1, \dots, E_n; F)$. Finally, if $x \in B(\mathbb{R})$, then $\mathbf{f}(x) = (f_i(x))$ belongs to $B(\mathbb{R}') \subset F_1 \times \dots \times F_m$ and we have:

$$g(\mathbf{f}(x)) = (g \circ \mathbf{f})(x).$$

3.1.10. Suppose that $E_i = K$ for $1 \leq i \leq n$. The space $\hat{\mathbf{P}}(\mathbf{K}^n; \mathbf{F})$ is then identified with the space of formal power series in n indeterminates $X_1, \dots, X_n$ with coefficients in $\mathbf{F}$ and an element of $\hat{\mathbf{P}}(\mathbf{K}^n; \mathbf{F})$ is written:

$$f = \sum_{\alpha} X^{\alpha} c_{\alpha} \quad \text{with } c_{\alpha} \in \mathbf{F}.$$

If $\mathbf{R} \in (\mathbf{R}_+^*)^n$ and if γ is a continuous semi-norm on $\mathbf{F}$, we have:

$$\|f\|_{\gamma, \mathbf{R}} = \|f\|_{\gamma, \mathbf{R}} = \sum_{\alpha} \mathbf{R}^{\alpha} \|c_{\alpha}\|_{\gamma}$$

One has $\mathbf{I}(f) = \tilde{\mathbf{I}}(f)$, $\mathbf{C}(f) = \tilde{\mathbf{C}}(f)$ and, when $n = 1$, $\rho(f) = \tilde{\rho}(f)$.

The space $\mathcal{H}(\mathbf{K}^n; \mathbf{K})$ of convergent series with coefficients in $\mathbf{K}$ is also denoted $\mathbf{K}\{\{\mathbf{X}_1, \dots, \mathbf{X}_n\}\}$; it is a subalgebra of $\mathbf{K}[[\mathbf{X}_1, \dots, \mathbf{X}_n]]$. The space $\mathcal{H}(\mathbf{K}^n; \mathbf{F})$ is a module over $\mathbf{K}\{\{\mathbf{X}_1, \dots, \mathbf{X}_n\}\}$ and if $\mathbf{F}$ is finite-dimensional, this module is identified with $\mathbf{K}\{\{\mathbf{X}_1, \dots, \mathbf{X}_n\}\} \otimes_{\mathbf{K}} \mathbf{F}$.

3.1.11. Let $\mathbf{f} \in \mathcal{H}(\mathbf{K}^n; \mathbf{K}^m)$, represented by a system of m convergent series $f_j(X_1, \dots, X_n)$, with coefficients in $\mathbf{K}$. Let likewise $\mathbf{g} \in \mathcal{H}(\mathbf{K}^m; \mathbf{K}^p)$, represented by a system of p convergent series $g_k(Y_1, \dots, Y_m) = \sum g_{k, \beta} Y^{\beta}$. The element $\mathbf{h} = \mathbf{g} \circ \mathbf{f}$ of $\mathcal{H}(\mathbf{K}^n; \mathbf{K}^p)$ (cf. 3.1.9) is represented by p formal series $h_k(X_1, \dots, X_n)$ determined as follows: for $\alpha \in \mathbb{N}^n$ and $\beta \in \mathbb{N}^m$, let $c_{\alpha, \beta}$ be the coefficient of X^{α} in the formal series $\mathbf{f}^{\beta} = \prod f_j^{\beta_j}$; then the family $(g_{k, \beta} c_{\alpha, \beta})_{\beta \in \mathbb{N}^m}$ is summable in $\mathbf{K}$ and has as its sum the coefficient of X^{α} in h_k .

3.1.12. Suppose $\mathbf{F}$ is quasi-complete and let $\hat{\mathbf{E}}_i$ be the completion of $\mathbf{E}_i$. A continuous polynomial on $\mathbf{E}_1 \times \dots \times \mathbf{E}_n$ taking values in $\mathbf{F}$ extends by continuity to a polynomial-continuous map on $\hat{\mathbf{E}}_1 \times \dots \times \hat{\mathbf{E}}_n$ with values in $\mathbf{F}$. From this we deduce a bijection j of $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ onto $\hat{\mathbf{P}}(\hat{\mathbf{E}}_1, \dots, \hat{\mathbf{E}}_n; \mathbf{F})$. If $f \in \mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$, then $j(f) \in \mathcal{H}(\hat{\mathbf{E}}_1, \dots, \hat{\mathbf{E}}_n; \mathbf{F})$ and conversely. The strict convergence indicators of f and of $j(f)$ are the same.

3.1.13. Suppose that $\mathbf{K} = \mathbf{R}$, but that $\mathbf{F}$ is equipped with a complex vector space structure compatible with its real vector space structure. Let $E_i^C = E_i \otimes_{\mathbf{R}} \mathbf{C}$. If $y \in E_i^C$, set:

$$\|y\| = \inf \sum_k |a_k| \cdot \|x_k\|,$$

the infimum being taken over all finite families of pairs $(x_k, a_k) \in E_i \times \mathbb{C}$ such that $y = \sum_k x_k \otimes a_k$. One thus obtains a norm on the complex vector space E_i^c ; this norm extends the given norm on E_i if one agrees to identify $x \in E_i$ with $x \otimes 1$. Let h be an $\mathbb{R}$ -polynomial-continuous map on $E_1 \times \cdots \times E_n$, taking values in F , and homogeneous of multidegree α ; there then exists a $\mathbb{C}$ -polynomial-continuous $\tilde{h}$ on $E_1^c \times \cdots \times E_n^c$, and a unique one, with values in F , extending h , and homogeneous of multidegree α . One has

$$\|\tilde{h}\|_\gamma = \|h\|_\gamma$$

for every continuous seminorm γ on the complex vector space F .

$$f = \sum_\alpha f_\alpha \in \mathcal{H}(E_1, \dots, E_n; F), \text{ then } \tilde{f} = \sum_\alpha \tilde{f}_\alpha \in \mathcal{H}(E_1^c, \dots, E_n^c; F).$$

The series f and $\tilde{f}$ have the same strict convergence indicator (and the same strict radius of convergence when $n = 1$).

Conversely, suppose $K = \mathbb{C}$. Let E_i^0 and F^0 the spaces over $\mathbb{R}$ obtained by restriction of scalars. If $f_\alpha \in P_\alpha(E_1, \dots, E_n; F)$, then $f_\alpha \in P_\alpha(E_1^0, \dots, E_n^0; F^0)$. If $f = \sum_\alpha f_\alpha \in \mathcal{H}(E_1, \dots, E_n; F)$ then the formal series $f^0 = \sum_\alpha f_\alpha \in \hat{P}(E_1^0, \dots, E_n^0; F^0)$ is a convergent series. The convergence indicators (resp. strict convergence indicators) of f and f^0 are identical, and we have $f(x) = f^0(x)$ for all $x \in \tilde{C}(f) = \tilde{C}(f^0)$.

3.2 Analytic functions

3.2.1. Let U be an open subset of E and f a mapping from U into F . We say that f is of class C^ω , or K -analytic (or simply analytic) in U if, for every point a in U , there exists a convergent series $f_a \in \mathcal{H}(E; F)$ such that $f(a + x) = f_a(x)$ for all x in E sufficiently close to zero. If $K = \mathbb{R}$ (resp. $\mathbb{C}$), one also says that f is real analytic (resp. complex analytic or holomorphic). The analytic maps from U into F form a vector subspace, denoted $\mathcal{C}^\omega(U; F)$, of the space of all mappings from U into F .

For $a \in U$, the formal series f_a is unique: it is called the power series expansion of f at the point a . If $f_a = \sum_{\alpha} (f_a)_{\alpha} (f_a)_{\alpha} \in P_{\alpha}(E_1, \dots, E_n; F)$, we set:

$$\Delta^{\alpha} f(a) = (f_a)_{\alpha}.$$

3.2.2. If $f \in \mathcal{C}^{\omega}(U; F)$, the map $\Delta^{\alpha} f : a \mapsto \Delta^{\alpha} f(a)$ from U into

$$P_{\alpha}(E_1, \dots, E_n; F)$$

is analytic. The map $\Delta^{\alpha} : f \mapsto \Delta^{\alpha} f$ is a K -linear map from $\mathcal{C}^{\omega}(U; F)$ to $\mathcal{C}^{\omega}(U; P_{\alpha}(E_1, \dots, E_n; F))$. For $a \in U$, we therefore have, for $\alpha, \beta \in \mathbb{N}^n$, $\Delta^{\beta}(\Delta^{\alpha} f)(a) \in P_{\beta}(E_1, \dots, E_n; P_{\alpha}(E_1, \dots, E_n; F))$. If $x = (x_i) \in E$, we therefore have $(\Delta^{\beta}(\Delta^{\alpha} f)(a))(x) \in P_{\alpha}(E_1, \dots, E_n; F)$ and $((\Delta^{\beta}(\Delta^{\alpha} f)(a))(x))(x) \in F$. This element of F is equal to $((\alpha, \beta))(\Delta^{\alpha+\beta} f(a))(x)$. This is expressed by writing:

$$\Delta^{\beta} \circ \Delta^{\alpha} = ((\alpha, \beta)) \Delta^{\alpha+\beta}.$$

3.2.3. Strict convergence indicators (and strict radii of convergence when $n = 1$) of power series expansions of f and of $\Delta^{\alpha} f$ at the same point a of U , are identical.

3.2.4. Let $f \in \mathcal{C}^{\omega}(U; F)$. Then f is strictly differentiable and infinitely differentiable in U (if $K = \mathbf{R}$, f is of class C^{∞} in U). The iterated derivatives of f are analytic, and their values at a point a are symmetric multilinear maps. One can then introduce the notation $D^{\alpha} f$ for iterated partial derivatives as in no. 2.4.2. We have:

$$\alpha! \Delta^{\alpha} f(a)(h) = D^{\alpha} f(a).(h, \dots, h)$$

for all $a \in U$ and $h \in E$, which we write:

$$\alpha! \Delta^{\alpha} f = D^{\alpha} f.$$

3.2.5. Let U be an open subset of E and f, g two analytic mappings from U into F . Let $a \in U$. For f and g to have contact of order $\geq k$ at a , it is necessary and sufficient that $\Delta^{\alpha} f(a) =$

$\Delta^\alpha g(a)$ for all α with $|\alpha| \leq k$. If f and g have contact of infinite order at a , they are equal in a neighborhood of a . The set of points of U where f and g have contact of infinite order is open and closed.

In particular, if U is connected and if there exists a nonempty open subset of U on which f and g are equal, then $f = g$ ("principle of analytic continuation").

3.2.6. Let U be an open subset of E and let f be an analytic map from U into F . If the derivative Df of f is zero, then f is locally constant.

3.2.7. Suppose F is quasi-complete and let G be a complete normed space. Let g be an analytic map from an open subset U of E into G , and let f be an analytic map from an open subset V of G , containing $g(U)$, into F . The composite map $f \circ g$ is analytic in U . Suppose further that $0 \in U$ and that $g(0) = 0$. The expansion of $f \circ g$ as a power series at 0 is then obtained by substituting, in the expansion of f at 0 , the power series expansion of g at 0 (3.1.9).

3.2.8. Let $F_1, \dots, F_m$ be separated polynormed spaces and u a continuous multilinear mapping from $F_1 \times \dots \times F_m$ into F . Let U be an open subset of E and $f_i \in \mathcal{C}^\omega(U; F_i)$. The function $u(f_1, \dots, f_m)$ is analytic, and its power series expansion at a point $a \in U$ is the series $u((f_1)_a, \dots, (f_m)_a)$ (3.1.8).

3.2.9. We assume F quasi-complete. Let $f \in \mathcal{H}(E_1, \dots, E_n; F)$; the function $x \mapsto f(x)$ (3.1.7) is analytic in the open set $C(f)$, the strict convergence domain of f . If $n = 1$ and if $\|a\| < \rho(f)$, the strict radius of convergence of the expansion of f as a power series in a is at least equal to $\rho(f) - \|a\|$. If $\rho(f) = +\infty$, one says that f is an entire function.

3.2.10. Keep the hypotheses of 3.2.9. If $K = \mathbb{C}$, the results of 3.2.9 remain exactly valid if one replaces $C(f)$ by $\tilde{C}(f)$ and $\rho(f)$ by $\tilde{\rho}(f)$ (for $n = 1$). If $K = \mathbb{R}$, the function $x \mapsto f(x)$ is analytic in $\tilde{C}(f)$.

3.2.11. Suppose that $E_i = K$ for $1 \leq i \leq n$. Let U be an open subset of K^n and $f \in \mathcal{C}^\omega(K^n; F)$. If $0 \in U$ and if $f_0 = \sum_\alpha X^\alpha c_\alpha$ is the power series expansion of f at 0 , the power series expansion of $\Delta^\alpha f$ at 0 is written (after identifying $P_\alpha(K^n; F)$ with F):

$$(\Delta^\alpha f)_0 = \sum_\beta ((\alpha, \beta)) X^\beta c_{\alpha+\beta}.$$

In particular, for $1 \leq i \leq n$ and for $x \in C(f_0)$:

$$\partial_i f(x) = \sum_{\alpha} (\alpha_i + 1) x^{\alpha} c_{\alpha + \varepsilon_i}.$$

3.3 Holomorphic Functions

In this section, we assume that $K = \mathbb{C}$.

3.3.1. Suppose F is quasi-complete. Let U be an open subset of E and f a mapping from U into F . The following conditions are equivalent:

- (i) f is holomorphic;
- (ii) f is differentiable;
- (iii) f is locally bounded and for any $a \in U$ and $h \in E$, the function $t \mapsto f(a + th)$ defined in an open neighborhood of 0 in $\mathbb{C}$ is holomorphic;
- (iv) f is locally bounded and for every continuous linear form u on F , the function $u \circ f$ with values in $\mathbb{C}$ is holomorphic;
- (v) f is continuous and locally bounded, and there exists a total set H in the dual of F such that $u \circ f$ is holomorphic for every $u \in H$.

When E is finite-dimensional (resp. when F is a Banach space), these conditions are still equivalent to conditions (iii'), (iv'), or (v') (resp. (iv') or (v')) obtained from (iii), (iv), or (v) (resp. (iv) or (v)) by removing the hypothesis “ f is locally bounded.”

3.3.2. Suppose F is quasi-complete. Let U be an open subset of E and (f_n) a sequence of holomorphic maps from U into F , having the following property:

- (W) Every point of U has a neighborhood in which the sequence (f_n) converges uniformly.

Then the limit f of the sequence (f_n) is holomorphic, the sequence of derivatives (Df_n) (with values in the quasi-complete space $\mathcal{L}(E; F)$) possesses property (W) and Df is the limit of (Df_n) .

3.3.3. Let U be an open subset of E and f a holomorphic map from U into F , with F assumed to be quasi-complete. Let $\mathbf{R} = (\mathbf{R}_i) \in (\mathbf{R}_+^*)^n$ and suppose that the polyball $\mathbf{B}(\mathbf{R})$ is contained

in U and f is bounded on $\mathbf{B}(\mathbf{R})$. We then have, for every $\alpha \in N^n$ and every $x = (x_i) \in \mathbf{B}(\mathbf{R})$:

$$\Delta^\alpha f(0)(x) = \int_0^1 \dots \int_0^1 f(\mathbf{e}(\theta_1)x_1, \dots, \mathbf{e}(\theta_n)x_n) \mathbf{e}(-\alpha_1\theta_1 - \dots - \alpha_n\theta_n) d\theta_1 \dots d\theta_n$$

(with $\mathbf{e}(\theta) = \exp 2\pi i\theta$) .

Additionally, let γ be a continuous seminorm on F and let M be the supremum of $\|f(x)\|_\gamma$ for $\|x_i\| = R_i$. One has $\|\Delta^\alpha f(0)(x)\|_\gamma \leq M$ for all $x \in B(R)$ and $\|\Delta^\alpha f(0)\|_\gamma \leq MR^{-\alpha}$. Finally, the domain of convergence of the series expansion of f at 0 contains the interior of the polydisc $B(R)$.

3.3.4. We keep the hypotheses of 3.3.3 and suppose moreover that $\mathbf{E}_i = \mathbf{C}$. Let $\sum X^\alpha c_\alpha$ be the series expansion of f at 0. We have:

$$c_\alpha = \mathbf{R}^{-\alpha} \int_0^1 \dots \int_0^1 f(\mathbf{e}(\theta_1)R_1, \dots, \mathbf{e}(\theta_n)R_n) \mathbf{e}(-\alpha_1\theta_1 - \dots - \alpha_n\theta_n) d\theta_1 \dots d\theta_n$$

and:

$$\|c_\alpha\|_\gamma \leq \mathbf{R}^{-\alpha} \sup_{x \in \mathbf{B}(\mathbf{R})} \|f(x)\|_\gamma$$

("Cauchy inequalities"). The strict domain of convergence of the series $\sum_\alpha X^\alpha c_\alpha$ contains the interior of $B(R)$.

3.3.5. Suppose E is finite-dimensional and F is quasi-complete. There then exists in $\mathcal{H}(\mathbf{E}; \mathbf{F})$ a series f_0 , one and only one, with infinite radius of convergence (for every norm on E), such that $f(x) = f_0(x)$ for all $x \in E$.

3.3.6. If f is a holomorphic mapping from E into F such that $f(E)$ is bounded, then the function f is constant ("Liouville's theorem").

3.3.7. We assume that $E \neq 0$. Let f be a holomorphic mapping from an open set U of E into F . Let a be a point of U and γ be a continuous seminorm on F . For every neighborhood V of a , contained in U , there exists $x \in V$, $x \neq a$, such that:

$$\|f(a)\|_\gamma \leq \|f(x)\|_\gamma.$$

If moreover $F = \mathbb{C}$ and if f is not constant in a neighborhood of a , we have $|f(a)| < \sup_{x \in V, x \neq a} |f(x)|$ and the map f is open in a neighborhood of a . Finally, let B be a bounded open set whose closure is contained in U , and let B' be its boundary. We have:

$$\sup_{x \in \overline{B}} |f(x)| = \sup_{x \in B'} |f(x)|$$

(“maximum principle”).

3.3.8. Suppose E is finite-dimensional. Let U be an open subset of E , and let S be a vector subspace of codimension ≥ 2 of E and f a holomorphic mapping from $U \cap (E - S)$ into F . Then f extends continuously to a holomorphic mapping from U into F .

Suppose that $E = \mathbb{C}$ and let $0 \leq \lambda < \mu \leq +\infty$. Let f be a holomorphic mapping from the open set $U = \{z \in \mathbb{C} \mid \lambda < |z| < \mu\}$ into F . There exists a unique family $(a_n(f))_{n \in \mathbb{Z}}$ of elements of F such that, for every compact H of U , the family $(a_n(f)z^n)_{n \in \mathbb{Z}}$ is uniformly summable, with sum $f(z)$ as z ranges over H (“Laurent expansion”).

Suppose further that $\lambda = 0$. We call the order of f at the point $x = 0$ the lower bound (finite or infinite) of the set of integers n such that $a_n(f) \neq 0$. If there exists a neighborhood V of 0 such that f is bounded in $V - \{0\}$, then f is of order 0 at the point $x = 0$ and extends by continuity to a holomorphic function on the open set $|z| < \mu$. Let p be an integer > 0 ; if f is of order $-p$ at the point $x = 0$, one says that 0 is a pole of order p of f .

3.3.10. Suppose that $E = \mathbb{C}$ and that F is normed. Let f be a holomorphic mapping from the open unit disk U of E into F , such that $f(0) = 0$ and let $M = \sup_{z \in U} \|f(z)\|$. One then has $\|f(z)\| \leq M \cdot |z|$ for all $z \in U$ (“Schwarz lemma”).

3.4 Real analytic functions

Suppose that $K = \mathbb{R}$.

3.4.1. Let U be an open subset of E and f a mapping from U into F , assumed to be quasi-

complete.

The following conditions are equivalent:

(i) f is analytic.

(ii) f is of class C^∞ and, for every $a \in U$, there exists a neighborhood V_a of a and a real number M such that, for every continuous semi-norm γ on F , there exists a constant A_γ such that one has

$$\|\Delta^\alpha f(x)\|_\gamma \leq A_\gamma M^{|\alpha|} \quad \text{for every } x \in V_a \text{ and every } \alpha \in \mathbb{N}^n.$$

3.4.2. Let U be an open subset of E and f a mapping from U into F . If F is quasi-complete, and if its strong dual F' is a Baire space, then f is analytic if and only if $u \circ f$ is analytic for every $u \in F'$.

4 Analytic functions (ultrametric case)

In this section, one assumes that the absolute value of K is ultrametric. One denotes by $(E_i)_{1 \leq i \leq n}$ a finite family of normed spaces over K , and by E the product space of the E_i , equipped with the norm:

$$\|x\| = \sup \|x_i\| \quad \text{if } x = (x_i).$$

One denotes by F a separated polynormed space over K .

4.1 Convergent series

4.1.1. Let $f = \sum_\alpha f_\alpha$ be a formal series belonging to $\hat{\mathbf{P}}(E_1, \dots, E_n; F)$, (cf. Appendix). If γ is a continuous semi-norm on F , and $\mathbf{R} = (R_i)$ is a system of n real numbers > 0 , we set

$$\|f\|_{\gamma, \mathbf{R}} = \sup_\alpha \|f_\alpha\|_\gamma \mathbf{R}^\alpha.$$

The definitions and results of no. 3.1.1 (second paragraph) and of no. 3.1.2 apply without

change; in particular, one defines the spaces

$$\mathcal{H}_{\mathbf{R}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F}) \quad \text{and} \quad \mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F}).$$

4.1.2. The canonical isomorphism j of $\hat{\mathbf{P}}(\mathbf{E}; \mathbf{F})$ onto $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ gives, by restriction, an isomorphism of topological vector spaces from $\mathcal{H}_{\mathbf{R}}(\mathbf{E}; \mathbf{F})$ onto $\mathcal{H}_{(\mathbf{R}, \dots, \mathbf{R})}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ for all $\mathbf{R} \in \mathbb{R}_+^*$; it also gives an isomorphism of $\mathcal{H}(\mathbf{E}; \mathbf{F})$ onto $\mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$. More precisely, if $f = \sum_m f_m \in \hat{\mathbf{P}}(\mathbf{E}; \mathbf{F})$ and if $j(f) = \sum_{\alpha} f_{\alpha}$, one has, for every continuous semi-norm γ on $\mathbf{F}$:

$$\begin{aligned} \|f_m\|_{\gamma} &= \sup_{|\alpha|=m} \|f_{\alpha}\|_{\gamma} \\ \|f\|_{\gamma, \mathbf{R}} &= \|j(f)\|_{\gamma, (\mathbf{R}, \dots, \mathbf{R})}. \end{aligned}$$

4.1.3. Let $f = \sum_{\alpha} f_{\alpha}$ be an element of $\mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$; let $I(f)$ be the set of $R \in (\mathbb{R}_+^*)^n$ such that, for every continuous semi-norm γ on $\mathbf{F}$, the product $\|f_{\alpha}\|_{\gamma} R^{\alpha}$ tends to zero when $|\alpha|$ tends to infinity. The set $I(f)$ is nonempty; it is called the strict convergence indicator of f . The set $\Omega(f)$ of points

$$(\log \mathbf{R}_1, \dots, \log \mathbf{R}_n) \quad \text{for} \quad \mathbf{R} \in I(f)$$

is a convex subset of $\mathbb{R}^n$.

When $n = 1$, the set $\mathbf{I}(f)$ is an interval of $\mathbb{R}$, open on the left and open or closed on the right; its upper bound (finite or $+\infty$) is denoted $\rho(f)$ and is called the strict radius of convergence of f .

With the notations of 4.1.2, one has $\mathbf{R} \in \mathbf{I}(f)$ if and only if

$$(\mathbf{R}, \dots, \mathbf{R}) \in \mathbf{I}(j(f)).$$

The set of points $x = (x_i)$ such that there exists $\mathbf{R} = (\mathbf{R}_i) \in \mathbf{I}(f)$ with $\|x_i\| \leq \mathbf{R}_i$ for $1 \leq i \leq n$ is called the strict domain of convergence of f and is denoted $C(f)$. It is an open subset of E , the union of the polyballs

$$\mathbf{B}(\mathbf{R}) = \{x \in E \mid \|x_i\| \leq R_i \text{ for } 1 \leq i \leq n\},$$

for $\mathbf{R} \in \mathbf{I}(f)$.

4.1.4. The results of 3.1.7 and 3.1.8 remain valid, replacing everywhere therein $\tilde{\mathbf{C}}(f)$ by $\mathbf{C}(f)$ and $\tilde{\mathcal{H}}_{\mathbb{R}}$ by $\mathcal{H}_{\mathbb{R}}$.

4.1.5. Let $\mathbf{F}_1, \dots, \mathbf{F}_m$ be complete normed spaces and suppose $\mathbf{F}$ quasi-complete. Let $\mathbf{f} = (f_i)_{1 \leq i \leq m}$, with $f_i \in \mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F}_i)$ and let $g \in \mathcal{H}(\mathbf{F}_1, \dots, \mathbf{F}_m; \mathbf{F})$, such that the point $(f_i(0))_{1 \leq i \leq m}$ of $\mathbf{E}$ belongs to the strict domain of convergence of g . Then, for every $\alpha \in \mathbb{N}^m$, the formal series $g_\alpha \circ \mathbf{f}$ belongs to $\mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ and the family of $g_\alpha \circ \mathbf{f}$ is summable in $\mathcal{H}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ (hence a fortiori in $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$). Its sum will be denoted $g \circ \mathbf{f}$.

More precisely, there exists $\mathbf{R} \in \bigcap_i \mathbf{I}(f_i)$ and $\mathbf{R}' \in \mathbf{I}(g)$ such that $\sup_{|\alpha| > 0} \|f_{i,\alpha}\| \mathbf{R}^\alpha < \mathbf{R}'_i$ (for $1 \leq i \leq m$). Under these conditions, the formal series $g_\alpha \circ \mathbf{f}$ belongs to $\mathcal{H}_{\mathbb{R}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ and the family of $g_\alpha \circ \mathbf{f}$ is summable in $\mathcal{H}_{\mathbb{R}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$. Finally, if $x \in \mathbf{B}(\mathbf{R})$, then $\mathbf{f}(x) = (f_i(x))$ belongs to $\mathbf{C}(g)$ and we have:

$$g(\mathbf{f}(x)) = (g \circ \mathbf{f})(x).$$

Suppose further that, for each i , there exists a family (e_j^i) of elements of $\mathbf{E}_i$ such that every element x of $\mathbf{E}_i$ is the sum of a summable family $(\lambda_j e_j^i)$ (with $\lambda_j \in \mathbf{K}$) such that $\|x\| = \sup_j |\lambda_j|$. One can then, in the preceding paragraph, replace the condition $\sup_{|\alpha| > 0} \|f_{i,\alpha}\| \mathbf{R}^\alpha < \mathbf{R}'_i$ by the condition $\|f_i\|_{\mathbf{R}} \leq \mathbf{R}'_i$.

4.1.6. Suppose that $E_i = K$ for $1 \leq i \leq n$. The space $\hat{\mathbf{P}}(\mathbf{K}^n; \mathbf{F})$ is then identified with the space of formal power series in n indeterminates $X_1, \dots, X_n$ with coefficients in $\mathbf{F}$, and an element f of $\hat{\mathbf{P}}(\mathbf{K}^n; \mathbf{F})$ is written:

$$f = \sum_{\alpha} X^\alpha c_\alpha \quad \text{with } c_\alpha \in \mathbf{F}.$$

$$\|f\|_{\gamma, \mathbf{R}} = \sup \|c_\alpha\|_{\gamma} \cdot \mathbf{R}^\alpha.$$

If $\mathbf{R} \in (\mathbf{R}_+^*)^n$ and if γ is a continuous semi-norm on F , we have:

The last paragraph of 3.1.10 remains valid, as does 3.1.11.

4.1.7. Suppose that K is a closed subfield of a complete valued field (necessarily ultrametric)

L. For $y \in E_i \otimes_K L$, set:

$$\|y\| = \inf(\sup_k |a_k| \cdot \|x_k\|)$$

the infimum being taken over all finite families of pairs $(x_k, a_k) \in E_i \times L$ such that $y = \sum_k x_k \otimes a_k$. We thus obtain a norm on the L-vector space $E_i \otimes_K L$, which induces on E_i the given norm. We denote by E_i^L the completion of $E_i \otimes_K L$ for this norm.

Now let F be a separated complete polynormed L-vector space. For every K-polynomial-continuous map f_α , homogeneous of multidegree α , on $E_1 \times \cdots \times E_n$, with values in the K-vector space polynormed F_K underlying F , there exists one and only one L-polynomial-continuous map $\tilde{f}_\alpha$, homogeneous of the same multidegree, on $E_1^L \times \cdots \times E_n^L$, with values in F , which extends f_α . For every continuous semi-norm on the L-vector space F , we have

$$\|\tilde{f}_\alpha\|_\gamma = \|f_\alpha\|_\gamma.$$

If $f = \sum_\alpha f_\alpha \in \mathcal{H}(E_1, \dots, E_n; F_K)$, then $\tilde{f} = \sum_\alpha \tilde{f}_\alpha \in \mathcal{H}(E_1^L, \dots, E_n^L; F)$. The series f and $\tilde{f}$ have the same strict convergence indicator (and the same strict radius of convergence when $n = 1$).

Conversely, let L be a non-discrete closed subfield of K and let E_i^0 and F^0 be the spaces over L obtained by restriction of scalars from the E_i and from F . If $f = \sum f_\alpha \in \mathcal{H}(E_1, \dots, E_n; F)$, then $f_\alpha \in P_\alpha(E_1^0, \dots, E_n^0; F^0)$; if we set $f^0 = \sum_\alpha f_\alpha \in \hat{P}(E_1^0, \dots, E_n^0; F^0)$, then $f^0 \in \mathcal{H}(E_1^0, \dots, E_n^0; F^0)$. One has $C(f) \subset C(f^0)$ and $f(x) = f^0(x)$ for all $x \in C(f)$.

4.2 Analytic functions

4.2.1. The definitions and results of 3.2.1 and 3.2.2 remain valid without change.

4.2.2. With the notation of 3.2.2, the strict convergence indicator of the power series expansion of $\Delta^a f$ at a point a of U contains that of the power series expansion of f at a .

4.2.3. The results of 3.2.4, 3.2.5, 3.2.7, 3.2.8 and 3.2.11 remain exact. That of 3.2.6 also, provided one assumes in addition that K is of characteristic zero.

4.2.4. Suppose F is quasi-complete and let $f \in \mathcal{H}(E_1, \dots, E_n; F)$. The function $x \mapsto f(x)$ is analytic in $C(f)$. For every $a \in C(f)$, the convergence indicator of the power series expansion of f at a is equal to that of f .

4.3 Some inequalities

4.3.1. We assume that K satisfies at least one of the following conditions: (a) the residue field of K is infinite;

(b) the image of K under the map $a \mapsto |a|$ is dense in $\mathbf{R}_+$. (In other words, we assume that K is not locally compact).

Let $f = \sum_{\alpha} f_{\alpha} \in \mathcal{H}(E_1, \dots, E_n; F)$ and let $R \in I(f)$. We have:

$$\sup_{x \in \mathbf{B}(\mathbf{R})} \|f(x)\|_{\gamma} = \sup_{\alpha} \sup_{x \in \mathbf{B}(\mathbf{R})} \|f_{\alpha}(x)\|_{\gamma}$$

for every continuous semi-norm γ on F ("Cauchy inequalities").

4.3.2. There exists a constant $a > 0$ such that for every continuous homogeneous polynomial $f_{\alpha} \in \mathbf{P}_{\alpha}(E_1, \dots, E_n; F)$ and every $\mathbf{R} \in (\mathbf{R}_+^*)^n$, we have:

$$a^{|\alpha|} \mathbf{R}^{\alpha} |\alpha|! \|f_{\alpha}\|_{\gamma} \leq \sup_{x \in \mathbf{B}(\mathbf{R})} \|f_{\alpha}(x)\|_{\gamma} \leq \|f_{\alpha}\|_{\gamma} \mathbf{R}^{\alpha}$$

for every continuous semi-norm γ on F . If K satisfies condition (b) of 4.3.1 or if the image of E_i under the map $x \mapsto \|x\|$ is contained in the image of K under the map $a \mapsto |a|$ and contains $\mathbf{R}_i$ (for $1 \leq i \leq n$), one may take $a = 1$.

4.3.3. If K has characteristic zero, the formal series $f = \sum_{\alpha} f_{\alpha}$ belongs to $\mathcal{H}(E_1, \dots, E_n; F)$ if and only if there exists $R \in (\mathbf{R}_+^*)^n$ such that

$$\sup_{\alpha} \sup_{x \in B(R)} \|f_{\alpha}(x)\|_{\gamma} < +\infty$$

for every continuous seminorm γ on F .

5 Manifolds

5.1 Charts and atlases. Manifolds.

5.1.1. Let X be a set. A chart on X is a triple $c = (U, \varphi, E)$, where U is a subset of X , E a Banach space and φ a bijection from U onto an open subset of E . We say that U is the domain of the chart c . If E is of finite dimension n , we say that c is of dimension n . Otherwise, we set $\dim c = +\infty$.

5.1.2. We say that two charts $c = (U, \varphi, E)$ and $c' = (U', \varphi', E')$ of X are C^r -compatible (or simply compatible when there can be no ambiguity about r) when the following conditions are satisfied:

- (a) $\varphi(U \cap U')$ (resp. $\varphi'(U \cap U')$) is open in E (resp. E');
- (b) the map $\varphi \circ \varphi'^{-1}$ (resp. $\varphi' \circ \varphi^{-1}$) of $\varphi'(U \cap U')$ onto $\varphi(U \cap U')$ (resp. of $\varphi(U \cap U')$ onto $\varphi'(U \cap U')$) is of class C^r (cf. 2.3.1, 3.2.1 and 4.2.1).

5.1.3. We call a C^r -atlas (or simply atlas) of a set X a set of charts of X that are pairwise C^r -compatible and whose domains have union X . Two atlases A and B of X are said to be C^r -equivalent if $A \cup B$ is an atlas. The relation of C^r -equivalence between atlases is an equivalence relation.

5.1.4. Let $\mathfrak{E}$ be a set of Banach spaces. We say that an atlas $\mathcal{A}$ is of type $\mathfrak{E}$ if one has $E \in \mathfrak{E}$ for every chart $c = (U, \varphi, E)$ of $\mathcal{A}$. Similarly, one says that an atlas $\mathcal{A}$ is of Hilbert type (resp. of Hilbert type with countable dimension) if E is a Hilbert space (resp. Hilbert space of countable type) for every chart (U, φ, E) of $\mathcal{A}$.

5.1.5. One calls a K -manifold of class C^r (or manifold of class C^r over K , or simply manifold when there can be no ambiguity about K and r) a set X equipped with a class of equivalent atlases (Sets, chap. II, § 6, no. 9) for the relation of C^r -equivalence. An atlas of this class is called an atlas of the manifold X . A chart belonging to an atlas of X is called a chart of the manifold X . A chart of X whose domain contains a point $a \in X$ is called a chart of X at a . A chart centered at a is a chart (U, φ, E) of X at a such that $\varphi(a) = 0$.

If X is a set and A an atlas of X , the set X equipped with the equivalence class of A is called

the manifold defined by A .

When $r \neq \omega$ (hence $K = \mathbb{R}$), a manifold of class C^r is sometimes called a differentiable manifold. A manifold of class C^ω is also called an analytic manifold over K (or also a K -analytic manifold).

When moreover $K = \mathbb{R}$ (resp. $\mathbb{C}$, $\mathbb{Q}_p$), one also says real analytic manifold (resp. complex analytic manifold, p -adic analytic manifold).

5.1.6. Let X be a manifold. The subsets of X that are unions of domains of charts of X form the set of open sets for a topology on X , called the topology underlying the manifold structure of X . For every chart $c = (U, \varphi, E)$ of X , the map φ is a homeomorphism from the open set U , equipped with the topology induced by that of X , onto the open set $\varphi(U)$ of E .

The topological space underlying X is a Baire space. When K is equal to $\mathbb{R}$ or $\mathbb{C}$, it is locally connected.

Let X be a manifold and let A be an atlas of X . For the topological space X to be Hausdorff, it is necessary and sufficient that the following condition be satisfied: for any charts (U, φ, E) and (V, ψ, F) belonging to the atlas A , the graph of the mapping $\psi \circ \varphi^{-1}$ of $\varphi(U \cap V)$ to $\psi(U \cap V)$ is closed in $\varphi(U) \times \psi(V)$.

Let X be a manifold; suppose that the topological space X is regular. Then for every point $a \in X$ there exists a chart (U, φ, E) of X at a possessing the following property: for a subset Y of U to be closed in X , it is necessary and sufficient that its image $\varphi(Y)$ be closed in E . If the space X is paracompact, there exists on X a distance compatible with the topology of X and making X into a complete metric space.

5.1.7. Let S be a set of Banach spaces. A manifold is said to be of type S (resp. of Hilbertian type, resp. of Hilbertian type of countable dimension) if it possesses an atlas of type S (resp. of Hilbertian type, resp. of Hilbertian type of countable dimension). If S is reduced to a single element E , a manifold of type S is also called a pure manifold of type E .

A pure manifold of dimension n is called a pure manifold of type K^n . A manifold is said to be locally of finite dimension if it is of type S with $S = \{K^n; n \in \mathbb{N}\}$.

5.1.8. Let X be a manifold and let $x \in X$. The dimension (finite or $+\infty$) of a chart of X at x depends only on x . It is called the dimension of X at x and is denoted $\dim_x X$. The dimension

of X , denoted $\dim X$, is the upper bound of the $\dim_x X$ for $x \in X$.

The function $x \mapsto \dim_x X$ is locally constant. The manifold X is locally of finite dimension (resp. pure of dimension n) if and only if one has $\dim_x X < +\infty$ (resp. $\dim_x X = n$) for all $x \in X$.

5.1.9. Suppose K is locally compact. Let X be a separated manifold. Then, for X to be locally of finite dimension, it is necessary and sufficient that X be locally compact.

5.1.10. Let X be a manifold and let $\xi^1, \dots, \xi^n$ be maps from a subset U of X into K . One says that $\xi = (\xi^1, \dots, \xi^n)$ is a coordinate system of X in U if the triple (U, ξ, K^n) is a chart of X ; this chart is also denoted $(U; \xi)$ or $(U; \xi^1, \dots, \xi^n)$. If $a \in U$, we also say that ξ is a coordinate system of X at a ; if moreover $\xi^i(a) = 0$ for $i = 1, 2, \dots, n$, one says that the coordinate system ξ is centered at a .

5.2 Examples of manifolds

5.2.1. Let X be a set; there exists on X one and only one manifold structure for which the underlying topological space is discrete; this structure is a pure manifold structure of dimension 0.

5.2.2. Let E be a Banach space. The triple $c = (E, \text{Id}_E, E)$ is a chart of E and $\mathcal{A} = \{c\}$ is an atlas of E , hence defines a pure manifold structure of type E on E ; the underlying topology is the given topology on E . In what follows, when we speak of the manifold structure on E , it will always be to the preceding structure that we refer.

In particular, this applies to every finite-dimensional vector space over K , equipped with the unique Hausdorff topology compatible with its vector structure (Esp. Vect. Top., chap. I, § 2, no. 3).

5.2.3. Let X be a manifold and U an open subset of X . There exists on U a manifold structure whose charts are the charts of the manifold X with domain contained in U . This structure is said to be induced by that of X (cf. no. 5.3.4); equipped with this structure, U is called an open submanifold of X .

In particular, every open subset of a Banach space E is equipped with a canonical structure of pure manifold of type E . Let X be a manifold; for a triple (U, φ, E) to be a chart of X ,

it is necessary and sufficient that U be open in X and that φ be an isomorphism of the open submanifold U of X onto an open submanifold of E .

5.2.4. Let X be a set, and $(X_i)_{i \in I}$ a covering of X . Suppose that on each X_i there is given a manifold structure such that the following condition is satisfied:

For any i and j in I , the set $X_i \cap X_j$ is open in X_i and X_j and the manifold structures induced on $X_i \cap X_j$ by those of X_i and X_j coincide.

There then exists on X one and only one manifold structure such that X_i is an open submanifold of X for every i in I . One says that the manifold X is obtained by gluing together the manifolds X_i .

5.2.5. Let X be a manifold. The set X_n of points x of X such that $\dim_x X = n$ (n integer ≥ 0) is an open submanifold of X , which is pure of dimension n .

5.2.6. Let E be a Banach space. One can equip the set $\mathbf{G}(E)$ of vector subspaces of E admitting a topological supplement with an analytic manifold structure in the following way: for every pair $(F_0, G_0) \in \mathbf{G}(E) \times \mathbf{G}(E)$ such that $E = F_0 \oplus G_0$, one denotes by U_{G_0} the set of $F \in \mathbf{G}(E)$ admitting G_0 as supplement, and one defines a bijection φ_{F_0, G_0} of U_{G_0} onto the Banach space $\mathcal{L}(F_0; G_0)$ by associating to every $F \in U_{G_0}$ the map from F_0 to G_0 having as graph the subspace F of $E = F_0 \times G_0$. The charts $(U_{G_0}, \varphi_{F_0, G_0}, \mathcal{L}(F_0; G_0))$ form an atlas of $\mathbf{G}(E)$. Equipped with the manifold structure defined by this atlas, $\mathbf{G}(E)$ is called the Grassmannian manifold of E .

The topological space $\mathbf{G}(E)$ is metrizable. If K is locally compact and E is finite-dimensional, $\mathbf{G}(E)$ is compact.

For every integer $n \geq 0$, the set $\mathbf{G}_n(E)$ (resp. $\mathbf{G}^n(E)$) of $F \in \mathbf{G}(E)$ of dimension (resp. codimension) n is open and closed in $\mathbf{G}(E)$ and is a pure open submanifold of $\mathbf{G}(E)$. If $K = \mathbf{R}$ or $\mathbf{C}$, $\mathbf{G}_n(E)$ is connected for every n .

The map which associates to each $F \in \mathbf{G}(E)$ its orthogonal in the dual E' of E is a morphism of $\mathbf{G}(E)$ to $\mathbf{G}(E')$ which induces an isomorphism of $\mathbf{G}^n(E)$ onto $\mathbf{G}_n(E')$ for every integer n .

If $K = \mathbf{R}$ or $\mathbf{C}$, or if E is finite-dimensional, $\mathbf{G}_1(E)$ is none other than the projective space associated with E , which is thus equipped with an analytic manifold structure.

5.3 Functions of class C^r and morphisms of manifolds

5.3.1. Let X be a manifold of class C^r , let F be a separated polynormed space, and let f be a map from X into F . We say that f is of class C^r if, for every chart (V, φ, E) of X , the map $f \circ \varphi^{-1}$ from $\varphi(V)$ into F is of class C^r . For this, it suffices that this condition be satisfied for the charts of an atlas of X . The set of maps of class C^r from X into F forms a vector subspace of the space of all maps from X into F . We denote it by $\mathcal{C}^r(X; F)$.

When $F = K$, we set $\mathcal{C}^r(X; K) = \mathcal{C}^r(X)$; it is a sub- K -algebra of the algebra of maps from X into K . The elements of $\mathcal{C}^r(X)$ are also called morphic functions.

When X is an open subset of a Banach space, this terminology agrees with that of nos. 2.3.1, 3.2.1 and 4.2.1.

5.3.2. Let X and Y be two manifolds of class C^r and let f be a map from X into Y . We say that f is of class C^r or is a morphism of manifolds (of class C^r) if it is continuous and if, for every chart (V, ψ, F)

of Y , the map $\psi \circ f$ from the open submanifold $f^{-1}(V)$ into the Banach space F is of class C^r . For this, it suffices that there exist an atlas $\mathcal{A}$ of Y such that, for every chart $(V, \psi, F) \in \mathcal{A}$, the set $f^{-1}(V)$ is open in X and the map $\psi \circ f$ from the open submanifold $f^{-1}(V)$ of X into F is of class C^r . The set of morphisms from X into Y is denoted by $\mathcal{C}^r(X; Y)$. When Y is a Banach space equipped with its canonical manifold structure, the definitions of 5.3.1 and 5.3.2 are consistent. A map of class C^ω is also called a K -analytic map (or simply analytic). When $K = \mathbb{C}$, we also say holomorphic map.

Let (U, φ, E) be a chart of X and (V, ψ, F) a chart of Y such that $f(U) \subset V$. The map $\psi \circ f \circ \varphi^{-1}$ of the open subset $\varphi(U)$ of E into the open subset $\psi(V)$ of F is called the expression of f in the given charts.

5.3.3. Suppose X and Y are of finite dimension; let $a \in X$, $b \in Y$ and f be a map from X into Y with $f(a) = b$. Suppose first that f is of class C^r and consider coordinate systems $(U; \xi^1, \dots, \xi^m)$ of X at a and $(V; \eta^1, \dots, \eta^n)$ of Y at b respectively, with $f(U) \subset V$. Let ξ be the map $(\xi^1, \dots, \xi^m)$ from U into K^m . There then exist functions w^j of class C^r on the open subset $\xi(U)$ of K^m , with values in K , such that the coordinates of a point $y = f(x)$ of Y , (for x in U), are given by the formulas:

$$\eta^j(y) = u^j(\xi^1(x), \dots, \xi^m(x)) \quad \text{for } 1 \leq j \leq n \quad (1)$$

which is equivalent to:

$$\eta^j \circ f = u^j(\xi^1, \dots, \xi^m) \quad \text{for } 1 \leq j \leq n. \quad (2)$$

We say that the preceding formulas constitute the expression of f by means of the chosen coordinate systems.

Conversely, if, for every point a of X , one can find a coordinate system of X at a and a coordinate system of Y at $b = f(a)$ satisfying the preceding conditions, then f is of class C^r .

5.3.4. Morphisms of manifolds satisfy the axioms of Ens., chap. IV, § 2:

- a) The composite of two morphisms is a morphism.
- b) For a bijection $f : X \rightarrow Y$ to be an isomorphism, it is necessary and sufficient that f and f^{-1} are morphisms.

5.3.5. Suppose that $r = \omega$. Let X be a manifold and f, g two analytic maps from X into a separated polynormed space or into a separated analytic manifold. The set of points in a neighborhood of which f and g are equal, is open and closed in X .

5.3.6. Suppose that $K = \mathbb{R}$ and $r \neq \omega$. A manifold X is said to admit partitions of unity of class C^r if, for every locally finite open covering of X , there exists a continuous partition of unity (Top. Gén., ch. IX, § 4, n° 3) subordinate to this covering and consisting of functions of class C^r .

Let E be a Banach space; consider the following property:

For any disjoint closed subsets A and B of E , there exists a function f of class C^r on E , with values in $\mathbb{R}$, such that $f(x) = 0$ for $x \in A$, $f(x) = 1$ for $x \in B$ and $0 \leq f(x) \leq 1$ for all $x \in E$.

If S is a set of Banach spaces having property (PU), then every paracompact manifold of type S admits partitions of unity of class C^r .

Every finite-dimensional space, every separable Hilbert space, has property (PU).

5.3.7. Suppose K is ultrametric. Let X be a paracompact manifold. For every open cover U

of X , there exists a partition of X into open and closed subsets subordinate to the cover U .

5.3.8. Suppose K is locally compact. Let X and Y be two pure manifolds of finite dimension, and let f be a morphism from X to Y . If $\dim X < \dim Y$ and if the topology of X admits a countable base of open sets, $f(X)$ is meager in Y .

5.4 Characterization of varieties by their sheaves of functions

5.4.1. Let X be a topological space, and let Y be a set. Suppose given, for every open set U of X , a set $\mathcal{L}(U)$ of maps from U to Y . We shall say that the family $\mathcal{L} = \{\mathcal{L}(U)\}$ is a sheaf of functions with values in Y if it satisfies the following condition:

Let $(U_i)_{i \in I}$ be a family of open subsets of X , with union U , and let f be a mapping from U to Y ; for f to belong to $\mathcal{L}(U)$ it is necessary and sufficient that $f|_{U_i}$ belongs to $\mathcal{L}(U_i)$ for all i in I .

5.4.2. Let X and Y be two varieties; for every open set $U \subset X$ let $\mathcal{L}(U)$ be the set of morphisms from U into Y ; then $\mathcal{L}$ is a sheaf of functions with values in Y .

When $Y = K$, we denote by C_X^r the sheaf thus defined.

5.4.3. Let X be a topological space and let S be a set of Banach spaces. For every $E \in S$ let F_E be a sheaf of functions on X with values in E . Suppose that the family of F_E , for $E \in S$, satisfies the following condition:

For every $x \in X$ there exists an open neighborhood U of x , a space

$E_0 \in S$, and a homeomorphism φ of U onto an open subset of E_0 , such that, for every open set $V \subset U$ and every $E \in S$, the set $\mathcal{F}_E(V)$ consists of the functions $g \circ \varphi$ where g ranges over the space $\mathcal{C}^r(\varphi(V); E)$.

There then exists on X a structure of a manifold of class C^r and of type S , and only one, compatible with the given topology on X and for which F_E is the sheaf of functions of class C^r on X with values in E .

5.4.4. Let X be a topological space and let F be a sheaf of functions with values in K , satisfying the following condition: for every point x of X , there exist an integer n , an open neighborhood U of x , and a homeomorphism φ of U onto an open subset of K^n , such that, for every open set V of U , the set $\mathcal{F}(V)$ consists of the functions $g \circ \varphi$ where g ranges over the set

of functions of class C^r with values in K on the open set $\varphi(V)$ of K^n .

There then exists on X a structure of a locally finite-dimensional manifold of class C^r and only one such that $F = C_X^r$.

5.4.5. Let X and X' be two manifolds of class C^r , locally finite-dimensional, and let f be a continuous map from X to X' . For f to be a morphism, it is necessary and sufficient that for every open set U' of X' and every function $g \in \mathcal{C}^r(U'; K)$, the function $g \circ f$ belongs to $\mathcal{C}^r(U; K)$, with $U = f^{-1}(U')$.

5.5 Tangent spaces, tangent linear maps

5.5.1. Let X be a manifold and let $a \in X$. Consider the pairs (c, h) where $c = (U, \varphi, E)$ is a chart of the manifold X at a and where h is an element of E . Two such pairs (c, h) and (c', h') are said to be equivalent if the derivative at $\varphi(a)$ of the map $\varphi' \circ \varphi^{-1}$ (which is defined on a neighborhood of $\varphi(a)$) transforms h into h' . One thus obtains an equivalence relation between pairs (c, h) and one calls a tangent vector at a to X a class of equivalent pairs (c, h) (Sets, Chap. II, § 6, no. 9).

The tangent vectors at a to X form a set denoted $T_a(X)$. If $c = (U, \varphi, E)$ is a chart of the manifold X at a , the map θ_c from E into $T_a(X)$ which associates to an element h of E the tangent vector represented by the pair (c, h) is a bijection. If we transport to $T_a(X)$ by means of θ_c the structure of topological K -vector space on E , the structure thus obtained does not depend on the choice of c and turns $T_a(X)$ into a Banach space, called the tangent space to X at a . The dimension (finite or $+\infty$) of $T_a(X)$ is equal to the dimension of X at a .

5.5.2. Let X, Y be two varieties, f be a morphism from X to Y and a be a point of X . Consider a chart $c = (U, \varphi, E)$ of X at a and a chart $c' = (V, \psi, F)$ of Y at b with $f(U) \subset V$; the map $\Phi = \psi \circ f \circ \varphi^{-1}$ of $\varphi(U)$ to F is of class C^r , and its derivative $D\Phi(\varphi(a))$ at the point $\varphi(a)$

is a continuous linear map from E into F . The continuous linear map $\theta_{c'} \circ D\Phi(\varphi(a)) \circ \theta_c^{-1}$ of $T_a(X)$ to $T_b(Y)$ does not depend on the charts c and c' chosen; it is denoted $T_a(f)$ and is called the tangent linear map to f at a . If g is a morphism from Y into a manifold Z , we have:

$$T_a(g \circ f) = T_{f(a)}(g) \circ T_a(f).$$

5.5.3. Let $f : X \rightarrow Y$ be a morphism of manifolds. If f is locally constant, we have $T_a(f) = 0$ for all a in X . Conversely, if $T_a(f) = 0$ for all $a \in X$ and if the field K has characteristic 0, then f is locally constant.

5.5.4. Let U be an open subset of a manifold X and let f be the canonical injection from U into X ; for every point a in U , the map $T_a(f)$ of $T_a(U)$ to $T_a(X)$ is an isomorphism of topological vector spaces, by which these two spaces will henceforth be identified.

5.5.5. Let U be an open subset of a Banach space E ; if ι is the canonical injection of U into E , the triple $c = (U, \iota, E)$ is a chart of U , and the atlas $\{c\}$ defines the manifold structure of U . Given a point a of U , the map θ_c is an isomorphism of topological vector spaces from E onto $T_a(U)$; the inverse isomorphism will be denoted $\zeta_E(a)$.

Let f be a map of class C^r from U into an open subset V of a Banach space F ; for every point a of U , we have:

$$Df(a) \circ \zeta_E(a) = \zeta_F(f(a)) \circ T_a(f)$$

where $Df(a)$ is the derivative of f at a (1.2.1). In other words, the diagram:

$$\begin{array}{ccc} T_a(U) & \xrightarrow{T_a(f)} & T_{f(a)}(V) \\ \downarrow \zeta_E(a) & & \downarrow \zeta_F(f(a)) \\ E & \xrightarrow{Df(a)} & F \end{array}$$

is commutative.

5.5.6. A cotangent vector to X at a , also called a tangent covector or simply a covector at a , is any continuous linear form on the topological vector space $T_a(X)$; these covectors are therefore the elements of the topological dual $T_a(X)'$ of the tangent space to X at a . This space will be endowed with the topology of uniform convergence on bounded subsets of $T_a(X)$; equipped with this topology, $T_a(X)'$ is a Banach space called the cotangent space to X at a . It is also denoted by $T'_a(X)$.

Let f be a morphism from X into a Banach space E . One calls the differential of f at a and denotes by $d_a f$ or df_a the continuous linear map $\zeta_E(f(a)) \circ T_a(f)$ of $T_a(X)$ into E . For $t \in T_a(X)$, one sometimes denotes $t(f)$ or $t.f$ the value $d_a f(t)$ of the linear map $d_a f$ at the point t . The map

$f \mapsto d_a f$ is linear.

If $E = K$, the differential $d_a f$ of f at a is a covector of X at a .

Let E, F, G be three Banach spaces and let $(u, v) \mapsto u \cdot v$ be a continuous bilinear map from $F \times G$ into E . Let f (resp. g) be a morphism from X to F (resp. G). Then the map $f \cdot g : x \mapsto f(x) \cdot g(x)$ is a morphism from X to E and we have, for every $t \in T_a(X)$:

$$d_a(f \cdot g)(t) = f(a) \cdot d_a g(t) + d_a f(t) \cdot g(a).$$

Taking in particular $G = E$ and $F = K$, the bilinear map in question being multiplication, we have:

$$d_a(fg) = f(a)d_a g + g(a)d_a f.$$

5.5.7. Let X be a locally finite-dimensional manifold and let $\xi^1, \dots, \xi^m$ be morphic functions defined in an open neighborhood U of a point a of X . The following conditions are equivalent:

- (i) there exists an open neighborhood V of a contained in U , such that the functions $\xi^i|_V$ (for $1 \leq i \leq m$) form a coordinate system of X in V ;
- (ii) the differentials $d_a \xi^i$ for $1 \leq i \leq m$ form a basis of $T_a(X)'$;
- (iii) the map $\xi = (\xi^1, \dots, \xi^m)$ from U into K^m is étale at a (see no. 5.7.6).

For the differentials $d_a \xi^1, \dots, d_a \xi^m$ to be linearly independent, it is necessary and sufficient that there exist a neighborhood V of a , contained in U , such that the functions $\xi^i|_V$ are part of a coordinate system of X in V .

5.5.8. Let X be a locally finite-dimensional manifold and let $\xi = (\xi^1, \dots, \xi^m)$ be a coordinate system of X in an open set U . Let $a \in U$: we denote by $(\partial_{1,a}, \dots, \partial_{m,a})$ the basis of the tangent space $T_a(X)$ dual to the basis $(d_a \xi^1, \dots, d_a \xi^m)$ of $T_a(X)'$. We therefore have:

$$\partial_{i,a}(\xi^j) = \delta_{i,j} \quad (\text{Kronecker index}).$$

The tangent vector $\partial_{i,a}$ is also denoted $(\partial/\partial \xi^i)_a$.

Let f be a function of class C^r on U , with values in a Banach space E . We denote by $\partial_i f$

or $\partial f / \partial \xi^i$ the function $a \mapsto \partial_{i,a}(f)$: this is a function of class C^{r-1} on U with values in E (a continuous function if $r = 1$). Its value at a point a of U is sometimes denoted $(\partial f / \partial \xi^i)_a$. For every $t \in T_a(X)$, we have:

$$d_a f(t) = \sum_{i=1}^m d_a \xi^i(t) \frac{\partial f}{\partial \xi^i}(a),$$

which can also be written:

$$d_a f = \sum_{i=1}^m \frac{\partial f}{\partial \xi^i}(a) d_a \xi^i.$$

These notations are consistent with those of 1.6.3.

5.5.9. Let X and Y be two manifolds and let f be a morphism from X to Y . We call the rank of f at a point a of X , and denote by $rg_a f$, the rank (finite or equal to $+\infty$) of the linear map $\overline{T}_a(f)$. The map $a \mapsto rg_a f$ is lower semicontinuous.

5.6 Products of manifolds

5.6.1. Let X and X' be two sets and let $c = (U, \varphi, E)$ (resp. $c' = (U', \varphi', E')$) be a chart of X (resp. X'). The triple

$$(U \times U', \varphi \times \varphi', E \times E')$$

is a chart of the product set $X \times X'$, denoted $c \times c'$.

5.6.2. Let X and X' be two manifolds of class C^r . On the product set $X \times X'$ there exists one and only one manifold structure of class C^r such that $c \times c'$ is a chart of $X \times X'$ for every chart c of X and every chart c' of X' . Equipped with this structure, $X \times X'$ is called the product manifold of the manifolds X and X' . The product of any finite number of manifolds is defined similarly. The topology of the manifold $X \times X'$ is the product topology of the topologies of X and X' . For $a \in X$ and $b \in X'$, we have:

$$\dim_{(a,b)}(X \times X') = \dim_a X + \dim_b X'.$$

5.6.3. Let X and X' be two manifolds, and let $X \times X'$ be their product. Let $a \in X$ and let $a' \in X'$. The canonical projections

$$\text{pr}_1 : X \times X' \rightarrow X \quad \text{and} \quad \text{pr}_2 : X \times X' \rightarrow X'$$

are morphisms. Let π_i ($i = 1, 2$) be their tangent linear maps at the point (a, a') . The map

$$(\pi_1, \pi_2) : T_{(a, a')}(X \times X') \rightarrow T_a(X) \times T_{a'}(X')$$

is an isomorphism, which allows one to identify the tangent space to $X \times X'$ at (a, a') with the product $T_a(X) \times T_{a'}(X')$.

The injection of $T_a(X)$ to $T_{(a, a')}(X \times X')$ resulting from this identification is the tangent linear map at a to the morphism $x \mapsto (x, a')$ of X to $X \times X'$; we have an analogous result for the injection of $T_{a'}(X')$ to $T_{(a, a')}(X \times X')$.

5.6.4. Let W , X and X' be three manifolds, and let $f : W \rightarrow X$, $f' : W \rightarrow X'$ be two maps. For the map

$$(f, f') : W \rightarrow X \times X'$$

to be a morphism, it is necessary and sufficient that f and f' be morphisms (this justifies the use of the term "product", cf. Ens., chap. IV, § 2, no. 4). In this case, if a is a point of W , we have:

$$T_a(f, f') = (T_a(f), T_a(f')),$$

taking into account the identification made above.

5.6.5. Let $f : X \rightarrow Y$ and $f' : X' \rightarrow Y'$ be morphisms of manifolds. Then $f \times f' : X \times X' \rightarrow Y \times Y'$ is a morphism. If $a \in X$ and $a' \in X'$, we have:

$$T_{(a, a')}(f \times f') = T_a(f) \times T_{a'}(f')$$

$$\text{rg}_{(a, a')}(f \times f') = \text{rg}_a f + \text{rg}_{a'} f'.$$

5.6.6. Let X_1 , X_2 and Z be three manifolds and f a morphism from $X_1 \times X_2$ to Z . Let $a \in X_1$

and $b \in X_2$. The tangent linear map to the partial map $x \mapsto f(x, b)$ (resp. $y \mapsto f(a, y)$) of X_1 (resp. X_2) into Z is denoted $T_{(a,b)}^1(f)$ (resp. $T_{(a,b)}^2(f)$). If one identifies $T_{(a,b)}(X_1 \times X_2)$ with $T_a(X_1) \times T_b(X_2)$, we have:

$$T_{(a,b)}(f).(u, v) = T_{(a,b)}^1(f).u + T_{(a,b)}^2(f).v$$

for every $u \in T_a(X_1)$ and every $v \in T_b(X_2)$.

5.6.7. ("Implicit function theorem"). With the hypotheses and notations of the preceding number, suppose moreover that $T_{(a,b)}^2(f)$ is bijective. Then there exists an open neighborhood U of a in X_1 and an open neighborhood V of b in X_2 with the following property: for every $x \in U$, there exists one and only one point $g(x)$ of V such that $f(x, g(x)) = f(a, b)$, and the map g is a morphism from U to V . For every $x \in U$, one has

$$T_x(g) = - \left(T_{(x,g(x))}^2(f) \right)^{-1} \circ T_{(x,g(x))}^1(f).$$

5.7 Immersions, étale morphisms

5.7.1. Let X and Y be two manifolds, f a morphism from X to Y and a a point of X . Set $b = f(a)$. The following conditions are equivalent:

- (i) The linear map $T_a(f)$ is injective and its image is a closed vector subspace of $T_b(Y)$ admitting a topological supplement ¹ ;
- (ii) There exist a Banach space F , a closed vector subspace E of F admitting a topological supplement, and charts (U, φ, E) of X at a and (V, ψ, F) of Y at b such that $f(U) \subset V$ and $\varphi = \psi \circ (f|U)$.
- (iii) There exist an open neighborhood U of a , an open neighborhood V of b containing $f(U)$, a manifold Z , a point c of Z , and an isomorphism of manifolds g from $U \times Z$ onto V such that $f(x) = g(x, c)$ for all $x \in U$.
- (iv) There exist an open neighborhood U of a , an open neighborhood V of b and a morphism q from V to X with $f(U) \subset V$, and $q(f(x)) = x$ for all $x \in U$.

When X and Y are of finite dimension, the preceding conditions are still equivalent to the following:

- (v) There exist an open neighborhood U of a , a coordinate system $(\eta^1, \dots, \eta^n)$ of Y in an open neighborhood V of b containing $f(U)$ and an integer $m \leq n$ such that $\eta^j \circ f = 0$ for $m < j \leq n$ and that the functions $\eta^1 \circ f, \dots, \eta^m \circ f$ form a coordinate system of X in U .

If properties (i) through (iv) are satisfied, one says that f is an immersion at a .

The set of points where f is an immersion is open in X ; if this open set is all of X , we say that f is an immersion.

For f to be an immersion, it is necessary and sufficient that X can be covered by open sets U_i such that, for all i , $f|U_i$ is an isomorphism of U_i onto a submanifold of Y (cf. no. 5.8.3).

5.7.2. Examples:

- (a) If X is an open subset of a manifold Y , the canonical injection of X into Y is an immersion.
- (b) Let E and F be two Banach spaces and u a continuous linear map from E into F . Then u is an immersion if and only if u is an isomorphism from E onto a closed vector subspace of F admitting a topological complement.

5.7.3. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms. If f and g are immersions, $g \circ f$ is an immersion. Conversely, if $g \circ f$ is an immersion, then f is an immersion. If $f : X \rightarrow Y$ and $f' : X' \rightarrow Y'$ are immersions, $f \times f'$ is an immersion.

5.7.4. Suppose that X and Y are finite-dimensional varieties over a field K of characteristic 0. If $f : X \rightarrow Y$ is an injective morphism, the set of points of X where f is an immersion is a dense open subset of X . If $K = \mathbb{R}$ or $\mathbb{C}$, this result remains true if one assumes only that X is finite-dimensional.

5.7.5. Let $f : X \rightarrow Y$ be an immersion and let g be a continuous map from a manifold Z to X . For g to be a morphism, it is necessary and sufficient that $f \circ g$ be a morphism.

5.7.6. Let $f : X \rightarrow Y$ be a morphism, and let a be in X . The following two properties are equivalent:

- (i) $T_a(f)$ is bijective.

- (ii) There exists an open neighborhood U of a and an open neighborhood V of $f(a)$ such that f induces an isomorphism from U onto V .

When these properties hold, we say that f is a local isomorphism at a , or that f is étale at a . If this holds for every $a \in X$, one says that f is étale, or is an étale map, or that X is étale over Y (by means of f). For a morphism to be étale, it is necessary and sufficient that it be both an immersion (n° 5.7.1) and a submersion (n° 5.9.1.).

5.7.7. Let $f : X \rightarrow Y$ be an immersion; suppose the variety X is pure of finite dimension. Then f is étale in each of the following two cases:

- (i) $\dim Y = \dim X$;
- (ii) f is surjective and the topology of X has a countable base of open sets.

5.7.8. For a morphism f to be an isomorphism onto an open submanifold (resp. an isomorphism), it is necessary and sufficient that f be étale and injective (resp. étale and bijective).

5.8 Inverse images of manifold structures, submanifolds

5.8.1. Let X be a topological space, Y be a manifold, and f be a map from X to Y . Consider the following conditions:

(QR) (resp. (R)) For every $a \in X$, there exists an open neighborhood U of a in X and a chart (V, φ, E) of the manifold Y at $f(a)$ such that $f(U) \subset V$ and that $\varphi \circ f$ induces a homeomorphism of U onto the intersection of $\varphi(V)$ with a closed vector subspace (resp. a closed vector subspace admitting a topological complement) F of E .

If condition (QR) is satisfied, there exists on X one and only one manifold structure for which the triples $(U, \varphi \circ f, F)$ (with the notations of (QR)) are charts of X . It is called the inverse image structure of the manifold structure of Y by f . Its topology is that of X .

For there to exist on X a manifold structure compatible with the given topology and for which f is an immersion, it is necessary and sufficient that condition (R) be satisfied. This structure is then unique: it is the inverse image under f of the manifold structure of Y .

5.8.2. The condition (R) above is satisfied in particular when for every $a \in X$ there exists an open neighborhood U of a such that $f|_U$ is a homeomorphism from U onto an open subset

of Y . In this case, the manifold X obtained is étale over Y (5.7.6).

5.8.3. Suppose now that X is a topological subspace of Y , with f being the canonical injection. If f satisfies condition (R) (resp. (QR)) of no. 5.8.1, then X , equipped with the inverse image structure of the manifold structure of Y under f , is said to be a submanifold (resp. quasi-submanifold) of Y . A submanifold is a quasi-submanifold.

Every quasi-submanifold is locally closed; every open set is a submanifold and its structure is that defined in No. 5.2.3.

When Y is locally of finite dimension, a topological subspace X of Y is a submanifold of Y if and only if, for every $a \in X$ there exists an open neighborhood U of a in Y , a coordinate system $(\xi^1, \dots, \xi^m)$ of Y in U , and an integer $k \leq m$ such that $U \cap X$ is the set of points of U where the functions ξ^i for $1 \leq i \leq k$ vanish simultaneously.

Let $f : X \rightarrow Y$ be a morphism of varieties. Suppose that f is an immersion, and let f induce a homeomorphism from X onto $f(X)$. Then $f(X)$ is a submanifold of Y , and f induces an isomorphism from X onto $f(X)$; one then says that f is an embedding of X into Y .

5.8.4. Let Y and W be two varieties, let X be a subvariety of Y , and let f be a map from X to W . For f to be a morphism, it is necessary and sufficient that, for every $a \in X$, there exist a neighbourhood U of a in Y and a morphism $g : U \rightarrow W$ such that g coincides with f on $U \cap X$.

5.8.5. Let X be a quasi-subvariety of a variety Y , and let g be a map from a variety Z into X . Suppose that K is of characteristic 0, or that X is a subvariety of Y . For g to be a morphism from Z to X , it is necessary and sufficient that it be a morphism from Z to Y : in other words, the variety structure of X is induced by that of Y (Ens., ch. IV, § 2, no. 4).

5.8.6. If X is a subvariety (resp. quasi-subvariety) of Y and if Y is a subvariety of a variety Z , then X is a subvariety (resp. quasi-subvariety) of Z .

5.8.7. Let X_i (for $i = 1, 2$) be a variety and let Y_i be a subvariety (resp. quasi-subvariety) of X_i (for $i = 1, 2$). Then $Y_1 \times Y_2$ is a subvariety (resp. quasi-subvariety) of $X_1 \times X_2$.

5.8.8. Let X be a quasi-subvariety of a variety Y ; let x be a point of X and denote by f the canonical injection of X into Y . The map $T_x(f) : T_x(X) \rightarrow T_x(Y)$ is injective and allows one to identify the space $T_x(X)$ with a closed vector subspace of $T_x(Y)$. The quotient topological vector space $T_x(Y)/T_x(X)$ is a Banach space, which is called the transversal space to X at x (in

Y). Its dimension (finite or $+\infty$) is called the codimension of X in Y at the point x.

If moreover X is a subvariety of Y, then the space $T_x(X)$ admits a topological supplement in $T_x(Y)$.

5.8.9. Let f be a map from a variety X to a variety Y, and let Γ be its graph. For f to be a morphism, it is necessary and sufficient that the following two conditions be satisfied:

(i) Γ is a subvariety of $X \times Y$.

(ii) For every $(x, y) \in \Gamma$, one has

$$T_{(x,y)}(X \times Y) = T_{(x,y)}(\Gamma) \oplus T_y(Y).$$

If this is so, the map pr_1 induces an isomorphism from Γ onto X, and $T_{(x,y)}(\Gamma)$ is identified with the graph of $T_x(f)$.

In particular, the diagonal of $X \times X$ is a subvariety of $X \times X$.

5.8.10. Let Y be a variety, and let $(f_i)_{i \in I}$ be a finite family of morphic functions on Y. Let X be the set of $x \in Y$ such that $f_i(x) = 0$ for every i. Assume the following hypothesis:

(J) For every $x \in X$, the differentials $d_x f_i$ are linearly independent in $T'_x(Y)$.

Then X is a closed subvariety of Y, the tangent space $T_x(X)$ is the subspace of $T_x(Y)$ formed by the α such that $\alpha f_i = 0$ for every i in I. Moreover, the codimension of X in Y is equal to Card (I) at each of its points.

5.8.11. (Simple points of an ideal). Let $\mathfrak{a}$ be an ideal of the polynomial algebra $K[X_1, \dots, X_n]$. A point $x = (x_1, \dots, x_n)$ of $\mathbf{K}^n$ is called a zero of $\mathfrak{a}$ if $f(x) = 0$ for all $f \in \mathfrak{a}$. If $x \in \mathbf{K}^n$, denote by $\mathbf{S}_x$ the subalgebra of $K(X_1, \dots, X_n)$ formed by the fractions f/g , with $f, g \in K[X_1, \dots, X_n]$, and $g(x) \neq 0$; denote by $\mathfrak{a}_x$ the ideal of $\mathbf{S}_x$ generated by $\mathfrak{a}$ in $\mathbf{S}_x$. A point x is called a simple zero of $\mathfrak{a}$ if it is a zero of $\mathfrak{a}$ and if the following condition is satisfied:

(S) There exists a finite sequence $(f_1, \dots, f_m)$ of elements of $\mathfrak{a}$ which generate the ideal $\mathfrak{a}_x$ and whose differentials at x are linearly independent. (This condition is equivalent to saying that the local ring $\mathbf{S}_x/\mathfrak{a}_x$ is regular (Alg. Comm., to appear).)

Let Z (resp. Z_s) be the set of zeros (resp. simple zeros) of $\mathfrak{a}$. The set Z is closed in K^n , Z_s

is open in Z , and Z_s is a subvariety of K^n . If $x \in Z_s$, the ideal $K[X_1, \dots, X_n] \cap \mathfrak{a}_x$ consists of the polynomials f vanishing on a neighborhood of x in Z_s .

Let $\bar{\mathfrak{a}}$ be the ideal of polynomials vanishing on Z . If K is algebraically closed, the set of simple zeros of $\mathfrak{a}$ is dense in Z .

5.8.12. Let X be a variety and L be the set of pairs (x, Z) where x is a point of X and Z a subvariety of X containing x . Given two pairs $\pi = (x, Z)$ and $\pi' = (x', Z')$, we denote by $R\{\pi, \pi'\}$ the relation:

”We have $x = x'$ and there exists a neighborhood U of x such that $U \cap Z = U \cap Z'$. Then R is an equivalence relation on L ; we denote by $\gamma_x(Z)$ the equivalence class of the pair $(x, Z) \in L$. On the set $J = L/R$, there exists one and only one variety structure satisfying the following condition:

For every subvariety Z of X , the map $x \mapsto \gamma_x(Z)$ from Z into J is an isomorphism of Z onto an open subvariety of J .

We say that J is the variety of germs of subvarieties of X (cf. Top. Gén., chap. I, 4^e ed., § 6, no. 10).

The map $\rho : J \rightarrow X$ defined by $\rho(\gamma_x(Z)) = x$ is an immersion; it is called the canonical immersion of J to X .

If X is a separated analytic variety of finite dimension, so is J .

5.9 Submersions and quotient varieties

5.9.1. Let $f : X \rightarrow Y$ be a morphism of varieties, a a point of X and let us set $b = f(a)$. The following conditions are equivalent:

- (i) The linear map $T_a(f)$ is surjective, and its kernel admits a topological complement in $T_a(X)$.
- (ii) There exists a chart (U, φ, E) of X at a , a chart (V, ψ, F) of Y at b and a surjective continuous linear map u from E to F such that

$$f(U) \subset V, \quad \psi \circ f = u \circ \varphi$$

and that the kernel of u admits a topological supplement in E .

- (iii) There exist an open neighborhood U of a , an open neighborhood V of b containing $f(U)$ and a morphism g from U into a variety Z such that the map (f, g) from U into $V \times Z$ is an isomorphism.
- (iv) There exists an open neighborhood V of b and a morphism s from V to X such that $s(b) = a$ and $f(s(y)) = y$ for all y in V ("local section").

When X and Y are finite-dimensional, the preceding conditions are equivalent to the following condition:

- (v) There exist an open neighborhood U of a , an open neighborhood V of b containing $f(U)$, and a coordinate system $(\eta^1, \dots, \eta^n)$ on V such that the functions $\eta^i \circ f$ on U be part of a coordinate system on U .

If properties (i) through (iv) are satisfied, one says that f is a submersion at a . The set of points where f is a submersion is open in X ; if this open set is all of X , then f is said to be a submersion.

5.9.2. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms. If f and g are submersions, the same holds for $g \circ f$. Conversely, if $g \circ f$ is a submersion and if f is surjective, then g is a submersion.

5.9.3. If $f : X \rightarrow Y$ and $f' : X' \rightarrow Y'$ are submersions, $f \times f'$ is a submersion.

5.9.4. A submersion $f : X \rightarrow Y$ is an open mapping (cf. Top. Gen., chap. I, 4^e ed., § 5, no. 1); in particular, the equivalence relation R defined by f is open, f defines, by passing to the quotient, a homeomorphism from X/R onto $f(X)$, and $f(X)$ is open in Y .

5.9.5. Let R be an equivalence relation on a variety X . We say that R is regular if there exists on the quotient space X/R a variety structure such that the canonical projection $p : X \rightarrow X/R$ is a submersion; this variety structure is then unique; it is a quotient of that of X (Sets, ch. IV, § 2, no. 6): in other words, for a morphism g from X/R into a variety Y to exist, it is necessary and sufficient that $g \circ p$ be a morphism from X to Y .

Let $C \subset X \times X$ be the graph of R . For R to be regular, it is necessary and sufficient that the following two conditions be satisfied:

(i) C is a submanifold of $X \times X$.

(ii) The map pr_1 from C into X is a submersion.

(ii) also means that if a and b are congruent modulo $\mathbf{R}$ there exists an open neighborhood U of a and a morphism s from U into X such that $s(a) = b$ and $s(x)$ is congruent to x modulo $\mathbf{R}$ for all $x \in U$.

Suppose R is regular. For the quotient variety X/R to be separated, it is necessary and sufficient that the graph of R be closed in $X \times X$.

5.9.6. Let X and X' be two varieties, R and R' regular equivalence relations on X and X' , and let $f : X \rightarrow X'$ be a morphism compatible with the relations R and R' . The map $\bar{f} : X/R \rightarrow X'/R'$ deduced from f by passing to quotients is then a morphism.

5.9.7. (“Transitivity of quotients”) Let R and S be two equivalence relations on a variety X such that R implies S , and let S/R be the quotient equivalence relation on X/R . Suppose that R is regular. Then, for S to be regular, it is necessary and sufficient that S/R be regular; if this is the case, the canonical bijection

$$(X/R)/(S/R) \rightarrow X/S$$

is an isomorphism of varieties.

5.9.8. (“Products of quotients”) Let $(X_i)_{i \in I}$ be a finite family of varieties, each equipped with a regular equivalence relation R_i . Let $X = \prod_{i \in I} X_i$ and let R be the equivalence relation on X that is the product of the R_i (cf. Top. Gen., chap. I, 4th ed., § 5, no. 3, cor. of prop. 8). Then R is regular, and the canonical bijection from X/R onto $\prod_{i \in I} (X_i/R_i)$ is an isomorphism of varieties.

5.10 Subimmersions

5.10.1. Let $f : X \rightarrow Y$ be a morphism of manifolds and let Γ be the graph of f . The map $j : x \mapsto (x, f(x))$ is an immersion of X into $X \times Y$ whose image is the submanifold Γ , and $f = pr_2 \circ j$ is the composite of the immersion j followed by the submersion pr_2 .

Let $a \in X$. We say that f is a subimmersion at a if there exists an open neighborhood U of a

, a manifold Z , a submersion s of U into Z and an immersion i of Z into Y such that $f|_U = i \circ s$. The set of points of X where f is a subimmersion is an open subset of X ; if this open set is X itself, we say that f is a subimmersion.

5.10.2. Immersions and submersions are subimmersions. If $f : X \rightarrow Y$ is a subimmersion, $g : Y \rightarrow Z$ an immersion, and $h : W \rightarrow X$ a submersion, then $g \circ f \circ h$ is a subimmersion.

If f and f' are subimmersions, $f \times f'$ is a subimmersion.

5.10.3. For $f : X \rightarrow Y$ to be a subimmersion at a point a of X , it is necessary and sufficient that there exist a chart (U, φ, E) of X at a , a chart (V, ψ, F) of Y at the point $f(a)$, and a continuous linear map g of E to F such that:

(i) $f(U) \subset V$, $g(\varphi(U)) \subset \psi(V)$ and $f|_U = \psi^{-1} \circ g \circ \varphi$;

(ii) the kernel (resp. image) of g is a closed subspace of E (resp. F) admitting a topological supplement.

5.10.4. Let X and Y be two finite-dimensional manifolds. For a morphism f from X to Y to be a subimmersion at a point a of X , it is necessary and sufficient that there exist a coordinate system $(\xi^1, \dots, \xi^m)$ of X at a , a coordinate system $(\eta^1, \dots, \eta^n)$ of Y at $f(a)$ and an integer $k \leq \inf(m, n)$ such that

$$\eta^i \circ f = \xi^i \quad \text{for } 1 \leq i \leq k$$

$$\eta^i \circ f = 0 \quad \text{for } k < i \leq n.$$

5.10.5. Let $f : X \rightarrow Y$ be a subimmersion. For every $b \in Y$, $f^{-1}(b)$ is a subvariety of X ; the subspace of $T_a(X)$ tangent to the submanifold $f^{-1}(b)$ at the point $a \in f^{-1}(b)$ is the kernel of $T_a(f)$.

5.10.6. ("Constant rank theorem") Let $f : X \rightarrow Y$ be a morphism of manifolds and let $a \in X$. If f is a subimmersion at a , one has $\text{rg}_x f = \text{rg}_a f$ for x near a .

Conversely, suppose the field K has characteristic zero. Let (U, φ, E) be a chart of X at a and let (V, ψ, F) be a chart of Y at $f(a)$, with $f(U) \subset V$. Set $g = \psi \circ f \circ \varphi^{-1}$. If there

exist a closed vector subspace E_1 of E and a closed vector subspace F_1 of F such that for every $x \in U$, the subspace E_1 (resp. F_1) is a topological supplement of the kernel (resp. image) of the derivative $Dg(\varphi(x))$ of g at the point $\varphi(x)$, then f is a subimmersion at a .

If K has characteristic zero (resp. $K = \mathbb{R}$ or $\mathbb{C}$) and if Y is finite-dimensional (resp. $rg_{af} < +\infty$), then f is a subimmersion at a if and only if $rg_{xf} = rg_{af}$ for every x sufficiently close to a .

If K has characteristic zero (resp. $K = \mathbb{R}$ or $\mathbb{C}$) and Y (resp. X) is finite-dimensional, the set of points $x \in X$ where f is a subimmersion is a dense open subset of X .

5.10.7. ("Canonical factorization of a subimmersion") Every subimmersion is the composite of a submersion and an immersion. Precisely, let $f : X \rightarrow Y$ be a subimmersion, and let J be the manifold of germs of submanifolds of Y (5.8.12). Let x be in X and $y = f(x)$; there exist open neighborhoods U of x such that $f|_U$ is a submersion of U onto a submanifold Z of Y ; the element $\gamma_y(Z)$ of J depends only on x and not on the chosen neighborhood U , and if we denote it $\lambda(x)$, the map λ is a submersion from X into J ; if we denote ρ the canonical immersion of J into Y , we have $f = \rho \circ \lambda$. If f is an immersion, the morphism λ from X into J is étale.

If g is a surjective submersion from X onto a variety Z and h is a morphism from Z to Y such that $f = h \circ g$, there exists a unique submersion μ from Z into J such that $\lambda = \mu \circ g$.

5.11 Fibered products and inverse images

5.11.1. Let F be a Banach space, $(E_i)_{i \in I}$ a finite family of Banach spaces and $\mathbf{f} = (f_i)_{i \in I}$ a family of continuous linear maps f_i of E_i into F . Let E be the product of the E_i and let f be the continuous linear map from E into F^I which is the product of the f_i . Finally, let D be the closed subspace of F^I formed by the $(y_i)_{i \in I}$ such that y_i is independent of i (the "diagonal" of F^I). The family $\mathbf{f}$ is said to be transversal if the continuous linear map obtained by composing f with the canonical projection of F^I onto the quotient F^I/D is surjective and if its kernel $f^{-1}(D)$ admits a topological complement.

If the spaces E_i and F are all finite-dimensional, the family $\mathbf{f}$ is transversal if and only if one has:

$$\text{codim} \left(\bigcap_i \text{Im } f_i \right) = \sum_i \text{codim} (\text{Im } f_i)$$

If $I = \{1, 2\}$, the pair (f_1, f_2) is transversal if and only if the map

$$f_1 + f_2 : \mathbf{E}_1 \oplus \mathbf{E}_2 \rightarrow \mathbf{F}$$

is surjective and if its kernel admits a topological supplement (if E_1 and E_2 are of finite dimension, it is equivalent to say that

$$\text{Im} f_1 + \text{Im} f_2 = \mathbf{F}).$$

5.11.2. Let S be a variety, $(X_i)_{i \in I}$ a finite family of varieties and $\mathbf{f} = (f_i)_{i \in I}$ a family of morphisms f_i of X_i into S . Let P be the subset of the product X of the X_i consisting of the points $(x_i)_{i \in I}$ such that $f_i(x_i)$ is independent of i . Let $x \in P$ and let $y = f_i(x_i) \in S$. One says that the family $\mathbf{f}$ is transversal at the point x of P if the maps $T_x(f_i)$ form a transversal family of continuous linear maps with values in the Banach space $T_y(S)$. We say that $\mathbf{f}$ is transversal if it is transversal at every point of P .

If $\mathbf{f}$ is transversal, then P is a submanifold of X , called the fibered product of the family of X_i above S (relative to the family $\mathbf{f}$), which we denote by $\prod_{i \in I} X_i$ (or more simply $X_1 \times_S X_2$ when $I = \{1, 2\}$ for example). For every point $x = (x_i)$ of P , the tangent space $T_x(\prod_S X_i)$ is the subspace of $\prod T_x(X_i)$ consisting of tangent vectors $t = (t_i)$ such that $T_{x_i}(f_i).t_i$ is independent of the index i .

If $f_1 : X_1 \rightarrow Y$ is a submersion and $f_2 : X_2 \rightarrow Y$ a morphism, the pair (f_1, f_2) is transversal.

5.11.3. ("Universal property of fibered products") Let $\mathbf{f} = (f_i)_{i \in I}$ be a transversal family of morphisms $f_i : X_i \rightarrow S$ and let P be the fibered product of the X_i over S ; for every $i \in I$, one denotes by π_i the morphism from P into X_i obtained by restricting to P the projection of X onto X_i ; then $f_i \circ \pi_i$ is a morphism from P to S independent of i . Let T be a variety, and let $g_i : T \rightarrow X_i$ be morphisms of varieties such that $f_i \circ g_i$ is a morphism from T to S independent of $i \in I$; there then exists a unique morphism h from T to P such that $g_i = \pi_i \circ h$ for all $i \in I$.

5.11.4. ("Associativity of the fibered product") Let $\mathbf{f} = (f_i)_{i \in I}$ be a finite family of morphisms of varieties $f_i : X_i \rightarrow S$, and let $(J_\lambda)_{\lambda \in \Lambda}$ be a partition of I . Suppose that, for every λ in Λ , the family $\mathbf{f}_\lambda = (f_i)_{i \in J_\lambda}$ is transversal, and we denote by P_λ the fiber product of this family;

for every point $x = (x_i)_{i \in J_\lambda}$ of P_λ , the element $f_i(x_i)$ of S is independent of $i \in J_\lambda$ and will be denoted by $u_\lambda(x)$;

then u_λ is a morphism from P_λ into S . For the family $\mathbf{u} = (u_\lambda)$ to be transversal, it is necessary and sufficient that the family $\mathbf{f}$ be so. The canonical bijection of $\prod_{\lambda \in \Lambda} \prod_{i \in J_\lambda} X_i$ onto $\prod_{i \in I} X_i$ then gives by restriction an isomorphism from the fibered product $\prod_{\lambda \in \Lambda} S^{P_\lambda}$ to the fibered product $\prod_{i \in I} S^{X_i}$.

5.11.5. Let S be a variety. One calls a variety over S a variety X equipped with a morphism $\lambda : X \rightarrow S$. Let (X, λ) be a variety over S and $f : S' \rightarrow S$ a morphism of varieties such that the pair (f, λ) is transversal. We will then denote $f^*(X)$ the variety $S' \times_S X$, endowed with the morphism $f^*(\lambda) : S' \times_S X$ defined by $f^*(\lambda)(s', x) = s'$. We say that $f^*(X)$ is deduced from X by base change from S to S' along f . If λ is a submersion (resp. an immersion, a subimmersion, an étale morphism), the same holds for $f^*(\lambda)$.

5.11.6. Let $f : X \rightarrow Y$ be a morphism of varieties and let W be a subvariety of Y ; let i be the canonical injection of W into Y . We say that f is transverse to W at a point $x \in f^{-1}(W)$ if the pair (f, i) is transversal at the point $(x, f(x))$ of $X \times W$. For this, it is necessary and sufficient that the following conditions be satisfied:

- (i) the tangent space $\mathbf{T}_{f(x)}(\mathbf{Y})$ is the sum of $\mathbf{T}_{f(x)}(\mathbf{W})$ and the image of $\mathbf{T}_x(f)$;
- (ii) the inverse image $\mathbf{T}_x(f)^{-1}(\mathbf{T}_{f(x)}(\mathbf{W}))$ of the tangent space to W at $f(x)$ admits a topological complement.

One says that f is transverse to W if it is transverse to W at every point of $f^{-1}(W)$.

For a morphism from X to Y to be a submersion, it suffices that it be transversal to every point of Y , and it is necessary that it be transversal to every submanifold of Y . For a finite family of morphisms $f_i : X_i \rightarrow S$ to be transversal, it is necessary and sufficient that the morphism g from $\prod_{i \in I} X_i$ to S^I defined by $g((x_i)_{i \in I}) = (f_i(x_i))_{i \in I}$ be transverse to the diagonal of S^I .

5.11.7. Suppose that the morphism $f : X \rightarrow Y$ is transverse to the subvariety W of Y . Then $f^{-1}(W)$ is a submanifold of X , and for x in $f^{-1}(W)$ the subspace of $\mathbf{T}_x(X)$ tangent to $f^{-1}(W)$ is the inverse image under $\mathbf{T}_x(f)$ of the subspace $\mathbf{T}_{f(x)}(\mathbf{W})$. By passing to the quotient, the linear map $\mathbf{T}_x(f)$ defines an isomorphism of topological vector spaces from the transverse space to

$f^{-1}(W)$ at x onto the transverse space to W at $y = f(x)$. If W has codimension d at y in Y , the submanifold $f^{-1}(W)$ of X has codimension d at every point of $f^{-1}(W)$. Finally, the map $x \mapsto (x, f(x))$ is an isomorphism of varieties from $f^{-1}(W)$ to the fibered product $X \times_Y W$.

5.11.8. Let Y_1 and Y_2 be two submanifolds of a variety X , and let ι_j be the injection of Y_j into X . One says that Y_1 and Y_2 are transverse if the pair (ι_1, ι_2) is transverse; this is equivalent to supposing that, for every point x of $Y_1 \cap Y_2$, the subspaces $T_x(Y_1)$ and $T_x(Y_2)$ of $T_x(X)$ have $T_x(X)$ as their sum, and that their intersection admits a topological supplement in $T_x(X)$. Under these conditions, $Y_1 \cap Y_2$ is a submanifold of X and for every x in $Y_1 \cap Y_2$, we have:

$$T_x(Y_1 \cap Y_2) = T_x(Y_1) \cap T_x(Y_2);$$

if moreover Y_i has codimension d_i at x , then $Y_1 \cap Y_2$ has codimension $d_1 + d_2$ at x .

5.11.9. Let f and g be two morphisms from a variety X to a variety Y . If the morphism $(f, g) : X \rightarrow Y \times Y$ is transverse to the diagonal of $Y \times Y$, the subset N of X consisting of the points x such that $f(x) = g(x)$ is a submanifold of X ; it is called the kernel of the double arrow.

$$f, g : X \rightarrow Y.$$

5.12 Varieties of groups

5.12.1. Let G be a group. A manifold structure on G is said to be compatible with the group structure of G if the map $(x, y) \mapsto xy$ of $G \times G$ into G is a morphism. The map $x \mapsto x^{-1}$ is then a morphism from G to itself. The set G , equipped with its group structure and its manifold structure, is called a group manifold ("of class C^r " if one wishes to be precise), or again a Lie group. If $r = \omega$ we also say analytic group. If $K = \mathbb{R}$ (resp. $\mathbb{C}$, $\mathbb{Q}_p$), one also says real (resp. complex, p-adic) Lie group. Every group variety is pure. A homomorphism of group varieties (or simply homomorphism) is any map from one group variety to another that is both a group homomorphism and a morphism of varieties.

If G is a group variety, the topological structure underlying the variety structure of G makes it a metrizable and complete topological group; it is locally compact if K is locally compact and G is of finite dimension.

5.12.2. Examples:

- (i) If V is a Banach space, the canonical manifold structure of V is compatible with its commutative group structure.
- (ii) Let A be a complete normed K -algebra with a unit element, and let A^* be the group of invertible elements of A . It is an open subspace of A , and the manifold structure induced on this open set by the canonical manifold structure of the K -vector space A is compatible with the group structure on A^* . In particular, taking A to be the algebra $\mathcal{L}(E)$ of continuous endomorphisms of a Banach space E ; the group A^* consists of the automorphisms of the topological vector space E ; we denote by $\mathbf{GL}(E)$ the group variety thus defined. When $A = M_n(K)$, we obtain a group variety structure on $\mathbf{GL}(n, K)$.
- (iii) If $G_1, \dots, G_n$ are group varieties, the product group $G = G_1 \times \dots \times G_n$ is a group variety, when endowed with the product variety structure of those of G_i .

5.12.3. Let G be a group variety, let H be a topological group, and let $f : H \rightarrow G$ be a continuous homomorphism satisfying condition (QR) of No. 5.8.1. The inverse image manifold structure of that of G by f then makes H a group variety. This applies in particular when H is a subgroup of G which is a subvariety (resp. quasi-subvariety); such a subgroup is called a group subvariety (resp. group quasi-subvariety) of G ; it is necessarily closed in G .

If $H_i (i = 1, \dots, n)$ is a group subvariety of G_i , then $H_1 \times \dots \times H_n$ is a group subvariety of $G_1 \times \dots \times G_n$.

5.12.4. Let G be a group variety and let H be a group subvariety of G . The equivalence relation $x^{-1}y \in H$ is regular, which allows one to endow the space G/H of left classes xH with a variety structure called the quotient of that of G . The canonical map $(g, x) \mapsto g.x$ of $G \times (G/H)$ into G/H is a morphism. Analogous results hold for the homogeneous space $H \backslash G$ of right classes Hx . If H is normal, the variety structure of G/H is compatible with its group structure.

5.12.5. Let G be a group variety and let X be a variety. A left action of the group G on the variety X is a morphism $(s, x) \mapsto sx$ of $G \times X$ into X such that

$$\begin{aligned} s(tx) &= (st)x & \text{if } s, t \in G, x \in X \\ ex &= x & \text{if } x \in X (e \text{ being the identity element of } G). \end{aligned}$$

One also says that the group variety G operates on the left on X ; right action laws are defined analogously.

Let $x \in X$ and let G_x be its stabilizer in G . Suppose that the map $g \mapsto g \cdot x$ is a subimmersion (a hypothesis automatically satisfied if the characteristic of K is 0 and X is of finite dimension). Then G_x is a group subvariety of G and the map from G/G_x to X obtained by passing to the quotient from $g \mapsto g \cdot x$ is an immersion. If moreover the orbit $G \cdot x$ of x is locally closed and if the topology of G admits a countable base, $G \cdot x$ is a subvariety of X and the map $G/G_x \rightarrow G \cdot x$ is an isomorphism of varieties.

5.13 Weakening of structure

Throughout this section, the letters r, s, r', s' denote either integers ≥ 1 , or one of the symbols ∞ and ω . One assumes that $K = \mathbb{R}$.

5.13.1. Let $r \leq s$, and X be a variety of class C^s . There exists on the topological space X a manifold structure of class C^r and only one such that every chart of X for the given structure is a chart of X for this new structure. Let X_r be the manifold of class C^r thus defined. It is said to be deduced from X by weakening the manifold structure of X , or again that its manifold structure is underlying that of X . The notion of weakening is transitive: if $r' \leq r$, one has $X_{r'} = (X_r)_{r'}$; it commutes with products: if Y is of class C^s , one has $(X \times Y)_r = X_r \times Y_r$; one has an analogous result for $X \times_s Y$ under the hypotheses of no. 5.11.2.

Let $a \in X$ and let c be a chart of X at a ; c is also a chart of X_r at a . One deduces from this (5.5.1) the isomorphisms

$$\theta_c : E \rightarrow T_a(X), \quad \theta'_c : E \rightarrow T_a(X_r),$$

whence an isomorphism $\theta'_c \circ \theta_c^{-1} : T_a(X) \rightarrow T_a(X_r)$. This isomorphism is independent of the choice of c ; it allows one to identify the tangent spaces to X and to X_r at a .

5.13.2. Let X (resp. X') be a manifold of class C^s (resp. $C^{s'}$) and let r satisfy $r \leq \inf(s, s')$

. A map $f : X \rightarrow X'$ is said to be of class C^r if it is a morphism of X_r to X'_r ; such a map is of class $C^{r'}$ for all $r' \leq r$. Moreover, the tangent linear map to $f : X_r \rightarrow X'_r$ at a point $a \in X$ coincides with the tangent linear map to f considered as a morphism of $X_{r'}$ to $X'_{r'}$.

One will most often denote by the same symbol the manifolds X and X_r ; thus, if X is of class C^s , the expression “a submanifold of X of class C^r ” ($r \leq s$) means “a submanifold of X_r ”.

5.14 Restriction of the base field

In this number, one is given two complete non-discrete valued commutative fields K and L , as well as an isomorphism σ of the valued field K onto a subfield of L . If E is a Banach space over L , one denotes by $\sigma_*(E)$ the vector space over K obtained by restriction of scalars (cf. Alg., chap. II, 3rd ed., § 8); the given topology on E is compatible with the structure of K -vector space, and $\sigma_*(E)$ is a Banach space over K .

5.14.1. Let X be an analytic manifold over L and let $c = (U, \varphi, E)$ be a chart of X . The map φ is a bijection of U onto an open subset of $\sigma_*(E)$ and the triple $c_\sigma = (U, \varphi, \sigma_*(E))$ is a chart of X . There exists on X a structure of analytic manifold over K and only one for which c_σ is a chart for every chart c of the L -analytic manifold X . One denotes by X_σ the analytic manifold over K thus obtained and one says that X_σ is obtained from X by restriction of scalars (from L to K by means of σ). The topological space underlying X_σ is the same as that underlying X .

5.14.2. Examples:

- (a) One takes $K = \mathbb{R}$, $L = \mathbb{C}$, σ being the canonical injection of $\mathbb{R}$ into $\mathbb{C}$. Thus, every complex analytic manifold is canonically equipped with a structure of real analytic manifold; this real analytic structure itself defines differential structures of class C^r for every r .
- (b) One takes $K = \mathbb{C}$, $L = \mathbb{C}$, σ being the conjugation $x \mapsto \bar{x}$. One thus associates to every complex analytic manifold X a complex analytic manifold $\bar{X}$, called the conjugate of X . If f is a complex-valued function, defined on an open subset U of X , f is analytic for the structure of $\bar{X}$ if and only if the conjugate function $\bar{f}$ is analytic for the structure of X . The manifolds X and $\bar{X}$ define by restriction of scalars the same real analytic manifold.

5.14.3. Let X be an analytic manifold over K , Y an analytic manifold over L . A map $f : X \rightarrow Y$ is said to be σ -analytic if it is a morphism of X into Y_σ . If $K \subset L$, σ is the injection of K into L , and X is an analytic manifold over L , one calls K -analytic map a map σ -analytic from X_σ into Y .

5.14.4. Let V be a Banach space over L . One has $V_\sigma = \sigma_*(V)$; the canonical analytic variety structure over K of $\sigma_*(V)$ is deduced by restriction of scalars from the canonical analytic variety structure over L of V .

5.14.5. Let X be an analytic variety over L , and let $a \in X$. Let c be a chart of X at a ; then c_σ is a chart of X_σ at a , and one deduces from it isomorphisms

$$\theta_c : E \rightarrow T_a(X), \quad \theta_{c_\sigma} : \sigma_*(E) \rightarrow T_a(X_\sigma)$$

whence an isomorphism $\theta_{c_\sigma} \circ \sigma_*(\theta_c)^{-1}$ of $\sigma_*(T_a(X))$ onto $T_a(X_\sigma)$; this isomorphism does not depend on the choice of c ; it allows one to identify $T_a(X_\sigma)$ with $\sigma_*(T_a(X))$ and even with $T_a(X)$ by abuse of notation.

If f is an analytic map on L from X into a manifold Y , the tangent linear map to f at a (f being regarded as a morphism of X_σ to Y_σ) is equal to $\sigma_*(T_a(f))$.

5.14.6. Let X and Y be two analytic varieties over L , and let $f : \mathbf{X}_\sigma \rightarrow \mathbf{Y}$ be a continuous σ -analytic map. Suppose that the characteristic of K is 0. Then, for f to be analytic over L , it is necessary and sufficient that, for every $a \in \mathbf{X}$, the map $T_a(f)$ be L -linear.

When $K = \mathbb{R}$, $L = \mathbb{C}$ (case (a) of no. 5.14.2), one has a more precise result: if X and Y are finite-dimensional and if $f : X \rightarrow Y$ is a map of class C^1 whose tangent map at every point of X is $\mathbb{C}$ -linear, then f is complex analytic.

5.14.7. Let X be a complex analytic manifold and g a map from X into itself satisfying the conditions:

(i) One has $g \circ g = \text{Id}_X$.

(ii) The map g is an isomorphism of the analytic variety X on the conjugate variety $\bar{X}$ (5.14.2).

The set X_0 of points x of X such that $g(x) = x$ is a closed analytic submanifold of the real analytic manifold underlying X . For $x \in X_0$, one has $\mathbf{T}_x(\mathbf{X}) = \mathbf{T}_x(\mathbf{X}_0) \oplus i\mathbf{T}_x(\mathbf{X}_0)$.

Let U be a connected open subset of X , and let f and g be two complex analytic maps from U into a complex separated locally convex space or into a complex separated analytic manifold. If f and g coincide on a nonempty subset of $U \cap X_0$, open in X_0 , then $f = g$.

Suppose X_0 is paracompact. If f is a real analytic map from X_0 into a separated locally convex space or into a separated complex analytic manifold, there exists an open neighborhood U of X_0 in X and a complex analytic map from U into the space of values of f , extending f . Any two such extensions coincide on a neighborhood of X_0 in X .

Suppose X is of finite dimension. For every point $a \in X_0$, there exists a system of (complex) coordinates $\zeta^1, \dots, \zeta^n$ in an open neighborhood U of a , such that $\zeta^i \circ g = \bar{\zeta}^i$ for $1 \leq i \leq n$; the restriction ξ^i of ζ^i to $U \cap X_0$ then takes real values and $(\xi^1, \dots, \xi^n)$ is a system of coordinates at a of the real analytic manifold X_0 .

5.14.8. For every paracompact real analytic manifold Y , there exists a pair (X, g) consisting of a complex analytic manifold X and a map g from X into itself satisfying conditions (i) and (ii) of 5.14.7, and an isomorphism f from Y onto X_0 . We say that X (endowed with f and g) is a complexification of Y .

6 Fibrations ¹

6.1 Fibrations

6.1.1. A fibration of class C^r , or simply fibration, is a triple (X, B, π) where B and X are varieties of class C^r and π is a morphism from X to B , having the following property:

- (F) For every $x \in B$, there exists an open neighborhood U of x , a manifold F , and an isomorphism φ of $\pi^{-1}(U)$ onto $U \times F$ such that $\pi(\varphi^{-1}(x, y)) = x$ for all $x \in U$ and every $y \in F$.

If $\lambda = (X, B, \pi)$ is a fibration, one calls X the space of λ , B the base of λ , and π the projection of λ . The map π is a submersion; in particular $\pi(X)$ is open in B and if R denotes the equivalence relation defined by π on X , the canonical map from X/R to $\pi(X)$ is an isomorphism. For every $x \in B$ the inverse image $\pi^{-1}(x)$ is a closed submanifold of X , called the fiber of x ,

and denoted X_x .

6.1.2. Examples:

- (a) If B and F are two manifolds, the triple $(B \times F, B, \text{pr}_1)$ is a fibration whose fibers are canonically isomorphic to F .
- (b) If $\lambda = (X, B, \pi)$ and $\lambda' = (X', B', \pi')$ are fibrations,

$$\lambda \times \lambda' = (X \times X', B \times B', \pi \times \pi')$$

is a fibration; it is called the product of λ and of λ' .

- (c) If $\lambda = (X, B, \pi)$ and $\lambda' = (X', B, \pi')$ are fibrations with the same base, $\lambda \times_B \lambda' = (X \times_B X', B, \pi \times_B \pi')$ is a fibration; it is called the product of λ and of λ' above B , or also the fibered product of λ and of λ' .

6.1.3. Let $\lambda = (X, B, \pi)$ and $\lambda' = (X', B', \pi')$ be two fibrations. One calls a morphism of λ to λ' any pair (f, g) , where f is a morphism of B to B' , and g a morphism of X to X' such that $\pi' \circ g = f \circ \pi$. When $B = B'$ and $f = \text{Id}_B$, one says that g is a B -morphism of λ to λ' ; if g is an isomorphism of X onto X' , g^{-1} is a B -morphism of λ' to λ , and one says that g is a B -isomorphism from λ onto λ' ; for this to be the case, it is necessary and sufficient that, for every $x \in B$, the map $g_x : X_x \rightarrow X'_x$ induced by g be an isomorphism.

6.1.4. The fibration $(B \times F, B, \text{pr}_1)$ is called the trivial fibration with base B and fiber F . An isomorphism of a fibration λ onto a trivial fibration is called a trivialization of λ .

6.1.5. Let $\lambda = (X, B, \pi)$ be a fibration, and let $f : B' \rightarrow B$ be a morphism. Let π' be the canonical morphism of $B' \times_B X$ to B' . The triple $(B' \times_B X, B', \pi')$ is a fibration, called the inverse image of λ by f or the fibration deduced from λ by base change of B to B' along f , and denoted $B' \times_B \lambda$, or again $f^* \lambda$. If f' denotes the canonical map of $B' \times_B X$ to X , the pair (f, f') is a morphism of $B' \times_B \lambda$ to λ ; it enjoys the following universal property: if (f, g) is a morphism of a fibration λ' with base B' into the fibration λ , there exists a unique B' -morphism $\varphi : \lambda' \rightarrow B' \times_B \lambda$ such that $(f, g) = (f, f') \circ \varphi$.

When B' is a submanifold of B , and f is the canonical injection of B' into B , $B' \times_B X$ is

identified with the submanifold $\pi^{-1}(B')$ of X , and π' is identified with the restriction of π to $\pi^{-1}(B')$; the inverse image of λ by f is then called the fibration induced by λ on B' .

6.1.6. If $\lambda = (X, B, \pi)$ is a fibration, one calls a morphic section (or simply section) of λ every morphism $s : B \rightarrow X$ such that $\pi \circ s = \text{Id}_B$.

6.2 Principal fibrations

6.2.1. Let B be a manifold and G a group manifold. A principal fibration with base B and structure group G is called a quadruple $\lambda = (P, G, B, \pi)$ where P is a variety on which G acts on the right by $(x, g) \mapsto x \cdot g$ (cf. no. 5.12.5), and where π is a morphism from P to B , these data being subject to the following axiom:

(P) For every $b \in B$ there exists an open neighborhood U of b and an isomorphism $f : U \times G \rightarrow \pi^{-1}(U)$ such that we have

$$\pi(f(u, g)) = u \text{ and } f(u, gg') = f(u, g) \cdot g' \text{ if } u \in U \text{ and } g, g' \in G.$$

6.2.2. Let $\lambda = (P, G, B, \pi)$ be a principal fibration. The triple (P, B, π) is a fibration; we have $\pi(P) = B$. The equivalence relation R defined by π on P coincides with the one defined by G ; its graph is none other than the product $P \times_B P$ (cf. no. 5.11.2); it is a submanifold of $P \times P$. The map $(x, g) \mapsto (x, x \cdot g)$ is an isomorphism of $P \times G$ onto $P \times_B P$; the map which assigns to every $(x, y) \in P \times_B P$ the unique element $g \in G$ such that $y = x \cdot g$ is a morphism of $P \times_B P$ to G .

The group G acts properly and freely on P (cf. Gen. Top., chap. III, 3rd ed., § 4). If $x \in P$ and if $b = \pi(x)$, the map $g \mapsto x \cdot g$ is an isomorphism from the variety G onto the fiber over b .

6.2.3. Conversely, let G be a group variety, and let P be a variety on which G acts on the right so as to satisfy the following two conditions:

- (a) G acts properly and freely on P .
- (b) For every $x \in P$, the map $g \mapsto x \cdot g$ is an immersion of G into P .

Then the equivalence relation defined by G in P is regular; if P/G denotes the quotient variety,

and π the canonical projection from P to P/G , the quadruple $(P, G, P/G, \pi)$ is a principal fibration.

When K has characteristic 0 and P is finite-dimensional, condition (b) above is a consequence of condition (a).

6.2.4. The conditions of the preceding number are satisfied in the following two cases:

- (a) P is a group variety and G a subgroup variety acting on P by right translations; the base of the principal fibration thus obtained is the homogeneous space P/G .
- (b) G is a discrete group acting properly and freely on P . The projection $\pi : P \rightarrow P/G$ is then an étale morphism (no. 5.7.6).

6.2.5. Examples:

- (a) Let B be a variety and let G be a group variety. Let G act on $B \times G$ by $(b, g) \cdot g' = (b, gg')$. The quadruple $(B \times G, G, B, pr_1)$ is a principal fibration.
- (b) Let $\lambda = (P, G, B, \pi)$ and $\lambda' = (P', G', B', \pi')$ be two principal fibrations. Let us make $G \times G'$ act on $P \times P'$ by the formula:

$$(x, x') \cdot (g, g') = (x \cdot g, x' \cdot g'), \quad x \in P, x' \in P', g \in G, g' \in G'.$$

The quadruple $\lambda \times \lambda' = (P \times P', G \times G', B \times B', \pi \times \pi')$ is a principal fibration; it is called the product of λ and of λ' .

- (c) Let $\lambda = (P, G, B, \pi)$ and $\lambda' = (P', G', B, \pi')$ two principal fibrations with the same base. The submanifold $P \times_B P'$ of $P \times P'$ is stable under the operations of $G \times G'$, and the quadruple

$$\lambda \times_B \lambda' = (P \times_B P', G \times G', B, \pi \times_B \pi')$$

is a principal fibration; it is called the product of λ and of λ' above B , or equivalently the fiber product of λ and of λ' .

6.3 Morphisms of principal fibrations

6.3.1. Let $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ and $\lambda' = (\mathbf{P}', \mathbf{G}', \mathbf{B}', \pi')$ be two principal fibrations. A morphism of λ to λ' is a triple (f, φ, h) , where $f : \mathbf{P} \rightarrow \mathbf{P}'$ and $h : \mathbf{B} \rightarrow \mathbf{B}'$ are morphisms, $\varphi : \mathbf{G} \rightarrow \mathbf{G}'$ is a homomorphism of group varieties, and where $\pi' \circ f = h \circ \pi$ and $f(x \cdot g) = f(x) \cdot \varphi(g)$

for $x \in \mathbf{P}, g \in \mathbf{G}$. Note that f determines h ; one will often say that (f, φ) , or even simply f is a morphism.

When $B = B'$ and $h = \text{Id}_B$ (resp. when $G = G'$ and $\varphi = \text{Id}_G$), a morphism is called a B-morphism compatible with φ (resp. a G-morphism compatible with h). A morphism that is both a B-morphism and a G-morphism is called a G-B-morphism (here again, one often says "morphism" when there can be no confusion). Every G-B-morphism $f : P \rightarrow P'$ is an isomorphism of the variety P onto the variety P' ; the inverse isomorphism f^{-1} is a G-B-morphism: f is a G-B-isomorphism of P onto P' .

Two principal fiber bundles P and P' with the same base B and the same structural group G are said to be G-B-isomorphic (or simply isomorphic) if there exists a G-B-isomorphism of P onto P' .

6.3.2. The principal fibration $(\mathbf{B} \times \mathbf{G}, \mathbf{G}, \mathbf{B}, \text{pr}_1)$ is called the trivial principal fibration with base B and structural group G . An isomorphism of a principal fibration $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ onto the trivial principal fibration with base B and structural group G is called a trivialization of λ . Every section s of λ defines a trivialization f_s of λ by the formula:

$$f_s^{-1}(b, g) = s(b) \cdot g \quad \text{for } b \in \mathbf{B} \text{ and } g \in \mathbf{G}.$$

Thus we obtain a bijection from the set of sections of λ onto the set of trivializations of λ . Moreover, if s_0 is a section of λ , every section s of λ can be written uniquely in the form $s(b) = s_0(b) \cdot r(b)$, where $r : B \rightarrow G$ is a morphism.

6.3.3. Let $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ be a principal fibration, and let $h : \mathbf{B}' \rightarrow \mathbf{B}$ be a morphism. Let π' (resp. h') be the canonical morphism from $\mathbf{B}' \times_{\mathbf{B}} \mathbf{P}$ to $\mathbf{B}'$ (resp. into $\mathbf{P}$). Let us make $\mathbf{G}$ act on $\mathbf{B}' \times_{\mathbf{B}} \mathbf{P}$ by the formula

$$(b', x) \cdot g = (b', x \cdot g), \quad (b', x) \in \mathbf{B}' \times_{\mathbf{B}} \mathbf{P}, \quad g \in \mathbf{G}.$$

The quadruple $(\mathbf{B}' \times_{\mathbf{B}} \mathbf{P}, \mathbf{G}, \mathbf{B}', \pi')$ is a principal fibration, called the inverse image of λ by h , and denoted $\mathbf{B}' \times_{\mathbf{B}} \lambda$ or also $h^* \lambda$. The map $h' : \mathbf{B}' \times_{\mathbf{B}} \mathbf{P} \rightarrow \mathbf{P}$ is a $\mathbf{G}$ -morphism compatible with h ; it enjoys the following universal property: if f is a $\mathbf{G}$ -morphism compatible with h of a principal fibration λ' with base $\mathbf{B}'$ into the fibration λ , there exists a unique $\mathbf{G}$ - $\mathbf{B}'$ -isomorphism $k : \lambda' \rightarrow \mathbf{B}' \times_{\mathbf{B}} \lambda$ such that $f = h' \circ k$.

When $\mathbf{B}'$ is a subvariety of $\mathbf{B}$, and h is the canonical injection of $\mathbf{B}'$ into $\mathbf{B}$, $\mathbf{B}' \times_{\mathbf{B}} \mathbf{P}$ is identified with the submanifold $\pi^{-1}(\mathbf{B}')$ of $\mathbf{P}$; it is called the principal fiber bundle induced by $\mathbf{P}$ over $\mathbf{B}'$, and it is denoted $\pi^{-1}(\mathbf{B}')$, or equivalently $\mathbf{P}|_{\mathbf{B}'}$. Every $x \in \mathbf{B}$ has an open neighborhood $\mathbf{U}$ such that $\mathbf{P}|_{\mathbf{U}}$ is trivial.

6.4 Construction of principal fibrations by means of cocycles

Let $\mathbf{B}$ be a manifold, $\mathbf{G}$ a group manifold, and let $\mathcal{U} = (\mathbf{U}_i)_{i \in \mathbf{I}}$ an open cover of $\mathbf{B}$.

6.4.1. One calls a cocycle of class C^r on $\mathbf{B}$ with values in $\mathbf{G}$, subordinate to $\mathcal{U}$, a family $(g_{i,j})_{(i,j) \in \mathbf{I} \times \mathbf{I}}$ possessing the following two properties: (1) for every pair $(i,j) \in \mathbf{I} \times \mathbf{I}$, $g_{i,j}$ is a map of class C^r of the open subset $\mathbf{U}_i \cap \mathbf{U}_j$ of $\mathbf{B}$ into $\mathbf{G}$;

(2) for every triple $(i,j,k) \in \mathbf{I}^3$, one has

$$g_{i,k}(x) = g_{i,j}(x).g_{j,k}(x) \quad \text{for every } x \in \mathbf{U}_i \cap \mathbf{U}_j \cap \mathbf{U}_k.$$

Two such cocycles $(g_{i,j})$ and $(g'_{i,j})$ are said to be cohomologous if there exists a family $(h_i)_{i \in \mathbf{I}}$ where, for every $i \in \mathbf{I}$, h_i is a map of class C^r of $\mathbf{U}_i$ into $\mathbf{G}$, such that:

$$g'_{i,j}(x) = h_i(x)^{-1}.g_{i,j}(x).h_j(x) \quad \text{for every } x \in \mathbf{U}_i \cap \mathbf{U}_j. \quad (3)$$

6.4.2. Let $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ be a principal fibration. For every $i \in \mathbf{I}$ let s_i be a section of λ over $\mathbf{U}_i$ (6.3.3). For every pair $(i,j) \in \mathbf{I}^2$ there then exists one and only one morphism $g_{i,j}$ of $\mathbf{U}_i \cap \mathbf{U}_j$ to $\mathbf{G}$ such that:

$$s_j(b) = s_i(b).g_{i,j}(b) \quad \text{for every } b \in \mathbf{U}_i \cap \mathbf{U}_j. \quad (4)$$

The family of $g_{i,j}$ is a cocycle on B with values in G , subordinate to the open covering $\mathcal{U}$. This cocycle is said to be associated with the object $(\lambda, \mathcal{U}, (s_i)_{i \in I})$ and the maps $g_{i,j}$ are called the transition functions of this object.

For $i \in I$ let $x \mapsto (\pi(x), f_i(x))$ be the trivialization defined by the section s_i of $\lambda|_{U_i}$ (6.3.2). For $x \in \pi^{-1}(U_i \cap U_j)$, we have:

$$f_i(x) = g_{i,j}(\pi(x)) \cdot f_j(x). \quad (5)$$

6.4.3. Conversely, let $g = (g_{i,j})$ be a cocycle on B with values in G , subordinate to the covering $\mathcal{U}$. There then exists a principal fibration $\lambda = (P, G, B, \pi)$ and a family of sections $(s_i)_{i \in I}$ of λ above the U_i such that relation (4) is satisfied. Then (5) also holds. If moreover $(\lambda', (s'_i))$ satisfies the same conditions, there exists a unique G - B -isomorphism f of λ onto λ' such that $s'_i = f \circ s_i$ for all $i \in I$. We express this result by saying that $(\lambda, (s_i))$ is determined up to unique isomorphism by the cocycle g .

6.4.4. Let $\lambda = (P, G, B, \pi)$ and $\lambda' = (P', G, B, \pi')$ be two principal fibrations. Let (s_i) (resp. (s'_i)) be a family of sections of λ (resp. λ') over the U_i and let g (resp. g') be the cocycle associated with $(\lambda, \mathcal{U}, (s_i))$ (resp. $(\lambda', \mathcal{U}, (s'_i))$). For λ and λ' to be G - B -isomorphic, it is necessary and sufficient that

the cocycles g and g' be cohomologous. More precisely, for every G - B -isomorphism f from λ onto λ' , there exists a family $(h_i)_{i \in I}$ of morphisms of U_i into G and exactly one such that relation (3) is satisfied and that we have $f \circ s'_i(x) = s_i(x) \cdot h_i(x)$ for all $i \in I$ and all $x \in U_i$; one thus obtains a bijection from the set of G - B -isomorphisms of λ onto λ' onto the set of families $(h_i)_{i \in I}$ satisfying (3).

6.4.5. We return to the notations of 6.4.2. and let $\mathcal{V} = (V_\alpha)_{\alpha \in A}$ be an open covering finer than the open covering $\mathcal{U}$. Let $\tau : A \rightarrow I$ be a map such that $V_\alpha \subset U_{\tau(\alpha)}$ for all $\alpha \in A$. Let s'_α be the restriction to V_α of the section $s_{\tau(\alpha)}$ and let $g' = (g'_{\alpha,\beta})$ be the cocycle subordinate to the open covering $\mathcal{V}$ associated with $(\lambda, \mathcal{V}, (s'_\alpha))$. The transition function $g'_{\alpha,\beta}$ is then the restriction to $V_\alpha \cap V_\beta$ of the transition function $g_{\tau(\alpha), \tau(\beta)}$.

6.5 Fiber bundles associated with a principal fibration

6.5.1. Let $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ be a principal fibration. Let F be a variety on which the group G acts on the left; we denote $(g, y) \mapsto g \cdot y$ the operation law of G on F . The group G acts on the right on $\mathbf{P} \times \mathbf{F}$ by the formula $(x, f) \cdot g = (x \cdot g, g^{-1} \cdot f)$; the equivalence relation defined by G in $\mathbf{P} \times \mathbf{F}$ is regular; the quotient $\mathbf{P} \times^G \mathbf{F} = (\mathbf{P} \times \mathbf{F})/G$ is endowed with a manifold structure.

Let E be a manifold. We say that E is equipped with a structure of fiber bundle associated with λ with typical fiber F when we are given a morphism $\rho : \mathbf{P} \times \mathbf{F} \rightarrow E$ satisfying the following property:

(As) We have $\rho(x \cdot g, g^{-1} \cdot f) = \rho(x, f)$ for $x \in \mathbf{P}, f \in \mathbf{F}, g \in \mathbf{G}$, and the map $\bar{\rho} : \mathbf{P} \times^G \mathbf{F} \rightarrow E$ deduced from ρ by passing to the quotient is an isomorphism of varieties.

It amounts to the same thing to say that $(\mathbf{P} \times \mathbf{F}, \mathbf{G}, E, \rho)$ is a principal fibration. The map ρ (or sometimes the map $\bar{\rho}$) is called the frame map of E , and is denoted $(x, f) \mapsto x \cdot f$; we have

$$(x \cdot g) \cdot f = x \cdot (g \cdot f), \quad \text{for } x \in \mathbf{P}, g \in \mathbf{G} \text{ and } f \in \mathbf{F}.$$

The data of λ and F determines E up to a unique isomorphism; in particular, one may take for E the variety $\mathbf{P} \times^G \mathbf{F}$ itself; it is said to be the associated fiber bundle to λ of fiber type F , and it is denoted $\lambda(F)$.

6.5.2. Let E be a fiber bundle associated with λ and with typical fiber F . There exists a unique morphism π_E from E into $\mathbf{B}$ such that $\pi_E(x, f) = \pi(x)$ if $x \in \mathbf{P}, f \in \mathbf{F}$; the triple $(E, \mathbf{B}, \pi_E)$ is a fibration; if $\mathbf{B}$ and F are separated, then E is separated.

Let $b \in \mathbf{B}$, and let $\mathbf{F}_b = \pi_E^{-1}(b)$; it is a closed submanifold of E . If $x \in \mathbf{P}$ is such that $\pi(x) = b$, let $\theta_x : \mathbf{F} \rightarrow \mathbf{F}_b$ be the map defined by $\theta_x(f) = x \cdot f$; it is an isomorphism of manifolds. Moreover, for every $g \in \mathbf{G}$, one has $\theta_{x \cdot g} = \theta_x \circ \rho_g$, where ρ_g denotes the automorphism $f \mapsto g \cdot f$ of F . Suppose

F is equipped with a structure s of arbitrary species Σ (cf. Ens., chap. IV, §1, n° 4) and suppose that s is invariant under G ; there then exists on $\mathbf{F}_b$ a structure s_b of species Σ and exactly one such that the $\theta_x : F \rightarrow \mathbf{F}_b$ are isomorphisms; one obtains it by transporting s by means of one of the θ_x (loc. cit., no. 5).

If s is a section of $\mathbf{P}$ above an open set $\mathbf{U}$ of $\mathbf{B}$, the map $(b, f) \mapsto s(b) \cdot f$ is an isomorphism

of $\mathbf{U} \times \mathbf{F}$ onto $\pi_{\mathbf{E}}^{-1}(\mathbf{U})$.

6.5.3. Examples:

(a) Let $\lambda = (\mathbf{B} \times \mathbf{G}, \mathbf{B}, \text{pr}_1)$, let $\mathbf{E} = \mathbf{B} \times \mathbf{F}$, and let

$$\rho : (\mathbf{B} \times \mathbf{G}) \times \mathbf{F} \rightarrow \mathbf{E}$$

be the map $(b, g, f) \mapsto (b, g \cdot f)$. We thus obtain on $\mathbf{B} \times \mathbf{F}$ a fiber bundle structure associated with λ with typical fiber $\mathbf{F}$, which is said to be trivial.

(b) Let $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ and $\lambda' = (\mathbf{P}', \mathbf{G}', \mathbf{B}', \pi')$ be two principal fibrations. Let $\mathbf{F}$ (resp. $\mathbf{F}'$) be a variety on which $\mathbf{G}$ (resp. $\mathbf{G}'$) acts on the left; and let $\mathbf{E}$ (resp. $\mathbf{E}'$) be a fiber space associated with λ (respectively to λ') with fiber type $\mathbf{F}$ (respectively $\mathbf{F}'$). The group $\mathbf{G} \times \mathbf{G}'$ acts on $\mathbf{F} \times \mathbf{F}'$ by $(g, g') \cdot (f, f') = (g \cdot f, g' \cdot f')$. The map $(\mathbf{P} \times \mathbf{P}') \times (\mathbf{F} \times \mathbf{F}') \rightarrow \mathbf{E} \times \mathbf{E}'$ is the product of the frame maps $\mathbf{P} \times \mathbf{F} \rightarrow \mathbf{E}$ and $\mathbf{P}' \times \mathbf{F}' \rightarrow \mathbf{E}'$ and endows $\mathbf{E} \times \mathbf{E}'$ with a structure of fiber bundle associated with $\lambda \times \lambda'$ with typical fiber $\mathbf{F} \times \mathbf{F}'$.

(c) With the notations of (b), if one supposes that $\mathbf{B}' = \mathbf{B}$, one defines in the same way on $\mathbf{E} \times_{\mathbf{B}} \mathbf{E}'$ a fiber bundle structure associated with $\lambda \times_{\mathbf{B}} \lambda'$ with typical fiber $\mathbf{F} \times \mathbf{F}'$.

6.5.4. The notations being those of no. 6.5.1, let $h : \mathbf{B}' \rightarrow \mathbf{B}$ be a morphism, let $\lambda' = \mathbf{B}' \times_{\mathbf{B}} \lambda$, and let $\mathbf{E}' = \mathbf{B}' \times_{\mathbf{B}} \mathbf{E}$. If $\mathbf{P}' = \mathbf{B}' \times_{\mathbf{B}} \mathbf{P}$, define a map $\mathbf{P}' \times \mathbf{F} \rightarrow \mathbf{E}'$ by setting $(b', x).f = (b', x.f)$; one thus endows $\mathbf{E}'$ with a structure of fiber bundle associated with λ' with fiber type $\mathbf{F}$; it is called the inverse image by h of the structure given on $\mathbf{E}$.

6.5.5. Let $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ be a principal fibration, and let $\mathbf{E}$ (resp. $\mathbf{E}'$) be a fiber space associated with λ with fiber type $\mathbf{F}$ (resp. $\mathbf{F}'$). Let $u : \mathbf{F} \rightarrow \mathbf{F}'$ be a morphism compatible with the operations of $\mathbf{G}$ (that is, such that $u(g.f) = g \cdot u(f)$ for $f \in \mathbf{F}$, $g \in \mathbf{G}$). There then exists a unique morphism $\bar{u} : \mathbf{E} \rightarrow \mathbf{E}'$ such that one has $\bar{u}(x.f) = x \cdot u(f)$ for $x \in \mathbf{P}$, $f \in \mathbf{F}$. If u is an immersion (resp. a submersion, a subimmersion), the same is true of $\bar{u}$.

In particular, suppose that a group variety $\mathbf{H}$ operates on the right on $\mathbf{F}$ in such a way that $g.(f.h) = (g.f).h$ for $g \in \mathbf{G}$, $f \in \mathbf{F}$, $h \in \mathbf{H}$. Every $h \in \mathbf{H}$ then defines an automorphism u_h of $\mathbf{F}$ compatible with the operations of $\mathbf{G}$, whence an automorphism $\bar{u}_h$ of $\mathbf{E}$; the group $\mathbf{H}$ operates

on the right on E by $(y, h) \mapsto \bar{u}_h(y)$.

6.5.6. Let $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ be a principal fibration, and let E be a fiber space associated with λ with fiber type $\mathbf{F}$. Let $s : \mathbf{B} \rightarrow E$ be a section of E . For every $x \in \mathbf{P}$, there exists a unique element $\sigma(x) \in \mathbf{F}$ such that $s(\pi(x)) = x \cdot \sigma(x)$. The map $\sigma : \mathbf{P} \rightarrow \mathbf{F}$ thus defined is a morphism of varieties, and satisfies the identity:

$$\sigma(x.g) = g^{-1} \cdot \sigma(x). \quad (*)$$

The map $s \mapsto \sigma$ is a bijection of the set of sections of E onto the set of morphisms of $\mathbf{P}$ into $\mathbf{F}$ which satisfy the identity (*).

6.5.7. Let $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ and $\lambda' = (\mathbf{P}', \mathbf{G}, \mathbf{B}, \pi')$ be two principal fibrations with the same base $\mathbf{B}$ and of the same structural group $\mathbf{G}$. The principal fibration $\lambda \times_{\mathbf{B}} \lambda' = (\mathbf{P} \times_{\mathbf{B}} \mathbf{P}', \mathbf{G} \times \mathbf{G}, \mathbf{B}, (\pi, \pi')_{\mathbf{B}})$ has as its structure group $\mathbf{G} \times \mathbf{G}$. Let us make $\mathbf{G} \times \mathbf{G}$ operate on the left on $\mathbf{G}$ by the formula:

$$(g, g') \cdot g_1 = g \cdot g_1 \cdot g^{-1},$$

and let E be the fiber bundle associated with $\lambda \times_{\mathbf{B}} \lambda'$ with fiber type $\mathbf{G}$ (endowed with the operation law defined above). Then the sections of E correspond bijectively to the isomorphisms from $\mathbf{P}$ to $\mathbf{P}'$. Precisely, if s is a section of E corresponding (cf. no. 6.5.6) to the morphism $\sigma : \mathbf{P} \times_{\mathbf{B}} \mathbf{P}' \rightarrow \mathbf{G}$, there exists a $\mathbf{G}$ - $\mathbf{B}$ -isomorphism $f_s : \mathbf{P} \rightarrow \mathbf{P}'$ such that $\sigma(x, f_s(x)) = e$ for all $x \in \mathbf{P}$; the map $s \mapsto f_s$ is a bijection from the set of sections of E onto the set of $\mathbf{G}$ - $\mathbf{B}$ -isomorphisms from $\mathbf{P}$ to $\mathbf{P}'$.

6.6 Extension and restriction of the structure group

6.6.1. Let $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi)$ be a principal fibration, and let φ be a homomorphism from $\mathbf{G}$ into a group variety $\mathbf{H}$. Let us make $\mathbf{G}$ operate on the left on $\mathbf{H}$ by $g \cdot h = \varphi(g) \cdot h$ and let $\mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ be the associated fiber bundle to λ with typical fiber $\mathbf{H}$. Since the right translations in $\mathbf{H}$ are compatible with the operations of $\mathbf{G}$, the group $\mathbf{H}$ operates on the right on $\mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ (cf. no. 6.5.5); if $\pi_{\mathbf{H}}$ denotes the projection of $\mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ onto $\mathbf{B}$, the quadruple $(\mathbf{P} \times^{\mathbf{G}} \mathbf{H}, \mathbf{H}, \mathbf{B}, \pi_{\mathbf{H}})$ is a principal

fibration, denoted $\varphi(\lambda)$; it is said to be deduced from λ by the homomorphism φ .

The map f of $\mathbf{P}$ to $\mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ which associates to $x \in \mathbf{P}$ the class of (x, e) is a $\mathbf{B}$ -morphism of $\mathbf{P}$ to $\mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ compatible with φ (cf. no. 6.3.1). Moreover, if f' is a $\mathbf{B}$ -morphism of $\mathbf{P}$ into a principal fiber bundle $\mathbf{P}'$ of the structural group $\mathbf{H}$ and if f' is compatible with φ , there exists a unique $\mathbf{H}$ - $\mathbf{B}$ -isomorphism θ of $\mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ onto $\mathbf{P}'$ such that $f' = \theta \circ f$.

6.6.2. Suppose that λ is defined by means of an open covering $\mathcal{V} = (U_i)$ of $\mathbf{B}$ and of a cocycle (g_{ij}) (6.4.2). Then $\varphi(\lambda)$ can be defined using the same covering and the cocycle (h_{ij}) , with $h_{ij} = \varphi \circ g_{ij}$.

6.6.3. Let $\mathbf{F}$ be a variety on which the group $\mathbf{H}$ acts on the left; we denote by $(h, y) \mapsto h \cdot y$ the action law of $\mathbf{H}$ on $\mathbf{F}$. The group $\mathbf{G}$ acts on $\mathbf{F}$ by $(g, y) \mapsto \varphi(g) \cdot y$. Let $\mathbf{E}$ be a fiber bundle associated with $\varphi(\lambda)$ with typical fiber $\mathbf{F}$. The map $(x, y) \mapsto f(x) \cdot y$ from $\mathbf{P} \times \mathbf{F}$ to $\mathbf{E}$ (the map f being that defined in no. 6.6.1) endows $\mathbf{E}$ with a structure of fiber bundle associated with λ of fiber type $\mathbf{F}$. In particular, $(\mathbf{P} \times^{\mathbf{G}} \mathbf{H}) \times^{\mathbf{H}} \mathbf{F}$ is identified with $\mathbf{P} \times^{\mathbf{G}} \mathbf{F}$.

6.6.4. Suppose that $\mathbf{G}$ is a Lie subgroup of $\mathbf{H}$, and that $\varphi : \mathbf{G} \rightarrow \mathbf{H}$ is the canonical injection of $\mathbf{G}$ into $\mathbf{H}$. One then says that $\varphi(\lambda)$ is deduced from λ by extension to $\mathbf{H}$ of the structural group. The morphism $f : \mathbf{P} \rightarrow \mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ of No. 6.6.1 is an isomorphism from $\mathbf{P}$ onto a closed submanifold of $\mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ (equal to $\mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ if $\mathbf{G} = \mathbf{H}$); this isomorphism is compatible with the operations of $\mathbf{G}$ (note that $\mathbf{G}$ acts on $\mathbf{P} \times^{\mathbf{G}} \mathbf{H}$ as a subgroup of $\mathbf{H}$).

6.6.5. Suppose further that $\mathbf{G}$ is a Lie subgroup of $\mathbf{H}$, and let $\mu = (Q, H, B, \pi)$ be a principal fibration with structure group $\mathbf{H}$ and base $\mathbf{B}$, and let $\mathbf{E}$ be a fiber bundle associated with μ of fiber type $\mathbf{H}/\mathbf{G}$. If γ denotes the canonical projection of $\mathbf{H}$ onto $\mathbf{H}/\mathbf{G}$, set $\delta(y) = y \cdot \gamma(e)$, for $y \in Q$ (where e denotes the identity element of $\mathbf{H}$); one thus obtains a morphism $\delta : Q \rightarrow \mathbf{E}$ and the quadruple (Q, G, E, δ) is a principal fibration. In particular, $Q/\mathbf{G}$ is canonically identified with $\mathbf{E}$, itself isomorphic to $Q \times^{\mathbf{H}} \mathbf{H}/\mathbf{G}^1$.

Now let $s : \mathbf{B} \rightarrow \mathbf{E}$ be a section of $\mathbf{E}$, and let λ_s be the inverse image under s of the fibration (Q, G, E, δ) that we have just defined. It is a principal fibration, with base $\mathbf{B}$ and with structural group $\mathbf{G}$, and its extension to $\mathbf{H}$ is isomorphic to μ ; any principal fibration enjoying these properties can be obtained in this way, up to isomorphism; two sections s_1 and s_2 of $\mathbf{E}$ define isomorphic fibrations if and only if they are transformed into one another by an

H-B-automorphism of μ .

6.7 Changes of structure

The structures and operations described in this paragraph are compatible with the changes of structure described in numbers 5.13 and 5.14.

7 Vector bundles

Throughout this paragraph, the letter B denotes a manifold of class C^r ($r \geq 1$) and the letter M denotes a set equipped with a map π from M to B . One says that B is the base of M and, for every $b \in B$, one denotes the fiber of M at b by M_b and calls it the subset $\pi^{-1}(b)$ of M .

7.1 Definition of vector bundles

7.1.1. A vector chart of M is a triple $t = (U, \varphi, F)$, where U is an open subset of B , F is a Banach space, and φ is a bijection from $\pi^{-1}(U)$ onto $U \times F$ such that $\pi(\varphi^{-1}(b, h)) = b$ for all $b \in U$ and all $h \in F$. We say that U is the domain of the vector chart t and that t is a vector chart of M at $b \in B$ if $b \in U$. For every $b \in U$, one denotes by t_b the bijection from F to M_b defined by $t_b(h) = \varphi^{-1}(b, h)$ for $h \in F$.

7.1.2. Two vector charts $t = (U, \varphi, F)$ and $t' = (U', \varphi', F')$ of M are C' -compatible (or simply compatible) if there exists a map λ of class C' from the manifold $U \cap U'$ into the Banach space $\mathcal{L}(F; F')$ such that:

$$t_b = t'_b \circ \lambda(b) \quad \text{for all } b \in U \cap U'.$$

7.1.3. A set of vector charts of M is said to be a C' -vector atlas (or simply vector atlas) of M if it consists of pairwise C' -compatible vector charts whose domains have union B . Two vector atlases A and B of M are said to be C' -equivalent (or equivalent) if $A \cup B$ is again a vector atlas of M . This relation is an equivalence relation.

7.1.4. A vector bundle structure of class C^r (basis B) on M is the data of a class of equivalent vector atlases (Sets, Chap. II, § 6, no. 9). A vector chart belonging to a vector atlas of this class

is called a vector chart of the vector bundle M .

Let M be a vector bundle with base B . For every $b \in B$ there exists on the fiber M_b a Banach space structure, and only one such that, for every vector chart $t = (U, \varphi, F)$ of the vector bundle M at b , the map t_b is an isomorphism from F onto M_b .

Let $c = (U, \psi, E)$ be a chart of the manifold B and let $t = (U, \varphi, F)$ be a vector chart of the vector bundle M , with the same domain U . For $x \in \pi^{-1}(U)$, set:

$$\alpha(x) = (\psi(\pi(x)), t_{\pi(x)}^{-1}(x)).$$

Then, the triple $(\pi^{-1}(U), \alpha, E \times F)$ is a chart of the set M . There exists on M a manifold structure of class C^r and only one (called the underlying structure of M) for which all the charts thus obtained are charts of the manifold M . The triple (M, B, π) is then a fibration (6.1.1).

7.1.5. Let F be a Banach space. Set $M = B \times F$, the map being the first projection. There exists on M a vector bundle structure (with base B), and only one, for which (B, Id_B, F) is a vector chart. We say that $B \times F$ endowed with this structure is the trivial vector bundle over the base B with fiber F , and it is sometimes denoted by F_B . The manifold structure of F_B is the product manifold structure, and for every $b \in B$, the map $h \mapsto (b, h)$ is a Banach space isomorphism from F onto the fiber of F_B at the point $b \in B$. One often denotes by 0 a trivial vector bundle whose fiber is reduced to 0 .

7.1.6. Let M be a vector bundle with base B . For $b \in B$ we call the dimension (finite or $+\infty$) of the Banach space M_b the rank of M at b , and denote it by $\text{rg}_b(M)$. One has $\dim_x M = \dim_b B + \text{rg}_b M$ for $b = \pi(x)$. The function $b \mapsto \text{rg}_b M$ is locally constant. We say that M is of finite rank if $\text{rg}_b M < +\infty$ for all $b \in B$.

7.2 Morphisms of vector bundles

7.2.1. Let B and B' be two manifolds and let f be a morphism from B to B' . Let M be a vector bundle with base B and M' a vector bundle with base B' . We say that a map g from M to M' is an f -morphism of vector bundles if the following condition is satisfied:

For every point $b_0 \in B$, there exists a vector chart $t = (U, \varphi, F)$ of M at b_0 , a vector chart $t' = (U', \varphi', F')$ of M' at $f(b_0)$ and a mapping λ of class C' from U into $\mathcal{L}(F; F')$ such that

$f(U) \subset U'$ and that $g_b \circ t_b = t'_{f(b)} \circ \lambda(b)$ for all $b \in U$, where g_b is the restriction of g to $\mathbf{M}_b$.

Under these hypotheses, g is a morphism of manifolds and for every $b \in B$, g induces a continuous linear map from M_b to $M'_{f(b)}$. We call the rank (finite or $+\infty$) of the linear map g_b the vector rank of g at $b \in B$, and denote it by $\text{rg}_b(g)$.

Conversely, if $r \geq \infty$ or if M is of finite rank, an f -morphism of fibrations from (M, B, π) to (M', B', π') which induces on each fiber M_b a linear map from M_b to $M'_{f(b)}$ (for every $b \in B$), is an f -morphism of vector bundles.

7.2.2. Let moreover f' be a morphism of B' into a manifold B'' . If g is an f -morphism of M to M' and if g' is an f' -morphism of M' into a vector bundle M'' with base B'' , the map $g' \circ g$ is an $(f' \circ f)$ -morphism of M to M'' . One has $(g' \circ g)_b = g'_{f(b)} \circ g_b$ for all b in B .

7.2.3. Let M and M' be two vector bundles with the same base B . One calls a B -morphism, or simply a morphism from M to M' , any Id_B -morphism. The composite of two morphisms is a morphism.

Let g be a morphism of vector bundles from M to M' ; if g is bijective, it is an isomorphism of the manifold M onto the manifold M' , the inverse mapping g^{-1} is a morphism of vector bundles from M' to M and one has $(g^{-1})_b = g_b^{-1}$ for all b in B . The map g is then an isomorphism of vector bundles.

7.2.4. Let f be a morphism of a manifold B' to B and let $\mathbf{M}$ be a vector bundle with base B . Set $\mathbf{M}' = B' \times_B \mathbf{M}$ and denote by π' (resp. g) the restriction to $\mathbf{M}'$ of the projection of $B' \times \mathbf{M}$ onto B' (resp. $\mathbf{M}$). On $\mathbf{M}'$ there exists a vector bundle structure with base B' (relative to π') and only one for which g is an f -morphism. One says that $\mathbf{M}'$ is the vector bundle with base B' inverse image of $\mathbf{M}$ by f and is denoted $f^*\mathbf{M}$; the f -morphism g is called the canonical f -morphism of $f^*\mathbf{M}$ to $\mathbf{M}$.

The manifold structure of $f^*\mathbf{M}$ is that of the fibered product of the manifolds B' and $\mathbf{M}$ over B (5.11.2); for every $b \in B'$, the map $x \mapsto (b, x)$ is an isomorphism of Banach spaces from $\mathbf{M}_{f(b)}$ onto $(f^*\mathbf{M})_b$.

The formation of inverse images of vector bundles is transitive.

Let N' be a vector bundle with base B' and let h be an f -morphism from N' to M . There exists one and only one B' -morphism $\tilde{h}$ from N' to f^*M such that $h = g \circ \tilde{h}$.

Let N be a vector bundle with base B and let v be a B -morphism from N to M . There exists one and only one B' -morphism, denoted f^*v , from f^*N to f^*M such that the diagram

$$\begin{array}{ccc} f^*N & \xrightarrow{f^*v} & f^*M \\ \downarrow & & \downarrow \\ N & \xrightarrow{v} & M \end{array}$$

is commutative.

7.2.5. Let B' be a submanifold of B and let i be the canonical injection from B' into B . If M is a vector bundle with base B , the inverse image i^*M is called the vector bundle induced on B' by M and is denoted $M|_{B'}$. If $t = (U, \varphi, F)$ is a vector chart of M , then $(U \cap B', \varphi|_{\pi^{-1}(U \cap B')}, F)$ is a vector chart of $M|_{B'}$. The canonical f -morphism from $M|_{B'}$ into M is an isomorphism of manifolds from $M|_{B'}$ onto the submanifold $\pi^{-1}(B')$ of M .

7.2.6. Let f be a morphism from a manifold B' into B and let M (resp. M') be a vector bundle with base B (resp. B'). One calls an f -comorphism from M into M' a B' -morphism from f^*M into M' . When $B = B'$ and $f = \text{Id}_B$, the data of an f -comorphism from M into M' is equivalent to the data of a B -morphism from M into M' .

Let g be an f -comorphism of M to M' . For every $b \in B'$, the map $g_b : x \mapsto g(b, x)$ from $M_{f(b)}$ to M'_b is a continuous linear map.

Additionally, let f' be a morphism of a manifold B'' to B' and let h be an f' -comorphism of M' into a vector bundle M'' with base B'' . The map $h \circ f'^*g$ of $f'^*(f^*M) = (f \circ f')^*M$ to M'' is then an $(f \circ f')$ -comorphism of M to M'' , denoted $h \circ g$. For $b \in B''$, one has

$$(h \circ g)_b = h_b \circ g_{f'(b)}.$$

7.3 Multilinear morphisms

7.3.1. Let $M_1, \dots, M_d$ and N be vector bundles with base B , and let u be a map from the set $M_1 \times_B \dots \times_B M_d$ into N . One says that u is a multilinear morphism (or d -linear) if the following condition is satisfied:

For every $b_0 \in \mathbf{B}$, there exists an open neighborhood U of b_0 in $\mathbf{B}$, vector charts $t^j = (U, \varphi^j, \mathbf{F}^j)$ of $\mathbf{M}_j$ (for $1 \leq j \leq d$) and $t = (U, \varphi, \mathbf{F})$ of $\mathbf{N}$, and a map λ of class $\mathbf{C}^r$ from U into the Banach space $\mathcal{L}(\mathbf{F}^1, \dots, \mathbf{F}^d; \mathbf{F})$ of continuous d -linear maps from $\mathbf{F}^1 \times \dots \times \mathbf{F}^d$ to $\mathbf{F}$, such that:

$$(t_b \circ \lambda(b))(x_1, \dots, x_d) = u(t_b^1(x_1), \dots, t_b^d(x_d))$$

for all $b \in U$ and all $x_j \in \mathbf{F}^j$.

Every multilinear morphism u is a morphism of manifolds from the fibered product $\mathbf{M}_1 \times_{\mathbf{B}} \dots \times_{\mathbf{B}} \mathbf{M}_d$ to $\mathbf{N}$ and induces for every $b \in \mathbf{B}$ a continuous d -linear map u_b from $(\mathbf{M}_1)_b \times \dots \times (\mathbf{M}_d)_b$ to $\mathbf{N}_b$.

If f is a morphism from $\mathbf{B}$ into a manifold $\mathbf{B}'$, a multilinear morphism from $\mathbf{M}_1 \times_{\mathbf{B}} \dots \times_{\mathbf{B}} \mathbf{M}_d$ into a vector bundle $\mathbf{M}'$ with base $\mathbf{B}'$ is by definition the composite of a multilinear morphism from $\mathbf{M}_1 \times_{\mathbf{B}} \dots \times_{\mathbf{B}} \mathbf{M}_d$ to $f^*\mathbf{M}'$ with the canonical f -morphism of $f^*\mathbf{M}'$ to $\mathbf{M}'$.

A bilinear morphism is also called a pairing. For $d = 1$, a linear morphism is a morphism in the sense of 7.2.1. For $d = 0$, a 0-linear morphism is identified with a section of $\mathbf{N}$ (7.4).

7.3.2. An algebra bundle over base $\mathbf{B}$ is a vector bundle $\mathbf{A}$ over base $\mathbf{B}$, equipped with a pairing of $\mathbf{A} \times_{\mathbf{B}} \mathbf{A}$ into $\mathbf{A}$. Each fiber A_b is then equipped with a $\mathbf{K}$ -algebra structure. If for every $b \in \mathbf{B}$ the algebra A_b has an identity element, denoted e_b , the map $b \mapsto e_b$ is a section of $\mathbf{A}$ (cf. 7.4). An algebra bundle $\mathbf{A}$ is said to be locally trivial if, for every point b_0 of $\mathbf{B}$, there exists a vector chart $t = (U, \varphi, \mathbf{E})$ from $\mathbf{A}$ at the point b_0 and a $\mathbf{K}$ -algebra structure on $\mathbf{E}$ such that t_b is an isomorphism of algebras from $\mathbf{E}$ onto A_b for all $b \in U$.

7.3.3. Let $\mathbf{A}$ be a bundle of associative algebras with unit element over base $\mathbf{B}$. One calls a bundle of $\mathbf{A}$ -modules over base $\mathbf{B}$ a vector bundle $\mathbf{M}$ over base $\mathbf{B}$ equipped with a pairing $m : \mathbf{A} \times_{\mathbf{B}} \mathbf{M} \rightarrow \mathbf{M}$ such that the map $m_b : A_b \times M_b \rightarrow M_b$ defines, for all $b \in \mathbf{B}$, a structure of A_b -module on the fiber M_b .

Let $\mathbf{M}$ and $\mathbf{M}'$ be two vector bundles in $\mathbf{A}$ -modules. We call an $\mathbf{A}$ -homomorphism from $\mathbf{M}$ to $\mathbf{M}'$ any morphism $g : \mathbf{M} \rightarrow \mathbf{M}'$ of vector bundles over the base $\mathbf{B}$ which induces for every b in $\mathbf{B}$ an A_b -linear map from M_b to M'_b .

Suppose $\mathbf{A}$ is locally trivial. We say that a bundle $\mathbf{M}$ of $\mathbf{A}$ -modules is locally trivial if for

every point b_0 of B , there exists a vector chart $t = (U, \varphi, E)$ from A at b_0 as in 7.3.2, a vector chart $t' = (U, \varphi', L)$ from M at b_0 , and a structure of E -module on L such that, for every $b \in U$, t'_b is a t_b -isomorphism of the A_b -module M_b onto the E -module L .

7.3.4. Let A be a Banach algebra over K (for example, a field equipped with a finite-dimensional K -algebra structure). The trivial bundle A_B is then a locally trivial algebra bundle. A locally trivial bundle M in A_B -modules is also called a vector bundle over A (with base B). The fibers M_b are then topological A -modules. An f -morphism u from M to another vector bundle over A is said to be A -linear if the maps u_b are A -linear for all $b \in B$.

7.4 Sections

7.4.1. Let M be a vector bundle with base B . For every open set U of B , we denote by $\mathcal{S}_M^r(U)$ the set of sections of class C^r of M over U , that is, morphisms s of class C^r from U into M such that $s(b) \in M_b$ for all $b \in U$. This set is equipped with a module structure over the ring $\mathcal{C}^r(U)$ of morphic functions by the rules:

(1)

$$(s + s')(b) = s(b) + s'(b) \quad (2)$$

$$(\varphi \cdot s)(b) = \varphi(b) \cdot s(b)$$

for s, s' in $\mathcal{S}_M^r(U)$ and φ in $\mathcal{C}^r(U)$. When the open set U varies, one obtains a sheaf $\mathcal{S}_M^r$ of maps from B into M (cf. no. 5.4.1), called the sheaf of sections of M .

7.4.2. Let $M_1, \dots, M_d$ and N be vector bundles over the base B , and let u be a multilinear morphism from $M_1 \times_B \dots \times_B M_d$ into N . For $1 \leq j \leq d$ let us give a section s_j of M_j on an open set U of B ; one defines a section $u(s_1, \dots, s_d)$ of N on U by the formula:

$$u(s_1, \dots, s_d)(b) = u_b(s_1(b), \dots, s_d(b)) \quad \text{for } b \in U.$$

The map $(s_1, \dots, s_d) \mapsto u(s_1, \dots, s_d)$ is $\mathcal{C}^r(U)$ -multilinear. It is sometimes denoted $\mathcal{S}(u)$.

7.4.3. Let f be a morphism of a manifold $\mathbf{B}'$ to $\mathbf{B}$ and let $\mathbf{M}$ be a vector bundle with base $\mathbf{B}$. For every open set $\mathbf{U}$ of $\mathbf{B}$ and every $s \in \mathcal{S}_{\mathbf{M}}^r(\mathbf{U})$, the map $x \mapsto (x, s(f(x)))$ is a section of class $\mathcal{C}^r$ of $f^*\mathbf{M}$ on the open subset $f^{-1}(\mathbf{U})$, denoted f^*s and called the inverse image of s by f . The map $s \mapsto f^*s$ from $\mathcal{S}_{\mathbf{M}}^r(\mathbf{U})$ to $\mathcal{S}_{f^*\mathbf{M}}^r(f^{-1}(\mathbf{U}))$ is semilinear with respect to the homomorphism $g \mapsto g \circ (f|_{f^{-1}(\mathbf{U})})$ from $\mathcal{C}^r(\mathbf{U})$ to $\mathcal{C}^r(f^{-1}(\mathbf{U}))$.

If moreover $\mathbf{N}$ is a vector bundle over $\mathbf{B}'$ and g is an f -comorphism from $\mathbf{M}$ to $\mathbf{N}$, one sometimes writes $\mathcal{S}(g)$ for the map $s \mapsto g \circ f^*s$ from $\mathcal{S}_{\mathbf{M}}^r(\mathbf{U})$ to $\mathcal{S}_{\mathbf{N}}^r(f^{-1}(\mathbf{U}))$.

7.4.4. Let $\mathbf{M}$ be a vector bundle over the base $\mathbf{B}$, of finite rank. A frame of $\mathbf{M}$ over an open set $\mathbf{U}$ of $\mathbf{B}$ is called a finite sequence $(s_1, \dots, s_n)$ of sections of $\mathbf{M}$ over $\mathbf{U}$ such that $(s_1(b), \dots, s_n(b))$ is a basis of the vector space M_b for all $b \in \mathbf{U}$. The sequence $(s_1, \dots, s_n)$ is then a basis of the $\mathcal{C}^r(\mathbf{U})$ -module $\mathcal{S}_{\mathbf{M}}^r(\mathbf{U})$. If f is a morphism from a manifold $\mathbf{B}'$ into $\mathbf{B}$, the sections f^*s_j form a frame of $f^*\mathbf{M}$ on $f^{-1}(\mathbf{U})$.

7.4.5. Let $\mathbf{L}$ be a field, equipped with a structure of finite-dimensional $\mathbf{K}$ -algebra, and let $(\mathbf{M}, \mathbf{B}, \pi)$ be a fibration. Suppose given on each fiber M_b a vector space structure over $\mathbf{L}$, of finite dimension. There then exists at most one vector bundle structure on $\mathbf{M}$ with base $\mathbf{B}$, compatible with the map π , the manifold structure of $\mathbf{M}$, and the $\mathbf{L}$ -vector structures on the fibers (7.3.4). For one to exist, it is necessary and sufficient that the following condition be satisfied:

(FV) For every $b_0 \in \mathbf{B}$, there exists an open neighborhood $\mathbf{U}$ of b_0 in $\mathbf{B}$ and an isomorphism of manifolds φ from $\pi^{-1}(\mathbf{U})$ onto the product $\mathbf{U} \times \mathbf{F}$ of $\mathbf{U}$ by a vector space $\mathbf{F}$ over $\mathbf{L}$ of finite dimension, such that for every $b \in \mathbf{U}$ the bijection φ_b from M_b onto $\mathbf{F}$ induced by φ is an isomorphism of vector spaces over the field $\mathbf{L}$.

The triples $(\mathbf{U}, \varphi, \mathbf{F})$ satisfying (FV) are then vector charts of the vector bundle $\mathbf{M}$.

Condition (FV) is equivalent to:

(FV') For every $b_0 \in \mathbf{B}$ there exists an integer n and n sections $s_1, \dots, s_n$ of $\mathbf{M}$ on an open neighborhood $\mathbf{U}$ of b_0 such that the map

$$(b, a_1, \dots, a_n) \mapsto a_1 s_1(b) + \dots + a_n s_n(b)$$

is an isomorphism from the manifold $\mathbf{U} \times \mathbf{L}^n$ to the manifold $\pi^{-1}(\mathbf{U})$.

7.4.6. Let $M_1, \dots, M_d$ and $\mathbf{N}$ be vector bundles with base $\mathbf{B}$, the M_j being of finite rank.

Suppose that for every open set U of B a $\mathcal{C}^r(U)$ -multilinear map φ_U from $\mathcal{S}_{M_1}^r(U) \times \cdots \times \mathcal{S}_{M_d}^r(U)$ to $\mathcal{S}_N^r(U)$ is given, such that for $V \subset U$ one has:

$$\varphi_U(s_1, \dots, s_d)|V = \varphi_V(s_1|V, \dots, s_d|V).$$

There then exists a multilinear morphism u of $M_1 \times_B \cdots \times_B M_d$ to N such that $\varphi_U(s_1, \dots, s_d) = u(s_1, \dots, s_d)$ for any sections s_j of M_j on the open subset U of B .

7.4.7. Let f be a morphism of a manifold B' to B and let M (resp. M') be a vector bundle with base B (resp. B') and of finite rank. Suppose that for every open set U of B a continuous $\mathcal{C}^r(U)$ -semilinear map φ_U from $\mathcal{S}_M^r(U)$ to $\mathcal{S}_{M'}^r(f^{-1}(U))$ is given, such that

$$\varphi_U(s)|f^{-1}(V) = \varphi_V(s|V)$$

for every open set $V \subset U$. There then exists a unique f -comorphism g from M to M' such that $\varphi_U(s) = \mathcal{S}(g)(s)$ for all $s \in \mathcal{S}_M^r(U)$.

*7.4.8. Let $\mathcal{F}$ be a sheaf of modules over the sheaf of rings $\mathcal{C}_B^r$. We say that $\mathcal{F}$ is locally free if for every $b \in B$ there exists an open neighborhood U of b and an integer n such that $\mathcal{F}|_U$ is isomorphic (as a sheaf of $\mathcal{C}_U^r$ -modules) to the sheaf $(\mathcal{C}_U^r)^n$.

If M is a vector bundle over the base B of finite rank, the sheaf $\mathcal{S}_M^r$ is locally free. Conversely, for every locally free sheaf $\mathcal{F}$ on B , there exists a vector bundle M and an isomorphism of sheaves of $\mathcal{S}_M^r$ onto $\mathcal{F}$. If M' is another finite-rank vector bundle with base B , the map $g \mapsto \mathcal{S}(g)$ is a bijection from the set of B -morphisms from M to M' onto the set of sheaf morphisms of $\mathcal{C}^r$ -modules from $\mathcal{S}_M^r$ to $\mathcal{S}_{M'}^r$.

7.5 Vector subbundles, quotient vector bundles, exact sequences

In this number, we call a vector bundle a vector bundle with base B and a morphism of vector bundles an Id_B -morphism.

7.5.1. Let M be a vector bundle. A subset M' of M is called a vector subbundle of M if, for every point $b \in B$, there exists a vector chart $t = (U, \varphi, E)$ from M at b and a closed vector subspace F of E admitting a topological complement, such that

$$\varphi(\pi^{-1}(U) \cap M') = U \times F.$$

Under these conditions, there exists on M' a vector bundle structure, and only one, for which the canonical injection of M' into M is a morphism. For each $b \in B$, the fiber M'_b of M' is the closed vector subspace $M' \cap M_b$ of M_b ; M' is a closed submanifold of M , and the manifold structure underlying the vector bundle structure of M' coincides with that induced by the manifold structure of M .

7.5.2. Let M' be a vector subbundle of M . Denote by R the following relation between points x, y of M :

there exists an element b of B such that $x \in M_b, y \in M_b$ and $x - y \in M'_b$. Then R is a regular equivalence relation on M (cf. No. 5.9.7). On the set M/R there exists one and only one vector bundle structure such that the canonical map from M onto M/R is a morphism. We denote it by M/M' and call the quotient of M by M' the vector bundle thus defined; for each point b of B , the bundle $(M/M')_b$ is the quotient topological vector space M_b/M'_b and the manifold structure on M/M' is the quotient of that on M .

7.5.3. We retain the hypotheses of 7.5.2. For every point b_0 of B , one can find an open neighborhood U of b_0 , a Banach space F , direct sum of two closed subspaces F' and F'' and an isomorphism of vector bundles ι from F_U onto $M|U$ with the following properties:

- (i) The restriction ι' of ι to $U \times F'$ is an isomorphism of F'_U onto $M'|U$.
- (ii) If ρ is the canonical map from M to $M'' = M/M'$ and if ι'' is the restriction of ι to $U \times F''$, the map $\rho \circ \iota''$ is an isomorphism of F''_U onto $M''|U$.

7.5.4. If a morphism of vector bundles $g : L \rightarrow M$ has its image contained in M' it is a morphism of vector bundles from L to M' .

Now consider a morphism of vector bundles $h : M \rightarrow N$ and suppose that, for every b in B , the restriction of h_b to M'_b is zero. If ρ is the canonical morphism from M to M/M' there exists one and only one morphism $\bar{h}$ of M/M' into N such that $h = \bar{h} \circ \rho$.

7.5.5. Let P and Q be two vector bundles, and g a morphism from P to Q ; for every point b of B , denote by N_b and I_b respectively the kernel and the image of the linear map $g_b : P_b \rightarrow Q_b$

. Set $N = \bigcup_{b \in B} N_b$ and

$I = \bigcup_{b \in B} I_b$. The morphism g is said to be locally direct if N is a vector subbundle of P and I a vector subbundle of Q . The morphism g then defines, by passing to the quotient, an isomorphism from P/N onto I . One says that N is the kernel of g and denotes it by $\text{Ker } g$. Likewise, the subbundle I is called the image of g and is denoted by $\text{Im } g$.

If $r \geq \infty$, the morphism g is locally direct if and only if g is a subimmersion. If P is of finite rank, the morphism g is locally direct if and only if the vector rank of g is locally constant, or equivalently if and only if g is a subimmersion.

7.5.6. Let $\mathbf{M} \xrightarrow{f} \mathbf{M}' \xrightarrow{g} \mathbf{M}''$ be two morphisms of vector bundles. One says that the sequence (f, g) is locally direct exact if the two morphisms f and g are locally direct and if $\text{Im } f = \text{Ker } g$. If $g \circ f = 0$, the set $\mathbf{D}$ of points $b \in \mathbf{B}$ such that the sequence $\mathbf{M}_b \xrightarrow{f_b} \mathbf{M}'_b \xrightarrow{g_b} \mathbf{M}''_b$ is direct exact (that is, such that $\text{Ker } f_b$ and $\text{Im } g_b$ admit topological supplements and that $\text{Im } f_b = \text{Ker } g_b$) is open and the sequence $\mathbf{M}|_{\mathbf{D}} \xrightarrow{f} \mathbf{M}'|_{\mathbf{D}} \xrightarrow{g} \mathbf{M}''|_{\mathbf{D}}$ is locally direct exact.

One defines in the same way locally direct exact sequences of arbitrary length. By abuse of language, one sometimes says direct exact sequence instead of locally direct exact sequence.

7.5.7. Let $0 \rightarrow \mathbf{M} \xrightarrow{f} \mathbf{M}' \xrightarrow{g} \mathbf{M}'' \rightarrow 0$ be a sequence of morphisms of vector bundles. For this sequence to be locally direct exact, it is necessary and sufficient that f be an isomorphism of $\mathbf{M}$ onto a vector subbundle $f(\mathbf{M})$ of $\mathbf{M}'$ and that g define, by passing to the quotient, an isomorphism of the quotient vector bundle $\mathbf{M}'/f(\mathbf{M})$ onto $\mathbf{M}''$.

7.6 Vector functors

In this number and in the three following numbers 7.7 to 7.9, the letter I denotes a finite set, the union of two disjoint subsets I_+ and I_- . We denote by $\mathcal{V} = (V_i)_{i \in I}$ (and analogously by $\mathcal{V}'$, $\mathcal{V}''$, ...) a family of Banach spaces indexed by I . We denote by $\text{Hom}(\mathcal{V}, \mathcal{V}')$ the Banach space $\prod_{i \in I_+} \mathcal{L}(V_i; V'_i) \times \prod_{i \in I_-} \mathcal{L}(V'_i; V_i)$. We denote by $\mathbf{f} = (f_i)$ an element of $\text{Hom}(\mathcal{V}, \mathcal{V}')$. We write $\text{Id}_{\mathcal{V}}$ for the element $(\text{Id}_{V_i})_{i \in I}$ of $\text{Hom}(\mathcal{V}, \mathcal{V})$. For $\mathbf{f} \in \text{Hom}(\mathcal{V}, \mathcal{V}')$ and $\mathbf{f}' \in \text{Hom}(\mathcal{V}', \mathcal{V}'')$, we denote by $\mathbf{f}' \circ \mathbf{f}$ the element of $\text{Hom}(\mathcal{V}, \mathcal{V}'')$ whose components are given by:

$$(\mathbf{f}' \circ \mathbf{f})_i = f'_i \circ f_i \quad \text{if } i \in I_+$$

$$(\mathbf{f}' \circ \mathbf{f})_i = f_i \circ f'_i \quad \text{if } i \in I_-.$$

7.6.1. One calls a vector functor (resp. finite-dimensional vector functor) of type I and of class $\mathbf{C}^r$ the data consisting, for every family $\mathcal{V} = (V_i)_{i \in I}$ of Banach spaces (resp. of finite-dimensional vector spaces over K), of a Banach space $\tau(\mathcal{V})$ and, for every $\mathbf{f} \in \text{Hom}(\mathcal{V}, \mathcal{V}')$, of an element $\tau(\mathbf{f}) \in \mathcal{L}(\tau(\mathcal{V}); \tau(\mathcal{V}'))$, these data being subject to the following two conditions:

(a) One has $\tau(\text{Id}_{\mathcal{V}}) = \text{Id}_{\tau(\mathcal{V})}$ and $\tau(\mathbf{f}' \circ \mathbf{f}) = \tau(\mathbf{f}') \circ \tau(\mathbf{f})$.

(b) The map $\tau : \text{Hom}(\mathcal{V}, \mathcal{V}') \rightarrow \mathcal{L}(\tau(\mathcal{V}); \tau(\mathcal{V}'))$ is of class $\mathbf{C}^r$.

7.6.2. Let $\mathcal{M} = (\mathbf{M}^i)_{i \in I}$ be a family of vector bundles with base B . For $b \in B$, set $\mathcal{M}_b = (\mathbf{M}_b^i)_{i \in I}$. Let τ be a vector functor and denote by $\tau(\mathcal{M})$ the disjoint union of the $\tau(\mathcal{M}_b)$ for $b \in B$.

There exists on $\tau(\mathcal{M})$ one vector bundle structure and only one (with base B relative to the map π of $\tau(\mathcal{M})$ into B) such that for every $b \in B$, one has $\tau(\mathcal{M}_b) = \pi^{-1}(b)$, possessing the following property:

Let U be an open subset of B and, for each i , let $t^i = (U, \varphi_i, F_i)$ be a vector chart of M^i with domain U ; set $\mathcal{F} = (F_i)_{i \in I}$ and let ψ_b be the element of $\text{Hom}(\mathcal{M}_b, \mathcal{F})$ defined by $(\psi_b)_i = (t_b^i)^{-1}$ for $i \in I_+$ and $(\psi_b)_i = t_b^i$ for $i \in I_-$; for $x \in \pi^{-1}(U)$, set $\psi(x) = (\pi(x), \tau(\psi_{\pi(x)})(x))$. Then the triple $(U, \psi, \tau(\mathcal{F}))$ is a vector chart of the vector bundle $\tau(\mathcal{M})$.

Equipped with this structure, $\tau(\mathcal{M})$ is called the vector bundle deduced from the family $\mathcal{M}$ by the vector functor τ .

7.6.3. Let f be a morphism of B into a manifold B' . Let $\mathcal{M} = (\mathbf{M}^i)$ (resp. $\mathcal{M}' = (\mathbf{M}'^i)$) be a family indexed by I of vector bundles over the base B (resp. B'). For every $i \in I_+$ let g_i be an f -morphism of $\mathbf{M}^i$ to $\mathbf{M}'^i$ and for every $i \in I_-$ let g_i be an f -comorphism of $\mathbf{M}'^i$ to $\mathbf{M}^i$. Set $\mathbf{g} = (g_i)_{i \in I}$ and for $b \in B$, set $\mathbf{g}_b = ((g_i)_b)_{i \in I}$ (cf. 7.2.1 and 7.2.6). There exists an f -morphism, and only one, denoted $\tau(\mathbf{g})$, from $\tau(\mathcal{M})$ to $\tau(\mathcal{M}')$ such that $\tau(\mathbf{g})_b = \tau(\mathbf{g}_b)$ for all $b \in B$.

If in particular $M^i = f^* M'^i$, the g_i being the canonical morphisms or comorphisms, then the

B-morphism from $\tau(\mathcal{M})$ to $f^*\tau(\mathcal{M}')$ defined by $\tau(\mathbf{g})$ (7.2.4) is an isomorphism: one expresses this fact by saying that τ commutes with inverse images.

In particular, let B' be a submanifold of B , and set $\mathcal{M}|B' = (M^i|B')_{i \in I}$. The vector bundles $\tau(\mathcal{M})|B'$ and $\tau(\mathcal{M}|B')$ are then canonically B' -isomorphic.

7.6.4. Let $\tau, \tau_1, \dots, \tau_d$ be vector functors (of type I and class C'). A d -linear morphism θ of $(\tau_1, \dots, \tau_d)$ to τ is the data consisting, for every family $\mathcal{V}$ of Banach spaces indexed by I , of a continuous d -linear map $\theta_{\mathcal{V}}$ from $\tau_1(\mathcal{V}) \times \dots \times \tau_d(\mathcal{V})$ to $\tau(\mathcal{V})$, this data satisfying the following condition: for every $f \in \text{Hom}(\mathcal{V}, \mathcal{V}')$ one has

$$\tau(\mathbf{f}) \circ \theta_{\mathcal{V}} = \theta_{\mathcal{V}'} \circ (\tau_1(\mathbf{f}) \times \dots \times \tau_d(\mathbf{f})).$$

For $d = 1$, one simply says a morphism of τ_1 to τ .

Now let $\mathcal{M}$ be a family indexed by I of vector bundles with base B . There exists one and only one B-morphism $\theta_{\mathcal{M}}$, which is d -linear, from $\tau_1(\mathcal{M}) \times_B \dots \times_B \tau_d(\mathcal{M})$ to $\tau(\mathcal{M})$ such that $(\theta_{\mathcal{M}})_b = \theta_{\mathcal{M}_b}$ for all $b \in B$.

With the notations of 7.6.3., we have

$$\tau(\mathbf{g}) \circ \theta_{\mathcal{M}} = \theta_{\mathcal{M}'} \circ (\tau_1(\mathbf{g}) \times \dots \times \tau_d(\mathbf{g})).$$

If $d = 1$ and if θ is an isomorphism (which means that $\theta_{\mathcal{V}}$ is an isomorphism for every family $\mathcal{V}$), then $\theta_{\mathcal{M}}$ is an isomorphism.

7.6.5. The definitions and results of nos. 7.6.2 to 7.6.4 extend to the case of vector functors in finite dimension, provided that the given vector bundles are everywhere assumed to be of finite rank.

They also extend to the following case: let L be a field equipped with a K -algebra structure of finite dimension; we take as τ a vector functor over L (i.e., satisfying the hypotheses of 7.6.1 with K replaced by L), and we consider only vector bundles over L in the sense of 7.3.4.

7.6.6. A functor is called a vector functor (resp. finite-dimensional) for isomorphisms if, for every Banach space V (resp. every finite-dimensional vector space over K), it assigns a Banach space $\tau(V)$ and for every isomorphism f from V onto a Banach space V' an isomorphism $\tau(f)$

from $\tau(V)$ onto $\tau(V')$, these data being subject to condition (a) of 7.6.1 and to the following condition:

(b') The map $f \mapsto \tau(f)$ from the open subset of $\mathcal{L}(V; V')$ consisting of the isomorphisms from V onto V' into $\mathcal{L}(\tau(V); \tau(V'))$ is of class C' .

The definitions and results of the preceding numbers extend to the case of vector functors for isomorphisms (by making $I_+ = \{1\}$ and $I_- = \emptyset$), except for those in the first paragraph of no. 7.6.3.

7.7 Direct sums, bundles of multilinear maps, dual

7.7.1. We assume that $I_- = \emptyset$. We define a vector functor σ called the direct sum functor by setting $\sigma(\mathcal{V}) = \bigoplus_{i \in I} V_i$ and $\sigma(f) = \bigoplus_{i \in I} f_i$. If $\mathcal{M} = (M^i)_{i \in I}$ is a family of vector bundles with base B , the vector bundle $\sigma(\mathcal{M})$ is called the direct sum of M^i and is denoted $\bigoplus_{i \in I} M^i$. For every $b \in B$ the fiber at b of $\bigoplus_{i \in I} M^i$ is the direct sum of the fibers of M^i at b . Let U be an open subset of B , and let $s_i \in \mathcal{S}_{M^i}^r(U)$ (for $i \in I$). The map $b \mapsto \sum_{i \in I} s_i(b)$ is then a section, denoted $\sum_i s_i$, of class C^r of $M = \bigoplus_{i \in I} M^i$ and the map $(s_i)_{i \in I} \mapsto \sum_i s_i$ is an isomorphism of $\mathcal{C}^r(U)$ -modules from $\bigoplus_i \mathcal{S}_{M^i}^r(U)$ onto $\mathcal{S}_M^r(U)$.

The underlying manifold of $\bigoplus_{i \in I} M^i$ identifies with the fibered product $\prod_B M^i$.

One denotes by pr_i the vector bundle morphism from $\bigoplus_{i \in I} M^i$ to M^i which on each fiber $\bigoplus_{i \in I} M_b^i$ is the i -th projection. One defines similarly the canonical injection j_i of M^i into $\bigoplus_{i \in I} M^i$.

Let f be a morphism from B into a manifold $\mathbf{B}'$; let H be a second finite set, and let $\mathcal{N} = (N^h)_{h \in H}$ be a family of vector bundles with base $\mathbf{B}'$. The map $u \mapsto (pr_h \circ u \circ j_i)_{(h,i) \in H \times I}$ is a bijection from the set of f -morphisms of $\bigoplus_{i \in I} M^i$ to $\bigoplus_{h \in H} N^h$ to the set of matrices $(u_{h,i})_{(h,i) \in H \times I}$, where $u_{h,i}$ is an f -morphism of M^i to N^h .

If $I = \{1, 2\}$, the sequence

$$0 \rightarrow \mathbf{M}^1 \xrightarrow{j_1} \mathbf{M}^1 \oplus \mathbf{M}^2 \xrightarrow{pr_2} \mathbf{M}^2 \rightarrow 0$$

is direct exact.

Conversely, let M be a vector bundle with base B and let M' be a vector subbundle of M .

Suppose that the manifold B is paracompact and that one of the following two conditions is satisfied:

- (i) K is different from $\mathbf{R}$ or $\mathbf{C}$;
- (ii) $K = \mathbf{R}$, $r \neq \omega$ and the manifold B admits partitions of unity of class C^r (5.3.6).

There then exists a vector subbundle M'' of M such that M is identified with the direct sum $M' \oplus M''$.

7.7.2. Suppose that $I_+ = \{0\}$ and that $I_- = \{1, 2, \dots, d\}$. We define a vector functor η_d of type I and of class C^r by setting $\eta_d(\mathcal{V}) = \mathcal{L}(V_1, \dots, V_d; V_0)$ and $\eta_d(\mathbf{f})(u) = f_0 \circ u \circ (f_1 \times \dots \times f_d)$ for $u \in \eta_d(\mathcal{V})$. If $\mathcal{M} = (M_i)_{i \in I}$ is a family of vector bundles with base B , the vector bundle $\eta_d(\mathcal{M})$ is denoted $\mathcal{L}(M_1, \dots, M_d; M_0)$.

Let u be a multilinear morphism from $\mathbf{M}_1 \times_B \dots \times_B \mathbf{M}_d$ to $\mathbf{M}_0$. The map $\hat{u} : b \mapsto u_b$ is then a section of $\mathcal{L}(\mathbf{M}_1, \dots, \mathbf{M}_d; \mathbf{M}_0)$ and the map $u \mapsto \hat{u}$ is bijective.

7.7.3. We keep the notation of 7.7.2 and suppose further that $d = 1$. The vector bundle $\mathcal{L}(\mathbf{M}_1; \mathbf{M}_0)$ is then called the bundle of homomorphisms of $\mathbf{M}_1$ to $\mathbf{M}_0$. Its sections correspond to the B -morphisms of $\mathbf{M}_1$ to $\mathbf{M}_0$.

If moreover M_0 is the trivial bundle K_B , the vector bundle $\mathcal{L}(\mathbf{M}_1; K_B)$ is called the dual of $M = M_1$ and is denoted M' : the fiber $(M')_b$ is the space of continuous linear forms on the fiber M_b of M at the point $b \in B$.

If s (resp. t) is a section of M (resp. M') over an open set U of B , the map $b \mapsto (b, \langle s(b), t(b) \rangle)$ is a section, denoted $\langle s, t \rangle$ of the trivial bundle K_B .¹

7.8 Bundles of alternating multilinear maps

In nos. 7.8.1 to 7.8.5, we assume that K has characteristic 0 or that the vector bundles under consideration have finite rank. It is not known whether the definition of the functor α_d given in 7.8.1 can be done without any restriction.

7.8.1. Let us take $I_+ = \{0\}$, $I_- = \{1\}$ and let d be an integer ≥ 1 . We define a

¹ When M is of finite rank, we write M^* in place of M' .

vector functor α_d by denoting by $\alpha_d(\mathcal{V})$ the Banach space of continuous alternating d -linear

maps from $\mathbf{V}_1^d$ into $\mathbf{V}_0$ and setting $\alpha_d(\mathbf{f})(u) = f_0 \circ u \circ f_1^d$ for $u \in \alpha_d(\mathcal{V})$. The vector bundle $\alpha_d((\mathbf{M}_1, \mathbf{M}_0))$ is denoted $\text{Alt}^d(\mathbf{M}_1; \mathbf{M}_0)$ and is called the vector bundle of alternating d -linear maps from $\mathbf{M}_1$ to $\mathbf{M}_0$.

The canonical injection of $\text{Alt}^d(\mathbf{M}_1; \mathbf{M}_0)$ to $\mathcal{L}(\mathbf{M}_1, \dots, \mathbf{M}_1; \mathbf{M}_0)$ is a morphism of vector bundles; $\text{Alt}^d(\mathbf{M}_1; \mathbf{M}_0)$ is a vector subbundle of $\mathcal{L}(\mathbf{M}_1, \dots, \mathbf{M}_1; \mathbf{M}_0)$.

One has $\text{Alt}^1(\mathbf{M}_1; \mathbf{M}_0) = \mathcal{L}(\mathbf{M}_1; \mathbf{M}_0)$. We set $\text{Alt}^0(\mathbf{M}_1; \mathbf{M}_0) = \mathbf{M}_0$.

If ω is a section¹ of $\text{Alt}^d(\mathbf{M}_1; \mathbf{M}_0)$ and if $s_1, \dots, s_d$ are sections of $\mathbf{M}_1$ there exists one and only one section of $\mathbf{M}_0$, denoted $\omega(s_1, \dots, s_d)$, such that

$$\omega(s_1, \dots, s_d)(b) = \omega(b)(s_1(b), \dots, s_d(b)) \quad \text{for all } b \in \mathbf{B}.$$

7.8.2. Let φ be a pairing of $N \times_B N'$ to N'' (cf. no. 7.3.1); for each b , we have a bilinear map φ_b of $N_b \times N'_b$ to N''_b which defines (cf. A, III, p. 142) a bilinear map of

$$\text{Alt}^d(\mathbf{M}; \mathbf{N})_b \times \text{Alt}^e(\mathbf{M}; \mathbf{N}')_b$$

to $\text{Alt}^{d+e}(\mathbf{M}; \mathbf{N}'')_b$. The collection of these bilinear maps defines a pairing u of

$$\text{Alt}^d(\mathbf{M}; \mathbf{N}) \times_{\mathbf{B}} \text{Alt}^e(\mathbf{M}; \mathbf{N}')$$

to $\text{Alt}^{d+e}(\mathbf{M}; \mathbf{N}'')$; if ω and ω' are respectively sections of $\text{Alt}^d(\mathbf{M}; \mathbf{N})$ and $\text{Alt}^e(\mathbf{M}; \mathbf{N}')$ over an open set U , the section $u(\omega, \omega')$ of $\text{Alt}^{d+e}(\mathbf{M}; \mathbf{N}'')$ over U will be denoted $\omega \wedge_{\varphi} \omega'$, and called the exterior product of ω and ω' .

We have the formula:

$$(1) \quad (\omega \wedge_{\varphi} \omega')(s_1, \dots, s_{d+e}) = \sum_{\sigma} \varepsilon_{\sigma} \varphi(\omega(s_{\sigma(1)}, \dots, s_{\sigma(d)}), \omega'(s_{\sigma(d+1)}, \dots, s_{\sigma(d+e)}))$$

where the s_i are sections of $\mathbf{M}$ over U , and where the summation is extended over the permutations σ of $\{1, 2, \dots, d+e\}$ such that

$$\sigma(1) < \dots < \sigma(d) \quad \text{and} \quad \sigma(d+1) < \dots < \sigma(d+e).$$

7.8.3. Let M be a vector bundle and A a bundle of algebras, with base B . Suppose that the fibres A_b of A are associative and commutative algebras, possessing an identity element, denoted e_b . For every open set U of B , we denote by $\Omega^d(U)$ the $\mathcal{C}^r(U)$ -module formed by the sections of the bundle $\text{Alt}^d(M; A)$ and $\Omega^*(U)$ the direct sum of the $\Omega^d(U)$ for $d \geq 0$. The multiplications on each fibre define a pairing of $A \times_B A$ into A , whence (7.8.2) a structure of graded algebra on $\Omega^*(U)$, which is associative and anticommutative. The subalgebra $\Omega^0(U)$ is the algebra of sections of A . An element ω of $\Omega^1(U)$ is identified with a U -morphism of $\mathbf{M}|U$ into $A|U$ (7.7.3): if $s \in \mathcal{S}_{\mathbf{M}}^r(U)$, we denote by $\langle \omega, s \rangle$ the section $\omega(s)$ of A (7.4.2). Let $s_j \in \mathcal{S}_{\mathbf{M}}^r(U)$ and $\omega_j \in \Omega^1(U)$ (for $1 \leq j \leq d$); we have:

¹ The reader should take care not to confuse this use of the letter ω with that defined on p. 10.

$$\omega(s_1, \dots, s_d) = \det(\langle \omega_i, s_j \rangle) \quad \text{for } \omega = \omega_1 \wedge \dots \wedge \omega_d. \quad (2)$$

7.8.4. Let $d \geq 1$. There exists a pairing i of $\mathbf{M} \times_B \text{Alt}^d(\mathbf{M}; A)$ to $\text{Alt}^{d-1}(\mathbf{M}; A)$ whose restriction to each fiber is given by the right inner product (cf. A, III, p. 156). If s is a section of $\mathbf{M}$ on the open subset U and if $\omega \in \Omega^d(U)$, one denotes by $i(s)\omega$ the section $i(s, \omega)$ of $\text{Alt}^{d-1}(\mathbf{M}; A)$ on U ; we set $i(s)\omega = 0$ for ω in $\Omega^0(U)$.

We thus associate to every section s of $\mathbf{M}$ on U an endomorphism of the $\mathcal{C}^r(U)$ -module $\Omega^*(U)$. We have the formulas:

$$(3) \quad (i(s)\omega)(s_1, \dots, s_{d-1}) = \omega(s, s_1, \dots, s_{d-1}) \quad \text{for } \omega \in \Omega^d(U), d \geq 1$$

$$(4) \quad i(s) \circ i(s) = 0$$

$$i(s)\omega = \langle \omega, s \rangle \quad \text{for } \omega \in \Omega^1(U) \quad (5)$$

$$(6) \quad i(s).(\omega \wedge \omega') = i(s)\omega \wedge \omega' + (-1)^d \omega \wedge i(s)\omega' \quad \text{for } \omega \in \Omega^d(\mathbf{U})$$

$$i(s)(\omega_1 \wedge \cdots \wedge \omega_p) = \sum_{i=1}^p (-1)^{i+1} \langle \omega_i, s \rangle \omega_1 \wedge \cdots \wedge \hat{\omega}_i \wedge \cdots \wedge \omega_p. \quad (7)$$

In the last formula, the ω_i are in $\Omega^1(\mathbf{U})$ and the sign indicates that the symbol it surmounts is to be omitted.

All the operations described above on sections are multilinear over the ring $\mathcal{C}^r(\mathbf{U}; \mathbf{K})$.

7.8.5. Let L be a Banach algebra over $\mathbf{K}$. The definitions and results of Sections 7.7 and 7.8 extend to the case of vector bundles over L : one defines analogously the bundles of L -multilinear or alternating L -multilinear maps.

7.9 Tensor products, tensor spaces, exterior algebra

We keep the notation of 7.6. Moreover, we denote by L a commutative field endowed with a structure of $\mathbf{K}$ -algebra of finite dimension, and we call vector bundle a vector bundle over L , with base B and locally finite rank.

7.9.1. Suppose that $\mathbf{I}_- = \emptyset$. If $\mathcal{V}$ and $\mathcal{V}'$ are two families indexed by I of finite-dimensional vector spaces over L , we denote by $\tau(\mathcal{V})$ the tensor product of the V_i for $i \in I$ (A, II, p. 71) and if $\mathbf{f} \in \text{Hom}(\mathcal{V}, \mathcal{V}')$, we set $\tau(\mathbf{f}) = \otimes f_i$. One thus defines a vector functor over L in finite dimension, and if $\mathcal{M} = (M_i)_{i \in I}$ is a family of vector bundles, we denote the vector bundle $\tau(\mathcal{M})$ by $\otimes_{i \in I} M_i$ and call it the tensor product (over L) of the M_i .

If s_i is a section of $\mathbf{M}_i$ over the open set U of B (for $i \in I$), the map $b \mapsto \otimes_{i \in I} s_i(b)$ is a section of $\otimes_{i \in I} \mathbf{M}_i$, denoted $\otimes_{i \in I} s_i$. The map $(s_i)_{i \in I} \mapsto \otimes_{i \in I} s_i$ is multilinear over the ring $\mathcal{C}^r(U; L)$.

7.9.2. The canonical isomorphisms defined in Alg., chap. II provide isomorphisms of vector functors. It follows from 7.6.4 that there are isomorphisms of vector bundles. For example, we have canonical isomorphisms:

$$(M_1 \oplus M_2) \otimes M_3 \rightarrow (M_1 \otimes M_3) \oplus (M_2 \otimes M_3)$$

$$M_1^* \otimes M_2 \rightarrow \mathcal{L}(M_1; M_2)$$

etc.

7.9.3. Let M be a vector bundle and let I and J be two disjoint finite sets. The tensor bundle $T_J^I(M)$ is defined as the tensor product $\bigotimes_{\alpha \in I \cup J} M_\alpha$, where $M_\alpha = M$ if $\alpha \in I$, and $M_\alpha = M^*$ if $\alpha \in J$, M^* denotes the dual of M (A, III, p. 63). The fiber $T_J^I(M)_b$ of this bundle at a point b is equal to the tensor space $T_J^I(M_b)$ defined in Alg., loc. cit. When $I = \{1, \dots, p\}$ and $J = \{p+1, \dots, p+q\}$, we write $T_q^p(M)$ instead of $T_J^I(M)$; we have

$$T_q^p(M) = (\bigotimes^p M) \otimes (\bigotimes^q M^*).$$

The data of a total order on I and on J defines a canonical isomorphism of $T_J^I(M)$ onto $T_q^p(M)$.

7.9.4. If I (resp. J) is the union of two disjoint subsets I' and I'' (resp. J' and J''), $T_J^I(M)$ is canonically identified with the tensor product $T_{J'}^{I'}(M) \otimes T_{J''}^{I''}(M)$. In particular, if s' (resp. s'') is a section of $T_{J'}^{I'}(M)$ (resp. of $T_{J''}^{I''}(M)$), the tensor product $s' \otimes s''$ is identified with a section of $T_J^I(M)$.

The dual of $T_J^I(M)$ is identified with $T_I^J(M)$.

7.9.6. Let $i \in I$ and $j \in J$. For every $b \in B$, the contraction homomorphism of indices i and j is defined (cf. Alg., loc. cit.); it is a homomorphism $(c_j^i)_b : T_J^I(M_b) \rightarrow T_{J-\{j\}}^{I-\{i\}}(M_b)$. The collection of the $(c_j^i)_b$ defines a morphism of vector bundles

$$c_j^i : T_J^I(M) \rightarrow T_{J-\{j\}}^{I-\{i\}}(M),$$

still called contraction of indices i and j . One defines similarly the contraction of the indices $i_1, \dots, i_k$ of I with the indices $j_1, \dots, j_k$ of J .

7.9.7. Let d be an integer ≥ 0 ; let V and V' be two finite-dimensional vector spaces over L and $f \in \text{Hom}_L(V, V')$; set $\lambda_d(V) = \bigwedge^d(V)$ and $\lambda_d(f) = \bigwedge^d(f)$. One thus defines a vector functor on L in finite dimension and, if M is a vector bundle, one denotes the vector bundle $\lambda_d(M)$ by $\bigwedge^d(M)$, calling it the d -th exterior power of M (over L).

The canonical map from $\otimes^d \mathbf{V}$ onto $\bigwedge^d(\mathbf{V})$ defines a morphism of vector functors, whence a morphism, said to be canonical, from $\otimes^d \mathbf{M}$ to $\bigwedge^d(\mathbf{M})$. This morphism is surjective.

The canonical isomorphisms of the space of alternating d-multilinear maps onto the space $\bigwedge^d(\mathbf{V}^*)$ or onto $(\bigwedge^d(\mathbf{V}))^*$ provide isomorphisms, said to be canonical, of the vector bundle $\text{Alt}^d(\mathbf{M}; \mathbf{L})$ of alternating d-multilinear maps over L, onto the vector bundle $\bigwedge^d(\mathbf{M}^*)$ or onto $(\bigwedge^d(\mathbf{M}))^*$.

7.9.8. Now set $\lambda(V) = \bigwedge(V)$ and $\lambda(f) = \bigwedge(f)$; one thus again defines a vector functor on L in finite dimension. The vector bundle $\lambda(M)$ is denoted $\bigwedge(M)$. Its fiber $\bigwedge(M)_b$ at $b \in B$ is the exterior algebra (over L) of the fiber M_b . The vector bundle $\bigwedge(M)$ is a locally trivial vector bundle of algebras.

The definitions and properties of interior products given in Alg., chap. III, 3rd ed., § 10 extend immediately to sections of vector bundles $\bigwedge(\mathbf{M})$ and $\bigwedge(\mathbf{M}^*)$ (cf. also formulas (1) to (7) of nos. 7.8.2 to 7.8.4).

7.9.9. Let M be a vector bundle. For every integer n, let B_n be the (open) set of points $b \in B$ such that the dimension (over L) of M_b is equal to n. For $b \in B_n$, set $N_b = \bigwedge^n(M_b)$ and let N be the disjoint union of the N_b for $b \in B$. There exists on N a vector bundle structure, and only one, such that the evident map from $N|B_n$ to $\bigwedge^n(M)|B_n$ is an isomorphism for every n. Endowed with this structure, the vector bundle N, of rank one at each point, is denoted $\det(M)$.

7.10 Vector bundles and principal bundles

7.10.1. Let F be a Banach space. A vector bundle M with base B is said to be pure of type F if all the fibers M_b of M (for $b \in B$) are isomorphic (as Banach spaces) to F.

Let M be a vector bundle with base B pure of type F, and let P be the open submanifold of the vector bundle $\mathcal{L}(F_B; M)$ composed of pairs (b, u) where $b \in B$ and where u is an isomorphism of $F_b = F$ onto M_b . The group $\mathbf{GL}(F)$ of automorphisms of F acts on the right on P by setting $(b, u).g = (b, u \circ g)$ for $(b, u) \in P$ and $g \in \mathbf{GL}(F)$. Denote by π_P the map $(b, u) \mapsto b$ from P into B. The quadruple $\lambda = (P, \mathbf{GL}(F), B, \pi_P)$ (where $\mathbf{GL}(F)$ is endowed with its canonical structure as a group manifold (5.12.2)) is a principal fibration (6.2.1): it is called the frame fibration of M. The map $((b, u), h) \mapsto u(h)$ from $P \times F$ into M endows M with a structure of fiber bundle

associated with λ with typical fiber F (6.5.1).

When $F = \mathbf{K}^n$ one can identify an isomorphism u of F onto $\mathbf{M}_b$ with the basis of $\mathbf{M}_b$ formed by the image under u of the canonical basis of $\mathbf{K}^n$. The frame bundle of $\mathbf{M}$ is then identified with the open submanifold of $\mathbf{M} \times_{\mathbf{B}} \dots \times_{\mathbf{B}} \mathbf{M}$ formed by the bases $(e_1, \dots, e_n)$ of the different fibers $\mathbf{M}_b$.

Let U be an open subset of B and let $t = (U, \varphi, E)$ be a vector chart of M with domain U , with $E = F$. The map $b \mapsto (b, t_b)$ is then a section of P , denoted $\tilde{t}$, and the map $t \mapsto \tilde{t}$ is a bijection from the set of vector charts of M of the form (U, φ, F) onto the set of sections of P over U .

7.10.2. Conversely, let $\lambda = (\mathbf{Q}, \mathbf{G}, \mathbf{B}, \pi_{\mathbf{Q}})$ be a principal fibration and let φ be a homomorphism of group manifolds from G into the group $\mathbf{GL}(F)$ of automorphisms of a Banach space F . Let G act on the left on F by setting $g \cdot h = \varphi(g)(h)$ ($h \in F, g \in G$). Let M be a fiber bundle associated with λ with typical fiber F , and let $\rho : \mathbf{Q} \times F \rightarrow \mathbf{M}$ be its frame map (6.5.1). Let π be the map from M to B defined by

$$\pi(\rho(q, h)) = \pi_{\mathbf{Q}}(q) \quad (q \in \mathbf{Q}, h \in F).$$

Let s be a section of Q over an open set U of B ; the map

$$\tilde{s} : (b, h) \mapsto (s(b), h)$$

is then a bijection from $U \times F$ onto $\pi^{-1}(U)$. On the pair (M, π) there exists a vector bundle structure with base B , and only one, for which the triples $t_s = (U, \tilde{s}^{-1}, F)$ are vector bundle charts (for every section s of Q). The manifold structure underlying this structure is that of the associated fiber bundle to λ .

Let $q \in \mathbf{Q}$; set $b = \pi_{\mathbf{Q}}(q)$ and let u be the isomorphism of F onto $\mathbf{M}_b$ defined by $u(h) = \rho(q, h)$ (for $h \in F$). The map $f : q \mapsto (b, u)$ is a B -morphism of principal fibrations, compatible with φ , from $(\mathbf{Q}, \mathbf{G}, \mathbf{B}, \pi_{\mathbf{Q}})$ into the frame bundle $(\mathbf{P}, \mathbf{GL}(F), \mathbf{B}, \pi_{\mathbf{P}})$ of the vector bundle $\mathbf{M}$.

7.10.3. Let us resume the notation of Section 7.6. For every $i \in I$, let

$$\lambda_i = (\mathbf{P}_i, \mathbf{G}_i, \mathbf{B}, \pi_i)$$

a principal fibration with base B, and suppose that G_i acts on the left on a Banach space V_i by means of a homomorphism

$$\varphi_i : \mathbf{G}_i \rightarrow \mathbf{GL}(V_i).$$

Let $\mathbf{M}_i$ be a fiber bundle associated with λ_i with typical fiber V_i . Set $\mathcal{M} = (\mathbf{M}_i)_{i \in I}$ and $\mathcal{V} = (V_i)_{i \in I}$ and let λ be the principal product fibration of the λ_i over B (6.2.5). Set $\lambda = (\mathbf{P}, \mathbf{G}, \mathbf{B}, \pi_{\mathbf{P}})$, with $\mathbf{G} = \prod_{i \in I} \mathbf{G}_i$.

Now let τ be a vector functor. For $g = (g_i) \in G$, let $\varphi(g)$ be the element of $\text{Hom}(\mathcal{V}, \mathcal{V})$ defined by:

$$\begin{aligned} \varphi(g)_i &= \varphi_i(g_i) & \text{if } i \in I_+ \\ \varphi(g)_i &= \varphi_i(g_i)^{-1} & \text{if } i \in I_-. \end{aligned}$$

The group G then acts on $\tau(\mathcal{V})$ by means of the morphism $g \mapsto \tau(\varphi(g))$ from G to $\mathbf{GL}(\tau(\mathcal{V}))$.

Let moreover $x = (x_i)$ be a point of P and let $b = \pi_{\mathbf{P}}(x)$. For each i , the map θ_{x_i} defined in no. 6.5.2 is an isomorphism from V_i onto $(\mathbf{M}_i)_b$. Let θ_x be the element of $\text{Hom}(\mathcal{V}, ((\mathbf{M}_i)_b)_{i \in I})$ defined by:

$$\begin{aligned} (\theta_x)_i &= \theta_{x_i} & \text{if } i \in I_+ \\ (\theta_x)_i &= \theta_{x_i}^{-1} & \text{if } i \in I_-. \end{aligned}$$

Let ρ be the map $(x, h) \mapsto (b, \tau(\theta_x)(h))$ of $\mathbf{P} \times \tau(\mathcal{V})$ into the vector bundle $\tau(\mathcal{M})$; the map ρ endows $\tau(\mathcal{M})$ with a structure of fiber bundle associated with λ with typical fiber $\tau(\mathcal{V})$.

These considerations generalize to the case of finite-dimensional vector functors, or of vector functors over a field L equipped with a structure of finite-dimensional K-algebra.

7.11 Change of structure

The structures and operations described in this paragraph are compatible with the changes of structure described in nos. 5.13 and 5.14.

Appendix A

Continuous polynomials and formal series

In this Appendix, F denotes a separated polynormed space over K , and E_i (for $1 \leq i \leq n$) denotes a normed space over K , and E denotes the topological vector space product of the E_i .

For $\alpha \in \mathbb{N}^n$, and $1 \leq j \leq |\alpha|$, one sets:

$$\alpha(j) = \inf \{k \in \mathbb{N} \mid 1 \leq k \leq n \text{ and } j > \alpha_1 + \dots + \alpha_{k-1}\}$$

(the sequence of $\alpha(j)$ is thus obtained by writing α_1 times 1, ..., α_n times n).

One denotes by E_α the topological vector space product of the family of the $E_{\alpha(j)}$ for $1 \leq j \leq |\alpha|$. One denotes by $\text{Hom}_\alpha(E_1, \dots, E_n; F)$ the space of $|\alpha|$ -multilinear maps from E_α into F and by $\mathcal{L}_\alpha(E_1, \dots, E_n; F)$ the subspace of $\text{Hom}_\alpha(E_1, \dots, E_n; F)$ formed by the continuous multilinear maps, equipped with the topology of uniform convergence on bounded subsets of E_α ; this is a separated polynormed space, whose topology can be defined by the family of seminorms $\|u\|_\gamma$ where, for a continuous seminorm γ on F , one denotes by $\|u\|_\gamma$ the infimum of the numbers $a \geq 0$ such that

$$\|u(x_1, \dots, x_{|\alpha|})\|_\gamma \leq a \|x_1\| \dots \|x_{|\alpha|}\|$$

where each x_j ranges over $E_{\alpha(j)}$.

One denotes by p_i the canonical projection of E onto E_i . For $\alpha \in \mathbb{N}^n$ with $\alpha \neq 0$, one denotes by p_α the map $x \mapsto (p_{\alpha(j)}(x))$ from E into E_α .

A.1. A map f from E into F is called a multihomogeneous polynomial of multidegree α

(with $\alpha \in \mathbf{N}^n$) on E with values in F if there exists an element

$$u \in \text{Hom}_\alpha(E_1, \dots, E_n; F)$$

such that $f = u \circ p_\alpha$.

A.2. One denotes by $P_\alpha(E_1, \dots, E_n; F)$ the image of $\mathcal{L}_\alpha(E_1, \dots, E_n; F)$ under the linear map $u \mapsto u \circ p_\alpha$ from $\text{Hom}_\alpha(E_1, \dots, E_n; F)$ into the space of mappings from E to F , equipped with the quotient topology induced by that of $\mathcal{L}_\alpha(E_1, \dots, E_n; F)$. An element of $P_\alpha(E_1, \dots, E_n; F)$ is called a multihomogeneous continuous polynomial of multidegree α on E with values in F . The topology of $P_\alpha(E_1, \dots, E_n; F)$ is defined by the family of seminorms:

$$\|f\|_\gamma = \inf_{u \in \mathcal{L}_\alpha(E_1, \dots, E_n; F), f = u \circ p_\alpha} \|u\|_\gamma$$

for γ ranging over the set of continuous seminorms on F . If F is a normed space, with norm γ , we write $\|f\|$ instead of $\|f\|_\gamma$. The space $P_\alpha(E_1, \dots, E_n; F)$ and its topology do not change if one substitutes for the given norms on each E_i equivalent norms. They can therefore be defined when the E_i are normable topological vector spaces.

In particular, an element of $P_k(E; F)$ is called a homogeneous continuous polynomial of total degree k on E with values in F . The space $P_k(E; F)$ is the topological direct sum of the spaces $P_\alpha(E_1, \dots, E_n; F)$ for $|\alpha| = k$. The space $P_0(E; F)$ is the space of constant maps from E to F and it is identified with F .

A.3. We denote by $P(E; F)$ or $P(E_1, \dots, E_n; F)$ the vector subspace of the vector space of all maps from E to F generated by the subspaces $P_k(E; F)$.

It is the direct sum of the subspaces $P_\alpha(E_1, \dots, E_n; F)$ for $\alpha \in \mathbf{N}^n$, and also of the subspaces $P_k(E; F)$ for $k \in \mathbf{N}$. An element of $P(E; F)$ is called a continuous polynomial on E with values in F .

A.4. Let G_j be normed spaces (for $1 \leq j \leq m$). Let $f_j \in P(E_1, \dots, E_n; G_j)$ (for $1 \leq j \leq m$) and let $g \in P(G_1, \dots, G_m; F)$. The map

$$h : x \mapsto g(f_1(x), \dots, f_m(x))$$

belongs to $\mathbf{P}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$. If moreover $f_j \in \mathbf{P}_\alpha(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{G}_j)$ for $1 \leq j \leq m$ (with $\alpha \in \mathbb{N}^n$) and $g \in \mathbf{P}_\beta(\mathbf{G}_1, \dots, \mathbf{G}_m; \mathbf{F})$ (with $\beta \in \mathbb{N}^m$), we have $h \in \mathbf{P}_{|\beta|_\alpha}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ and

$$\|h\|_\gamma \leq \|g\|_\gamma \cdot \|f\|^\beta$$

for every continuous seminorm γ on $\mathbf{F}$ (setting $\|f\|^\beta = \prod_{1 \leq j \leq m} \|f_j\|^{\beta_j}$).

A.5. We denote by $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ the product vector space of the $\mathbf{P}_\alpha(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ for $\alpha \in \mathbb{N}^n$, equipped with the product topology of the discrete topologies on each of the factors. This topology makes $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ a separated and complete topological group. The linear map from $\mathbf{P}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ to $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ extending the canonical injections of the $\mathbf{P}_\alpha(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ is injective and its image is dense in $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$: one generally identifies a continuous polynomial on $\mathbf{E}$ with values in $\mathbf{F}$ with its image in $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$.

The identity map of $\mathbf{P}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F}) = \mathbf{P}(\mathbf{E}; \mathbf{F})$ extends by continuity to an isomorphism from $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ onto $\hat{\mathbf{P}}(\mathbf{E}; \mathbf{F})$.

An element of $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ is called a formal series with continuous components (or, by abuse of language, a formal series) on the product of $\mathbf{E}_i$ with values in $\mathbf{F}$.

A.6. Let $\mathbf{G}_j$ (for $1 \leq j \leq m$) be normed spaces and let $g \in \mathbf{P}(\mathbf{G}_1, \dots, \mathbf{G}_m; \mathbf{F})$. The map $\mathbf{f} \mapsto g \circ \mathbf{f}$ from $\prod_{1 \leq j \leq m} \mathbf{P}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{G}_j)$ to $\mathbf{P}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ extends by continuity to a map (still denoted $\mathbf{f} \mapsto g \circ \mathbf{f}$) from $\prod_{1 \leq j \leq m} \hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{G}_j)$ to $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$.

Let $f_j = (f_{j,\alpha})_{\alpha \in \mathbb{N}^n} \in \hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{G}_j)$, with $f_{j,0} = 0$. Set $\mathbf{f} = (f_j)$: the map $g \mapsto g \circ \mathbf{f}$ of $\mathbf{P}(\mathbf{G}_1, \dots, \mathbf{G}_m; \mathbf{F})$ to $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$ extends by continuity to a map (still denoted $g \mapsto g \circ \mathbf{f}$) of $\hat{\mathbf{P}}(\mathbf{G}_1, \dots, \mathbf{G}_m; \mathbf{F})$ to $\hat{\mathbf{P}}(\mathbf{E}_1, \dots, \mathbf{E}_n; \mathbf{F})$.

INDEX OF NOTATIONS

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Part II

Paragraphs 8 to 15 (99 pages)

8 First-order differential calculus

From no. 8.3 on, one assumes

- either that K is of characteristic zero,
- or that the manifolds considered are locally of finite dimension.

In the latter case, the expressions "Banach space, vector bundle, vector functor" mean respectively "finite-dimensional vector space, finite-rank vector bundle, finite-dimensional vector functor with finite-dimensional values".

8.1 Tangent bundle

8.1.1. Let $r \in \mathbb{N}_K$ and let X be a K -manifold of class C^r . One denotes by $T(X)$ the disjoint union of the tangent spaces $T_x(X)$ (5.5.1) for $x \in X$, and π the canonical map from $T(X)$ to X . Let $c = (U, \varphi, E)$ be a chart of the manifold X ; for $x \in U$ and $t \in T_x(X)$, set $\zeta_c(x, t) = (x, \theta_c^{-1}(t))$, the map θ_c being that defined in 5.5.1. The triple $c' = (U, \zeta_c, E)$ is a vector chart (7.1.1) of $T(X)$ (endowed with the map $\pi : T(X) \rightarrow X$). There exists on $T(X)$ a structure of vector bundle with base X and of class C^{r-1} , unique, such that, for every chart c of X , the corresponding vector chart c' is a vector chart of $T(X)$ ⁽¹⁾. Endowed with this structure, $T(X)$ is called the tangent bundle of X . In particular, $T(X)$ is endowed with a manifold structure of class C^{r-1} , and the triple $(T(X), X, \pi)$ is a fibration (7.1.4).

Let U be an open subset of a Banach space E and let i be the canonical injection of U into E . The vector chart $c' = (U, \zeta_c, E)$ associated with the chart $c = (U, i, E)$ from U defines an isomorphism from $T(U)$ onto the trivial bundle E_U (7.1.5); we identify these two bundles by means of this isomorphism.

If X is a pure manifold of type E (5.1.7), then $T(X)$ is a pure manifold of type $E \times E$.

8.1.2. Let $f : X \rightarrow Y$ be a morphism of manifolds of class C^r . We define an f -morphism $T(f) : T(X) \rightarrow T(Y)$ of vector bundles of class C^{r-1} by $T(f)(x, t) = (f(x), T_x(f)(t))$ for $x \in X$ and $t \in T_x(X)$.

If $f = \text{Id}_X$, one has $T(f) = \text{Id}_{T(X)}$. For $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, one has $T(g \circ f) = T(g) \circ T(f)$.

8.1.3. Let $f : X \rightarrow Y$ be a morphism of manifolds of class C^r , and let $f' : T(X) \rightarrow f^*T(Y)$

¹ When $K = \mathbb{R}$ and $r = 1$, $T(X)$ is a topological vector bundle (cf. Notations and conventions).

be the unique X -morphism such that $T(f) = g \circ f'$, where g is the canonical morphism $f^*T(Y) \rightarrow T(Y)$ (cf. 7.2.4).

For f to be an immersion (resp. a submersion), it is necessary and sufficient that $0 \rightarrow T(X) \xrightarrow{f'} f^*T(Y)$ (resp. $T(X) \xrightarrow{f'} f^*T(Y) \rightarrow 0$) be a locally direct exact sequence (7.5.6). If f is an immersion, the cokernel bundle of f' is called the normal or transverse bundle of f . If f is a submersion, the kernel bundle of f' is called the tangent bundle to the fibers of f , or the relative tangent bundle of X over Y , and is denoted $T(X/Y)$. Its fiber $T_x(X/Y)$ at $x \in X$ is the tangent space at x to the submanifold $f^{-1}(f(x))$ of X . The sequence

$$0 \rightarrow T_x(X/Y) \xrightarrow{i} T_x(X) \xrightarrow{T_x(f)} T_{f(x)}(Y) \rightarrow 0$$

where i is the canonical injection, is exact.

For f to be étale, it is necessary and sufficient that f' be an isomorphism.

When K is of characteristic 0, for f to be a subimmersion, it is necessary and sufficient that f' be locally direct.

8.1.4. Let X_1 and X_2 be two manifolds. We have $T(X_1 \times X_2) = p_1^*T(X_1) \oplus p_2^*T(X_2)$, where p_1 and p_2 are the canonical projections (5.6.3).

8.1.5. Let n be an integer ≥ 0 ; the tangent bundle $T(\mathbf{K}^n)$ of $\mathbf{K}^n$ is identified with the trivial bundle with fiber $\mathbf{K}^n$ (8.1.1). The constant sections defined by the elements of the canonical basis of $\mathbf{K}^n$ form a frame of $T(\mathbf{K}^n)$.

If X is a manifold and $(u^1, \dots, u^n)$ a coordinate system of X on an open set U of X (5.1.10), the preceding frame defines, by transport of structure, a frame of $T(X)$ on U ; it is called the tangent frame defined by $(u^1, \dots, u^n)$ and is denoted $(\partial/\partial u^1, \dots, \partial/\partial u^n)$ (cf. 5.5.8).

8.2 Vector fields

Let X be a manifold of class C^r , and let U be an open set of X .

8.2.1. A section (not necessarily continuous, cf. Notations and Conventions) of the tangent bundle $T(X)$ over U is called a vector field on U .

8.2.2. The dual bundle $T'(X)$ (7.7.3) of $T(X)$ is called the cotangent bundle of X . Its sections

are called covector fields. If f is a function of class C^k on X (with $1 \leq k \leq r$), with values in K , its differential $df: x \mapsto d_x f$ (5.5.6) is a covector field of class C^{k-1} . More generally, if f is a function of class C^k on X ($1 \leq k \leq r$), with values in a Banach space E , its differential df is a section of class C^{k-1} of the bundle $\mathcal{L}(T(X); E_X)$.

8.2.3. Let ξ be a vector field of class C^{r-1} on X , and let $f \in \mathcal{C}^r(X; F)$ be a function of class C^r with values in a Banach space F . We denote by $\langle \xi, df \rangle$, or $\mathbf{D}_\xi(f)$, or again $\xi(f)$ the function $x \mapsto d_x f(\xi(x))$; it is an element of $\mathcal{C}^{r-1}(X; F)$. For f fixed, the map $\xi \mapsto \mathbf{D}_\xi(f)$ is $\mathcal{C}^{r-1}(X)$ -linear. If $\mathbf{D}_\xi f = 0$ for every function f of class C^r with values in a Banach space, defined on an open set of X , we have $\xi = 0$.

Let E, F, G be Banach spaces and let $(u, v) \mapsto u \cdot v$ be a continuous bilinear map from $E \times F$ to G . Let $f \in \mathcal{C}^r(X; E)$, $g \in \mathcal{C}^r(X; F)$ and let $f \cdot g \in \mathcal{C}^r(X; G)$ be their product. If ξ is a vector field of class C^{r-1} on X , one has

$$\mathbf{D}_\xi(f \cdot g) = \mathbf{D}_\xi(f) \cdot g + f \cdot \mathbf{D}_\xi(g).$$

In particular, the map $\mathbf{D}_\xi: \mathcal{C}^r(X) \rightarrow \mathcal{C}^{r-1}(X)$ is a derivation of the algebra $\mathcal{C}^r(X)$ into the $\mathcal{C}^r(X)$ -module $\mathcal{C}^{r-1}(X)$.

8.2.4. Suppose that X is locally of finite dimension and that one of the following two hypotheses holds:

- (i) $r = \infty$ and X is Hausdorff;
- (ii) $r = \omega$ and X is isomorphic to an open polydisc of K^n .

Then, for every derivation D of the algebra $\mathcal{C}^r(X)$, there exists one and only one vector field ξ of class C^r such that $D = D_\xi$.

8.2.5. Let $(u^1, \dots, u^n)$ be a coordinate system of X on U , and ξ a vector field on U . The i -th coordinate of ξ in the tangent frame (8.1.5) defined by $(u^1, \dots, u^n)$ is $\mathbf{D}_\xi(u^i)$. In other words, we have

$$\xi = \sum_{1 \leq i \leq n} \mathbf{D}_\xi(u^i) \cdot \partial / \partial u^i.$$

8.2.6. Let $\varphi : X \rightarrow Y$ be a morphism of manifolds of class C^r , ξ a vector field on X and η a vector field on Y . We say that ξ is φ -related to η if, for all $x \in X$, one has $\eta(\varphi(x)) = T_x(\varphi)(\xi(x))$. Suppose that this is so, and that ξ and η are of class C^{r-1} . Then, for every Banach space E , the diagram

$$\begin{array}{ccc} \mathcal{C}^r(Y; E) & \xrightarrow{\varphi^*} & \mathcal{C}^r(X; E) \\ \downarrow D_\eta & & \downarrow D_\xi \\ \mathcal{C}^{r-1}(Y; E) & \xrightarrow{\varphi^*} & \mathcal{C}^{r-1}(X; E) \end{array}$$

where $\varphi^*(f) = f \circ \varphi$, is commutative.

When φ is étale, for every vector field η on Y , there exists one and only one vector field on X that is φ -related to η ; we denote it $\varphi^*\eta$.

8.2.7. Let $\varphi : X \rightarrow Y$ be a morphism of manifolds of class C^r . If ω is a field of covectors⁽¹⁾ on Y , one defines a field of covectors $\varphi^*\omega$ on X by

$$(\varphi^*\omega)(x) = {}^t(T_x(\varphi))(\omega(\varphi(x))) \quad (x \in X).$$

More generally, let τ be a vector functor (7.6) of class C^{r-1} , of type $(\mathbf{I}_+, \mathbf{I}_-)$ with $\mathbf{I}_+ = \emptyset$ and $\mathbf{I}_-$ reduced to a single element (such a functor is said to be contravariant). If ω is a section of class C^{r-1} of the vector bundle $\tau(T(Y))$ with base Y we define a section $\varphi^*\omega$ of $\tau(T(X))$ by $(\varphi^*\omega)(x) = \tau(T_x(\varphi))(\omega(\varphi(x)))$ ($x \in X$).

8.3 Differential forms, exterior differentiation

Recall that, in this issue and the following ones, we assume either that K has characteristic 0, or that the varieties under consideration are locally of finite dimension.

8.3.1. Let X be a manifold of class C^r and let F be a vector bundle of class C^k with base X , with $k+1 \in \mathbb{N}_K$ and $k+1 \leq r$. Let p be an integer ≥ 0 and let $\text{Alt}^p(T(X); F)$ be the vector bundle of class C^k of p -alternating multilinear maps from $T(X)$ into F (7.8.1 and Errata to no. 7.8). A section of this bundle over an open set U of X is called an alternating differential form (or simply differential form) of degree p on U , with values in F . Those of these sections that are of class C^k form a $\mathcal{C}^k(U)$ -module, denoted ${}^k\Omega^p(U; F)$. If E is a Banach space, one writes

${}^k\Omega^p(U; E)$ instead of ${}^k\Omega^p(U; E_x)$ and one speaks of differential forms with values in E . In these notations, one sometimes omits k (resp. E) when it is equal to $r - 1$ (resp. K).

A differential form of degree 0 (resp. 1) with scalar values is a function (resp. a field of covectors).

Let $\varphi: Y \rightarrow X$ be a morphism of manifolds of class C^r , and let $\omega \in {}^k\Omega^p(X; F)$. There then exists a form $\varphi^*\omega \in {}^k\Omega^p(Y; \varphi^*F)$ and only one such that

$$(\varphi^*\omega)_y(v_1, \dots, v_p) = \omega_{\varphi(y)}(T_y(\varphi).v_1, \dots, T_y(\varphi).v_p) \quad (1)$$

for all $y \in Y$ and every family $v_1, \dots, v_p$ of elements of $T_y(Y)$.

If $\psi: Z \rightarrow Y$ is a morphism of manifolds of class C^r , one has

$$(\varphi \circ \psi)^*\omega = \psi^*(\varphi^*\omega). \quad (2)$$

When Y is a submanifold of X and φ is the canonical injection, one sometimes writes $\omega|Y$ instead of $\varphi^*\omega$ and one says that $\omega|Y$ is induced on Y by ω .

8.3.2. The definitions and results of no. 7.8 apply to differential forms. In particular, a pairing of vector bundles $F' \times_X F'' \rightarrow F$ gives rise to an exterior product ${}^k\Omega^{p'}(U; F') \times {}^k\Omega^{p''}(U; F'') \rightarrow {}^k\Omega^p(U; F)$, where $p = p' + p''$ (7.8.2); it is denoted $(\omega', \omega'') \mapsto \omega' \wedge \omega''$. The direct sum of the ${}^k\Omega^p(U; K)$ for $p \geq 0$ is thus endowed with a structure of associative and alternating graded algebra (A, III, p. 53).

Let ξ be a vector field on X and ω a differential form of degree $p \geq 1$ on X , with values in a vector bundle F ; the interior product of ξ and ω is the differential form $i(\xi, \omega)$ of degree $p - 1$ on X , with values in F , defined by

$$i(\xi, \omega)_x(v_1, \dots, v_{p-1}) = \omega_x(\xi(x), v_1, \dots, v_{p-1}) \quad (3)$$

for all $x \in X$ and every family $v_1, \dots, v_{p-1}$ of elements of $T_x(X)$. If ω is a differential form of degree 0, we set $i(\xi, \omega) = 0$. We also write $i(\xi)\omega$ or $i_\xi\omega$ in place of $i(\xi, \omega)$. When $F = K_X$, the definition given above coincides with that of 7.8.4.

Let $F' \times_X F'' \rightarrow F$ a pairing of vector bundles. For $\omega' \in {}^k\Omega^{p'}(U; F')$ and $\omega'' \in {}^k\Omega^{p''}(U; F'')$

, one has

$$i(\xi)(\omega' \wedge \omega'') = (i(\xi)\omega') \wedge \omega'' + (-1)^{p'} \omega' \wedge i(\xi)\omega''. \quad (4)$$

8.3.3. Let $(u^1, \dots, u^n)$ be a coordinate system in the open set U of X . Every element ω of ${}^k\Omega^p(U; F)$ is written uniquely

$$\omega = \sum_{i_1 < \dots < i_p} f_{i_1, \dots, i_p} du^{i_1} \wedge \dots \wedge du^{i_p}$$

where the $f_{i_1, \dots, i_p}$ are sections of class $\mathbf{C}^k$ of F on U ; in this formula, the sign $\wedge$ is relative to the canonical pairing $\mathbf{K}_X \times_X \mathbf{K}_X \rightarrow \mathbf{K}_X$ and the point denoting the product is relative to the canonical pairing $F \times_X \mathbf{K}_X \rightarrow F$.

8.3.4. Let p be an integer ≥ 0 and r an element of $\mathbf{N}_K$. Let E and F be Banach spaces, U an open subset of E and $\alpha \in {}^r\Omega^p(U; F)$. Let $\tilde{\alpha} \in \mathcal{C}^r(U; \text{Alt}^p(E; F))$ be the element corresponding to α . If $x \in U$, the differential $d_x \tilde{\alpha}$ of $\tilde{\alpha}$ at x (5.5.6) is an element of $\mathcal{L}(E; \text{Alt}^p(E; F))$; if $t \in E$, one has $(d_x \tilde{\alpha})(t) \in \text{Alt}^p(E; F)$, and if $t_1, \dots, t_p$ are in E , one has $(d_x \tilde{\alpha})(t)(t_1, \dots, t_p) \in F$. There exists one and only one element $d\alpha$ of ${}^{r-1}\Omega^{p+1}(U; F)$ such that one has

$$(d\alpha)_x(t_0, \dots, t_p) = \sum_{i=0}^p (-1)^i (d_x \tilde{\alpha})(t_i)(t_0, \dots, \hat{t}_i, \dots, t_p)^{(1)} \quad (5)$$

for all $x \in U$ and $t_0, \dots, t_p$ in E .

8.3.5. Let p be an integer ≥ 0 and r an element of $\mathbf{N}_K$. Let X be a manifold of class $\mathbf{C}^{r+1}$, F a Banach space, and $\omega \in {}^r\Omega^p(X; F)$. There exists a differential form $\pi \in {}^{r-1}\Omega^{p+1}(X; F)$ and only one such that, for every chart $c = (U, \varphi, E)$ of X , if ω_c is the differential form on $\varphi(U)$ such that $\omega|_U = \varphi^*(\omega_c)$, one has $\pi_c = d\omega_c$, where $d\omega_c$ is defined by formula (5) of 8.3.4.

The differential form π is called the exterior differential of ω ; it is denoted $d\omega$. It has the following properties:

- (6) If $\psi : Y \rightarrow X$ is a morphism of varieties of class $\mathbf{C}^{r+1}$ and if $\omega \in {}^r\Omega^n(X; F)$, one has $\psi^*(d\omega) = d(\psi^*\omega)$; in particular, d commutes with the operation of restriction to a submanifold.

(7) For $p = 0$, the map $d : {}^r\Omega^0(\mathbf{X}; \mathbf{F}) \rightarrow {}^{r-1}\Omega^1(\mathbf{X}; \mathbf{F})$ coincides with the one defined in 8.2.2.

(8) If $r \geq 2$ and if $\omega \in {}^r\Omega^p(\mathbf{X}; \mathbf{F})$, one has $d(d\omega) = 0$.⁽²⁾

(9) For every continuous linear map $u : \mathbf{F} \rightarrow \mathbf{F}'$ of Banach spaces, we have $d(u(\omega)) = u(d(\omega))$ for $\omega \in {}^r\Omega^p(\mathbf{U}; \mathbf{F})$.

(10) For every continuous bilinear mapping $F' \times F'' \rightarrow F$ of Banach spaces, we have

$$d(\omega' \wedge \omega'') = (d\omega') \wedge \omega'' + (-1)^{p'} \omega' \wedge d\omega''$$

for $\omega' \in {}^r\Omega^{p'}(\mathbf{X}; \mathbf{F}')$ and $\omega'' \in {}^r\Omega^{p''}(\mathbf{X}; \mathbf{F}'')$.

8.3.6. Let $(u^1, \dots, u^n)$ be a coordinate system of $\mathbf{X}$ in $\mathbf{U}$. Let

$$\omega = \sum_{i_1 < \dots < i_p} f_{i_1, \dots, i_p} du^{i_1} \wedge \dots \wedge du^{i_p}$$

¹ The sign $\wedge$ indicates that the symbol it surmounts is to be omitted (cf. 7.8.4).

² If $\omega \in {}^1\Omega^p(X; F)$ and if $d\omega \in {}^1\Omega^{p+1}(X; F)$, we still have $d(d\omega) = 0$.

a differential form on $\mathbf{U}$ (with $f_{i_1, \dots, i_p} \in \mathcal{C}^r(\mathbf{U}; \mathbf{F})$). We then have

$$d\omega = \sum_{i_1 < \dots < i_p} df_{i_1, \dots, i_p} \wedge du^{i_1} \wedge \dots \wedge du^{i_p}. \quad (11)$$

8.3.7. Suppose $\mathbf{K}$ has characteristic zero. Let $\omega \in {}^r\Omega^p(\mathbf{X}; \mathbf{F})$ be a differential form of degree $p \geq 1$ such that $d\omega = 0$. For every $x \in \mathbf{X}$ there exists an open neighborhood $\mathbf{U}$ of x and a differential form $\pi \in {}^r\Omega^{p-1}(\mathbf{U}; \mathbf{F})$ such that $d\pi = \omega$ on $\mathbf{U}$.

8.4 Infinitesimal transformations

Let $r \in \mathbf{N}_\kappa$ and let X be a manifold of class $\mathbf{C}^{r+1}$.

8.4.1. Let τ be a vector functor for isomorphisms (7.6.6), of class $\mathbf{C}^r$. If $\mathbf{E}$ is a Banach space, we denote by $\mathbf{GL}(\mathbf{E})$ the open subset of $\mathcal{L}(\mathbf{E}; \mathbf{E})$ consisting of the automorphisms of $\mathbf{E}$. We denote by $\tau'_\mathbf{E}$ the linear tangent map at the identity element $\text{Id}_\mathbf{E}$ of the morphism $\tau : \mathbf{GL}(\mathbf{E}) \rightarrow$

$\mathbf{GL}(\tau(\mathbf{E}))$; it is a continuous linear map from $\mathcal{L}(\mathbf{E}; \mathbf{E})$ to $\mathcal{L}(\tau(\mathbf{E}); \tau(\mathbf{E}))$.

8.4.2. Let E be a Banach space and set $F = \tau(E)$. Let U be an open subset of E , ξ a vector field on U of class C^r , $\tilde{\xi}: U \rightarrow E$ the corresponding map (obtained by identifying $T(U)$ with $U \times E$ (8.1.1)), and $f: U \rightarrow F$ a map of class C^r . For $x \in U$ we define an element $(\theta_{\xi}f)(x)$ of F by the formula

$$(\theta_{\xi}f)(x) = d_x f(\tilde{\xi}(x)) - \tau'_E(d_x \tilde{\xi})(f(x)). \quad (1)$$

(Note that we have, on the one hand, $\tilde{\xi}(x) \in E$ and $d_x f \in \mathcal{L}(E; F)$, whence $d_x f(\tilde{\xi}(x)) \in F$; on the other hand, $d_x \tilde{\xi} \in \mathcal{L}(E; E)$, $\tau'_E(d_x \tilde{\xi}) \in \mathcal{L}(F; F)$, and $f(x) \in F$, whence $\tau'_E(d_x \tilde{\xi})(f(x)) \in F$. The function $\theta_{\xi}f: x \mapsto (\theta_{\xi}f)(x)$ is of class C^{r-1} in U .)

8.4.3. Let F denote the vector bundle $\tau(T(X))$ (7.6.2 and 7.6.6). Let ξ be a vector field on X and f a section of F , both of class C^r . Let x be a point of X and $c = (U, \varphi, E)$ a chart of X at x . Let $c' = (U, \zeta_c, E)$ be the corresponding vector chart of $T(X)$ (8.1.1) and $\tau(c') = (U, \psi_c, \tau(E))$ the corresponding vector chart of F (7.6.2). Set $x_c = \varphi(x)$; let ξ_c be the map from $\varphi(U)$ into E such that $\zeta_c(\xi(u)) = (u, \xi_c(\varphi(u)))$ and f_c the map from $\varphi(U)$ into $\tau(E)$ such that $\psi_c(f(u)) = (u, f_c(\varphi(u)))$ for all $u \in U$. Denote by a_c the element $(\theta_{\xi_c}f_c)(x_c)$ of $\tau(E)$ defined in 8.4.2. There exists an element a of F_x and only one such that $\psi_c(a) = (x, a_c)$ for every chart c of X at x . We denote it by $(\theta_{\xi}f)(x)$. It depends only on the germs of ξ and f at x (and even only on their first-order jet (12.1.2)).

The map $\theta_{\xi}f: x \mapsto (\theta_{\xi}f)(x)$ is a section of class C^{r-1} of F ; it depends K -bilinearly on the pair (ξ, f) .

8.4.4. We keep the previous hypotheses and notations. There exists (7.6.4) a morphism of vector bundles τ' and one and only one from $\mathcal{L}(T(X); T(X))$ to $\mathcal{L}(F; F)$ such that, for all $x \in X$, the restriction of τ' to the fiber $\mathcal{L}(T(X); T(X))_x = \mathcal{L}(T_x(X); T_x(X))$ is equal to $\tau'_{T(X)}$ (8.4.1). If g is a function of class C^r on X , with values in K , one has

$$\theta_{g\xi}f = g\theta_{\xi}f - \tau'(dg \otimes \xi)(f)$$

(in this formula, one identifies $dg \otimes \xi$, which is a section of $T'(X) \otimes T(X)$, to a section of

$\mathcal{L}(T(X); T(X))$; its image $\tau'(dg \otimes \xi)$ is a section of $\mathcal{L}(F; F)$ and maps f to a section of F).

8.4.5. Let I be an open subset of K containing 0, U an open subset of X , and φ a map of class C^r of $I \times U$ in X . For $(t, x) \in I \times U$, one denotes by $\varepsilon_{t,x}$ the vector $(1, 0) \in K \times T_x(X) = T_{(t,x)}(I \times U)$. For $t \in I$, one denotes by φ_t the map from U to X defined by $\varphi_t(x) = \varphi(t, x) (x \in U)$. We assume that the following three conditions are satisfied:

- a) φ_0 is the canonical injection from U into X ;
- b) for all $t \in I$, the map φ_t is an isomorphism of manifolds of class C^r from U onto an open subset of X ;
- c) the section $(t, x) \mapsto T_{(t,x)}(\varphi)(\varepsilon_{t,x})$ of the bundle $\varphi^*T(X)$ is of class C^r . Furthermore, we set:

$$d)\xi(x) = T_{(0,x)}(\varphi)(\varepsilon_{0,x}) \quad \text{for all } x \in U.$$

The map $\xi: x \mapsto \xi(x)$ is a vector field of class C^r on U . We say that it is the initial (or starting) field of φ .

Let τ be a vector functor for isomorphisms, of class C^r ; set $F = \tau(T(X))$ and let f be a section of class C^r of F on X . For $x \in U$ and $t \in I$, set $\varphi_t^*f(x) = \tau(T_x(\varphi_t)^{-1})(f(\varphi_t(x)))$; it is an element of F_x . The map $t \mapsto \varphi_t^*f(x)$ of I to F_x is of class C^{r-1} for all $x \in U$ and one has

$$\frac{d}{dt} (\varphi_t^*f(x))_{t=0} = (\theta_\xi.f)(x) \quad \text{for all } x \in U. \quad (2)$$

8.4.6. Given a vector field ξ of class C^r on X and a point x_0 of X , there exist an open interval I of K containing 0, an open set U of X containing x_0 and a mapping $\varphi: I \times U \rightarrow X$ of class C^r satisfying the conditions a, b) and c) of 8.4.5 and such that the initial field of φ is the restriction $\xi|U$ of ξ to U .

8.4.7. Example. — Let F be a Banach space and p an integer ≥ 0 . If V is a Banach space, set $\alpha_p(V) = \text{Alt}^p(V; F)$; if u is an isomorphism of V onto a Banach space V' , denote by $\alpha_p(u)$ the isomorphism of $\alpha_p(V)$ onto $\alpha_p(V')$ deduced from u by transport of structure. One thus obtains a vector functor for isomorphisms, denoted α_p (7.8.1 and Errata to no. 7.8). One may apply the

preceding to it; one has

$$\alpha_p(T(X)) = \text{Alt}^p(T(X); F).$$

The sections of $\alpha_p(T(X))$ are the differential forms of degree p on X , with values in F ; if ω is such a form and if ξ is a vector field on X , both of class C^r , $\theta_\xi.\omega$ is a differential form of degree p on X , with values in F and of class C^{r-1} . One has

$$\theta_\xi.\omega = d(i(\xi)\omega) + i(\xi)d\omega \quad (3)$$

and, when $r \geq 2$,

$$\theta_\xi.d\omega = d(\theta_\xi.\omega). \quad (4)$$

When $p = 0$, the functor α_p is the constant vector functor defined by F ; the bundle $\alpha_0(T(X))$ is identified with the trivial bundle F_X , and the sections of this bundle are identified with the functions on X with values in F . If ξ is a vector field of class C^r on X and if $f \in \mathcal{C}^r(X; F)$, one has

(5)

$$\theta_\xi.f = \langle \xi, df \rangle = D_\xi(f) \quad (8.2.3);$$

it is an element of $\mathcal{C}^{r-1}(X; F)$.

8.4.8. Let τ and τ_1 be two vector functors for isomorphisms. A morphism of vector functors $h: \tau_1 \rightarrow \tau$ defines a morphism of vector bundles, still denoted h , of $\tau_1(T(X))$ to $\tau(T(X))$ (7.6.4), and associates to a section f of $\tau_1(T(X))$ a section $h(f)$ of $\tau(T(X))$. If ξ is a vector field on X and if f and ξ are of class C^r , one has $\theta_\xi.h(f) = h(\theta_\xi.f)$.

More generally, let $\tau, \tau_1, \dots, \tau_d$ be vector functors for isomorphisms and let $h: (\tau_1, \dots, \tau_d) \rightarrow \tau$ be a d -linear morphism (7.6.4). Let $f_i \in \mathcal{S}_{\tau_i(T(X))}^r(X)$ (for $i = 1, \dots, d$), and let ξ be a vector field of class C^r on X . We have

$$\theta_\xi.h(f_1, \dots, f_d) = \sum_{1 \leq i \leq d} h(f_1, \dots, f_{i-1}, \theta_\xi.f_i, f_{i+1}, \dots, f_d). \quad (6)$$

For example, if f is a section of $\tau(T(X))$ and g is a function on X , with values in K , both of class C^r , one has

$$\theta_\xi.(gf) = (\theta_\xi.g)f + g(\theta_\xi.f). \quad (7)$$

If ω_1 and ω_2 are two differential forms of class C^r , with values in Banach spaces F_1 and F_2 respectively, we have

$$\theta_\xi.(\omega_1 \wedge \omega_2) = (\theta_\xi.\omega_1) \wedge \omega_2 + \omega_1 \wedge (\theta_\xi.\omega_2), \quad (8)$$

the exterior product being taken with respect to a continuous bilinear map from $F_1 \times F_2$ into a Banach space F .

8.4.9. Let τ be a vector functor for isomorphisms, contravariant (8.2.7), of class C^r , $\varphi : X \rightarrow Y$ a morphism of manifolds of class C^{r+1} , ξ (resp. η) a vector field of class C^r on X (resp. Y) such that ξ be φ -related to η (8.2.6). For every section f of $\tau(T(Y))$, of class C^r , one has

$$\varphi^*(\theta_\eta.f) = \theta_\xi.\varphi^*f.$$

8.5 The bracket

Let $r \in N_K$ and let X be a manifold of class C^{r+1} .

8.5.1. We apply the definitions of no. 8.4, taking for τ the identity functor: $\tau(V) = V$ for every Banach space V and $\tau(u) = u$ for every isomorphism u of Banach spaces. We then have $\tau(T(X)) = T(X)$. If ξ and η are two vector fields of class C^r on X , we set

$$[\xi, \eta] = \theta_\xi.\eta; \quad (1)$$

it is a vector field of class C^{r-1} on X , called the bracket of ξ and η .

8.5.2. The map $(\xi, \eta) \mapsto [\xi, \eta]$ is K -bilinear and alternating. If f and g are functions of class C^r on X , with values in K , we have:

$$[f\xi, g\eta] = fg[\xi, \eta] + (fD_\xi g)\eta - (gD_\eta f)\xi. \quad (2)$$

8.5.3. Let F be a Banach space. We have

$$\mathbf{D}_{[\xi, \eta]} = \mathbf{D}_\xi \circ \mathbf{D}_\eta - \mathbf{D}_\eta \circ \mathbf{D}_\xi \quad (3)$$

in the space of maps from $\mathcal{C}^{r+1}(X; F)$ to $\mathcal{C}^{r-1}(X; F)$.

If $r \geq 2$ and if ζ is a third vector field of class C^r on X , we have

$$[[\xi, \eta], \zeta] = [\xi, [\eta, \zeta]] - [\eta, [\xi, \zeta]]^{(1)} \quad (4)$$

or also

$$[\xi, [\eta, \zeta]] + [\eta, [\zeta, \xi]] + [\zeta, [\xi, \eta]] = 0.$$

More generally, if $r \geq 2$ and if τ is a vector functor for isomorphisms, we have

$$\theta_{[\xi, \eta]} = \theta_\xi \circ \theta_\eta - \theta_\eta \circ \theta_\xi \quad (5)$$

in the space of maps from $\mathcal{C}^{r+1}(X; F)$ to $\mathcal{C}^{r-1}(X; F)$ (with $F = \tau(T(X))$).

If $r \geq \infty$ the vector fields of class C^r on X form a K-Lie algebra for the bracket (LIE, I, § 1, no. 2).

8.5.4. Let E be a Banach space, U an open subset of E , ξ and η two vector fields of class C^r on U , identified with elements of $\mathcal{C}^r(U; E)$. Their bracket is given by the formula:

$$[\xi, \eta] = \mathbf{D}_\xi \eta - \mathbf{D}_\eta \xi. \quad (6)$$

8.5.5. Let $(u^1, \dots, u^n)$ be a coordinate system of X in an open set U . Let $\xi = \sum a^i \frac{\partial}{\partial u^i}$ and $\eta = \sum b^i \frac{\partial}{\partial u^i}$ be two vector fields of class C^r on U . Set $[\xi, \eta] = \sum c^i \frac{\partial}{\partial u^i}$. One has

$$c^i = \sum_{1 \leq j \leq n} \left(a^j \frac{\partial b^i}{\partial u^j} - b^j \frac{\partial a^i}{\partial u^j} \right). \quad (7)$$

8.5.6. Let $\varphi : X \rightarrow Y$ be a morphism of manifolds of class C^{r+1} , ξ_1 and ξ_2 two vector fields on X , η_1 and η_2 two vector fields on Y , all of class C^r . If ξ_1 and ξ_2 are φ -related to η_1

and η_2 respectively, $[\xi_1, \xi_2]$ is φ -related to $[\eta_1, \eta_2]$.

8.5.7. Let ω be a differential form of degree p on X , with values in a Banach space F and of class C^r , and let $\xi_0, \dots, \xi_p$ be vector fields of class C^r on X . We have

$$(\theta_{\xi_0} \cdot \omega)(\xi_1, \dots, \xi_p) = D_{\xi_0} \omega(\xi_1, \dots, \xi_p) - \sum_{i=1}^p \omega(\xi_1, \dots, [\xi_0, \xi_i], \dots, \xi_p) \quad (8)$$

and

$$\begin{aligned} d\omega(\xi_0, \dots, \xi_p) &= \sum_{i=0}^p (-1)^i D_{\xi_i} \omega(\xi_0, \dots, \hat{\xi}_i, \dots, \xi_p) \\ &\quad + \sum_{i < j} (-1)^{i+j} \omega([\xi_i, \xi_j], \xi_0, \dots, \hat{\xi}_i, \dots, \hat{\xi}_j, \dots, \xi_p). \end{aligned} \quad (9)$$

If ξ and η are vector fields of class C^r , one has

$$\theta_{\xi} \circ i(\eta) - i(\eta) \circ \theta_{\xi} = i([\xi, \eta]) \quad (10)$$

in the space of maps from ${}^r\Omega^p(X; F)$ to ${}^{r-1}\Omega^{p-1}(X; F)$.

For $p = 1$, formula (9) becomes:

$$d\omega(\xi, \eta) = D_{\xi} \langle \eta, \omega \rangle - D_{\eta} \langle \xi, \omega \rangle - \langle [\xi, \eta], \omega \rangle. \quad (11)$$

8.6 Liftings

Let $r \in \mathbb{N}_{\mathbb{K}}$ and let $g : X \rightarrow Y$ be a morphism of manifolds of class C^{r+1} .

8.6.1. A lifting of g in $T(Y)$ is a map $\psi : X \rightarrow T(Y)$ such that the diagram

$$\begin{array}{ccc} & & T(Y) \\ & \nearrow \psi & \downarrow \pi \\ X & \xrightarrow{g} & Y \end{array}$$

is commutative, π denoting the canonical projection of $T(Y)$ onto Y . The data of ψ is equivalent to that of a section of the vector bundle $g^*T(Y)$, the inverse image under g of $T(Y)$.

The notion of lifting generalizes that of a vector field (to which it reduces when $g = \text{Id}_X$); an important part of the definitions and results concerning θ_ξ and $i(\xi)$ extends to liftings; we shall limit ourselves to briefly indicating a few of them.

8.6.2. Let $\psi : X \rightarrow T(Y)$ be a lifting of $g : X \rightarrow Y$, and let ω be a differential form of degree p on Y , with values in a vector bundle F with base Y . When $p \geq 1$, one denotes by $i(\psi, \omega)$ or $i(\psi)\omega$ or $i_\psi(\omega)$ the differential form on X , of degree $p - 1$, with values in g^*F , such that

$$i(\psi, \omega)_x(v_1, \dots, v_{p-1}) = \omega_{g(x)}(\psi(x), T_x(g).v_1, \dots, T_x(g).v_{p-1}) \quad (1)$$

for all $x \in X$ and every family $v_1, \dots, v_{p-1}$ of elements of $T_x(X)$. When $p = 0$, we set $i(\psi, \omega) = 0$. We say that $i(\psi, \omega)$ is the interior product of ψ and of ω ; if ψ and ω are of class C^r , then so is $i(\psi, \omega)$. Let $F' \times_Y F'' \rightarrow F$ be a pairing of vector bundles with base Y . For $\omega' \in {}^\tau\Omega^{p'}(Y; F')$ and $\omega'' \in {}^\tau\Omega^{p''}(Y; F'')$, one has

$$i(\psi)(\omega' \wedge \omega'') = (i(\psi)\omega') \wedge g^*\omega'' + (-1)^{p'} g^*\omega' \wedge i(\psi)\omega''. \quad (2)$$

8.6.3. Examples

a) A vector field ξ on X defines a lifting $T(g) \circ \xi$ of g in $T(Y)$, and we have:

$$i(T(g) \circ \xi) = i(\xi) \circ g^*. \quad (3)$$

b) A vector field η on Y defines a lifting $\eta \circ g$ of g in $T(Y)$, and we have:

$$i(\eta \circ g) = g^* \circ i(\eta). \quad (4)$$

c) More generally, let $h : Y \rightarrow Z$ be a morphism of manifolds of class C^{r+1} . If ψ is a lifting of g to $T(Y)$, then $T(h) \circ \psi$ is a lifting of $h \circ g$ to $T(Z)$ and we have:

$$i(T(h) \circ \psi) = i(\psi) \circ h^*. \quad (5)$$

If φ is a lifting of h to $T(Z)$, then $\varphi \circ g$ is a lifting of $h \circ g$ to $T(Z)$ and we have:

$$i(\varphi \circ g) = g^* \circ i(\varphi). \quad (6)$$

8.6.4 (“Infinitesimal transformations”). Let ψ be a lifting of class C^r of g and let ω be a differential form of degree p on Y , taking values in a Banach space F , and of class C^r . The differential form on X

$$d(i(\psi)\omega) + i(\psi)d\omega \quad (7)$$

is of degree p and of class C^{r-1} ; it is denoted $\theta_\psi.\omega$.

If f is a function of class C^r on X , we have

$$\theta_{f\psi}.\omega = df \wedge i(\psi)\omega + f\theta_\psi.\omega. \quad (8)$$

8.6.5 (“Characterization of $\theta_\psi.\omega$ as derivative”). Keep the hypotheses of 8.6.4. Let $x \in X$. There then exists an open neighborhood U of x , an open neighborhood I of 0 in K , and a morphism $G: I \times U \rightarrow Y$ of class C^r having the following two properties:

- a) For every $x \in U$, one has $G(0, x) = g(x)$.
- b) For every $x \in U$, the image under $T_{(0,x)}(G)$ of the tangent vector $(1, 0)$ is equal to $\psi(x)$.

Suppose that (U, I, G) satisfies these conditions; for every $t \in I$, denote by G_t the map $x \mapsto G(t, x)$ from U to Y ; set $\omega_t = G_t^*(\omega)$. For every $x \in X$, the map $t \mapsto \omega_t(x)$ from I to $\text{Alt}^p(T_x(X), F)$ is of class C^r , and its derivative at the origin is given by the formula

$$\frac{d}{dt}(\omega_t(x))_{t=0} = (\theta_\psi.\omega)(x). \quad (9)$$

8.7 Weakening of structure

In this section, we assume $K = \mathbb{R}$.

8.7.1. Let X be a manifold of class C^s , and let F be a vector bundle over X , of class C^k , with $0 \leq k \leq s$. If $0 \leq k' \leq k$, there exists on F a unique structure $F_{k'}$ of vector bundle with base X and of class $C^{k'}$ such that every vector chart of F is a vector chart of $F_{k'}$. We say that $F_{k'}$ is obtained from F by weakening the bundle structure of F .

8.7.2. Let $r \in \mathbb{N}_{\mathbb{R}}$ with $r \leq s$, and let X_r be the manifold of class C^r underlying X (5.13.1). The tangent bundle $T(X_r)$ is of class C^{r-1} ; it is the bundle obtained by weakening of structure from the tangent bundle $T(X)$, which is of class C^{s-1} .

Let F be a vector bundle over X , of class C^k , for $0 \leq k \leq r-1$. The differential forms of class C^k on X with values in F are canonically identified with the corresponding differential forms on X_r . The various operations on these forms described in the rest of this paragraph are compatible with this identification.

In the remainder of this booklet, we will often leave it to the reader to spell out the other results of this kind.

8.8 Almost complex manifolds and complex manifolds

In this section, we assume $K = \mathbb{R}$ and denote by X a (real) manifold of class C^r , with $r \in \mathbb{N}_{\mathbb{R}}$.

8.8.1 (“Complex vector bundles”). A vector bundle over $\mathbb{C}$ with base X (7.3.4) is also called a complex vector bundle with base X . If M is such a vector bundle, one calls a complex vector chart of M a triple (U, φ, E) , where U is an open subset of X , E a complex Banach space, and φ an isomorphism of complex vector bundles from $M|_U$ onto the trivial bundle $E_U = U \times E$. A family (U_i, φ_i, E_i) of complex vector charts on M whose domains U_i cover X is called a complex vector atlas of M ; every complex vector bundle possesses a complex vector atlas (7.3.3).

One sometimes says $\mathbb{C}$ -vector instead of complex vector. Similarly, when one wishes to speak of a vector bundle over $\mathbb{R}$, of an ordinary vector chart of this bundle, etc., one sometimes says “real vector” or “ $\mathbb{R}$ -vector” instead of “vector”.

Let M be a complex vector bundle with base X . There exists one and only one automorphism j of the (real) vector bundle underlying M (7.3.3) whose restriction to each fibre is multiplication

by the element i of $\mathbb{C}$; the square of j is equal to -1 . Conversely, if M is a (real) vector bundle with base X and j is an automorphism of M such that $j^2 = -1$, there exists on M a unique structure of complex vector bundle admitting M as underlying (real) vector bundle and for which j is multiplication by i .

8.8.2 ("Complexification of a vector bundle"). One defines a vector functor τ by setting $\tau(E) = E \otimes \mathbb{C}$ for every (real) Banach space E , and $\tau(u) = u \otimes Id_{\mathbb{C}}$ for every morphism $u : E \rightarrow E'$ of (real) Banach spaces.

If F is a vector bundle with base X and of class C^k ($0 \leq k \leq r$), its image under the vector functor τ is denoted $F \otimes \mathbb{C}$. There exists on $F \otimes \mathbb{C}$ a unique structure of complex vector bundle with base X and of class C^k , compatible with its structure of (real) vector bundle and inducing on each fibre $(F \otimes \mathbb{C})_x = F_x \otimes \mathbb{C}$ its canonical structure of complex Banach space (EVT, II, § 8, n° 1, Example). Endowed with this structure, one says that $F \otimes \mathbb{C}$ is the complexification of F .

Let U be an open subset of X ; the canonical map

$$\mathcal{S}_{\mathbb{R}}^k(U) \otimes \mathbb{C} \rightarrow \mathcal{S}_{\mathbb{F} \otimes \mathbb{C}}^k(U)$$

is an isomorphism, by which one identifies these two spaces. If $s \in \mathcal{S}_{\mathbb{F} \otimes \mathbb{C}}^k(U)$, the elements σ, τ of $\mathcal{S}_{\mathbb{R}}^k(U)$ such that $s = \sigma + i\tau$ are called the real part and the imaginary part of s ; the section $\bar{s} = \sigma - i\tau$ is called the conjugate of s .

In particular, a section of the bundle $T(X) \otimes \mathbb{C}$ is called a complex vector field on X .

Let H be a complex vector bundle over the base X . The canonical isomorphisms $\text{Alt}_{\mathbb{R}}^p(T_x(X); H) \rightarrow \text{Alt}_{\mathbb{C}}^p(T_x(X) \otimes \mathbb{C}; H)$ define a canonical isomorphism of complex vector bundles, called the canonical one, from $\text{Alt}_{\mathbb{R}}^p(T(X); H)$ onto $\text{Alt}_{\mathbb{C}}^p(T(X) \otimes \mathbb{C}; H)$ (7.8.5), by means of which we will identify these two bundles. Let ω be a section of this bundle, in other words a differential form of degree p on X , with values in H , and let ζ be a complex vector field, with real part ξ and imaginary part η . We then set:

$$i(\zeta, \omega) = i(\xi, \omega) + i.i(\eta, \omega).$$

If H is a trivial bundle, we set

$$\theta_{\zeta}.\omega = \theta_{\xi}.\omega + i\theta_{\eta}.\omega.$$

Similarly, we extend the bracket by linearity to complex vector fields. The formulas of the preceding numbers remain valid.

8.8.3. An almost complex structure of class C^k ($k \leq r - 1$) on X is the data of a complex vector bundle structure on $T(X)$, of class C^k , whose underlying real structure is the natural structure of $T(X)$. Such a structure is equivalent to the data of an automorphism j of class C^k of $T(X)$ such that $j^2 = -1$.

Let us take such a structure. Extend j to a $\mathbf{C}$ -automorphism of $T(X) \otimes \mathbf{C}$. There exists a unique decomposition of $T(X) \otimes \mathbf{C}$ as a direct sum of two complex subbundles:

$$T(X) \otimes \mathbf{C} = T'(X) \oplus T''(X) \quad (1)$$

such that j coincides with multiplication by i on $T'(X)$ and by $-i$ on $T''(X)$. The projector p' of $T(X) \otimes \mathbf{C}$ onto $T'(X)$ is equal to $\frac{1}{2}(1 - ij)$; the projector $p'' = 1 - p'$ of $T(X) \otimes \mathbf{C}$ onto $T''(X)$ is equal to $\frac{1}{2}(1 + ij)$. The composites

$$\pi' : T(X) \rightarrow T(X) \otimes \mathbf{C} \xrightarrow{p'} T'(X)$$

$$\pi'' : T(X) \rightarrow T(X) \otimes \mathbf{C} \xrightarrow{p''} T''(X)$$

are $\mathbf{R}$ -isomorphisms; the first maps j to multiplication by i , the second to multiplication by $-i$.

8.8.4 (“Forms of type (p, q) ”). We keep the hypotheses and notations of the preceding number. Let F be a complex vector bundle with base X , and let p and q be two integers ≥ 0 , and let $n = p + q$. Let ω be a differential form of degree n on an open set U of X , with values in F ; identify (8.8.2) ω with a section of $\text{Alt}_{\mathbf{C}}^n(T(X) \otimes \mathbf{C}; F)$. We say that ω is of type (p, q) if, for all $x \in U$, one has

$$\omega_x(v_1, \dots, v_n) = 0$$

as soon as $p+1$ vectors v_i belong to $T'_x(X)$ or $q+1$ vectors v_i belong to $T''_x(X)$. If one identifies $\bigwedge^n T_x(X) \otimes C$ with the direct sum of the spaces $\bigwedge^{m'} T'_x(X) \otimes \bigwedge^{m''} T''_x(X)$ for $m' + m'' = n$ (A, III, p. 85) and ω_x with a C -linear form on $\bigwedge^n T_x(X)$, it is equivalent to say that ω_x vanishes on $\bigwedge^{m'} T'_x(X) \otimes \bigwedge^{m''} T''_x(X)$ for $(m', m'') \neq (p, q)$.

For every pair (p, q) of integers ≥ 0 with $p + q = n$ there exists a complex vector subbundle $\text{Alt}^{p,q}(T(X); F)$ of $\text{Alt}^n(T(X); F)$ and only one such that its sections over an open set U of X are the differential forms of type (p, q) on X , with values in F . The vector bundle $\text{Alt}^n(T(X); F)$ is the direct sum of the vector bundles $\text{Alt}^{p,q}(T(X); F)$ for $p \geq 0$, $q \geq 0$ and $p + q = n$. Every differential form ω of degree n with values in F decomposes uniquely into

$$\omega = \sum_{p+q=n} \omega_{p,q}$$

where $\omega_{p,q}$ is a form of type (p, q) , called the component of type (p, q) of ω . If ω is of class C^h , with $0 \leq h \leq k$, the same holds for its components.

Examples

- a) A differential form of degree n is of type $(n, 0)$ if and only if it is C -multilinear.
- b) Let F' , F'' , and F be three complex vector bundles and let $F' \times_X F'' \rightarrow F$ be a C -bilinear pairing. If ω' (resp. ω'') is a differential form of type (p', q') (resp. (p'', q'')) with values in F' (resp. F''), the differential form $\omega' \wedge \omega''$ is of type $(p' + p'', q' + q'')$.
- c) Suppose that F is the complexification of a (real) vector bundle (8.8.2). If ω is a differential form of type (p, q) with values in F , its conjugate $\overline{\omega}$ (8.8.2) is of type (q, p) .

8.8.5. In addition to the preceding hypotheses, we assume $k \geq 1$. The following three conditions are then equivalent:

- (i) For every open set U of X and for every pair (ξ, η) of complex vector fields on U , of class C^k such that $\xi(x)$ and $\eta(x)$ belong to $T'_x(X)$ for all $x \in U$, one has $[\xi, \eta](x) \in T'_x(X)$ for all $x \in U$.

(ii) For every open set U of X and every pair (ξ, η) of vector fields (real) on U , of class C^k , one has

$$[\xi, \eta] + j[j\xi, \eta] + j[\xi, j\eta] - [j\xi, j\eta] = 0.$$

(iii) For every open set U of X and for every complex differential form ω of type $(1, 0)$ on U , of class C^k , the component of type $(0, 2)$ of $d\omega$ is zero.

Suppose these conditions are satisfied. Let ω a differential form of type (p, q) on an open set U of X , of class C^h with $1 \leq h \leq k$ with values in a complex Banach space F . We denote by $d'\omega$ (resp. $d''\omega$) the component of type $(p+1, q)$ (resp. $(p, q+1)$) of $d\omega$; the other components of $d\omega$ are then zero and one has

$$d\omega = d'\omega + d''\omega. \quad (2)$$

If $k \geq 2$, one has

$$d'(d'\omega) = 0, d''(d''\omega) = 0, d'(d''\omega) + d''(d'\omega) = 0. \quad (3)$$

With the notation of 8.8.4, example b), we have

(4)

$$d'(\omega' \wedge \omega'') = d'(\omega') \wedge \omega'' + (-1)^{p'+q'} \omega' \wedge d'(\omega'') \quad (4)$$

$$d''(\omega' \wedge \omega'') = d''(\omega') \wedge \omega'' + (-1)^{p'+q'} \omega' \wedge d''(\omega'') \quad (5)$$

Together with those of 8.8.4, example c), we have

$$d'(\overline{\omega}) = \overline{d''(\omega)}. \quad (6)$$

In formulas (4) and (5) (resp. (6)), the differential forms ω' and ω'' (resp. ω) are assumed

to be of class $\mathbf{C}^1$.

8.8.6 (“Almost complex structure of a complex manifold”). Let X^c be a complex analytic manifold, admitting X as its underlying real analytic manifold (5.14.2.a)). The bundle $T(X)$ is identified with the (real) vector bundle underlying the complex vector bundle $T(X^c)$, which endows X with an almost complex structure of class C^ω , said to be defined by the given complex analytic manifold structure on X . This almost complex structure satisfies conditions (i) to (iii) of 8.8.5; in particular, the operators d' and d'' of 8.8.5 are defined. If $f : X^c \rightarrow Y^c$ is a morphism of complex analytic manifolds, f^* commutes with the operators d' and d'' .

8.8.7. We retain the hypotheses and notation of 8.8.6. Let $\bar{X}^c$ be the conjugate of X^c (5.14.2.b)). Embed X by the diagonal map $x \mapsto (x, x)$ into the complex manifold $X^c \times \bar{X}^c$, and let g be the anti-automorphism $(x, y) \mapsto (y, x)$ of $X^c \times \bar{X}^c$. The pair $(X^c \times \bar{X}^c, g)$ is a complexification of X (5.14.8). In particular, for every $x \in X$, the tangent space $T_x(X^c \times \bar{X}^c)$ is identified with $T_x(X) \otimes \mathbf{C}$; in this identification, $T_x(X^c \times \{x\})$ corresponds to $T'_x(X)$ and $T_x(\{x\} \times \bar{X}^c)$ corresponds to $T''_x(X)$.

8.8.8. Suppose $r = \omega$; any almost complex structure of class C^ω on X satisfying the equivalent conditions (i) to (iii) of 8.8.5 is the almost complex structure defined by one and only one complex analytic manifold structure on X . ⁽¹⁾

8.8.9 (“Holomorphic forms”). Let us resume the hypotheses and notations of 8.8.6, and let F be a complex Banach space. The morphic differential forms (in other words, complex analytic ones) on X^c with values in F are still called holomorphic differential forms; the identification of $T(X^c)$ with $T(X)$ identifies them with (real analytic) differential forms on X . For a differential form ω of degree p on X , with values in F and of class $\mathbf{C}^k$, with $k \geq 1$, for it to be holomorphic, it is necessary and sufficient that it be of type $(p, 0)$ and that $d''\omega = 0$.

Let ξ be a vector field on X . We say that ξ is holomorphic if it is a complex analytic mapping from X^c into $T(X^c) = T(X)$, in other words, a complex analytic vector field on X^c . If this is the case, and if ω is a holomorphic differential form on X with values in F , the form $\theta_\xi \cdot \omega$ (resp. $i(\xi)\omega$) is the same, whether one considers ξ as a complex analytic vector field on X^c and ω as an element of $\Omega^p(X^c; F)$, or ξ as a vector field on X and ω as an element of $\Omega^p(X; F)$. Moreover, let $\xi' = \pi'(\xi)$ and $\xi'' = \pi''(\xi)$ be the components of ξ in $T'(X)$ and $T''(X)$

respectively (8.8.3); we have

(7)

$$\theta_{\xi}.\omega = \theta_{\xi'}.\omega, \quad \theta_{\xi''}.\omega = 0, \quad (7)$$

$$i(\xi)\omega = i(\xi')\omega, i(\xi'')\omega = 0. \quad (8)$$

8.8.10. Example. — Suppose that X^c is an open subset of C^n and denote by $z^1, \dots, z^n$ the coordinate functions on X^c . For $1 \leq k \leq n$, set $z^k = x^k + iy^k$, where x^k and y^k take real values. Consider the tangent frame $(\partial/\partial z^k)$ of $T(X^c)$ defined by the coordinate system (z^k) and the tangent frame $(\partial/\partial x^k, \partial/\partial y^k)$ of $T(X)$ defined by the coordinate system (x^k, y^k) (8.1.5). We have $j(\partial/\partial x^k) = \partial/\partial y^k$ and the image of $\partial/\partial z^k$ under the map $\pi' : T(X) = T(X^c) \rightarrow T'(X)$ (8.8.3) is equal to

$$\frac{1}{2}(\partial/\partial x^k - i\partial/\partial y^k);$$

one generally still denotes it by $\partial/\partial z^k$. ⁽²⁾ The complex conjugate vector field of this field $\partial/\partial z^k$ is denoted $\partial/\partial \bar{z}^k$; we have

$$\partial/\partial \bar{z}^k = \frac{1}{2}(\partial/\partial x^k + i\partial/\partial y^k). \quad (9)$$

If f is a function of class C^1 on X , with values in a complex Banach space F , the differential forms $d'f$ and $d''f$ (8.8.5) are given by

$$d'f = \sum_{1 \leq k \leq n} \frac{\partial f}{\partial z^k} dz^k, d''f = \sum_{1 \leq k \leq n} \frac{\partial f}{\partial \bar{z}^k} d\bar{z}^k. \quad (10)$$

⁽¹⁾ If X is of finite dimension, this remains true for an almost complex structure of class C^k with $k = \dim X$ (cf. A. Newlander and L. Nirenberg, "Complex analytic coordinates in almost complex manifolds," Ann. of Math. LXX (1957), pp. 391–404).

⁽²⁾ Care must be taken: the canonical identification of $T(X^c)$ with $T(X)$ does not transform the vector field $\partial/\partial z^k$ on X^c into the vector field $\partial/\partial z^k$ on X , but into the vector field $\partial/\partial z^k +$

$\partial/\partial\bar{z}^k$. However, the two vector fields denoted $\partial/\partial z^k$ have the same value on holomorphic differential forms (8.8.9), which are in fact the only forms to which one can apply a vector field of X^c .

Recall that we have

$$df = d'f + d''f$$

$$dz^k = dx^k + idy^k \quad d\bar{z}^k = dx^k - idy^k.$$

If $\mathbf{A} = (i_1, \dots, i_p)$ is a strictly increasing sequence of integers belonging to the interval $[1, n]$, we set

$$dz^{\mathbf{A}} = dz^{i_1} \wedge \dots \wedge dz^{i_p}$$

and denote by $d\bar{z}^{\mathbf{A}}$ the conjugate of $dz^{\mathbf{A}}$. Every differential form ω of type (p, q) with values in F is written uniquely as

$$\omega = \sum_{\mathbf{A}, \mathbf{B}} f_{\mathbf{A}, \mathbf{B}} dz^{\mathbf{A}} \wedge d\bar{z}^{\mathbf{B}} \quad (11)$$

where $\mathbf{A}$ (resp. $\mathbf{B}$) ranges over the set of strictly increasing sequences of p (resp. q) elements of $[1, n]$, the $f_{\mathbf{A}, \mathbf{B}}$ being functions with values in F . If ω is of class $\mathbf{C}^k$, with $k \geq 1$, the same holds for the functions $f_{\mathbf{A}, \mathbf{B}}$, and one has

(12)

$$d' \omega = \sum_{\mathbf{A}, \mathbf{B}} d' f_{\mathbf{A}, \mathbf{B}} \wedge dz^{\mathbf{A}} \wedge d\bar{z}^{\mathbf{B}} \quad (12)$$

$$d'' \omega = \sum_{\mathbf{A}, \mathbf{B}} d'' f_{\mathbf{A}, \mathbf{B}} \wedge dz^{\mathbf{A}} \wedge d\bar{z}^{\mathbf{B}}. \quad (13)$$

9 Differential equations and foliations

In nos. 9.1, 9.2, and 9.3, one assumes that K has characteristic 0. In no. 9.4, one assumes that K has characteristic $p \neq 0$.

9.1 Integral curves

In this section, X denotes a manifold of class C^r and ξ is a vector field of class C^s on X , with r, s in N_K , $s \leq r - 1$.

9.1.1. Let I be an open subset of K , and let $f : I \rightarrow X$ a map of class C^k ($k \in N_K$, $k \leq r$) from I into X . For $t \in I$, one denotes by $f'(t)$ the vector $T_t(f)(1) \in T_{f(t)}(X)$; this vector is sometimes called the velocity of f at time t . When X is a Banach space, $f'(t)$ is the derivative of f at t .

One says that f is an integral curve of ξ if one has

$$f'(t) = \xi(f(t)) \quad (1)$$

for all $t \in I$. Such a curve is of class C^{s+1} .

If x is a point of X , one calls integral curve of ξ with origin x an integral curve $f : I \rightarrow X$ of ξ such that $0 \in I$ and $f(0) = x$; for all $x \in X$, there exists an integral curve of ξ with origin x , defined in a suitable open neighborhood of 0 in K .

If $f : I \rightarrow X$ is an integral curve of ξ and if $a \in K$, the map $t \mapsto f(t - a)$ of $I + a$ to X is an integral curve of ξ .

Let $f_1 : I_1 \rightarrow X$ and $f_2 : I_2 \rightarrow X$ two integral curves of ξ , and $t \in I_1 \cap I_2$. If $f_1(t) = f_2(t)$ the maps f_1 and f_2 coincide in a neighborhood of t .

9.1.2 ("Flows"). Let W be an open subset of $X \times K$. For every $x \in X$, denote by W_x the set of $t \in K$ such that $(x, t) \in W$. We call an integral flow of ξ with domain W a map f of class C^k ($k \leq r$) from W to X such that, for every $x \in X$, the map $f_x : W_x \rightarrow X$ defined by $f_x(t) = f(x, t)$ is an integral curve of ξ , and that whenever $(x, 0) \in W$, one has $f(x, 0) = x$.

For every point $x \in X$, there exists an integral flow of ξ of class C^s whose domain is a neighborhood of $(x, 0)$ in $X \times K$. Two integral flows of ξ defined in a neighborhood of $(x, 0)$

coincide on a neighborhood of $(x, 0)$.

Let $x \in X$ and let f be an integral flow of ξ whose domain W is a neighborhood of $(x, 0)$.
There exists a neighborhood V of $(x, 0, 0)$ in $X \times K \times K$ such that, if $(x, t_1, t_2) \in V$, one has

$$(x, t_1) \in W, \quad (f(x, t_1), t_2) \in W, \quad (x, t_1 + t_2) \in W$$

and

$$f(f(x, t_1), t_2) = f(x, t_1 + t_2). \quad (2)$$

If X is paracompact, or if X is separated and $K = \mathbb{R}$ or $\mathbb{C}$, there exists an integral flow of ξ whose domain is a neighborhood of $X \times \{0\}$.

9.1.3 ("The real case"). We assume that $K = \mathbb{R}$ and that X is separated. We call an integral arc of ξ any integral curve of ξ whose domain is an open interval of $\mathbb{R}$. Two integral arcs f_1 and f_2 of ξ , defined in intervals I_1 and I_2 , and coinciding at a point of $I_1 \cap I_2$, coincide in $I_1 \cap I_2$.

For every point $x \in X$, there exists an integral arc with origin x and one and only one $f_x : I_x \rightarrow X$ such that, for every integral arc $f : I \rightarrow X$ with origin x , one has $I \subset I_x$ and $f = f_x|I$. The arc f_x is called the maximal integral arc of ξ with origin x , and its domain I_x is sometimes called the lifetime interval of the point x for the field ξ .

If $I_x =] - a_-, a_+[$ (and if a_+ is finite, the restriction of f_x to the interval $[0, a_+[$ is a proper map from $[0, a_+[$ into X . In particular, $f_x(t)$ has no limit when t tends to a_+ .

9.1.4. With the notations and hypotheses of 9.1.3, the set Ω of the pairs $(x, t) \in X \times \mathbb{R}$ such that $t \in I_x$ is open in $X \times \mathbb{R}$ and the map $f : \Omega \rightarrow X$ defined by $f(x, t) = f_x(t)$ is an integral flow of ξ with domain Ω .

The functions α_+ and $\alpha_- : X \rightarrow]0, +\infty]$ defined by $I_x =] - \alpha_-(x), \alpha_+(x)[$ are lower semi-continuous. For $(x, t) \in \Omega$ and $y = f(x, t)$, one has $\alpha_+(y) = \alpha_+(x) - t$ and $\alpha_-(y) = \alpha_-(x) + t$.

If $t_1 \in I_x$ and if $t_2 \in I_{f(x, t_1)}$, one has $t_1 + t_2 \in I_x$ and

$$f(f(x, t_1), t_2) = f(x, t_1 + t_2). \quad (3)$$

If the function α_+ (resp. α_-) is bounded below on X by a constant > 0 , it is constant and equal to $+\infty$.

One has $\alpha_+ = \alpha_- = +\infty$ (in other words, one has $\Omega = X \times \mathbf{R}$) in each of the following cases:

- (i) the field ξ has compact support (for example if X is compact);
- (ii) there exists a transitive group of automorphisms of X preserving ξ ;
- (iii) there exists on X a Riemannian structure for which X is complete and ξ bounded.

If $\Omega = X \times \mathbf{R}$, the map $f : X \times \mathbf{R} \rightarrow X$ is a right action law of class C^s of the group manifold $\mathbf{R}$ on the manifold X , cf. 5.12.5.

9.1.5 ("The complex case"). Suppose that $K = \mathbf{C}$ and that X is separated. For every $a \in]0, +\infty]$, denote by D_a the open disk with center 0 and radius a in $\mathbf{C}$. For every $x \in X$, there exists one and only one number $\rho(x) \in]0, +\infty]$ and one and only one integral curve $f_x : D_{\rho(x)} \rightarrow X$ with origin x such that, for every $a \in]0, +\infty]$ and every integral curve $f : D_a \rightarrow X$ with origin x , one has $a \leq \rho(x)$ and $f = f_x|_{D_a}$.

For every $\lambda \in \mathbf{C}$ let $\alpha_\lambda(x)$ be the upper bound of the lifetime interval of x for the vector field $\lambda\xi$ on the real manifold deduced from X by restriction of scalars. One has $\rho(x) = \inf_{|\lambda|=1} \alpha_\lambda(x)$.

The set Δ of the pairs $(x, t) \in X \times \mathbf{C}$ such that $t \in D_{\rho(x)}$ is open in $X \times \mathbf{C}$ and the map $f : \Delta \rightarrow X$ defined by $f(x, t) = f_x(t)$ is an integral flow of ξ . The function $\rho : X \rightarrow]0, +\infty]$ is lower semicontinuous. For $(x, t) \in \Delta$ and $y = f(x, t)$, one has $\rho(y) \geq \rho(x) - |t|$. If ρ is bounded below on X by a constant > 0 , one has $\rho = +\infty$ and $\Delta = X \times \mathbf{C}$.

When $\Delta = X \times \mathbf{C}$, the map $f : X \times \mathbf{C} \rightarrow X$ is a right action law of the group manifold $\mathbf{C}$ on the manifold X .

9.1.6. Let $\varphi : X \rightarrow Y$ be a morphism of manifolds and η a vector field on Y such that ξ is φ -related to η (cf. 8.2.6). If $f : I \rightarrow X$ is an integral curve of ξ , the map $\varphi \circ f : I \rightarrow Y$ is an integral curve of η . If $K = \mathbf{R}$ and if X and Y are separated, for every $x \in X$, the lifetime interval of x for ξ is contained in the lifetime interval of $\varphi(x)$ for η . If moreover φ is proper, these intervals are equal.

9.1.7 ("Time-dependent equations"). Let W be an open subset of $X \times K$ and let $\Xi: W \rightarrow T(X)$ be a lifting of class C^s of $\text{pr}_1: W \rightarrow X$, that is, a map of class C^s from W into $T(X)$ such that $\Xi(x, t)$ belongs to $T_x(X)$ for all $(x, t) \in W$. One calls an integral curve of Ξ a map f of class C^k (with $k \in \mathbb{N}_K$ and $k \leq r$), from an open subset I of K into X such that, for all $t \in I$, one has

$$(f(t), t) \in W, \quad f'(t) = \Xi(f(t), t).$$

Let η be the vector field of class C^s on W defined by $\eta(x, t) = (\Xi(x, t), (t, 1))$. For a map f from an open subset I of K into X to be an integral curve of Ξ , it is necessary and sufficient that $t \mapsto (f(t), t)$ be an integral curve of η in the sense of 9.1.1.

Let $g: I \rightarrow W$ be an integral curve of η and t_0 a point of I such that $\text{pr}_2(g(t_0)) = t_0$. The set of elements $t \in I$ such that $\text{pr}_2(g(t)) = t$ is a neighborhood of t_0 that coincides with I if I is connected.

9.1.8 ("Parameters and initial conditions"). Let Z be a manifold of class C^r , let W be an open subset of $X \times K \times Z$, and let Ξ be a lifting of class C^s of $\text{pr}_1: W \rightarrow X$. If $z \in Z$, one denotes by W_z the set of $(x, t) \in X \times K$ such that $(x, t, z) \in W$, and one denotes Ξ_z as the map $(x, t) \mapsto \Xi(x, t, z)$ from W_z into $T(X)$.

Let (x_0, t_0, z_0) be a point of W . There then exists an open neighborhood $X_1 \times I_1 \times Z_1$ of (x_0, t_0, z_0) in W and a map of class C^s

$$f: X_1 \times I_1 \times I_1 \times Z_1 \rightarrow X$$

such that, for any $(x_1, t_1, z_1) \in X_1 \times I_1 \times Z_1$, the map $t \mapsto f(x_1, t_1, t, z_1)$ is an integral curve of Ξ_{z_1} (in the sense of 9.1.7) satisfying the relation $f(x_1, t_1, t_1, z_1) = x_1$.

Two maps

$$f_1: X_1 \times I_1 \times I_1 \times Z_1 \rightarrow X \quad \text{and} \quad f_2: X_2 \times I_2 \times I_2 \times Z_2 \rightarrow X$$

satisfying the above conditions coincide in a neighborhood of (x_0, t_0, t_0, z_0) .

9.2 Foliations

In this section, X denotes a manifold of class C^r , with $r \in \mathbb{N}_K$. Unless otherwise stated, all manifolds and morphisms considered are assumed to be of class C^r .

9.2.1. Let S be a manifold, and let $p : X \rightarrow S$ be a submersion. For every $s \in S$, we endow $X_s = p^{-1}(s)$ with the manifold structure induced by that of X (5.10.5); the set X is the disjoint union of the X_s . Denote by X_p the manifold structure on X obtained by gluing the X_s , for $s \in S$ (5.2.4); it is the unique manifold structure on X for which the X_s are open submanifolds. The topological space X_p is the sum of the spaces X_s .

Let V be a manifold and f a map from V into X . For f to be a morphism from V to X_p , it is necessary and sufficient that f be a morphism from V to X and that $p \circ f$ be locally constant.

9.2.2. A foliation of X is a manifold Y having the same underlying set as X and satisfying the following condition:

- (F) For every $x \in X$, there exists an open submanifold U of X containing x , a manifold S , and a submersion $p : U \rightarrow S$ such that the manifold U_p is an open submanifold of Y .

We also say that the pair (X, Y) is a foliated manifold. If (X, Y) and (X', Y') are foliated manifolds, a morphism from (X, Y) to (X', Y') is any map $f : X \rightarrow X'$ which is both a morphism from the manifold X to the manifold X' and a morphism from the manifold Y to the manifold Y' .

Let (X, Y) be a foliated manifold. The identity map $Y \rightarrow X$ is a bijective immersion. A subset U of X is called a leaf if it is an open set of Y ; in this case, U is endowed with the topological structure and the manifold structure induced by those of Y ; for example, U is said to be a connected leaf if it is an open and connected subset of Y . The leaves that are submanifolds of X form a basis for the topology of Y . When $K = \mathbb{R}$ or $\mathbb{C}$, the connected components of Y are leaves, called the maximal connected leaves of (X, Y) .

9.2.3. If $s \in \mathbb{N}_K$, $s \leq r$, we call a foliation of class C^s of X a foliation of the manifold of class C^s underlying X .

9.2.4. Examples

- a) The manifold X is a foliation of X , called the coarse foliation of X .

b) If $p : \mathbf{X} \rightarrow \mathbf{S}$ is a submersion, $\mathbf{X}_p$ is a foliation of $\mathbf{X}$ called the foliation of $\mathbf{X}$ defined by p .

The foliation of $\mathbf{X}$ defined by the identity map is the set $\mathbf{X}$ endowed with its structure of a pure manifold of dimension 0 (5.2.1); it is called the discrete foliation of $\mathbf{X}$.

c) Let E be a Banach space, F a closed vector subspace of E admitting a topological complement, and p the canonical projection $E \rightarrow E/F$. The foliation E_p of E is called the foliation of E defined by F .

d) Let Γ be a discrete group acting properly and freely on the topological space X (TG, III, § 4, no. 4). Let Y be a foliation of X , and suppose that, for every $s \in \Gamma$, the map $x \mapsto sx$ is an automorphism of (X, Y) . The equivalence relation on X (resp. Y) whose classes are the orbits of Γ is regular (5.9.5). If we denote by X/Γ (resp. Y/Γ) the corresponding quotient manifold, the pair $(X/\Gamma, Y/\Gamma)$ is a foliated manifold, called the quotient of (X, Y) by Γ .

9.2.5. Let Y be a foliation of X and U an open subset of X . Then U is open in Y , and U , endowed with the manifold structure induced by that of Y , is a foliation of X , called the foliation induced by Y .

More generally, let $f : X' \rightarrow X$ be a morphism such that f and the identity map $Y \rightarrow X$ form a transversal pair (5.11.1). The fibered product $Y' = X' \times_X Y$ is canonically identified with a foliation of X' called the pullback foliation of Y by f .

9.2.6. Let $p : \mathbf{X} \rightarrow \mathbf{S}$ and $p' : \mathbf{X}' \rightarrow \mathbf{S}'$ be two submersions, and let f be a morphism from $\mathbf{X}$ to $\mathbf{X}'$. For f to be a morphism from $\mathbf{X}_p$ to $\mathbf{X}_{p'}$, it is necessary and sufficient that, for every $x \in \mathbf{X}$, there exists an open neighborhood U of x in $\mathbf{X}$ and a morphism g from $p(U)$ to $\mathbf{S}'$ such that $g \circ p = p' \circ f$ on U .

9.2.7. Let Y be a foliation of X . A foliated chart of (X, Y) is a quadruple (U, φ, E, F) such that (U, φ, E) is a chart of X , F is a closed vector subspace of the Banach space E admitting a topological complement, and φ is an isomorphism from the foliation of U induced by Y to the foliation of $\varphi(U)$ defined by the canonical mapping from E onto E/F . For every point $x \in \mathbf{X}$ there exists a foliated chart (U, φ, E, F) of (X, Y) such that $x \in U$. Let $n = \dim_x \mathbf{X}$ and let $m = \dim_x \mathbf{Y}$. If n is finite, there exists an open neighborhood U of x and a coordinate system

$\zeta^1, \dots, \zeta^n$ of X over U such that the foliation of U induced by Y coincides with the foliation defined by the map $(\zeta^{m+1}, \dots, \zeta^n)$ from U into $\mathbf{K}^{n-m}$.

9.2.8. Let Y be a foliation of X . When x ranges over X , the spaces $T_x(Y)$ are the fibers of a vector subbundle of class C^{r-1} of $T(X)$, which is called the subbundle of $T(X)$ tangent to the foliation Y and is denoted $T(X, Y)$. If the foliation Y is defined by a submersion $p : X \rightarrow S$, then $T(X, Y) = \text{Ker } T(p)$ is the relative tangent bundle $T(X/S)$ of X over S (8.1.3).

Let (X, Y) and (X', Y') be two foliated manifolds, and let f be a morphism from X to X' . For f to be a morphism of foliated manifolds, it is necessary and sufficient that $T(f)$ map $T(X, Y)$ into $T(X', Y')$. In particular, let $f : Z \rightarrow X$ be a morphism of manifolds; for f to be a morphism from Z to Y , it is necessary and sufficient that $T(f)$ map $T(Z)$ into $T(X, Y)$. For a submanifold Z of X to be a leaf of (X, Y) , it is necessary and sufficient that one have

$$T_z(Z) = T_z(X, Y) \quad \text{for all } z \in Z.$$

Let Y be a foliation of X and let U be a leaf of Y , admitting a countable base of open sets. Let Z be a manifold and f a map from Z to U . For f to be a morphism of manifolds from Z to U , it is necessary and sufficient that the composite of f and the canonical injection of U into X be a morphism of manifolds from Z to X . If X is locally compact and countable at infinity, every connected leaf of (X, Y) admits a countable base of open sets (cf. TG, I, 3rd ed., §11, no. 7, cor. 2 of th. 1).

9.2.9. Suppose that $K = \mathbf{R}$ or $\mathbf{C}$. Let Y be a foliation of X . The following conditions are equivalent:

- a) There exist a manifold S and a submersion $p : X \rightarrow S$ such that $Y = X_p$.
- b) For every $x \in X$, there exists a submanifold S_x of X possessing the following two properties:

b_1) The tangent space $T_x(S_x)$ to S_x at x is a topological supplement of $T_x(X, Y)$ in $T_x(X)$.

b_2) Every connected leaf of (X, Y) intersects S_x at at most one point.

Suppose these conditions are satisfied, and let R be the relation on X “ x and y belong to the same connected leaf”; then R is a regular equivalence relation (5.9.5) on X , and, if p denotes the canonical projection $X \rightarrow X/R$, we have $Y = X_p$.

Example. — Take $K = \mathbb{R}$ and $X = \mathbb{R}^2$. Let the discrete group $\Gamma = \mathbb{Z}^2$ act on X by translations. Let $m \in \mathbb{R}$ and let (X, Y_m) be the foliation defined by the submersion $p: (x_1, x_2) \mapsto x_2 - mx_1$ from X onto $\mathbb{R}$. Set $X' = X/\Gamma$ and $Y_m' = Y_m/\Gamma$; then Y_m' is a foliation of the torus X' (cf. 9.2.4, example d)). If m is rational, there exists a submersion p' from X' onto $\mathbb{R}/\mathbb{Z}$ such that $Y_m' = X_{p'}$. If m is irrational, every maximal connected leaf of (X', Y_m') is dense in X' , and there does not exist a submersion p' from X' into a manifold S such that $Y_m' = X_{p'}$.

9.2.10. Let Y be a foliation of X , and let $\pi : X \rightarrow S$ be a morphism of manifolds such that the composite $Y \rightarrow X \xrightarrow{\pi} S$ is étale. If S' is an open subset of S , a section $\sigma : S' \rightarrow X$ of π above S' is said to be horizontal (with respect to the foliation Y) if it is a morphism from S' into Y ; it is equivalent to say that $\sigma(S')$ is a leaf of (X, Y) , or again that $T(\sigma)$ maps $T(S')$ to $T(X, Y)$. For every $s_0 \in S$ and for every $x_0 \in \pi^{-1}(s_0)$, there exists a horizontal section defined in a neighborhood of s_0 and taking the value x_0 at s_0 ; two such sections coincide on a neighborhood of s_0 .

More generally, let $s_0 \in S$, let T be a manifold, let f be a morphism from T into the submanifold $\pi^{-1}(s_0)$ of X , and let $t_0 \in T$. There then exists an open neighborhood T' (resp. S') of t_0 in T (resp. of s_0 in S), and a morphism

$$F : S' \times T' \rightarrow X$$

satisfying the following condition: for every $t \in T'$, the map $s \mapsto F(s, t)$ is a horizontal section of π above S' taking the value $f(t)$ at the point s_0 . Two such maps F coincide on a neighborhood of (s_0, t_0) .

9.3 Integrable subbundles

In this section, X denotes a manifold of class C^r and F a vector subbundle of class C^s of $T(X)$, with r, s in N_K , $s \leq r - 1$.

9.3.1. Let V be a manifold of class C^k ($k \in N_K, k \leq r$) and let f be a morphism of class C^k

from V into X . We say that f is an integral of F if $T(f)$ maps $T(V)$ into F .

A submanifold Z of class C^k of X is called an integral submanifold of F if the injection $Z \rightarrow X$ is an integral of F in the above sense, i.e. if one has $T_x(Z) \subset F_x$ for all $x \in Z$.

Let $x \in X$. If Z_1 and Z_2 are two integral submanifolds of F containing x and if $T_x(Z_1) = F_x$, there exists a neighborhood U of x such that $U \cap Z_2 \subset Z_1$. If moreover $T_x(Z_2) = F_x$, the germs of Z_1 and Z_2 at x coincide.

9.3.2. We say that F is integrable if there exists a foliation Y of X of class C^s such that $F = T(X, Y)$ (9.2.8). Such a foliation is then unique; it is called the integral foliation of F . For a morphism $f: V \rightarrow X$ of class C^s to be an integral of F , it is necessary and sufficient that f be a morphism from V into Y .

Example. — If F has rank 1 at every point, F is integrable, and defines on X a pure foliation of dimension 1. Suppose moreover that $K = \mathbb{R}$, that X is separated, and that F admits as a frame a vector field ξ nowhere zero. Then, for every $x \in X$, the maximal connected leaf containing x is the image of the maximal integral arc of ξ with origin x (9.1.3).

9.3.3 ("Integrability criteria"). The following conditions are equivalent:

- (i) The vector subbundle F of $T(X)$ is integrable.
- (ii) For every $x \in X$, there exists an integral submanifold Z of class C^s of F such that $x \in Z$ and $T_x(Z) = F_x$.
- (iii) For every open set U of X and vector fields ξ and η belonging to $\mathcal{S}_F^s(U)$ (7.4.1), one has $[\xi, \eta] \in \mathcal{S}_F^{s-1}(U)$.

Let $(\xi_i)_{i \in I}$ be a family of sections of F of class C^s such that, for every $x \in X$, the set of the $\xi_i(x)$ is a total subset of the Banach space F_x (EVT, I, § 2, no. 1). The conditions (i), (ii), and (iii) are then equivalent to:

- (iv) for every pair (i, j) of elements of I and every $x \in X$, one has $[\xi_i, \xi_j](x) \in F_x$.

When I is finite and the family (ξ_i) is a frame of F , the preceding conditions are still equivalent to:

- (v) there exists a family $(c_{ij}^k)_{(i,j,k) \in I \times I \times I}$ of functions on X with values in K such that $[\xi_i, \xi_j] =$

$\sum_k c_{ij}^k \xi_k$ for all i, j in I .

(If this is the case, the functions $c_{i,j}^k$ are of class C^{s-1} .)

9.3.4. Let $s' \in \mathbf{N}_{\mathbb{K}}$, with $s' \leq s$, and let $\mathbf{F}'$ be the vector bundle of class $C^{s'}$ deduced from $\mathbf{F}$ by weakening of structure (8.7.1). For $\mathbf{F}$ to be integrable, it is necessary and sufficient that $\mathbf{F}'$ be integrable. In this case, the integral foliation of $\mathbf{F}'$ is deduced from that of $\mathbf{F}$ by weakening of structure.

9.3.5. Let E be a Banach space and p be an integer ≥ 0 . For every $x \in X$, denote by

$N(p, E)_x$ the vector subspace of $\text{Alt}^p(T_x(X); E)$ consisting of the elements u such that $u(v_1, \dots, v_p) = 0$ for $v_1, \dots, v_p$ in F_x . The spaces $N(p, E)_x$, for $x \in X$, are the fibers of a vector subbundle of class C^s of $\text{Alt}^p(T(X); E)$. Denote it by $N(p, E)$. If F is integrable, we have:

(vi) for every open set U of X and every form $\omega \in \mathcal{S}_{N(p, E)}^s(U)$, one has $d\omega \in \mathcal{S}_{N(p+1, E)}^{s-1}(U)$.

Conversely, if (vi) holds for $p = 1$ and for every Banach space E , the bundle F is integrable; when $K = \mathbb{R}$ or $\mathbb{C}$, it even suffices to verify (vi) for $p = 1$ and $E = K$.

Suppose that the dual of $T(X)/F$ admits a frame $(\omega_1, \dots, \omega_n)$. The integrability of F is then equivalent to the following condition:

(vii)

$$d\omega_i \wedge \omega_1 \wedge \dots \wedge \omega_n = 0 \quad \text{for } 1 \leq i \leq n.$$

If moreover there exists a vector subbundle G of class C^s of $T(X)$ such that $T(X)$ is the direct sum of F and G , condition (vii) is equivalent to:

(viii) there exist differential forms α_i^j (i, j in I) of degree 1 on X , of class C^{s-1} , such that

$$d\omega_i = \sum_j \alpha_i^j \wedge \omega_j \quad \text{for all } i \in I.$$

9.3.6. Let L be a Banach space and let ω be a differential form of degree 1 on X with values in L , of class C^s . We make the following two hypotheses:

a) for all $x \in X$, ω_x is a surjective homomorphism of $T_x(X)$ to L , and the kernel of ω_x is

$$F_x;$$

b) there exists a vector subbundle G of $T(X)$ such that $T(X)$ is the direct sum of F and G .

The integrability of F is then equivalent to:

(ix) there exists a differential form α of degree 1 on X with values in $\text{End}(\mathbf{L})$ such that $d\omega = \alpha \wedge \omega$, the exterior product being defined by the canonical pairing $\text{End}(\mathbf{L}) \times \mathbf{L} \rightarrow \mathbf{L}$ (7.8.2 and 8.3.2).

9.3.7 ("Equations in total differentials"). Suppose that X is the product of two manifolds A and B of class C^r ; one denotes $p_1 : X \rightarrow A$ and $p_2 : X \rightarrow B$ the two projections. Let f be a morphism of class C^s of $p_1^*T(A)$ to $p_2^*T(B)$. The graphs of the maps $f_{(a,b)} : T_a(A) \rightarrow T_b(B)$ are the fibers of a vector subbundle of class C^s of $T(X)$; denote it by F^f .

Let A' be an open subset of A , and let $\varphi : A' \rightarrow B$ be a morphism of class C^k ($k \in \mathbb{N}_{\mathbb{K}}$, $k \leq r$). One says that φ is an integral of f if, for all $a \in A'$, one has $T_a(\varphi) = f_{a,\varphi(a)}$; this is equivalent to saying that the map $a \mapsto (a, \varphi(a))$ of A' into X is an integral of F^f (9.3.1). Such a map is of class C^{s+1} . If φ_1 and φ_2 are two integrals of f and take the same value at a point $a \in A$, they coincide in a neighborhood of a .

More generally, let Z be a manifold of class C^k , let A' be an open subset of A , let $a \in A'$ and let Φ_1, Φ_2 be morphisms of class C^k from $Z \times A'$ into B .

Φ_1 and Φ_2 coincide on $Z \times \{a\}$ and that, for all $z \in Z$, the morphisms

$$a \mapsto \Phi_1(z, a) \quad \text{and} \quad a \mapsto \Phi_2(z, a)$$

are integrals of f . Then Φ_1 and Φ_2 coincide in a neighborhood of $Z \times \{a\}$.

Suppose that F^f is integrable. Let Z be a manifold of class C^k ($k \in \mathbb{N}_{\mathbb{K}}$, $k \leq s$), (z_0, a_0) a point of $Z \times A$, and ρ be a morphism of class C^k from Z into B . There exists an open neighborhood $Z' \times A'$ of (z_0, a_0) in $Z \times A$ and a morphism $\Phi : Z' \times A' \rightarrow B$ of class C^k such that, for all $z \in Z'$, the map $a \mapsto \Phi(z, a)$ from A' into B admits f as its tangent map and takes the value $\rho(z)$ at the point a_0 .

9.3.8. We retain the hypotheses and notation of 9.3.7 and suppose that A (resp. B) is an open submanifold of a Banach space E (resp. M). The map f then identifies with a morphism of class C^s from $X = A \times B$ into the Banach space $\mathcal{L}(E; M)$. Let $D_1 f$ (resp. $D_2 f$) denote

the first (resp. second) partial derivative of f (1.6.2); it is a morphism of class C^{s-1} from X into $\mathcal{L}(E; \mathcal{L}(E; M))$ (resp. into $\mathcal{L}(M; \mathcal{L}(E; M))$), a space identified in the obvious way with $\mathcal{L}_2(E; M)$ (respectively with $\mathcal{L}(M, E; M)$). For F to be integrable, it is necessary and sufficient that, for every $x \in X$, the bilinear map

$$\Delta_x : (h_1, h_2) \mapsto D_1 f(x)(h_1, h_2) + D_2 f(x)(f(x)h_1, h_2)$$

from $E \times E$ to M be symmetric. Under this condition, if φ is an integral of f defined in an open subset A' of A , the second derivative of φ at a point a of A' is $\Delta(a, \varphi(a))$.

If $E = K^n$, and if one denotes $(x^1, \dots, x^n)$ the coordinate functions on K^n , the map f is defined by a family $(f_1, \dots, f_n)$ of maps from $A \times B$ to M , and the integrability condition is written:

$$\frac{\partial f_i}{\partial x^j} + (D_2 f_i) \cdot f_j = \frac{\partial f_j}{\partial x^i} + (D_2 f_j) \cdot f_i$$

for all integers i, j in $[1, n]$. A map φ of an open subset A' of A into B is an integral of f if and only if one has

$$\frac{\partial \varphi}{\partial x^i} = f_i(x, \varphi(x)) \quad \text{for all } x \in A' \text{ and all } i \in [1, n],$$

in other words, if one has

$$d\varphi = \sum_{1 \leq i \leq n} f_i(x, \varphi(x)) dx^i.$$

9.4 Integrable bundles in characteristic $p \neq 0$

In this number, we assume K has characteristic $p \neq 0$. We denote by X a K -analytic manifold locally of finite dimension.

9.4.1 ("p-th powers"). Let ξ be a vector field on an open subset U of X . There exists a vector field ξ^p and only one on U such that we have

$$D_{\xi^p}(f) = (D_{\xi})^p(f) = \underbrace{D_{\xi}(D_{\xi}(\dots(D_{\xi}(f))\dots))}_{p \text{ times}} \quad (1)$$

for every analytic function f defined in an open subset of U .

If φ is an analytic function on U , we have

$$(\varphi \xi)^p = \varphi^p \xi^p + (D_{\varphi \xi})^{p-1}(\varphi) \cdot \xi \quad (2)$$

and if η is a vector field on U , we have

$$[\xi^p, \eta] = \text{ad}(\xi)^p(\eta) = [\xi, [\xi, \dots, [\xi, \eta] \dots]]. \quad (3)$$

9.4.2 ("Jacobson's identity"). Let L be the free F_p -Lie algebra (LIE, II, § 2, no. 2) on a set $\{x, y\}$ with two elements. There exists one and only one element $\Lambda_p(x, y)$ of L such that we have

$$(x + y)^p = x^p + y^p + \Lambda_p(x, y) \quad (4)$$

in the enveloping algebra of L . Examples:

$$\Lambda_2(x, y) = [x, y]; \quad \Lambda_3(x, y) = [x, [x, y]] - [y, [x, y]].$$

If ξ and η are vector fields on X , we have

$$(\xi + \eta)^p = \xi^p + \eta^p + \Lambda_p(\xi, \eta). \quad (5)$$

9.4.3. Example. — Take $X = K$ (resp. K^*) and denote by x the canonical map $X \rightarrow K$. Let ξ (resp. η) be the vector field $\partial/\partial x$ (resp. $x \cdot \partial/\partial x$); it is invariant under translations of the additive (resp. multiplicative) group X . We have

$$\xi^p = 0 \quad \text{and} \quad \eta^p = \eta. \quad (6)$$

9.4.4 ("Integrable bundles"). Let F be a vector subbundle of $T(X)$. We say that F is integrable if, for every $x \in X$, there exists a coordinate system $(\zeta^1, \dots, \zeta^n)$ of X at x and an integer $m \leq n$

such that $(\partial/\partial\zeta^1, \dots, \partial/\partial\zeta^m)$ is a frame of F in a neighborhood of x .

For F to be integrable, it is necessary and sufficient that it satisfy the following condition:

For every open set U of X , $\mathcal{S}_{\mathbb{F}}^{\omega}(U)$ is a Lie subalgebra of $\mathcal{S}_{T(X)}^{\omega}(U)$ stable under the operation $\xi \mapsto \xi^p$.

10 Measures defined by differential forms

In this paragraph, we assume that K is locally compact. From Section 10.2 onward, we assume $K = \mathbb{R}$. In Sections 10.1 and 10.4, all manifolds considered are assumed to be locally of finite dimension; when they are Hausdorff, they are locally compact.

10.1 Modulus measure of a differential form

Notation. — We denote μ a Haar measure (INT, VII, §1, no. 2) on the additive group of K . We denote $\mu^{\otimes n}$ the measure $\mu \otimes \dots \otimes \mu$ onto K^n and $\mu_U^{\otimes n}$ its restriction to an open set U of K^n . For $a \in K$ we denote by $\text{mod}(a)$ the modulus of a (INT, VII, §1, no. 10 and AC, VI, §9, no. 1). If $K = \mathbb{R}$, we have $\text{mod}(a) = |a|$; if $K = \mathbb{C}$, we have $\text{mod}(a) = |a|^2$; if K is ultrametric, we have $\text{mod}(a) = q^{-v(a)}$, where q denotes the number of elements in the residue field of K and v the normalized valuation of K (AC, VI, §9, no. 1, prop. 1).

10.1.1. Let U and V be open subsets of K^n and let $f: U \rightarrow V$ a map of class C^r , with $r \in \mathbb{N}_K$. Let $x \in U$; we call the Jacobian of f at x , and denote by $\text{Jac}_x(f)$, the determinant of the linear map $Df(x)$ derivative of f at x . We denote $\text{Jac}(f)$ the function $x \mapsto \text{Jac}_x(f)$ the function $\text{mod}(\text{Jac}(f))$ is a continuous function in U with positive real values.

10.1.2 ("Change of variable in integrals"). Under the hypotheses of 10.1.1, suppose that f let be a class isomorphism C^r of the manifold U onto the manifold V . Then the image under f of the measure $\text{mod}(\text{Jac}(f)) \cdot \mu_U^{\otimes n}$ is $\mu_V^{\otimes n}$; for every continuous function with compact support $\varphi: V \rightarrow \mathbb{C}$, one has

$$\int_V \varphi(y) \mu^{\otimes n}(y) = \int_U \varphi(f(x)) \cdot \text{mod}(\text{Jac}_x(f)) \mu^{\otimes n}(x). \quad (1)$$

10.1.3. Let X be a variety of class C^r and let A be a subset of X . One says that A is locally

negligible if, for every chart (U, φ, K^n) of the manifold X , the set $\varphi(A \cap U)$ is $\mu^{\otimes n}$ -negligible (INT, IV, § 2, no. 2); this condition does not depend on the choice of μ , and it suffices to verify it for a family of charts whose domains cover A . Every set contained in the union of a countable family of locally negligible sets is locally negligible.

Examples

- a) Every submanifold of X that has codimension ≥ 1 at every point (i.e. has empty interior) is locally negligible.
- b) Let g be a function of class C^r on X , and let A_g the set of points $x \in X$ such that $g(x) = 0$; suppose that the interior of A_g is empty, and that $r = \omega$; then A_g is locally negligible.
- c) Let $f : Y \rightarrow X$ be a morphism of manifolds of class C^r , the manifold Y being a countable union of compact sets. It is assumed that one has $\dim_y Y \leq \dim_{f(y)} X$ for all $y \in Y$. Its image under f of every locally negligible subset of Y is a locally negligible subset of X . The same is true of the image under f of the set of points where f is not étale (“first Sard theorem”); in particular, if $\dim_y Y < \dim_{f(y)} X$ for all $y \in Y$, $f(Y)$ is a locally negligible subset of X .
- d) Suppose K has characteristic zero. Let $f : Y \rightarrow X$ be a morphism of manifolds of class C^r ($r \in N_K, r \geq \infty$), Y being a countable union of compact sets. Let C be the set of points of Y where f is not a submersion (“critical points”). Then $f(C)$ is a locally negligible subset of X (“second Sard theorem”). ⁽¹⁾

10.1.4. Let X be a paracompact manifold of class C^r . On X there exists a class M of equivalent measures (INT, V, § 5, n° 6), and only one, such that, for every $\nu \in M$ and for every chart $c = (U, \varphi, K^n)$ of X , the image under φ of the restriction of ν to U is equivalent to $\mu_{\varphi(U)}^{\otimes n}$. One says that M is the canonical class of measures on X . For a subset A of X to be locally negligible (10.1.3), it is necessary and sufficient that it be locally negligible (INT, IV, § 2, n° 2) for one (resp. every) measure $\nu \in M$.

If $K = \mathbb{R}$ and $r \neq \omega$ (resp. $K = \mathbb{R}$ or $\mathbb{C}$ and $r = \omega$, resp. K is ultrametric), there exists $\nu \in M$ such that, for every chart $c = (U, \varphi, K^n)$ of X , the density of $\varphi(\nu_U)$ with respect to $\mu_{\varphi(U)}^{\otimes n}$ is

everywhere >0 and of class C^{r-1} (resp. of class C^∞ , resp. locally constant). If ν and ν' are two such measures, the density of ν' with respect to ν is everywhere >0 and of class C^{r-1} (resp. of class C^∞ , resp. locally constant).

10.1.5. Let X be a manifold of class C^r ; let $T'(X)$ the cotangent bundle of X (8.2.2) and set $\Omega = \det(T'(X))$ (7.9.9). The vector bundle Ω has rank 1 at each point; when X is pure of dimension n , it is identified with the vector bundle $\text{Alt}^n(T(X); K_X)$. Let ω a section of Ω on X ; one says, by abuse of language, that ω is a differential form of maximum degree on X . Let $c = (U, \varphi, K^n)$ be a chart of X ; denote by $u^1, \dots, u^n$ the coordinate functions on K^n . There exists a function f_c and only one on $\varphi(U)$ such that $\omega|_U = \varphi^*(f_c \cdot du^1 \wedge \dots \wedge du^n)$. We say that ω is locally of integrable modulus if, for every chart $c = (U, \varphi, K^n)$ of X , the real function $\text{mod}(f_c) : \varphi(U) \rightarrow \mathbf{R}$ is locally integrable for $\mu_{\varphi(U)}^{\otimes n}$ (INT, IV, § 4, n° 1); it suffices to verify this property for the charts of an atlas of X . If ω is continuous, and in particular if ω is of class C^s , with $s \in \mathbb{N}_K, s \leq r-1$, then ω is locally of integrable modulus.

10.1.6 (“Positive measure defined by a differential form of maximum degree”). Keep the notations of 10.1.5 and suppose moreover that X is separated and that ω is locally of integrable modulus. If $c = (U, \varphi, K^n)$ is a chart of X , denote by ν_c the measure on $\varphi(U)$ product of the measure $\mu_{\varphi(U)}^{\otimes n}$ by the function $\text{mod}(f_c)$ (INT, V, § 5, n° 2, def. 2) and let α_c the image of ν_c by φ^{-1} . There exists on X a measure α and only one such that, for every chart $c = (U, \varphi, K^n)$ of X , the restriction of α to U is equal to α_c . One says that the measure α is the modulus of ω and one denotes it $\text{mod}(\omega)_\mu$. It is a positive measure. When $K = \mathbf{R}$ and μ is Lebesgue measure, one writes $|\omega|$ instead of $\text{mod}(\omega)_\mu$.

If X is purely of dimension n , and if a is a real number >0 , one has $\text{mod}(\omega)_{a\mu} = a^n \text{mod}(\omega)_\mu$.

Let A be a subset of X . For A to be locally negligible (10.1.3), it is necessary and sufficient that, for every open set U of X and every differential form ω of maximum degree on U which is locally of integrable modulus, the set $A \cap U$ be locally negligible for the measure $\text{mod}(\omega)_\mu$.

Suppose moreover X is paracompact and let ν be a measure on X belonging to the canonical equivalence class M (10.1.4). The measure $\text{mod}(\omega)_\mu$ is based on ν (INT, V, § 5, no. 2, def. 2). For $\text{mod}(\omega)_\mu$ to belong to M , it is necessary and sufficient that the set of $x \in X$ such that

$\omega(x) = 0$ be locally negligible.

10.1.7. Example. — Let G be a Lie group over K , of finite dimension n , and let ω be a differential form of degree n on G , invariant under left translations, and nonzero. The corresponding measure $\text{mod}(\omega)_\mu$ is then a left Haar measure on the locally compact group G .

When G is the multiplicative group K^* , one may take for ω the form dx/x , and one has $\text{mod}(\omega)_\mu = (\text{mod})^{-1} \cdot \mu$, cf. INT, VII, § 1, no. 10, prop. 14.

(7)

10.2 Orientations

Recall that, in this issue and the following ones, we assume $K = \mathbb{R}$.

10.2.1 ("Orientation of a real vector space"). Let E be a real vector space of finite dimension n . We denote by $\text{Or}(E)$ the set of orientations of E (A, VI, § 2, no. 7); the elements of $\text{Or}(E)$ are the two closed half-lines of the vector space $\det(E) = \bigwedge^n E$. If $\xi \in \text{Or}(E)$, the other element of $\text{Or}(E)$ is called the orientation opposite to ξ , and denoted $-\xi$.

Let $0 \rightarrow E' \xrightarrow{\alpha} E \xrightarrow{\beta} E'' \rightarrow 0$ be a short exact sequence of finite-dimensional (real) vector spaces; set $p' = \dim E'$, $p'' = \dim E''$, and let ξ' (resp. ξ'') be an orientation of E' (resp. E''). We denote by $\xi' \xi''$ (resp. $\xi'' \xi'$) the orientation of E containing a nonzero $(p' + p'')$ -vector $u' \wedge u''$ (resp. $u'' \wedge u'$), where u' is the image under $\bigwedge^{p'} \alpha$ of a p' -vector of E' belonging to ξ' and where u'' is a p'' -vector of E whose image under $\bigwedge^{p''} \beta$ belongs to ξ'' (cf. A, VI, § 2, no. 7). If $E = E' \times E''$, α and β being the canonical maps, we say that $\xi' \xi''$ is the product of the orientation ξ' and of the orientation ξ'' .

10.2.2 ("Space of orientations"). Let B be a topological space and M a real vector bundle over B (in the topological sense — cf. § 6, p. 61, Note (*)), of finite rank (7.1.6). Let Or_M be the disjoint union of the $\text{Or}(M_b)$, for $b \in B$, and let $\pi : \text{Or}_M \rightarrow B$ be the map such that $\pi(\text{Or}(M_b)) = \{b\}$. There exists on Or_M one and only one topological space structure such that:

- a) π is continuous.
- b) If s is a continuous section of $\det(M)$ that is nowhere zero on an open set U of B , and if $\xi(s(b))$ is the orientation of M_b defined by the element $s(b)$ of $\det(M_b)$, the map $b \mapsto$

$\xi(s(b))$ from U into Or_M is continuous.

The space Or_M is called the space of orientations of M . The group $\{\pm 1\}$ acts on Or_M by $\xi \mapsto \pm \xi$ (10.2.1). The quadruple $(Or_M, \{\pm 1\}, B, \pi)$ is a principal fibration (6.2.1) with base B and structural group $\{\pm 1\}$; the projection $\pi : Or_M \rightarrow B$ defines an isomorphism of $Or_M/\{\pm 1\}$ onto B , cf. no. 6.2.

When B is endowed with a manifold structure, we endow Or_M with the manifold structure pulled back from that of B by π (5.8.1); the fibration defined above is then a fibration of manifolds, and the morphism π is étale.

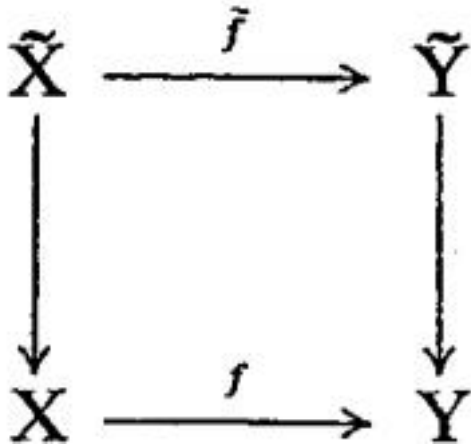
10.2.3. We retain the hypotheses of 10.2.2. An orientation of the vector bundle M is called a continuous section $\xi : B \rightarrow Or_M$ of the projection $\pi : Or_M \rightarrow B$. We say that M is orientable if it possesses an orientation; this is equivalent to saying that the bundle Or_M is trivializable, i.e. isomorphic to $B \times \{\pm 1\}$. If B is connected and nonempty, every orientable vector bundle over B has two orientations, opposite to each other.

When M is reduced to 0, the bundle $\det(M)$ is the trivial bundle R_B and Or_M is identified with $Or(R) \times B$; the bundle M possesses a canonical orientation, corresponding to the positive half-line of R .

10.2.4 ("Orientation of a manifold"). Let X be a locally finite-dimensional manifold over R of class C^τ and let $T(X)$ be its tangent bundle. We denote by $\tilde{X}$ the manifold $Or_{T(X)}$; the group $\{\pm 1\}$ acts properly and freely on $\tilde{X}$ and $\tilde{X}/\{\pm 1\}$ is identified with X ; the fibers of the projection $\pi : \tilde{X} \rightarrow X$ are the sets $Or(T_x(X))$, for $x \in X$. An orientation of X is called an orientation of the bundle $T(X)$, in other words, a continuous section of the bundle $\tilde{X}$; such a section is of class C^τ . We say that X is orientable if it possesses an orientation. Every manifold of dimension zero is orientable and possesses a canonical orientation (10.2.3).

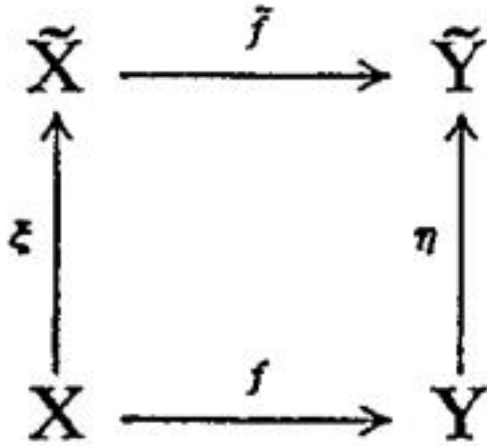
Let $\xi \in \tilde{X}$ and let $x = \pi(\xi)$ be its image in X ; the map $T_\xi(\pi) : T_\xi(\tilde{X}) \rightarrow T_x(X)$ is an isomorphism; it allows one to identify ξ with an element $\tilde{\xi}$ of $Or(T_x(X))$. The map $\xi \mapsto \tilde{\xi}$ is an orientation of $\tilde{X}$, said to be canonical; in particular, $\tilde{X}$ is orientable.

10.2.5 ("Orientation of a morphism"). Let X and Y be two locally finite-dimensional manifolds and let $f : X \rightarrow Y$ be a morphism of manifolds. One calls an orientation of f a morphism $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ making the diagram commutative



and compatible with the action of the group $\{\pm 1\}$.

If $\tilde{f}$ is an orientation of f and if η is an orientation of Y , there exists one and only one orientation ξ of X such that the diagram



is commutative. We say that ξ is associated with η by f .

Examples

- a) If Y is reduced to a point, an orientation of f is equivalent to an orientation of X .
- b) Suppose f is étale. For every $x \in X$, the map $T_x(f)$ is an isomorphism of $T_x(X)$ onto $T_{f(x)}(Y)$, and defines (by transport of structure) a bijection

$$\tilde{f}_x : \text{Or}(T_x(X)) \rightarrow \text{Or}(T_{f(x)}(Y)).$$

The family of $\tilde{f}_x$ defines an orientation $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ of f , said to be canonical.

- c) More generally, suppose that f is a submersion, and let α be an orientation of the bundle $T(X/Y)$ (8.1.3). Let $x \in X$ and ξ be an orientation of $T_x(X/Y)$. Set $y = f(x)$; we have an

exact sequence (8.1.3)

$$0 \rightarrow T_x(X/Y) \rightarrow T_x(X) \xrightarrow{T_x(f)} T_y(Y) \rightarrow 0$$

and there therefore exists one and only one orientation $\tilde{f}_\alpha(\xi)$ of $T_y(Y)$ such that $\xi = \tilde{f}_\alpha(\xi)\alpha(x)$ (10.2.1). The map $\tilde{f}_\alpha: \tilde{X} \rightarrow \tilde{Y}$ is an orientation of f , said to be associated with α . The map $\alpha \mapsto \tilde{f}_\alpha$ is a bijection from the set of orientations of $T(X/Y)$ onto the set of orientations of f .

d) If $f: X \rightarrow Y$ is an immersion, the orientations of f correspond to the orientations of the normal bundle to f .

10.2.6 ("Orientation of a product"). Let $\mathbf{X}_1$ (resp. $\mathbf{X}_2$) be a manifold locally of finite dimension and ξ_1 (resp. ξ_2) an orientation of $\mathbf{X}_1$ (resp. $\mathbf{X}_2$). Let $\mathbf{X} = \mathbf{X}_1 \times \mathbf{X}_2$. The map $(x_1, x_2) \mapsto \xi_1(x_1)\xi_2(x_2)$ (10.2.1) is an orientation of $\mathbf{X}_1 \times \mathbf{X}_2$, called the product of the orientations ξ_1 and ξ_2 , and denoted $\xi_1\xi_2$, or $\xi_1 \otimes \xi_2$.

Suppose that X_i ($i = 1, 2$) are pure of dimension n_i . The canonical isomorphism $X_1 \times X_2 \rightarrow X_2 \times X_1$ transforms $\xi_1 \otimes \xi_2$ into $\xi_2 \otimes \xi_1$ (resp. into $-\xi_2 \otimes \xi_1$) if one of the n_i is even (resp. if n_1 and n_2 are odd).

10.2.7 ("Complex case"). Let F be a finite-dimensional complex vector space and let F_R be the underlying real vector space. Let $\{e_1, \dots, e_m\}$ be a basis of F ; then $\{e_1, ie_1, e_2, ie_2, \dots, e_m, ie_m\}$ is a basis of F_R and the orientation of F_R defined by the $2m$ -vector

$$e_1 \wedge ie_1 \wedge e_2 \wedge ie_2 \wedge \dots \wedge e_m \wedge ie_m$$

is independent of the choice of the basis $\{e_1, \dots, e_m\}$ of F ; this is called the orientation of F_R defined by the given complex structure. If F is the direct sum of two subspaces G and H , the orientation of $F_R = G_R \oplus H_R$ is the product of the orientations of G_R and H_R .

Let X be a real manifold locally of finite dimension, equipped with an almost complex structure (8.8.3). For every $x \in X$ let $\xi(x)$ be the orientation of $T_x(X)$ defined by its complex structure. The map $x \mapsto \xi(x)$ is an orientation of X , called the orientation defined by the al-

most complex structure. In particular, we shall equip the real analytic manifold X underlying a complex analytic manifold X^c , locally of finite dimension, with the orientation defined by the associated almost complex structure (8.8.6). If one replaces the given complex analytic manifold structure on X by its conjugate (5.4.12.b)), this orientation is multiplied by the function $x \mapsto (-1)^{\dim_x X^c}$.

10.2.8. Examples

- a) Let E be a finite-dimensional real vector space. The real analytic manifold defined by E (5.2.2) is orientable. More precisely, the map $\xi \mapsto \xi(0)$ is a bijection from the set of orientations of this manifold onto $\text{Or}(E)$.

In particular, the manifold $\mathbf{R}^n$ has a canonical orientation ξ^n defined by the half-line $\mathbf{R}_+ e_1 \wedge \cdots \wedge e_n$ of $\text{Or}(\mathbf{R}^n)$.

- b) Let X be a sphere with center 0 and radius ρ in $\mathbf{R}^n$; it is a submanifold of $\mathbf{R}^n$. For every $x \in X$, the space $\mathbf{T}_x(\mathbf{R}^n) = \mathbf{R}^n$ is the direct sum of the line $\mathbf{R}x$ and of the hyperplane $\mathbf{T}_x(X)$. Let η_x be the orientation $\mathbf{R}_+ x$ of $\mathbf{R}x$ (“outward normal”) and let ξ_x be the unique orientation of $\mathbf{T}_x(X)$ such that the product of η_x and ξ_x is the orientation ξ^n of $\mathbf{R}^n$. The map $x \mapsto \xi_x$ is an orientation of X , said to be canonical.
- c) Let X be a connected manifold, and let G be a discrete group acting properly and freely on X . For X/G to be orientable, it is necessary and sufficient that X be orientable and that the action of G on the set of orientations of X be trivial.
- d) Real projective space $\mathbf{P}_{n-1}(\mathbf{R})$ is identified with the quotient of the sphere S_{n-1} by the group $\{\pm 1\}$ acting by $x \mapsto \pm x$; it is orientable if n is even, and non-orientable if n is odd.
- e) Every quotient of a Lie group by a connected Lie subgroup is orientable.

10.3 M-twisted differential forms

10.3.1. Let X be a real manifold of class C^r and let M be a vector bundle with base X , of finite rank and of class C^k , with $0 \leq k \leq r$. Let $\lambda_M = (\text{Or}_M, \{\pm 1\}, X, \pi)$ be the principal fibration associated with the manifold of orientations of M (10.2.2). Let the structural group

$\{\pm 1\}$ act on $\mathbf{R}$ by multiplication. We denote by $\tilde{\mathbf{R}}_M$ the bundle associated with λ_M of fiber type $\mathbf{R}$ (endowed with the operation law defined above) (6.5.1); it is equipped (7.10.2) with a vector bundle structure over the base X , of rank 1 and class C^k ; it is called the bundle of M -twisted scalars.

10.3.2. We have a commutative diagram:

$$\begin{array}{ccc}
 \text{Or}_M \times \mathbf{R} & \xrightarrow{\rho} & \tilde{\mathbf{R}}_M \\
 \text{pr}_1 \downarrow & \nearrow \varphi & \downarrow p \\
 \text{Or}_M & \xrightarrow{\pi} & X
 \end{array}$$

where p denotes the canonical projection of $\tilde{\mathbf{R}}_M$ onto X , ρ the frame map (6.5.1) of $\tilde{\mathbf{R}}_M$, and where $\varphi(\xi) = \rho(\xi, 1)$ for $\xi \in \text{Or}_M$. In particular, the inverse image of the bundle $\tilde{\mathbf{R}}_M$ by π is identified with the trivial bundle $\text{Or}_M \times \mathbf{R}$. One also identifies an element ξ of Or_M with its image in $\tilde{\mathbf{R}}_M$ by φ ; with this convention, one has $\rho(\xi, a) = a\xi$ for all $(\xi, a) \in \text{Or}_M \times \mathbf{R}$; we have $a\xi = b\eta$ if and only if there exists $e \in \{\pm 1\}$ such that $b = ea$ and $\eta = e\xi$. An orientation ξ of $\mathbf{M}$ thus defines a section $x \mapsto \xi(x)$ of $\tilde{\mathbf{R}}_M$.

There exists one and only one isomorphism j from $\tilde{\mathbf{R}}_M \otimes \tilde{\mathbf{R}}_M$ onto the trivial bundle $\mathbf{R}_X = X \times \mathbf{R}$ such that $j(\xi \otimes \xi) = (\pi(\xi), 1)$ for all $\xi \in \text{Or}_M$; this allows one to identify the vector bundle $\tilde{\mathbf{R}}_M$ with its dual.

10.3.3. Let F moreover be a vector bundle over the base X and of class C^{r-1} . A differential form on X with values in $\tilde{\mathbf{R}}_M \otimes F$ is called an M -twisted differential form with values in F . Such a form of degree p is a section of the bundle $\text{Alt}^p(T(X); \tilde{\mathbf{R}}_M \otimes F)$, which one identifies in the obvious way with $\tilde{\mathbf{R}}_M \otimes \text{Alt}^p(T(X); F)$. When F is the trivial bundle defined by a Banach space E , one speaks simply of an M -twisted form with values in E ; when $E = \mathbb{C}$ (resp. $\mathbb{R}$), one says complex M -twisted form (resp. real M -twisted form, or M -twisted form).

Let ω be an M -twisted form of degree p with values in F . The fact that $\pi: \text{Or}_M \rightarrow X$ is étale and that $\pi^*(\tilde{\mathbf{R}}_M) = \text{Or}_M \times \mathbf{R}$ allows one to identify $\pi^*(\tilde{\mathbf{R}}_M \otimes \text{Alt}^p(T(X); F))$ with

$\text{Alt}^p(T(\text{Or}_{\mathbb{M}}); \pi^*(F))$, and ω defines a section $\tilde{\omega}$ of $\text{Alt}^p(T(\text{Or}_{\mathbb{M}}); \pi^*(F))$ (7.4.3), in other words, a differential form of degree p on $\text{Or}_{\mathbb{M}}$ with values in $\pi^*(F)$. This yields a bijection from the space of M -twisted forms of degree p on X with values in F onto the space of forms $\tilde{\omega}$ of degree p on $\text{Or}_{\mathbb{M}}$ with values in $\pi^*(F)$ such that

$$\tilde{\omega}(-\xi) = -\tilde{\omega}(\xi) \quad \text{for every } \xi \in \text{Or}_{\mathbb{M}}.$$

When M is equipped with an orientation ξ , the map $\omega \mapsto \xi \otimes \omega$ makes it possible to identify the sections of $\text{Alt}^p(T(X); F)$ (in other words, the usual differential forms of degree p) with M -twisted forms of degree p with values in F .

10.3.4. Let ω be an M -twisted differential form of degree p on X , with values in a Banach space E , and of class C^s ($1 \leq s \leq \inf(k, r-1)$). There exists one and only one M -twisted differential form $d\omega$, of degree $p+1$ on X , with values in E , such that, for every open set U of X , every orientation ξ of $M|_U$, and every differential form α on U such that $\omega|_U = \xi \otimes \alpha$, one has $d\omega|_U = \xi \otimes d\alpha$. The form $d\omega$ is of class C^{s-1} ; it is called the exterior differential of ω .

If one denotes by $\tilde{\omega}$ (resp. $\widetilde{d\omega}$) the differential form on $\text{Or}_{\mathbb{M}}$ corresponding to ω (respectively to $d\omega$) as in 10.3.3, one has $d(\tilde{\omega}) = \widetilde{d\omega}$.

If ζ is a vector field of class C^s onto X , one defines in an analogous manner the M -twisted form $\theta_{\zeta}.\omega$. One has

$$\theta_{\zeta}.\omega = d(i(\zeta)\omega) + i(\zeta)d\omega$$

and $\theta_{\zeta}.\omega$ is of class C^{s-1} .

10.4 Measure associated with a twisted differential form

Recall that, in this section, we assume $K = \mathbb{R}$ and that all manifolds considered are locally finite-dimensional.

10.4.1. Let X be a manifold of class C^r . One can apply the definitions and results of 10.3 by taking as vector bundle M the tangent bundle $T(X)$. One then has $\text{Or}_{\mathbb{M}} = \tilde{X}$ (10.2.4); we denote

it by $\tilde{\mathbf{R}}_X$ and call simply the bundle of twisted scalars the bundle $\tilde{\mathbf{R}}_{T(X)}$ (10.3.1)¹; a $T(X)$ -twisted differential form with values in a vector bundle F is called simply a twisted (or odd) differential form with values in F . When X is equipped with an orientation, the map $\omega \mapsto \xi \otimes \omega$ allows one to identify the usual differential forms (sometimes called "even") with the twisted forms.

10.4.2. Let $f: X \rightarrow Y$ be a morphism of manifolds of class C^r and let $\tilde{f}: \tilde{X} \rightarrow \tilde{Y}$ be an orientation of f (10.2.5). There exists one and only one isomorphism j from $f^*(\tilde{\mathbf{R}}_Y)$ onto $\tilde{\mathbf{R}}_X$ such that $\tilde{x} = j(\pi(\tilde{x}), \tilde{f}(\tilde{x}))$ for all $\tilde{x} \in \tilde{X}$. We identify these two bundles by means of j . If ω is a twisted form of degree p on Y with values in a vector bundle F , the inverse image $f^*(\omega)$ is identified with a twisted differential form of degree p on X with values in $f^*(F)$. If one denotes by $\tilde{\omega}$ (resp. $\tilde{f}^*(\omega)$) the differential form on $\tilde{Y}$ (resp. $\tilde{X}$) corresponding to ω (resp. $f^*(\omega)$) as in 10.3.3, one has $\tilde{f}^*(\tilde{\omega}) = \tilde{f}^*(\omega)$.

When F is the trivial bundle defined by a Banach space, the operation f^* commutes with exterior differentiation (10.3.4).

10.4.3. Let X be a pure manifold of dimension n , and let ω be a twisted differential form of degree n on X , with values in a Banach space E . Let $c = (U, \varphi, \mathbb{R}^n)$ be a chart of X and let ξ^n be the canonical orientation of $\mathbb{R}^n$; denote by $u^1, \dots, u^n$ the coordinate functions on $\mathbb{R}^n$. There exists one and only one function f_c on $\varphi(U)$, with values in E , such that

$$\omega|_U = \varphi^*(\xi^n \otimes f_c \cdot du^1 \wedge \dots \wedge du^n).$$

One says that ω is locally integrable if, for every chart $c = (U, \varphi, \mathbb{R}^n)$ of X , the corresponding function f_c is locally integrable with respect to $\lambda_{\varphi(U)}^{\otimes n}$ (where λ denotes Lebesgue measure on $\mathbb{R}$). Suppose that this is the case and that X is separated. There exists on X a vector measure $\alpha(\omega)$ with values in E and one and only one possessing the following property: for every chart $c = (U, \varphi, \mathbb{R}^n)$ of X , the image under φ of the restriction to U of the measure $\alpha(\omega)$ is the measure $f_c \lambda_{\varphi(U)}^{\otimes n}$ (INT, VI, § 2, no. 4). One says that $\alpha(\omega)$ is the measure defined by ω and one denotes it most often simply by ω . The support of $\alpha(\omega)$ is contained in the support $\text{Supp } \omega$ of ω (calling the support of a differential form ω the closure of the set of $x \in X$ such that $\omega(x) \neq 0$).

¹Note that this notation conflicts with the notation $\tilde{\mathbf{R}}_M$ when X is regarded as endowed with its vector-bundle structure of rank 0 over itself.

If f is a real function on X , locally integrable with respect to the measure $\alpha(\omega)$, then the differential form $f\omega$ is locally integrable and one has $\alpha(f\omega) = f.\alpha(\omega)$.

If g is a real function on X essentially integrable for $\alpha(\omega)$, its integral is generally denoted $\int_X g\omega$, or $\int g\omega$, or $\int g(x)\omega(x)$; one avoids denoting it $\int g d\omega$, because of the risk of confusion with the exterior differential $d\omega$ of ω , which is defined (and moreover equal to 0) if ω is of class C^1 . If A is a subset of X whose characteristic function φ_A is essentially integrable for $\alpha(\omega)$, the integral of φ_A is denoted $\int_A \omega$. If the constant 1 is essentially integrable for $\alpha(\omega)$, one says that ω is integrable.

Now let p be an integer, with $0 \leq p \leq n$, and let ω be a twisted differential form of degree p on X , with values in a Banach space E . Let moreover Y be a pure submanifold of dimension p of X ; suppose that the canonical injection $i : Y \rightarrow X$ is equipped with an orientation. The inverse image $i^*(\omega)$ (10.4.2) is then sometimes denoted $\omega|Y$. If $\omega|Y$ is integrable, we set

$$\int_Y \omega = \int_Y \omega|Y.$$

10.4.4. Let X be a pure manifold of dimension n , equipped with an orientation ξ . The identification $\omega \mapsto \xi \otimes \omega$ between usual differential forms and twisted differential forms makes it possible to apply the preceding results to forms of degree n on X ; thus, to every locally integrable form ω of degree n , there is associated a measure on X , still denoted ω . If $E = \mathbf{R}$, such a form ω is locally of integrable modulus (10.1.5) and the measure $\text{mod}(\omega)_\mu$ corresponding to it (10.1.6) is none other than the absolute value $|\omega|$ of the measure ω (INT, III, § 1, no. 6).

10.4.5. Example. — Let X be a pure real manifold of dimension 1, oriented, separated, and let $z : X \rightarrow \mathbf{C}$ be a differentiable map from X to $\mathbf{C}$. The complex differential form dz defines on X a complex measure also denoted dz . This applies in particular when X is a pure differentiable submanifold of dimension 1 of $\mathbf{C}$, z being the injection of X into $\mathbf{C}$.

More specifically, let X be a circle with center 0 and radius $\rho > 0$; orient X as indicated in 10.2.8, b), given the usual identification of $\mathbf{C}$ with $\mathbf{R}^2$. If $x \in X$, the tangent space $T_x(X)$ is identified with the line $\mathbf{R}ix$ of $T_x(\mathbf{C}) = \mathbf{C}$, and the chosen orientation is the half-line $\mathbf{R}_+ix$. If f is a continuous function on X , with values in a complex Banach space, the integral of f for the measure dz is given by the formula

$$\int_{\mathbb{X}} f(z) dz = i\rho \int_0^{2\pi} f(\rho e^{i\alpha}) e^{i\alpha} d\alpha.$$

For example, if n is an integer, we have

$$\int_{\mathbb{X}} z^n dz = i\rho^{n+1} \int_0^{2\pi} e^{(n+1)i\alpha} d\alpha = \begin{cases} 0 & \text{if } n \neq -1 \\ 2i\pi & \text{if } n = -1. \end{cases}$$

10.4.6 (“Complex case”). Let X^c be a separated complex analytic manifold, pure of dimension m , and let ω be a holomorphic differential form of degree m on X^c . Let X be the real analytic manifold underlying X^c ; the form ω identifies with a form of type $(m, 0)$ on X (8.8.9); let $\bar{\omega}$ be its conjugate (8.8.2). The form $i^{m^2} \omega \wedge \bar{\omega}$ is a real differential form of degree $2m$ on X ; taking into account the canonical orientation of X (10.2.7), this form identifies with a measure on X , which is none other than the positive measure $\text{mod}(\omega)_\mu$ defined in 10.1.6, where μ is the measure defined by the differential form $i dz \wedge \bar{dz}$, in other words twice the Lebesgue measure $\lambda^{\otimes 2}$ on $\mathbf{C}$ (identified with $\mathbf{R}^2$ in the usual way).

Let H be the space of holomorphic forms ω of degree m on $\mathbf{X}^c$ such that the measure $\omega \wedge \bar{\omega}$ is bounded. For α, β in H , the complex measure $\alpha \wedge \bar{\beta}$ is bounded; setting

$$(\alpha|\beta) = i^{m^2} \int_{\mathbf{X}} \alpha \wedge \bar{\beta},$$

one obtains a Hermitian form on H , which makes H into a Hilbert space.

11 Stokes’ formula

In this paragraph, we assume $K = \mathbf{R}$.

11.1 Pieces

11.1.1. Let E be a Banach space. A subset S of E is called a closed half-space if there exists a continuous linear functional $h \neq 0$ and a real number k such that $S = \{x|h(x) \leq k\}$ (cf. EVT, II, §2, no. 6); the boundary of S is then the closed hyperplane $\{x|h(x) = k\}$; it is also called the

boundary of S , and is denoted by ∂S .

11.1.2. Let X be a manifold of class C^r and A a subset of X . One says that A is a piece of X if, for every $a \in A$, there exists a chart $c = (U, \varphi, E)$ of X at a such that $\varphi(A \cap U)$ is an open subset of a closed half-space of E . We set $\partial A = A - \mathring{A}$ (the set of non-interior points of A). It is a submanifold of A , called the boundary of A .² If $a \in \partial A$, there exists a chart $c = (U, \varphi, E)$ of X centered at a and a closed hyperplane H of E such that $\varphi(A \cap U)$ is an open neighborhood of 0 in one of the closed half-spaces S_c of E defined by H (EVT, II, § 2, no. 6), and $\varphi(\partial A \cap U) = H \cap \varphi(A \cap U)$. Such a chart is said to be adapted to A at a . The closed half-space S_c is then the unique closed half-space of E for which $\varphi(A \cap U)$ is an open set.

For a closed subset A of X to be a piece of X , it is necessary and sufficient that $\mathring{A}$ be dense in A and that $A - \mathring{A}$ be a submanifold of X of codimension 1 at each of its points.

11.1.3. Examples

- a) In $\mathbf{R}^n$ a closed ball of radius > 0 is a piece; its boundary is the corresponding sphere.
- b) If A is a piece of the manifold X and if B is a closed subset of ∂A then $A - B$ is a piece of X .
- c) Let $\varphi : X \rightarrow X'$ be a morphism of manifolds of class C^r , and let A' a piece of X' . If φ is transverse to $\partial A'$ (5.11.6), $\varphi^{-1}(A')$ is a piece of X whose boundary is $\varphi^{-1}(\partial A')$.

In particular, let $h : \mathbf{X} \rightarrow \mathbf{R}$ be a function of class C^r , and let $a \in \mathbf{R}$; suppose that there exists no $x \in \mathbf{X}$ such that $h(x) = a$ and $d_x h = 0$. The set $\{x | h(x) \leq a\}$ is a closed piece of $\mathbf{X}$ with boundary $h^{-1}(a)$.

- d) If $r = \infty$ and if X is locally compact, every compact subset of X admits a fundamental system of neighborhoods consisting of compact pieces.

11.1.4. Let X be a manifold of class C^r and let A be a piece of X . Let $a \in \partial A$ and $v \in T_a(X)$. Let $c = (U, \varphi, E)$ be a chart of X adapted to A at a (11.1.2), and let $h = \theta_c^{-1}(v)$ be the element

²This terminology comes from the fact that A is naturally endowed with a structure of a “manifold with boundary”, with boundary ∂A . For the definition of this category, as well as for the more general category of “manifolds with corners”, the reader may consult H. Cartan, *Séminaire 1961/62, Topologie Différentielle*, lectures 1–2–3 (by A. Douady), Benjamin, New York, 1969. Let us mention that one can show that every “manifold with boundary” whose boundary is paracompact is isomorphic to a closed piece of a manifold.

of E corresponding to ν (5.5.1). Let S_c be the closed half-space of E of which $\varphi(A \cap U)$ is an open subset (11.1.2). One says that ν is an inward (resp. strictly inward, resp. outward, resp. strictly outward) vector for A at a if h belongs to S_c (resp. $\dot{S}_c$, $-S_c$, $-\dot{S}_c$), a condition which is independent of the choice of the adapted chart c . We denote by $T_a^+(A)$ (resp. $T_a^-(A)$) the set of outward (resp. inward) vectors of $T_a(X)$ for A . These are closed half-spaces of $T_a(X)$, whose boundary contains 0. One has

$$T_a(\partial A) = T_a^+(A) \cap T_a^-(A).$$

11.2 Stokes' formula for pieces

In this section, X denotes a manifold of class C^r ($r \geq 2$) pure of finite dimension n and separated. A denotes a piece of X and i the canonical injection of ∂A into X . E denotes a Banach space.

11.2.1. Let $x \in \partial A$ and let ξ be an orientation of $T_x(\partial A)$; one denotes $\tilde{i}_x(\xi)$ the orientation of $T_x(X)$ containing the elements $\nu \wedge u$, where ν is a strictly outward vector for A at x (11.1.4) and where u is a nonzero element of $\bigwedge^{n-1} T_x(\partial A)$ belonging to the orientation ξ . The map $\tilde{i}_x$ is a bijection of $\text{Or}(T_x(\partial A))$ onto $\text{Or}(T_x(X))$. The maps $\tilde{i}_x$, for $x \in \partial A$, define a morphism $\tilde{i}: \partial A \rightarrow \tilde{X}$ which is an orientation of i (10.2.5). If ξ is an orientation of X , the orientation of ∂A associated with ξ by $\tilde{i}$ (10.2.5) is said to be defined by ξ .

Example. — When $X = R^n$, ξ is the usual orientation of R^n , and A is a closed ball of radius $\rho > 0$, the orientation of the sphere ∂A defined by ξ is the canonical orientation (10.2.8, b)).

11.2.2. Let ω be a twisted differential form of degree p on X with values in a Banach space E (10.4.1); the inverse image $i^*(\omega)$ of ω by the oriented morphism $i: \partial A \rightarrow X$ is denoted $\omega|_{\partial A}$ and is called the form induced by ω on ∂A (cf. 10.4.3).

11.2.3. Let us make one of the following two assumptions:

- (i) ω is a twisted differential form of degree $n - 1$ on X , with values in E ;
- (ii) X is oriented, ∂A is equipped with the corresponding orientation (11.2.1), and ω is a differential form of degree $n - 1$ on X with values in E .

Suppose moreover that ω is of class C^1 and that the intersection of A and the support of ω

is compact. The exterior differential $d\omega$ of ω is continuous (8.3.5 and 10.3.4).

The characteristic function of A is essentially integrable with respect to the vector measure defined by $d\omega$ on X , and the differential form $\omega|_{\partial A}$ of degree $n - 1$ on ∂A is integrable (10.4.3 and 10.4.4). We have

$$\int_A d\omega = \int_{\partial A} \omega \quad (\text{Stokes' formula})$$

11.2.4. Let α be a twisted differential form of degree n on X with values in a Banach space E , with compact support and of class C^1 . For there to exist a twisted differential form ω of degree $n - 1$ on X , with values in E , with compact support and of class C^1 , such that $\alpha = d\omega$, it is necessary that $\int_X \alpha = 0$. If X is connected, this condition is sufficient; if moreover α is of class C^k ($k \leq r - 1$, $k \neq \omega$), one can choose ω of class C^k .

11.2.5. Suppose that X is the real manifold underlying $\mathbb{C}$, that E is a complex Banach space, and that A is a compact piece of X . Endow X with the orientation defined by its complex structure (10.2.7). Let f be a continuous map from A into E whose restriction to the interior $\mathring{A}$ of A is holomorphic. Denote by dz the differential of the injection $z: \partial A \rightarrow \mathbb{C}$. The form $f \cdot dz$, the product of f and dz , is a differential form of degree 1 on ∂A , with values in E , of class C^0 , and one has:

$$\int_{\partial A} f dz = 0 \quad (\text{Cauchy's formula})$$

When f extends to a holomorphic map, still denoted f , from an open neighborhood U of A into E , the differential form $f \cdot dz$ (where z now denotes the canonical injection of U into $\mathbb{C}$) is of class C^ω on U and its exterior differential is zero.

11.2.6 (“Derivative of an integral”). Suppose X is oriented. Let Y be a manifold of class C^r , and let α be a differential form of degree n on Y , with values in E , of class C^1 , and with compact support. Let I be an open subset of $\mathbb{R}$ containing 0, and let $g: I \times X \rightarrow Y$ be a morphism of class C^r ; for $t \in I$, denote by g_t the map $x \mapsto g(t, x)$ from X into Y . Denote by ψ the morphism from X into $T(Y)$ defined by $\psi(x) = T_{(0,x)}(g)(1, 0)$; it is a lifting of g_0 (8.6.5). Suppose that the restriction of $\text{pr}_1: I \times A \rightarrow I$ to the intersection of $I \times A$ and the support of $g^*(\alpha)$ is proper

(TG, I, § 10). For every $t \in I$, the intersection of A and the support of $g_t^*(\alpha)$ is then compact; the map $t \mapsto \int_A g_t^*(\alpha)$ is of class C^1 on I and its derivative at the origin is given by the formula

$$\frac{d}{dt} \left(\int_A g_t^*(\alpha) \right)_{t=0} = \int_A \theta_\psi \cdot \alpha = \int_A i(\psi) d\alpha + \int_{\partial A} i(\psi) \alpha,$$

cf. no. 8.6.

11.3 Stokes' formula for locally polyhedral sets

In this number, X denotes a pure real manifold of finite dimension n of class C^r ($r \geq 2$); we assume X is separated.³

11.3.1. Let A be a subset of a finite-dimensional real vector space. We say that A is polyhedral if it is a finite union of finite intersections of closed half-spaces.

A subset A of X is said to be locally polyhedral if, for every $x \in X$, there exists a chart $c = (U, \varphi, E)$ of X at x and a polyhedral subset A_c of E such that $\varphi(A \cap U) = \varphi(U) \cap A_c$. A piece of X is a locally polyhedral subset.

11.3.2. Let A be a closed subset of X , and let $\text{Fr}(A) = A - \mathring{A}$ be its frontier. A point $x \in \text{Fr}(A)$ is said to be regular if there exists an open neighborhood U of x such that $A \cap U$ is a piece of the manifold U (in which case x belongs to the boundary of $A \cap U$). We denote by ∂A the set of regular points of $\text{Fr}(A)$ and call it the regular boundary (or simply the boundary) of A . The set $A' = \mathring{A} \cup \partial A$ is a piece of X , with boundary ∂A .

11.3.3. * Example. — Let $\mathcal{H}$ be a locally finite set of hyperplanes of a finite-dimensional real affine space E of dimension n , and let C be a chamber of E relative to $\mathcal{H}$ (LIE, V, § 1, no. 3). The closure $\overline{C}$ of C is a locally polyhedral subset of the manifold E ; the regular boundary of $\overline{C}$ is the union of the faces of C (loc. cit., no. 4). In particular, if C is an open simplex (loc. cit., no. 6) with vertices $a_0, \dots, a_n$, the boundary of $\overline{C}$ is the union of the $\overline{C_{\{i\}}}$ ($0 \leq i \leq n$), where $C_{\{i\}}$ is the open simplex with vertices $a_0, \dots, a_{i-1}, a_{i+1}, \dots, a_n$ in the affine space generated by these vertices.* 11.3.4. Let A be a locally polyhedral subset of X , and let ω be a twisted differential form of degree $n-1$ on X with values in a Banach space E ; suppose that ω is of class C^1 and that

³The reader interested in more general cases may consult H. Whitney, *Geometric Integration Theory*, Chap. III, § 18 (Princeton Univ. Press, 1957).

the intersection of its support with A is compact. Then, the characteristic function of A (resp. A' , $\mathring{A}$) is essentially integrable for the vector measure defined by $d\omega$ on X , and one has

$$\int_A d\omega = \int_{A'} d\omega = \int_{\mathring{A}} d\omega.$$

The differential form $\omega|_{\partial A}$ (for the canonical orientation of the canonical injection of ∂A , considered as the boundary of the piece A' , into X) is integrable on ∂A , and one has

$$\int_A d\omega = \int_{\partial A} \omega \quad (\text{Stokes' formula}).$$

When X is equipped with an orientation, and ∂A is equipped with the corresponding orientation (11.2.1), this formula remains valid if ω is an ordinary differential form of degree $n - 1$ with values in E , of class C^1 , and such that $A \cap \text{Supp } \omega$ is compact.

11.4 Relative Stokes' formula (integration over the fibers)

In this number, X and S denote two (real) manifolds of class C^r , and $\pi : X \rightarrow S$ a submersion. We denote by n an integer and suppose that, for every $s \in S$, the fiber $X_s = \pi^{-1}(s)$ of π at s (which is a submanifold of X (5.10.5)) is pure of dimension n .

11.4.1. Let E and H be two vector spaces and let F be a vector subspace of finite dimension n of H .

Let p be an integer ≥ 0 , u an alternating $(n + p)$ -linear map from H^{n+p} into E , and $t_1, \dots, t_p$ elements of H/F ; let $t'_1, \dots, t'_p$ be representatives of $t_1, \dots, t_p$ in H . The map

$$(x_1, \dots, x_n) \mapsto u(t'_1, \dots, t'_p, x_1, \dots, x_n)$$

is an n -linear alternating map from F^n into E , which depends only on u and on the t_i ; it is denoted $u \perp (t_1, \dots, t_p)$ (cf. A, III, p. 158 for the particular case $E = \mathbf{K}$). The map

$$(t_1, \dots, t_p) \mapsto u \perp (t_1, \dots, t_p)$$

is p -linear alternating.

11.4.2 (“ π -twisted”). Consider the vector bundle $T(X/S)$ on X (8.1.3); it has finite rank n at each point of X . We set $\tilde{R}_\pi = \tilde{R}_{T(X/S)}$ (10.3.1) and $\tilde{X}_\pi = \text{Or}_{T(X/S)}$ (10.2.2). Let $s \in S$; given the natural identification $T(X/S)|_{X_s}$ with $T(X_s)$ (8.1.3), the fiber at s of the map $\tilde{\pi} : \tilde{X}_\pi \rightarrow S$, composite of π and the canonical map $\tilde{X}_\pi \rightarrow X$, is $\tilde{X}_s$ (10.2.4). A π -twisted differential form on X is called a $T(X/S)$ -twisted differential form (10.3.3).

11.4.3 (“S-orientation of an S-morphism”). Let X' be a variety equipped with a submersion $\pi' : X' \rightarrow S$ whose fibers X'_s are pure of finite dimension n' , and let $\varphi : X' \rightarrow X$ be a morphism such that $\pi' = \pi \circ \varphi$. An S-orientation of φ is a morphism $\tilde{\varphi} : \tilde{X}'_{\pi'} \rightarrow \tilde{X}_\pi$, commuting with the action of the group $\{\pm 1\}$ and such that the diagram

$$\begin{array}{ccc} \tilde{X}'_{\pi'} & \xrightarrow{\tilde{\varphi}} & \tilde{X}_\pi \\ \downarrow & & \downarrow \\ X' & \xrightarrow{\varphi} & X \end{array}$$

is commutative. The datum of an S-orientation of φ allows, as in 10.4.2, one to identify the bundles $\tilde{R}_{\pi'}$ and $\varphi^*(\tilde{R}_\pi)$ and to define the pullback of a π -twisted differential form by the S-oriented morphism φ ; it is a π' -twisted differential form on X' .

11.4.4. Let A be a piece of X such that the restriction of π to ∂A is a submersion. For every $s \in S$, the fiber $X_s = \pi^{-1}(s)$ is a submanifold of X transverse to ∂A , and $A \cap X_s$ is a piece of X_s , denoted A_s , with boundary $\partial A \cap X_s$.

Let i be the canonical injection of ∂A into X . There exists one and only one S-orientation $\tilde{i}$ of i such that, for every $s \in S$, the restriction of $\tilde{i}$ to $(\partial A_s)^\sim$ (identified with the fiber at s of the submersion $\partial \tilde{A}_{\pi|_{\partial A}} \rightarrow S$) is the orientation defined in 11.2.1 of the canonical injection of ∂A_s into X_s . If ω is a π -twisted differential form on X , the inverse image of ω under the morphism i thus oriented is denoted $\omega|_{\partial A}$ (cf. 11.2.2).

11.4.5 (“Interior product”). Let p be an integer ≥ 0 and let ω be a π -twisted differential form of degree $n + p$ on X , with values in a vector bundle E with base X . Let $s \in S$ and $x \in X_s$; we

have $\omega(x) = \xi \otimes u$, where ξ is an orientation of $T_x(X_s) = T(X/S)_x$, identified with an element of $(\tilde{\mathbf{R}}_{X_s})_x = (\tilde{\mathbf{R}}_n)_x$ (10.3.2), and where u is an alternating $(n + p)$ -linear map from $T_x(X)$ into E_x . Let $t_1, \dots, t_p$ be tangent vectors to S at s . Put

$$\theta(x) = \xi \otimes (u_\perp(t_1, \dots, t_p)) \in (\tilde{\mathbf{R}}_{X_s})_x \otimes (\text{Alt}^n(T(X); E))_x$$

(cf. 11.4.1). One thus defines a twisted differential form $\theta : x \mapsto \theta(x)$ of degree n on X_s , with values in $E|_{X_s}$. We denote it by $\omega \perp (t_1, \dots, t_p)$.

In particular, let M be a vector bundle of class C^{r-1} with base S and take $E = \pi^*(M)$. The bundle $E|_{X_s}$ is naturally identified with the trivial bundle with base X_s defined by the Banach space M_s , and the form $\omega \perp (t_1, \dots, t_p)$ is identified with a twisted differential form on X_s , with values in M_s .

11.4.6. We keep the preceding notation (in particular $E = \pi^*(M)$) and assume moreover that X is separated and ω is continuous. Let A be a piece of X such that the restriction of π to ∂A is a submersion and the restriction of π to the intersection of A and the support of ω is proper. For $s \in S$ and $t_1, \dots, t_p \in T_s(S)$, the form $\omega \perp (t_1, \dots, t_p)$ (11.4.5) is a twisted differential form of degree n on X_s , continuous, with values in the Banach space M_s , and its support meets A_s in a compact set. There exists one and only one differential form α of degree p on S , with values in the vector bundle M , and such that

$$\alpha(s)(t_1, \dots, t_p) = \int_{A_s} \omega \perp (t_1, \dots, t_p)$$

for all $s \in S$ and $t_1, \dots, t_p \in T_s(S)$. The form α is denoted $\int_{\pi|A} \omega$ (or simply $\int_{\pi} \omega$ when $A = X$); one says that it is obtained by integrating ω over A along the fibers of π . If ω is of class C^k ($0 \leq k \leq r - 1$, $k \leq \omega$), the same is true of $\int_{\pi|A} \omega$.

When ω is a form of degree $< n$, we agree that $\int_{\pi|A} \omega = 0$.

11.4.7. We retain the preceding hypotheses and notation.

a) For every continuous scalar differential form β on S , we have

$$\int_{\pi|A} (\pi^* \beta) \wedge \omega = \beta \wedge \int_{\pi|A} \omega.$$

- b) Let η be a continuous vector field on X and ζ a continuous vector field on S , such that η is π -related to ζ (8.2.6). We have

$$i(\zeta) \int_{\pi|A} \omega = \int_{\pi|A} i(\eta) \omega.$$

- c) Suppose moreover that η , ζ , and ω are of class C^1 , that $\eta(x) \in T_x(\partial A)$ for all $x \in \partial A$, and that $\mathbf{M}$ is the trivial bundle defined by a Banach space E . We have

$$\theta_\zeta \cdot \int_{\pi|A} \omega = \int_{\pi|A} \theta_\eta \cdot \omega,$$

(cf. 8.4.2 and 10.3.4).

- d) If ω is of degree $n + p$ and of class C^1 , one has

$$d \left(\int_{\pi|A} \omega \right) = \int_{\pi|A} d\omega + (-1)^p \int_{\pi|\partial A} \omega|_{\partial A}.$$

- e) Suppose that S is reduced to a point. The forms π -twisted on X are then the ordinary twisted forms (10.4.1). If ω is of degree n , the form $\int_{\pi|A} \omega$ of degree 0 on S is the constant $\int_A \omega$ (10.4.3).

11.4.8. Let $\pi' : S \rightarrow S'$ be a submersion such that the fibers of π' are pure submanifolds of constant finite dimension n' . Set $\pi'' = \pi' \circ \pi$; it is a submersion of X onto S' whose fibers are pure submanifolds of dimension $m = n + n'$.

Let $x \in X$. The sequence

$$0 \rightarrow T(X/S)_x \xrightarrow{\text{Id}} T(X/S')_x \xrightarrow{T_x(\pi)} T(S/S')_{\pi(x)} \rightarrow 0$$

is exact. There exists one and only one isomorphism j from the vector bundle $\tilde{\mathbf{R}}_\pi \otimes \pi^*(\tilde{\mathbf{R}}_{\pi'})$ onto $\tilde{\mathbf{R}}_{\pi''}$ such that, if ξ (resp. η) is an orientation of $T(X/S)_x$ (resp. of $(\pi^*T(S/S'))_x$, identified with $T(S/S')_{\pi(x)}$), one has $j(\xi \otimes \eta) = \eta\xi$ (the product being defined by the preceding exact sequence (10.2.1)).

Let $\mathbf{M}'$ be a vector bundle with base S' ; set $\mathbf{M} = \pi'^*(\mathbf{M}')$, and $\mathbf{E} = \pi^*(\mathbf{M}) = \pi''^*(\mathbf{M}')$.

If ω is a differential form that is π'' -twisted on X with values in E , the isomorphism j allows it to be identified with a form that is π -twisted with values in $\pi^*(\tilde{\mathbf{R}}_{\pi'}) \otimes E$, or equivalently with values in $\pi^*(\tilde{\mathbf{R}}_{\pi'} \otimes \mathbf{M})$.

Let A also be a piece of X such that $\pi|_{\partial A} : \partial A \rightarrow S$ is a submersion; the same is then true of $\pi''|_{\partial A} : \partial A \rightarrow S'$. Suppose further that ω is continuous and that the restriction of π'' to $A \cap \text{Supp } \omega$ is proper; then the same holds for the restriction of π to $A \cap \text{Supp } \omega$. On the one hand, one can consider the differential form $\int_{\pi''|_A} \omega$ on S' , which is a form with values in M' ; on the other hand, the differential form $\int_{\pi|_A} \omega$ on S , which is a form with values in $\tilde{\mathbf{R}}_{\pi'} \otimes M$, or equivalently a π' -twisted form with values in $\mathbf{M} = \pi'^*(M')$. Moreover, $\pi(A)$ is an open submanifold of S and the restriction of π' to $\pi(A) \cap \text{Supp } \int_{\pi|_A} \omega$ is proper. The differential form $\int_{\pi'|_{\pi(A)}} \int_{\pi|_A} \omega$ is defined; it is a differential form on S' , with values in M' . One has

$$\int_{\pi' \circ \pi|_A} \omega = \int_{\pi'|_{\pi(A)}} \int_{\pi|_A} \omega.$$

If $A = X$, we have

$$\int_{\pi' \circ \pi} \omega = \int_{\pi'} \int_{\pi} \omega.$$

12 Jets

In this paragraph, we denote by X and Y two varieties of class $\mathbf{C}^r$, where $r \in \mathbb{N}_{\mathbb{K}}$, and let k be an integer such that $0 \leq k \leq r$.

From no. 12.3 onward, we assume:

either that K has characteristic zero;

or that the manifolds, Banach spaces, and vector bundles under consideration are locally finite-dimensional.

12.1 Jets of maps

12.1.1. Let $x \in X$ and let f, g be two continuous maps defined in a neighborhood of x and with values in Y . We say that f and g have contact of order $\geq k$ at x if $f(x) = g(x)$ and if there

exist charts (U, φ, E) of X at x and (V, ψ, F) of Y at $f(x)$ such that the maps $\psi \circ f \circ \varphi^{-1}$ and $\psi \circ g \circ \varphi^{-1}$, defined in a neighborhood of $\varphi(x)$ in E and with values in F , have contact of order $\geq k$ at $\varphi(x)$ (1.1.2). This property is then verified by all charts of X at x and of Y at $f(x)$.

12.1.2. Let $x \in X$, $y \in Y$. In the set of maps of class C^r defined in a neighborhood of x , with values in Y , and mapping x to y , the relation « f and g have contact of order $\geq k$ » is an equivalence relation; the class of f for this relation is denoted $j_x^k(f)$ and is called the jet of order k of f , with source x and target y . The source of a jet j is denoted $s(j)$ and its target $b(j)$.

The set of jets of order k from X to Y (resp. with source x , resp. with target y) is denoted $J^k(X, Y)$ (resp. $J_x^k(X, Y)$, resp. $J^k(X, Y)_y$) and one sets

$$J_x^k(X, Y)_y = J_x^k(X, Y) \cap J^k(X, Y)_y.$$

12.1.3. If U and V are open subsets of X and Y , respectively, we identify in the obvious way $J^k(U, V)$ with the inverse image of $U \times V$ under the map

$$(s, b) : J^k(X, Y) \rightarrow X \times Y.$$

12.1.4. Let Z be a manifold of class C^r , and let $(x, y, z) \in X \times Y \times Z$. Let $j \in J_x^k(X, Y)_y$ and let $j' \in J_y^k(Y, Z)_z$. The maps $f' \circ f$, with $f \in j$ and $f' \in j'$, have the same k th-order jet at x ; this jet is called the composite of j and j' and is denoted $j' \circ j$; we have $s(j' \circ j) = s(j)$ and $b(j' \circ j) = b(j')$. If T is a manifold of class C^r and if $j'' \in J_z^k(Z, T)$, one has

$$j'' \circ (j' \circ j) = (j'' \circ j') \circ j.$$

12.1.5. Let $j \in J_x^k(X, Y)$, with $x \in X$, and let k' be an integer such that $0 \leq k' \leq k$. The jets $j_x^{k'}(f)$, for $f \in j$, are equal; the jet thus defined is denoted $r^{k, k'}(j)$; the map $r^{k, k'} : J^k(X, Y) \rightarrow J^{k'}(X, Y)$ is surjective.

12.1.6. Suppose that r is equal to ∞ or to ω . Let f and g be two maps of class C^r defined in a neighborhood of a point $x \in X$ and with values in Y . If $j_x^m(f) = j_x^m(g)$ for every integer $m \geq 0$, one says that f and g have contact of infinite order at x . One thus obtains an equivalence relation whose classes are called jets of infinite order at x ; the jet of infinite order of f is denoted $j_x^\infty(f)$

or $j_x^\omega(f)$. The definitions and results above extend without modification to jets of infinite order.

12.2 Jets of maps of Banach spaces

In this section, E and F denote two Banach spaces. If m is an integer ≥ 0 , one denotes by $P_m(E; F)$ the Banach space of continuous homogeneous polynomials of degree m on E with values in F (Part 1, Appendix, A.2).

12.2.1. Let U be an open subset of E , let $f: U \rightarrow F$ be a map of class C^r , and let $a \in U$. There exists one and only one continuous polynomial

$$\tilde{f} = f_0 + \cdots + f_k \quad (\text{with } f_m \in P_m(E; F))$$

of degree $\leq k$, having the same jet of order k at the origin as $x \mapsto f(x - a)$. If $0 \leq m \leq k$, the m -th component f_m of $\tilde{f}$ is denoted $\Delta^m f(a)$ or $\Delta_a^m(f)$. When $r = \omega$, this notation coincides with that of 3.2.1 and 4.2.1; when $r \leq \infty$, one has

$$m! \Delta^m f(a)(h) = D^m f(a)(h, \dots, h) = D^m f(a).h^m.$$

The map $f \mapsto \tilde{f}$ defines, by passing to the quotient, a bijection from $J_a^k(E, F)$ onto $\prod_{0 \leq m \leq k} P_m(E; F)$ by means of which one identifies these two spaces; in particular, $J_a^k(E, F)$ is equipped with a Banach space structure over K . If $j \in J_a^k(E, F)$, one denotes by $\Delta_a^m(j)$ the m -th component of j ; we have

$$\Delta_a^m(j) \in P_m(E; F) \quad \text{for } 0 \leq m \leq k, \quad \text{and} \quad \Delta_a^0(j) = b(j) \in F.$$

12.2.2. Let U (resp. V) be an open subset of E (resp. F). Denote by $Q^k(E, F)$ the product Banach space of the $P_m(E; F)$ for $1 \leq m \leq k$. The map

$$j \mapsto (s(j), b(j), \Delta_{s(j)}^1(j), \dots, \Delta_{s(j)}^k(j))$$

is a bijection of $J^k(U, V)$ onto $U \times V \times Q^k(E, F)$, by means of which these two sets are identified. In particular, $J_0^k(E, F)_0$ is identified with $Q^k(E, F)$.

12.3 Jet manifolds

12.3.1. Let $c = (U, \varphi, E)$ and $c' = (V, \psi, F)$ be charts of X and Y , respectively. The maps φ and ψ define, by transport of structure, a bijection π of $\mathbf{J}^k(U, V)$ onto

$$\mathbf{J}^k(\varphi(U), \psi(V)) = \varphi(U) \times \psi(V) \times Q^k(E, F),$$

cf. 12.2.2. If one sets $\mathbf{W} = \mathbf{J}^k(U, V)$ and $\mathbf{G} = E \times F \times Q^k(E, F)$, the triple $(\mathbf{W}, \pi, \mathbf{G})$ is a chart of $\mathbf{J}^k(X, Y)$. The charts thus obtained form a $\mathbf{C}^{r-k}$ -atlas and this endows $\mathbf{J}^k(X, Y)$ with a K -manifold structure of class $\mathbf{C}^{r-k}$.⁴

If $(x, y) \in X \times Y$, the sets $J_x^k(X, Y)$, $J^k(X, Y)_y$, and $J_x^k(X, Y)_y$ are closed submanifolds of $J^k(X, Y)$.

12.3.2. If X and Y are pure (resp. finite-dimensional, resp. separated, resp. connected), then so is $J^k(X, Y)$.

If U (resp. V) is open in X (resp. in Y), $J^k(U, V)$ is an open submanifold of $J^k(X, Y)$, cf. 12.1.3.

12.3.3. The maps $s : \mathbf{J}^k(X, Y) \rightarrow X$, $b : \mathbf{J}^k(X, Y) \rightarrow Y$ and $(s, b) : \mathbf{J}^k(X, Y) \rightarrow X \times Y$ are fibrations (6.1.1) of class $\mathbf{C}^{r-k}$. When $k = 0$, (s, b) is an isomorphism.

12.3.4. Denote by T_X and T_Y the vector bundles $\text{pr}_1^*T(X)$ and $\text{pr}_2^*T(Y)$ on $X \times Y$, and let $\mathcal{L}(T_X; T_Y)$ be the bundle of homomorphisms of T_X to T_Y (7.7.3); if $(x, y) \in X \times Y$, one has

$$\mathcal{L}(T_X; T_Y)_{(x,y)} = \mathcal{L}(T_x(X); T_y(Y)).$$

When $k = 1$, the map

$$\mathbf{T} : \mathbf{J}^1(X, Y) \rightarrow \mathcal{L}(T_X; T_Y)$$

thus defined is an isomorphism of manifolds of class $\mathbf{C}^{r-1}$.

For $X = K$, this isomorphism identifies $J_0^1(K, Y)$ with $T(Y)$; for $Y = K$, it identifies $J^1(X, K)_0$ with the dual $T'(X)$ of $T(X)$.

⁴When $k = r$ (which is possible only if $K = \mathbf{R}$), $\mathbf{J}^k(X, Y)$ is endowed with a topological manifold structure (cf. Notations and Conventions), and the morphisms and fibrations considered below are to be understood in the purely topological sense (cf. the footnote to § 6).

12.3.5. Let k' be an integer such that $0 \leq k' \leq k$. The map

$$r^{k,k'} : J^k(X, Y) \rightarrow J^{k'}(X, Y) \quad (\text{cf. 12.1.5})$$

is a fibration of class C^{r-k} .

12.3.6. Let Z be a manifold of class C^r , and let T be the set of pairs

$$(j, j') \in J^k(X, Y) \times J^k(Y, Z)$$

such that $b(j) = s(j')$. Then T is a submanifold of $J^k(X, Y) \times J^k(Y, Z)$, and the map $(j, j') \mapsto j' \circ j$ is a morphism of class C^{r-k} from T to $J^k(X, Z)$.

12.3.7. If $f : \mathbf{X} \rightarrow \mathbf{Y}$ is a morphism of class C^r , the map $x \mapsto j_x^k(f)$ is a morphism $j^k(f)$ of class C^{r-k} from $\mathbf{X}$ to $\mathbf{J}^k(\mathbf{X}, \mathbf{Y})$.

12.3.8. Let k' and k'' be positive integers with sum k . Let $x \in \mathbf{X}$, let U be an open neighborhood of x , and let $f : U \rightarrow \mathbf{Y}$ be a morphism of class C^r . The map

$$x \mapsto j_x^{k'}(f) : U \rightarrow J^{k'}(X, Y)$$

is of class $C^{r-k'}$, and its jet of order k'' at x depends only on $j_x^k(f)$. We thus obtain a canonical map

$$\alpha : J^k(X, Y) \rightarrow J^{k''}(X, J^{k'}(X, Y))$$

which is of class C^{r-k} ; if K is of characteristic zero, it is an embedding.

12.3.9. Let $\mathbf{X}', \mathbf{Y}'$ be varieties of class C^r , let $f : \mathbf{X} \rightarrow \mathbf{X}'$ and $g : \mathbf{Y}' \rightarrow \mathbf{Y}$ be morphisms, and let $(x, y') \in \mathbf{X} \times \mathbf{Y}'$, $x' = f(x)$, $y = g(y')$. If $u \in J_x^k(\mathbf{X}', \mathbf{Y}')_{y'}$, set

$$J_{x'}^k(f, g)_y(u) = j_{y'}^k(g) \circ u \circ j_x^k(f).$$

One thus obtains a map

$$J^k(f, g) : J^k(\mathbf{X}', \mathbf{Y}') \times_{\mathbf{X}'} \mathbf{X} \rightarrow J^k(\mathbf{X}, \mathbf{Y})$$

which is of class C^{r-k} .

12.3.10. Let A be a compact subset of X . Let $\mathcal{C}^k(X; Y)$ be the set of maps of class C^k from X to Y . For every $f \in \mathcal{C}^k(X; Y)$, the map $j^k(f)|_A$ belongs to the set $\mathcal{C}(A; J^k(X, Y))$ of continuous maps from A to $J^k(X, Y)$. We have thus defined a map λ from $\mathcal{C}^k(X; Y)$ to $\mathcal{C}(A; J^k(X, Y))$. The inverse image under λ of the topology of compact convergence on $\mathcal{C}(A; J^k(X, Y))$ (TG, X, § 3, def. 1) is a topology on $\mathcal{C}^k(X; Y)$; it is called the topology of C^k -uniform convergence on A .

12.4 Frames and principal fibrations

12.4.1. Let $j \in \mathbf{J}^k(\mathbf{X}, \mathbf{Y})$ and let $x = s(j)$, $y = b(j)$. We say that j is invertible if there exists $j' \in \mathbf{J}_y^k(\mathbf{Y}, \mathbf{X})_x$ such that $j' \circ j = j_x^k(\text{Id}_{\mathbf{X}})$ and $j \circ j' = j_y^k(\text{Id}_{\mathbf{Y}})$; the jet j' is then uniquely determined and is denoted j^{-1} . If $k = 0$, every jet is invertible. If $k \geq 1$, the following conditions are equivalent:

- a) j is invertible;
- b) the map $T(j) : T_x(X) \rightarrow T_y(Y)$ (cf. 12.3.4) is an isomorphism;
- c) there exists an isomorphism g of class C^r from an open neighborhood of x onto an open neighborhood of y such that $j_x^k(g) = j$.

12.4.2. Let E be a Banach space. We denote by $\mathbf{GL}^k(E)$ the set of jets of order k from E to E that are invertible and have 0 as source and target; this is an open subset of $J_0^k(E, E)_0$. Equipped with the composition law of jets and with the manifold structure induced by that of the Banach space $J_0^k(E, E)_0 = Q^k(E, E)$, it is a group manifold of class C^ω .

One has $\mathbf{GL}^0(E) = \{e\}$. The group $\mathbf{GL}^1(E)$ identifies, by means of T , with the group $\mathbf{GL}(E)$ of automorphisms of E .

If $k' \leq k$, the map $r^{k,k'} : \mathbf{GL}^k(E) \rightarrow \mathbf{GL}^{k'}(E)$ is a homomorphism of class C^ω , and it is a surjective submersion. The map $f \mapsto \text{Id}_E + f$ is an isomorphism of group manifolds from $P_k(E, E)$ onto the kernel of $r^{k,k-1}$.

12.4.3. Let E be a Banach space. For every $x \in X$, one calls an E -frame of order k of X at x every invertible element of $J_0^k(E, X)_x$. The set of E -frames of order k of X is an open

submanifold $R^k(E, X)$ of $J_0^k(E, X)$; the map b has as restriction a morphism, still denoted b , of $R^k(E, X)$ into X ; likewise, if $k' \leq k$, we still denote by $r^{k,k'}$ the morphism from $R^k(E, X)$ to $R^{k'}(E, X)$, restriction of the map $r^{k,k'}$ of 12.1.5.

12.4.4. Suppose that X is pure of type E (5.1.7). The group $\mathbf{GL}^k(E)$ acts on the right on $R^k(E, X)$ by the law $(\rho, u) \mapsto \rho \circ u$, and the quadruple $\lambda_X = (R^k(E, X), \mathbf{GL}^k(E), X, b)$ is a principal fibration (6.2.1) with structural group $\mathbf{GL}^k(E)$, base X , and class C^{r-k} .

If $k' \leq k$, let H be the kernel of $r^{k,k'} : \mathbf{GL}^k(E) \rightarrow \mathbf{GL}^{k'}(E)$; the quadruple $(R^k(E, X), H, R^{k'}(E, X), r^{k,k'})$ is a principal fibration of class C^{r-k} .

The manifold $J^k(X, Y)$ is equipped with a structure of fiber bundle associated with λ_X (6.5.1): the typical fiber is $J_0^k(E, Y)$, on which $\mathbf{GL}^k(E)$ acts on the left by the law $(u, j) \mapsto j \circ u^{-1}$; the frame map $R^k(E, X) \times J_0^k(E, Y) \rightarrow J^k(X, Y)$ takes (ρ, j) to $j \circ \rho^{-1}$. The projection $J^k(X, Y) \rightarrow X$ corresponding to this fiber-bundle structure is s .

12.4.5. Let F be a Banach space, and suppose Y is pure of type F . Then $J^k(X, Y)$ is equipped with a structure of fiber bundle associated with λ_Y : the typical fiber is $J^k(X, F)_0$, on which $\mathbf{GL}^k(F)$ acts on the left by the law $(v, j) \mapsto v \circ j$; the frame map is $(\sigma, j) \mapsto \sigma \circ j$; the projection $J^k(X, Y) \rightarrow Y$ is b .

12.4.6. The hypotheses being those of 12.4.4 and 12.4.5, let μ be the principal fibration

$$(\mathbf{R}^k(E, X) \times \mathbf{R}^k(F, Y), \mathbf{GL}^k(E) \times \mathbf{GL}^k(F), X \times Y, b \times b),$$

the product of λ_X and λ_Y . Then $J^k(X, Y)$ is equipped with a structure of fiber bundle associated with μ : the typical fiber is $J_0^k(E, F)_0$, on which $\mathbf{GL}^k(E) \times \mathbf{GL}^k(F)$ acts on the left by the law $((u, v), j) \mapsto v \circ j \circ u^{-1}$; the frame map is $((\rho, \sigma), j) \mapsto \sigma \circ j \circ \rho^{-1}$; the projection $J^k(X, Y) \rightarrow X \times Y$ is (s, b) .

12.5 Jets of sections

12.5.1. Let $\pi : Y \rightarrow X$ be a submersion, let $x \in X$, and let $s \in J_x^k(X, Y)$. We say that s is a jet of section (of order k) of π if s is of the form $j_x^k(f)$, where f is a section of class C^r of π above an open neighborhood of x ; this condition is equivalent to

$$j_{b(s)}^k(\pi) \circ s = j_x^k(\text{Id}_X).$$

We denote by $\mathbf{P}_x^k(\pi)$ (resp. $\mathbf{P}^k(\pi)$) the set of jets $s \in J_x^k(X, Y)$ (resp. $J^k(X, Y)$) which are jets of a section of π ; it is a submanifold of class $\mathbf{C}^{r-k}$ of $J^k(X, Y)$. The maps

$$s: \mathbf{P}^k(\pi) \rightarrow X \quad \text{and} \quad b: \mathbf{P}^k(\pi) \rightarrow Y$$

are submersions; if π is a fibration, they are fibrations. For $k = 0$, b is an isomorphism, by which one identifies $\mathbf{P}^0(\pi)$ with Y .

If $k' \leq k$, the map $r^{k,k'}: J^k(X, Y) \rightarrow J^{k'}(X, Y)$ maps $\mathbf{P}^k(\pi)$ to $\mathbf{P}^{k'}(\pi)$; the map from $\mathbf{P}^k(\pi)$ to $\mathbf{P}^{k'}(\pi)$ deduced from $r^{k,k'}$ is a fibration of class $\mathbf{C}^{r-k}$.

12.5.2. Let Z be a manifold of class $\mathbf{C}^r$; suppose that $Y = X \times Z$ and let $\pi = pr_1: Y \rightarrow X$. The restriction of $J^k(\text{Id}_X, pr_2)$ (cf. 12.3.9) to $\mathbf{P}^k(\pi)$ is an isomorphism of $\mathbf{P}^k(\pi)$ onto $J^k(X, Z)$, by which one identifies these two manifolds.

12.5.3. Let $\mathbf{Y}'$ be a manifold of class $\mathbf{C}^r$, let $\pi: \mathbf{Y} \rightarrow \mathbf{X}$ and $\pi': \mathbf{Y}' \rightarrow \mathbf{X}$ be submersions, and let $g: \mathbf{Y} \rightarrow \mathbf{Y}'$ be a morphism such that $\pi' \circ g = \pi$. The map $\mathbf{J}^k(\text{Id}_{\mathbf{X}}, g)$ has as its restriction a map

$$\mathbf{P}^k(g): \mathbf{P}^k(\pi) \rightarrow \mathbf{P}^k(\pi')$$

which is of class $\mathbf{C}^{r-k}$.

Let $\mathbf{X}'$ be a manifold of class $\mathbf{C}^r$, and let $f: \mathbf{X}' \rightarrow \mathbf{X}$ be a morphism; set $(\mathbf{Y}', \pi') = f^*(\mathbf{Y}, \pi)$, cf. 5.11.5. The map $\mathbf{J}^k(f, \text{Id}_{\mathbf{Y}})$ (cf. 12.3.9) defines a map of class $\mathbf{C}^{r-k}$ from $f^*\mathbf{P}^k(\pi)$ to $\mathbf{P}^k(\pi')$.

12.6 Jets of sections of a vector bundle

12.6.1. Let E be a vector bundle of class $\mathbf{C}^r$ with base X , and let π be its projection. Let $c_0 = (U, \varphi, F_0)$ be a vector chart of E and $c_1 = (U, \psi, F_1)$ a chart of the manifold X , with the same domain U . These charts define a bijection θ of $\mathbf{P}^k(\pi)|U$ onto $J^k(\psi(U), F_0) = \psi(U) \times F_0 \times Q^k(F_1, F_0)$, cf. 12.2.2 and 12.3.1, whence a vector chart

$$d = (U, \theta, G), \quad \text{with} \quad G = F_0 \times Q^k(F_1, F_0) = \prod_{m=0}^k P_m(F_1; F_0)$$

of $\mathbf{P}^k(\pi)$. The charts thus obtained form a $\mathbf{C}^{r-k}$ -atlas, which endows $\mathbf{P}^k(\pi)$ with a vector-bundle structure of class $\mathbf{C}^{r-k}$, with base X . This vector bundle is denoted $\mathbf{P}^k(\mathbf{E})$; the underlying manifold structure is the one defined in 12.5.1.

Let U be an open subset of X , and let $f \in \mathcal{S}_E^r(U)$ be a section of class $\mathbf{C}^r$ of E over U . The map $j^k(f) : x \mapsto j_x^k(f)$ is a section of class $\mathbf{C}^{r-k}$ of $\mathbf{P}^k(E)$ over U . The map $j^k : \mathcal{S}_E^r(U) \rightarrow \mathcal{S}_{\mathbf{P}^k(E)}^{r-k}(U)$ is K -linear.

12.6.2. If F is a Banach space, the identification (12.5.2) of $\mathbf{P}^k(F_X)$ with $J^k(X, F)$ endows the latter manifold with a vector-bundle structure of class $\mathbf{C}^{r-k}$, with base X . When $F = K$, one writes $\mathbf{P}^k(X)$ instead of $\mathbf{P}^k(K_X)$.

Example. — Take $X = K^n$, and denote by $u_1, \dots, u_n$ the coordinate functions on K^n ; then the fiber $\mathbf{P}_0^k(X)$ of $\mathbf{P}^k(X)$ at 0 has as basis the family of jets of order k

of the monomials $u_1^{m_1} \dots u_n^{m_n}$, with $m_i \geq 0$, $\sum_{i=1}^n m_i \leq k$.

12.6.3. Let d be an integer ≥ 0 , $E_1, \dots, E_d$, $\mathbf{F}$ vector bundles of class $\mathbf{C}^r$ with base $\mathbf{X}$, and let $u : E_1 \times_{\mathbf{X}} \dots \times_{\mathbf{X}} E_d \rightarrow \mathbf{F}$ be a multilinear morphism (7.3.1) of class $\mathbf{C}^r$. Then there exists a unique multilinear morphism

$$\mathbf{P}^k(u) : \mathbf{P}^k(E_1) \times_{\mathbf{X}} \dots \times_{\mathbf{X}} \mathbf{P}^k(E_d) \rightarrow \mathbf{P}^k(\mathbf{F})$$

such that

$$\mathbf{P}^k(u) (j^k(s_1), \dots, j^k(s_d)) = j^k(u(s_1, \dots, s_d))$$

for every open subset U of X and every sequence of sections $s_i \in \mathcal{S}_{E_i}^r(U)$, $1 \leq i \leq d$.

If A is a bundle of algebras (7.3.2) with base X , the morphism from $\mathbf{P}^k(A) \times_{\mathbf{X}} \mathbf{P}^k(A)$ to $\mathbf{P}^k(A)$ deduced from the multiplication $A \times_X A \rightarrow A$ makes $\mathbf{P}^k(A)$ into a bundle of algebras; if A is a bundle of associative algebras, $\mathbf{P}^k(A)$ is a bundle of associative algebras; if moreover M is a bundle of A -modules (7.3.3), $\mathbf{P}^k(M)$ is a bundle of $\mathbf{P}^k(A)$ -modules. In particular, $\mathbf{P}^k(X)$ is a bundle of associative, commutative, and unital algebras; if E is a vector bundle, $\mathbf{P}^k(E)$ is a bundle

of $\mathbf{P}^k(\mathbf{X})$ -modules. If $E \rightarrow F$ is a morphism of vector bundles, then $\mathbf{P}^k(u) : \mathbf{P}^k(\mathbf{E}) \rightarrow \mathbf{P}^k(\mathbf{F})$ is a $\mathbf{P}^k(\mathbf{X})$ -homomorphism.

12.6.4. If $E \xrightarrow{u} F \xrightarrow{v} G$ is a locally split exact sequence of vector bundles with base X , then so is the sequence

$$\mathbf{P}^k(\mathbf{E}) \rightarrow \mathbf{P}^k(\mathbf{F}) \rightarrow \mathbf{P}^k(\mathbf{G})$$

where the homomorphisms considered are $\mathbf{P}^k(u)$ and $\mathbf{P}^k(v)$.

12.6.5. Let k' and k'' be two positive integers with sum k , and let E be a vector bundle of class C^r and base X . The morphism

$$\alpha : J^k(X, E) \rightarrow J^{k''}(X, J^{k'}(X, E)) \quad (\text{cf. 12.3.8})$$

induces a morphism of vector bundles

$$\beta : \mathbf{P}^k(\mathbf{E}) \rightarrow \mathbf{P}^{k''}(\mathbf{P}^{k'}(\mathbf{E}))$$

which is of class C^{r-k} . If U is an open subset of X and $f \in \mathcal{S}_E^r(U)$, one has

$$\beta(j^k(f)) = j^{k''}(j^{k'}(f)).$$

When K is of characteristic zero, β is an isomorphism of $\mathbf{P}^k(\mathbf{E})$ onto a vector subbundle of $\mathbf{P}^{k''}(\mathbf{P}^{k'}(\mathbf{E}))$.

12.6.6. Let k' be an integer such that $0 \leq k' \leq k$, and let E be a vector bundle of class C^r and base X . The map

$$r^{k,k'} : \mathbf{P}^k(\mathbf{E}) \rightarrow \mathbf{P}^{k'}(\mathbf{E})$$

is a locally split surjective morphism of class C^{r-k} . Its kernel $\mathbf{N}^{k,k'}(\mathbf{E})$ consists of the jets of sections of E having contact of order $\geq k'$ with the zero section.

12.6.7 ("The vector functor P_m "). Using the notation of 7.6, 7.7, and 7.8, set $I_+ = \{0\}$, $I_- = \{1\}$, and let m be an integer ≥ 0 . If $\mathcal{V} = (V_0, V_1)$ is a pair of Banach spaces, denote by

$\tau_m(\mathcal{V})$ the Banach space $P_m(V_1; V_0)$ of continuous homogeneous polynomials of degree m on V_1 with values in V_0 (A.2). Likewise, if $f = (f_0, f_1)$, where $f_0 : V_0 \rightarrow V'_0$ and $f_1 : V'_1 \rightarrow V_1$ are morphisms of Banach spaces, denote by $\tau_m(f)$ the morphism $p \mapsto f_0 \circ p \circ f_1$ from $P_m(V_1; V_0)$ to $P_m(V'_1, V'_0)$. One thus obtains a vector functor τ_m of class $\mathbf{C}^\omega$. If $\mathbf{E}_0$ and $\mathbf{E}_1$ are two vector bundles with base $\mathbf{X}$, one denotes by $\mathbf{P}_m(\mathbf{E}_1; \mathbf{E}_0)$ the vector bundle deduced from $(\mathbf{E}_0, \mathbf{E}_1)$ by τ_m (7.6.2).

12.6.8. Let us resume the notations and hypotheses of 12.6.1. There exists one and only one morphism of vector bundles

$$\iota : P_k(T(\mathbf{X}); \mathbf{E}) \rightarrow P^k(\mathbf{E}) \quad (\text{cf. 12.6.6}),$$

such that, whatever the vector chart $c_0 = (U, \varphi, F_0)$ of $\mathbf{E}$ and the chart $c_1 = (U, \psi, F_1)$ of $\mathbf{X}$, the following diagram is commutative:

$$\begin{array}{ccc} P_k(T(\mathbf{X}), \mathbf{E})|_U & \xrightarrow{\iota|_U} & P^k(\mathbf{E})|_U \\ \downarrow \eta & & \downarrow \theta \\ U \times P_k(F_1; F_0) & \xrightarrow{i} & U \times \prod_{m=0}^k P_m(F_1; F_0) \end{array}$$

where: 1) $\iota|_U$ is the restriction of ι to $P_k(T(\mathbf{X}), \mathbf{E})|_U$;

2) θ is the bijection defined in 12.6.1;

3) i is the map $(u, p) \mapsto (u, 0, \dots, 0, p)$;

4) η is deduced, by means of the vector functor τ_k of the vector chart c'_1 of $T(\mathbf{X})$ (cf. 8.1.1) and of the vector chart c_0 of $\mathbf{E}$.

The morphism ι is an isomorphism of $P_k(T(\mathbf{X}); \mathbf{E})$ onto the vector subbundle $\mathbf{N}^{k, k-1}(\mathbf{E})$ of $P^k(\mathbf{E})$ (cf. 12.6.6); the sequence

$$0 \rightarrow P_k(T(\mathbf{X}); \mathbf{E}) \xrightarrow{\iota} P^k(\mathbf{E}) \xrightarrow{r^{k, k-1}} P^{k-1}(\mathbf{E}) \rightarrow 0$$

is a locally split exact sequence of vector bundles.

More generally, let us set $N_0 = P^k(\mathbf{E})$, $N_m = \mathbf{N}^{k, m-1}(\mathbf{E})$ for $m \geq 1$, so that the N_m form a

decreasing sequence of vector subbundles of $\mathbf{P}^k(\mathbf{E})$:

$$\mathbf{P}^k(\mathbf{E}) = \mathbf{N}_0 \supset \mathbf{N}_1 \supset \cdots \supset \mathbf{N}_k \supset \mathbf{N}_{k+1} = 0.$$

For $0 \leq m \leq k$, the projection $r^{k,m} : \mathbf{P}^k(\mathbf{E}) \rightarrow \mathbf{P}^m(\mathbf{E})$ defines an isomorphism of $\mathbf{N}_m/\mathbf{N}_{m+1}$ onto $\mathbf{N}^{m,m-1}(\mathbf{E})$; in view of what precedes, one thus obtains isomorphisms

$$\iota_m : \mathbf{N}_m/\mathbf{N}_{m+1} \rightarrow \mathbf{P}_m(\mathcal{T}(\mathbf{X}); \mathbf{E}) \quad (0 \leq m \leq k).$$

In particular, one has $\mathbf{N}_0/\mathbf{N}_1 \simeq \mathbf{E}$ and $\mathbf{N}_1/\mathbf{N}_2 \simeq \mathcal{L}(\mathcal{T}(\mathbf{X}); \mathbf{E})$. For $k = 1$, this gives an exact sequence:

$$0 \rightarrow \mathcal{L}(\mathcal{T}(\mathbf{X}); \mathbf{E}) \rightarrow \mathbf{P}^1(\mathbf{E}) \rightarrow \mathbf{E} \rightarrow 0.$$

12.7 Weakening of structure

We assume $\mathbf{K}=\mathbf{R}$. Let $r' \in \mathbf{N}_K$ with $k \leq r' \leq r$, and let $\mathbf{X}'$ and $\mathbf{Y}'$ be the manifolds of class $C^{r'}$ obtained from $\mathbf{X}$ and $\mathbf{Y}$ by weakening of structure (5.13.1). Let $x \in \mathbf{X}$ and $j \in \mathbf{J}_x^k(\mathbf{X}, \mathbf{Y})$; let $\mathbf{U}$ be an open subset of $\mathbf{X}$ containing x , and let f be a map of class C^k from $\mathbf{U}$ into $\mathbf{Y}$ such that $j_x^k(f) = j$. Consider f as a germ of a morphism from $\mathbf{X}'$ to $\mathbf{Y}'$; its jet $j' \in \mathbf{J}_x^k(\mathbf{X}', \mathbf{Y}')$ depends only on j . The map $j \mapsto j'$ is an isomorphism of class $C^{r'-k}$ of $\mathbf{J}^k(\mathbf{X}, \mathbf{Y})$ onto $\mathbf{J}^k(\mathbf{X}', \mathbf{Y}')$; it allows us to identify $\mathbf{J}^k(\mathbf{X}', \mathbf{Y}')$ with the manifold of class $C^{r'-k}$ deduced from the manifold $\mathbf{J}^k(\mathbf{X}, \mathbf{Y})$ of class C^{r-k} by weakening of structure. An analogous result applies to the manifolds $\mathbf{P}^k(\pi)$ and $\mathbf{P}^k(\mathbf{E})$ of nos. 12.5 and 12.6.

13 Point distributions

13.1 Symmetric tensors and Banach spaces

13.1.1. Let $\mathbf{E}$ be a module over a commutative ring $\mathbf{A}$. If n is an integer ≥ 0 , we denote by $\mathbf{TS}^n(\mathbf{E})$ the module of symmetric tensors of degree n of $\mathbf{E}$ ($\mathbf{A}$, III, p. 71 and $\mathbf{A}$, IV, § 5, no. 2); the direct sum of the $\mathbf{TS}^n(\mathbf{E})$ is denoted $\mathbf{TS}(\mathbf{E})$. If r is either an integer or one of the symbols

∞ , ω , we set

$$\text{TS}^{(r)}(\mathbf{E}) = \bigoplus_{n \leq r} \text{TS}^n(\mathbf{E}) \quad \text{and} \quad \text{TS}^{(r)+}(\mathbf{E}) = \bigoplus_{1 \leq n \leq r} \text{TS}^n(\mathbf{E});$$

consequently, if $r \geq 0$, $\text{TS}^{(r)}(\mathbf{E})$ is the direct sum of $\text{TS}^{(0)}(\mathbf{E}) = \mathbf{A}$ and $\text{TS}^{(r)+}(\mathbf{E})$. One has $\text{TS}^{(1)+}(\mathbf{E}) = \mathbf{E}$ and $\text{TS}^{(\infty)}(\mathbf{E}) = \text{TS}^{(\omega)}(\mathbf{E}) = \text{TS}(\mathbf{E})$.

If n is an integer ≥ 0 and if $x \in E$, one denotes by $\gamma_n(x)$ the element $x \otimes \cdots \otimes x$ of $\text{TS}^n(\mathbf{E})$; for $n = 0$, $\gamma_0(x) = 1$.

13.1.2 ("Polynomial maps"). Assume that A is an infinite integral domain and that E is a free A -module. Let F be an A -module. A map $f: E \rightarrow F$ is called a homogeneous polynomial of degree n if it satisfies the following equivalent conditions (cf. A, IV, § 5, no. 9):

- a) There exists a linear map $\tilde{f}: \text{TS}^n(\mathbf{E}) \rightarrow F$ such that $f(x) = \tilde{f}(\gamma_n(x))$ for all $x \in E$.
- b) There exists a multilinear map $u: E^n \rightarrow F$ such that $f(x) = u(x, \dots, x)$ for all $x \in E$.

Suppose that this is the case. The map $\tilde{f}$ satisfying a) is then unique; if $t \in \text{TS}^n(\mathbf{E})$, one denotes by $\langle f, t \rangle$ the element $\tilde{f}(t)$. If $u: E^n \rightarrow F$ satisfies b), the corresponding linear map from $\otimes^n E$ to F coincides with $\tilde{f}$ on the submodule $\text{TS}^n(\mathbf{E})$. When $n!$ is invertible in A , one can choose u symmetric, and do so uniquely.

13.1.3. Let E be a Banach space over K and F a separated polynormed space over K . Let U be an open neighborhood of 0 in E , $f: U \rightarrow F$ a map of class C^r ($r \in \mathbb{N}_{\mathbb{K}}$), and t an element of $\text{TS}^{(r)}(\mathbf{E})$, cf. 13.1.1 (applied to the ring $A=K$). Let k be an integer $\leq r$ such that $t \in \text{TS}^{(k)}(\mathbf{E})$. One can decompose t and f in a unique way into

$$t = t_0 + \cdots + t_k, \quad \text{with} \quad t_i \in \text{TS}^i(\mathbf{E})$$

and

$$f = f_0 + \cdots + f_k + h,$$

where f_i is a continuous homogeneous polynomial of degree i ($0 \leq i \leq k$) and where h has contact of order $\geq k$ with 0 at the point 0. One then sets

$$\langle f, t \rangle = \sum_{i=0}^k \langle f_i, t_i \rangle;$$

this is an element of F that does not depend on the choice of k .

13.1.4. Let E and E' be two Banach spaces, U an open neighborhood of 0 in E , and $\varphi: U \rightarrow E'$ a map of class C^r such that $\varphi(0) = 0$. Let $t \in TS^{(r)}(E)$. There exists one and only one element $t' \in TS^{(r)}(E')$ such that, for any separated polynormed space F , any open neighborhood U' of 0 in E' , and any map $f: U' \rightarrow F$ of class C^r , one has

$$\langle f, t' \rangle = \langle f \circ \varphi, t \rangle. \quad (*)$$

This element t' is denoted $\varphi_*(t)$. The map

$$\varphi_*: TS^{(r)}(E) \rightarrow TS^{(r)}(E')$$

thus defined is linear.

13.2 Point distributions

In this section, X denotes a manifold of class C^r , and x is a point of X .

13.2.1. Let $\mathcal{A}$ be the set of pairs (c, t) , where $c = (U, \varphi, E)$ is a chart of X centered at x , and where $t \in TS^{(r)}(E)$. Two elements $((U, \varphi, E), t)$ and $((U', \varphi', E'), t')$ of $\mathcal{A}$ are said to be equivalent if $t' = \gamma_*(t)$, where $\gamma = \varphi' \circ \varphi^{-1}$. One thus obtains an equivalence relation on $\mathcal{A}$; an equivalence class for this relation is called a point distribution at x on X .

Let $T_x^{(r)}(X)$ be the set of point distributions at x on X . If c is a chart of X centered at x , the map

$$\theta_c: TS^{(r)}(E) \rightarrow T_x^{(r)}(X)$$

which maps t to the class of (c, t) , is a bijection. In all that follows, we equip $T_x^{(r)}(X)$ with the structure of a K -vector space obtained by transporting that of $TS^{(r)}(E)$ by θ_c ; this structure does not depend on the choice of c . Similarly, if $k \leq r$, we denote by $T_x^{(k)}(X)$ and $T_x^{(k)+}(X)$

the images of $TS^{(k)}(E)$ and $TS^{(k)+}(E)$ under θ_c ; they do not depend on the choice of c . An element of $T_x^{(r)}(X)$ is said to be of order $\leq k$ if it belongs to $T_x^{(k)}(X)$. One has

$$T_x^{(k)}(X) = T_x^{(0)}(X) \oplus T_x^{(k)+}(X).$$

There exists one and only one element ε_x of $T_x^{(0)}(X)$ such that we have $\theta_c(1) = \varepsilon_x$ for every chart c of X centered at x . If t is a point distribution in x , we call the constant term of t the element λ of K such that $t - \lambda \varepsilon_x \in T_x^{(r)+}(X)$; we say that t is without constant term if its constant term is zero.

13.2.2. Let F be a separated polynomial space, and let f be a function of class C^r with values in F , defined on a neighborhood of x . Let t be a point distribution at x on X .

Consider a map $c = (U, \varphi, E)$ of X centered at x , and set $f_c = f \circ \varphi^{-1}$ and $t_c = \theta_c^{-1}(t)$; the element $\langle f_c, t_c \rangle$ of F defined in 13.1.3 does not depend on the choice of c ; we denote it $\langle f, t \rangle$. One has $\varepsilon_x(f) = f(x)$. ⁽¹⁾ The constant term of t is $\langle 1, t \rangle$.

If F' is a separated polynormed space and if $u : F \rightarrow F'$ is continuous and linear, we have $\langle u \circ f, t \rangle = u(\langle f, t \rangle)$.

Suppose that F is a Banach space, and that t is of order $\leq k$, with k finite. Let $j \in J_x^k(X, F)$, cf. 12.1.2, and let $f \in j$. Then $\langle f, t \rangle$ depends only on the jet j ; it is denoted $\langle j, t \rangle$. The map from $T_x^{(k)}(X) \times J_x^k(X, F)$ to F thus defined is bilinear. When $F = K$, we have set $J_x^k(X, F) = P_x^k(X)$, cf. 12.6.2, and we obtain a bilinear form on $T_x^{(k)}(X) \times P_x^k(X)$.

13.2.3. Let $\varphi : X \rightarrow Y$ be a morphism of manifolds of class C^r and let $y = \varphi(x)$. Let $c = (U, \psi, E)$ be a chart of X centered at x and let $c' = (U', \psi', E')$ be a chart of Y centered at y ; let $\bar{\varphi}$ be the expression of φ in these charts (5.3.2), and let $\bar{\varphi}_*$ be the corresponding map from $TS^{(r)}(E)$ to $TS^{(r)}(E')$, cf. 13.1.4. There exists one and only one linear map, denoted $T_x^{(r)}(\varphi)$ or φ_* , from $T_x^{(r)}(X)$ to $T_y^{(r)}(Y)$ making the diagram commutative

$$\begin{array}{ccc}
\text{TS}^{(r)}(\mathbf{E}) & \xrightarrow{\tilde{\varphi}_*} & \text{TS}^{(r)}(\mathbf{E}') \\
\downarrow \theta_c & & \downarrow \theta_{c'} \\
\text{T}_x^{(r)}(\mathbf{X}) & \xrightarrow{\varphi_*} & \text{T}_y^{(r)}(\mathbf{Y}).
\end{array}$$

This map does not depend on the choice of c and c' . If $t \in \text{T}_x^{(r)}(\mathbf{X})$, one says that $\varphi_*(t)$ is the image of t by φ . One has $\varphi_*(\varepsilon_x) = \varepsilon_y$. If $k \leq r$, one has

$$\varphi_*(\text{T}_x^{(k)}(\mathbf{X})) \subset \text{T}_y^{(k)}(\mathbf{Y}) \quad \text{and} \quad \varphi_*(\text{T}_x^{(k)+}(\mathbf{X})) \subset \text{T}_y^{(k)+}(\mathbf{Y}).$$

If $\varphi' : X \rightarrow Y$ is a morphism of class C^r such that $\varphi'(x) = y$, one has $\varphi_* = \varphi'_*$ if and only if $j_x^r(\varphi) = j_x^r(\varphi')$, cf. 12.1.

If φ is an immersion (resp. a submersion) at x , φ_* is injective (resp. surjective); the converse is true if $\dim_y Y < +\infty$.

If $\varphi' : Y \rightarrow Z$ is a morphism of manifolds of class C^r and if $t \in \text{T}_x^{(r)}(\mathbf{X})$, one has $(\varphi' \circ \varphi)_*(t) = \varphi'_*(\varphi_*(t))$.

Let F be a separated polynormed space, and let f be a function of class C^r with values in F , defined in a neighborhood of y . We then have

$$\langle f, \varphi_*(t) \rangle = \langle f \circ \varphi, t \rangle \quad \text{for all } t \in \text{T}_x^{(r)}(\mathbf{X}). \quad (1)$$

For t given, the relations (1) (for f and F variables) characterize $\varphi_*(t)$; when $\dim_y Y < +\infty$, one may restrict oneself to $F = K$.

13.2.4. Suppose that X is an open submanifold of a Banach space E . Denote by φ_x the map $y \mapsto y - x$ from X into E ; the chart $c = (X, \varphi_x, E)$ is centered at x . We then identify $\text{T}_x^{(r)}(\mathbf{X})$ with $\text{TS}^{(r)}(\mathbf{E})$ by means of θ_c^{-1} . In particular, we have $\text{T}_x^{(\infty)}(\mathbf{X}) = \text{TS}(\mathbf{E})$.

Let F be a Banach space, let $u : E \rightarrow F$ a continuous linear map, and let $x \in E$, $y \in F$ such

that $u(x) = y$. Identify as above $T_x^{(\infty)}(E)$ with $TS(E)$ and $T_y^{(\infty)}(F)$ with $TS(F)$. The map

$$u_* : TS(E) \rightarrow TS(F) \quad (\text{cf. 13.2.3})$$

coincides with the map $TS(u)$ induced by the canonical extension of u to the tensor algebra.

13.2.5. If $k \leq r$, one denotes by $T^{(k)}(X)$ (resp. $T^{(k)+}(X)$) the set sum of $T_a^{(k)}(X)$ (resp. of $T_a^{(k)+}(X)$) for $a \in X$. A map $t : X \rightarrow T^{(k)}(X)$ such that $t(a) \in T_a^{(k)}(X)$ for all $a \in X$ is called a field of point distributions of order $\leq k$.

Suppose k is finite and X is locally of finite dimension. For every $a \in X$, the bilinear form $(j, t) \mapsto \langle j, t \rangle$ (cf. 13.2.2) defines an isomorphism i_a of $T_a^{(k)}(X)$ onto the dual $P_a^k(X)^*$ of $P_a^k(X)$. The i_a define a bijection $i : T^{(k)}(X) \rightarrow P^k(X)^*$, where $P^k(X)^*$ denotes the dual of the vector bundle $P^k(X)$; by transport of structure via i^{-1} , one endows $T^{(k)}(X)$ with a vector bundle structure with base X and class $\mathbf{C}^{r-k}$. A field of point distributions of order $\leq k$ is said to be of class $\mathbf{C}^s$, with $s \leq r - k$, if it is a section of class $\mathbf{C}^s$ of the bundle $T^{(k)}(X)$.

Let $\varphi : X \rightarrow Y$ be a morphism of manifolds of class $\mathbf{C}^r$ and suppose that Y , like X , is locally finite-dimensional. Then $\varphi_* : T^{(k)}(X) \rightarrow T^{(k)}(Y)$ is a φ -morphism of vector bundles of class $\mathbf{C}^{r-k}$.

13.2.6. Let k be an integer such that $0 \leq k \leq r$. If U is an open set of a finite-dimensional Banach space E , the vector bundle $T^{(k)}(U)$ is identified, by virtue of 13.2.4, with the trivial bundle with fibre $TS^{(k)}(E)$.

Take in particular $E = K^n$, with $n \geq 0$, and let $(e_1, \dots, e_n)$ be the canonical basis of E . If $\alpha = (\alpha_1, \dots, \alpha_n)$ is an element of N^n , denote by Δ^α the element $\gamma_{\alpha_1}(e_1) \dots \gamma_{\alpha_n}(e_n)$ of $TS(E)$ (the product used being the symmetric product of symmetric tensors, cf. A, IV, § 5, no. 3). ⁽¹⁾ The Δ^α , for $|\alpha| \leq k$, form a basis of $TS^{(k)}(E)$; if $a \in K^n$, one denotes by Δ_a^α the corresponding elements of $T_a^{(k)}(E)$. Let F be a separated polynormed space, and f a function of class $\mathbf{C}^r$, with values in F , defined in a neighbourhood of a ; if $\alpha \in N^n$, the element $\langle f, \Delta_a^\alpha \rangle$ coincides with the element $(\Delta^\alpha f)(a)$ defined in 2.5.3, 3.2.1 and 4.2.1.

Suppose X is locally finite-dimensional, and let $\xi = (\xi^1, \dots, \xi^n)$ be a coordinate system of X in an open set U . If $a \in U$ and if α is a multi-index such that $|\alpha| \leq k$, one denotes by $(\Delta_\xi^\alpha)_a$ the point distribution at a whose image under ξ is the point distribution $\Delta_{\xi(a)}^\alpha$ onto K^n ; the

distribution fields

$$\Delta_{\xi}^{\alpha} : a \mapsto (\Delta_{\xi}^{\alpha})_a \quad (|\alpha| \leq k)$$

form a frame on U of the vector bundle $\mathbf{T}^{(k)}(\mathbf{X})$. If $f : U \rightarrow \mathbf{F}$ is of class $\mathbf{C}^r$, we set $\langle f, (\Delta_{\xi}^{\alpha})_a \rangle = \Delta_{\xi}^{\alpha} f(a)$ and denote by $\Delta_{\xi}^{\alpha} f$ the function $a \mapsto \Delta_{\xi}^{\alpha} f(a)$; it is a function of class $\mathbf{C}^{r-k}$ in U .

¹ If $m = |\alpha|$ and if Σ is the set of maps σ from $\{1, \dots, m\}$ to $\{1, \dots, n\}$ such that $\text{Card } \sigma^{-1}(i) = \alpha_i$, one has

$$\Delta^{\alpha} = \gamma_{\alpha_1}(e_1) \dots \gamma_{\alpha_n}(e_n) = \sum_{\sigma \in \Sigma} e_{\sigma(1)} \otimes \dots \otimes e_{\sigma(n)}, \quad \text{cf. A, IV, §5, no. 4.}$$

13.3 Point distributions and tangent spaces

In this section, X denotes a manifold of class $\mathbf{C}^r$.

13.3.1. Let $x \in X$ and let k be an integer $\leq r$. Let $c = (U, \varphi, E)$ a chart of X centered at x ; the isomorphism $\theta_c : \text{TS}^{(r)}(E) \rightarrow T_x^{(r)}(X)$ defined in 13.2.1 maps $\text{TS}^{(k)}(E)$ onto $T_x^{(k)}(X)$ and $\text{TS}^{(k-1)}(E)$ onto $T_x^{(k-1)}(X)$; by restriction and passage to the quotient, it induces an isomorphism

$$\theta_{c,k} : \text{TS}^k(E) = \text{TS}^{(k)}(E) / \text{TS}^{(k-1)}(E) \rightarrow T_x^{(k)}(X) / T_x^{(k-1)}(X).$$

On the other hand, c defines an isomorphism, already denoted θ_c , from E onto the tangent space $T_x(X)$, cf. 5.5.1. Let i_k denote the composite

$$T_x^{(k)}(X) / T_x^{(k-1)}(X) \xrightarrow{\theta_{c,k}^{-1}} \text{TS}^k(E) \xrightarrow{\text{TS}^k(\theta_c)} \text{TS}^k(T_x(X)).$$

The isomorphism i_k is independent of the choice of c ; for $k = 1$, it is used to identify $T_x^{(1)+}(X) = T_x^{(1)}(X) / T_x^{(0)}(X)$ with the tangent space $T_x(X)$. If $t \in T_x^{(k)}(X)$, one may write $i_k(t)$ for the image under i_k of the class of t modulo $T_x^{(k-1)}(X)$.

Set

$$\mathrm{gr}_k T_x^{(r)}(X) = T_x^{(k)}(X) / T_x^{(k-1)}(X) \quad \text{and} \quad \mathrm{gr} T_x^{(r)}(X) = \bigoplus_{0 \leq k \leq r} \mathrm{gr}_k T_x^{(r)}(X).$$

One says that $\mathrm{gr} T_x^{(r)}(X)$ is the graded object associated with the increasing filtration $(T_x^{(k)}(X))_{k \leq r}$ of $T_x^{(r)}(X)$. The i_k define an isomorphism of graded vector spaces

$$i: \mathrm{gr} T_x^{(r)}(X) \rightarrow \mathrm{TS}^{(r)}(T_x(X)).$$

If $r = \infty$ or ω , i is an isomorphism of $\mathrm{gr} T_x^{(r)}(X)$ onto $\mathrm{TS}(T_x(X))$. When one wishes to specify x (or X , or both), one writes i_x (or i_X , or $i_{X,x}$) in place of i .

13.3.2. In addition to the above hypotheses, suppose that X is locally of finite dimension. The isomorphisms i_k relative to the various points of X define an isomorphism of vector bundles

$$i_k: T^{(k)}(X) / T^{(k-1)}(X) \rightarrow \mathrm{TS}^k(T(X))$$

where $\mathrm{TS}^k(T(X))$ denotes the vector bundle of class C^{r-1} deduced from $T(X)$ by the finite-dimensional vector functor TS^k (cf. 7.6.5). For $k \geq 1$, i_k is an isomorphism of class C^{r-k} ; for $k = 0$, it is the identity map of the trivial bundle K_X .

13.3.3. Example. — With hypotheses as above, let $\xi = (\xi^1, \dots, \xi^n)$ be a coordinate system of X at x . Let $(\partial_{i,x})_{1 \leq i \leq n}$ denote the basis of $T_x(X)$ defined by ξ (cf. 5.5.8), and let $((\Delta_\xi^\infty)_x)_{|\alpha| \leq k}$ denote the basis of $T_x^{(k)}(X)$ defined in 13.2.6. One has:

$$\begin{aligned} i_k((\Delta_\xi^\alpha)_x) &= 0 & \text{if } |\alpha| < k \\ i_k((\Delta_\xi^\alpha)_x) &= \gamma_{\alpha_1}(\partial_{1,x}) \dots \gamma_{\alpha_n}(\partial_{n,x}) & \text{if } |\alpha| = k. \end{aligned} \quad (1)$$

¹ Here again, the product of the $\gamma_{\alpha_i}(\partial_{i,x})$ is a symmetric product in $\mathrm{TS}(T_x(X))$.

13.3.4. Suppose that K has characteristic zero or that X is locally of finite dimension. Let k be an integer such that $0 \leq k \leq r$ and let $x \in X$; let F be a Banach space. By 12.6.8, one has an exact sequence

$$0 \rightarrow P_k(T_x(X); F) \xrightarrow{i} P_x^k(F_X) \xrightarrow{\sigma} P_x^{k-1}(F_X) \rightarrow 0, \quad (i)$$

with $\sigma = r^{k,k-1}$. On the other hand, by 13.3.1, one has an exact sequence

$$0 \leftarrow TS^k(T_x(X)) \xleftarrow{i} T_x^{(k)}(X) \xleftarrow{s} T_x^{(k-1)}(X) \leftarrow 0, \quad (\text{ii})$$

where s is the inclusion of $T_x^{(k-1)}(X)$ to $T_x^{(k)}(X)$ and $i = i_k$. These exact sequences are "paired in F ". More precisely, write them in the form:

(i)

$$0 \rightarrow A \xrightarrow{i} B \xrightarrow{\sigma} C \rightarrow 0 \quad (\text{ii})$$

$$0 \leftarrow A' \xleftarrow{i} B' \xleftarrow{s} C' \leftarrow 0.$$

If $a \in \mathbf{A}$ and $a' \in \mathbf{A}'$, the element $\langle a, a' \rangle$ of F is defined by 13.1.2; if $b \in \mathbf{B}$ and $b' \in \mathbf{B}'$, (resp. if $c \in \mathbf{C}$ and $c' \in \mathbf{C}'$), the element $\langle b, b' \rangle$ (resp. $\langle c, c' \rangle$) of F is defined by 13.2.2; one then has:

$$\langle ia, b' \rangle = \langle a, ib' \rangle \quad \text{and} \quad \langle \sigma b, c' \rangle = \langle b, sc' \rangle \quad \text{if} \quad a \in \mathbf{A}, \quad b \in \mathbf{B}, \quad b' \in \mathbf{B}', \quad c' \in \mathbf{C}'.$$

13.3.5. Let Y be a manifold of class C^r let $\varphi : X \rightarrow Y$ be a morphism of class C^r and let $x \in X$, $y \in Y$ such that $y = \varphi(x)$. The map $\varphi_* : T_x^{(r)}(X) \rightarrow T_y^{(r)}(Y)$ defined in 13.2.3 is compatible with the filtrations of these spaces; it defines, by passage to the associated graded objects, a linear map

$$\text{gr}(\varphi_*) : \text{gr } T_x^{(r)}(X) \rightarrow \text{gr } T_y^{(r)}(Y).$$

Let us also denote by $TS^{(r)}(T_x(\varphi))$ the map from $TS^{(r)}(T_x(\mathbf{X}))$ to $TS^{(r)}(T_y(\mathbf{Y}))$ induced by the canonical extension $TS(T_x(\varphi))$ of $T_x(\varphi)$. The diagram

$$\begin{array}{ccc}
\text{gr } T_x^{(r)}(X) & \xrightarrow{\text{gr}(\varphi_*)} & \text{gr } T_y^{(r)}(Y) \\
\downarrow i_{\mathbf{X},x} & & \\
\text{TS}^{(r)}(T_x(X)) & \xrightarrow{\text{TS}^{(r)}(T_x(\varphi))} & \text{TS}^{(r)}(T_y(Y))
\end{array}$$

is commutative.

13.3.6. Suppose that K is of characteristic zero. Using the isomorphisms $\varphi_{\mathbb{M}} : S^k(\mathbf{M}) \rightarrow \text{TS}^k(\mathbf{M})$ defined in A, IV, § 5, no. 8, one may replace, in all that precedes, the $\text{TS}^k(\mathbf{T}_x(\mathbf{X}))$ by the symmetric powers k th $S^k(\mathbf{T}_x(\mathbf{X}))$ of the tangent space $\mathbf{T}_x(\mathbf{X})$.

13.4 Tensor product of point distributions

13.4.1. Let $\mathbf{X}_1$ and $\mathbf{X}_2$ be two manifolds of class $\mathbf{C}^r$, let $x_1 \in \mathbf{X}_1$, $x_2 \in \mathbf{X}_2$ and $t_1 \in T_{x_1}^{(k_1)}(\mathbf{X}_1)$, $t_2 \in T_{x_2}^{(k_2)}(\mathbf{X}_2)$ with $k_1 + k_2 \leq r$. Set $\mathbf{X} = \mathbf{X}_1 \times \mathbf{X}_2$, $x = (x_1, x_2)$ and $k = k_1 + k_2$. For $i = 1, 2$ let $c_i = (U_i, \varphi_i, E_i)$ be a chart of $\mathbf{X}_i$ centered at x_i and denote by $\tilde{t}_i$ the element of $TS(E_i)$ such that $\theta_{c_i}(\tilde{t}_i) = t_i$, cf. 13.2.1. Let σ be the canonical isomorphism of $TS(E_1) \otimes TS(E_2)$ onto $TS(E_1 \times E_2)$, cf. A, IV, § 5, no. 5. The element $\sigma(\tilde{t}_1 \otimes \tilde{t}_2)$ is the symmetric product of $\tilde{t}_1$ and $\tilde{t}_2$ (identified with elements of $TS(E_1 \times E_2)$ by means of the canonical injections $TS(E_i) \rightarrow TS(E_1 \times E_2)$); it belongs to $TS^{(k)}(E_1 \times E_2)$. Set $c = c_1 \times c_2$; it is a chart of $\mathbf{X}$ centered at x . The image under θ_c of $\sigma(\tilde{t}_1 \otimes \tilde{t}_2)$ is an element of $T_x^{(k)}(\mathbf{X})$, which does not depend on the choice of charts c_i . It is called the direct product, or the symmetric tensor product (or even simply the tensor product) of t_1 and t_2 and is denoted $t_1 \times t_2$ or $t_1 \otimes t_2$.

One has $\varepsilon_{x_1} \otimes \varepsilon_{x_2} = \varepsilon_x$. The constant term of $t_1 \otimes t_2$ is the product of the constant terms of t_1 and of t_2 .

The isomorphism $(y_1, y_2) \mapsto (y_2, y_1)$ of $\mathbf{X}_1 \times \mathbf{X}_2$ onto $\mathbf{X}_2 \times \mathbf{X}_1$ transforms $t_1 \otimes t_2$ into $t_2 \otimes t_1$.

13.4.2 (“Associativity and functoriality”). Let X_i ($i = 1, 2, 3$) be manifolds of class $\mathbf{C}^r$, and let $x_i \in X_i$, $t_i \in T_{x_i}^{(k_i)}(\mathbf{X}_i)$ with $k_1 + k_2 + k_3 \leq r$. One has

$$(t_1 \otimes t_2) \otimes t_3 = t_1 \otimes (t_2 \otimes t_3);$$

this point distribution is denoted $t_1 \otimes t_2 \otimes t_3$. One defines in the same way arbitrary finite tensor products.

Let $\varphi_1: \mathbf{X}_1 \rightarrow \mathbf{Y}_1$ and $\varphi_2: \mathbf{X}_2 \rightarrow \mathbf{Y}_2$ be morphisms of manifolds of class $\mathbf{C}^r$, and let $t_1 \in \mathbf{T}^{(k_1)}(\mathbf{X}_1)$, $t_2 \in \mathbf{T}^{(k_2)}(\mathbf{X}_2)$ with $k_1 + k_2 \leq r$. One has

$$(\varphi_1 \times \varphi_2)_*(t_1 \otimes t_2) = \varphi_{1*}(t_1) \otimes \varphi_{2*}(t_2).$$

13.4.3. With the notation of 13.4.1, let F_1, F_2 and F be separated polynormed spaces, and let $(u_1, u_2) \mapsto u_1.u_2$ be a continuous bilinear map from $F_1 \times F_2$ to F . Let

$$f_i: \mathbf{X}_i \rightarrow F_i \quad (i = 1, 2)$$

be maps of class $\mathbf{C}^r$, and define $f_1 \otimes f_2: \mathbf{X} \rightarrow F$ by

$$(f_1 \otimes f_2)(y_1, y_2) = f_1(y_1).f_2(y_2).$$

The map $f_1 \otimes f_2$ is of class $\mathbf{C}^r$, and we have

$$\langle f_1 \otimes f_2, t_1 \otimes t_2 \rangle = \langle f_1, t_1 \rangle . \langle f_2, t_2 \rangle.$$

13.4.4. With the notation as in 13.4.1, let f be a map of class $\mathbf{C}^r$ from $\mathbf{X}$ into a separated polynormed space F . If $y_1 \in \mathbf{X}_1$, denote by f_{y_1} the map $y_2 \mapsto f(y_1, y_2)$ from $\mathbf{X}_2$ to F and set $g(y_1) = \langle f_{y_1}, t_2 \rangle$. The function $g: \mathbf{X}_1 \rightarrow F$ thus defined is of class $\mathbf{C}^{r-k_1}$, and we have

$$\langle f, t_1 \otimes t_2 \rangle = \langle g, t_1 \rangle,$$

in other words

$$\langle f, t_1 \otimes t_2 \rangle = \langle y_1 \mapsto \langle y_2 \mapsto f(y_1, y_2), t_2 \rangle, t_1 \rangle.$$

We likewise have

$$\langle f, t_1 \otimes t_2 \rangle = \langle y_2 \mapsto \langle y_1 \mapsto f(y_1, y_2), t_1 \rangle, t_2 \rangle.$$

13.4.5. With the notation as in 13.4.1, let

$$\alpha_{k_1, k_2} : T_{x_1}^{(k_1)}(X_1) \otimes T_{x_2}^{(k_2)}(X_2) \rightarrow T_x^{(k)}(X)$$

the linear map defined by the bilinear map $(t_1, t_2) \mapsto t_1 \otimes t_2$; this map is injective; it is bijective if k_1 and k_2 are infinite.

If $n \leq r$, denote by $T_x^{(n)}(X_1, X_2)$ the vector subspace of $T_{x_1}^{(n)}(X_1) \otimes T_{x_2}^{(n)}(X_2)$ generated by the subspaces $T_{x_1}^{(k_1)}(X_1) \otimes T_{x_2}^{(k_2)}(X_2)$ for $k_1 + k_2 \leq n$. There exists a linear map

$$\alpha_n : T_x^{(n)}(X_1, X_2) \rightarrow T_x^{(n)}(X)$$

and exactly one that extends the mappings α_{k_1, k_2} defined above; it is an isomorphism. In what follows, we identify $T_x^{(n)}(X)$ by means of α_n^{-1} to the vector subspace $T_x^{(n)}(X_1, X_2)$ of $T_{x_1}^{(n)}(X_1) \otimes T_{x_2}^{(n)}(X_2)$.

13.4.6. We keep the notations of 13.4.5 and 13.3.1. By passing to the quotient, α_{k_1, k_2} defines a linear map

$$\varepsilon_{k_1, k_2} : \text{gr}_{k_1} T_{x_1}^{(r)}(X_1) \otimes \text{gr}_{k_2} T_{x_2}^{(r)}(X_2) \rightarrow \text{gr}_k T_x^{(r)}(X).$$

The maps ε_{k_1, k_2} ($k_1 + k_2 \leq r$) are the components of a graded linear map of degree zero

$$\varepsilon : \bigoplus_{k_1 + k_2 \leq r} (\text{gr}_{k_1} T_{x_1}^{(r)}(X_1) \otimes \text{gr}_{k_2} T_{x_2}^{(r)}(X_2)) \rightarrow \bigoplus_{k \leq r} \text{gr}_k T_x^{(r)}(X)$$

which is an isomorphism. The diagram

$$\begin{array}{ccc} \bigoplus_{k_1 + k_2 \leq r} (\text{gr}_{k_1} T_{x_1}^{(r)}(X_1) \otimes \text{gr}_{k_2} T_{x_2}^{(r)}(X_2)) & \xrightarrow{\varepsilon} & \text{gr} T_x^{(r)}(X) \\ \downarrow i_{12} & & \\ \bigoplus_{k_1 + k_2 \leq r} (\text{TS}^{k_1}(T_{x_1}(X_1)) \otimes \text{TS}^{k_2}(T_{x_2}(X_2))) & \xrightarrow{\sigma} & \bigoplus_{k \leq r} \text{TS}^k(T_{x_1}(X_1) \times T_{x_2}(X_2)) \end{array}$$

is commutative (in this diagram, i_{12} denotes the homomorphism induced by $i_{x_1} \otimes i_{x_2}$ and σ is the isomorphism defined in A, IV, § 5, no. 5). If r is infinite, this diagram is simply written:

$$\begin{array}{ccc}
\text{gr } T_{x_1}^{(\infty)}(X_1) \otimes \text{gr } T_{x_2}^{(\infty)}(X_2) & \xrightarrow{\varepsilon} & \text{gr } T_x^{(\infty)}(X) \\
\downarrow i_{x_1} \otimes i_{x_2} & & \\
\text{TS}(T_{x_1}(X_1)) \otimes \text{TS}(T_{x_2}(X_2)) & \xrightarrow{\sigma} & \text{TS}(T_x(X)).
\end{array}$$

13.5 Coproducts

In this section, X denotes a manifold of class C^r and x is a point of X .

13.5.1. Let $k \leq r$. If Δ is the diagonal map $y \mapsto (y, y)$ from X into $X \times X$, then Δ_* maps $T_x^{(k)}(X)$ into $T_{(x,x)}^{(k)}(X \times X)$, which is a vector subspace of $T_x^{(k)}(X) \otimes T_x^{(k)}(X)$ (cf. 13.4.5). We thus obtain a linear map, again denoted by Δ_* (or c):

$$T_x^{(k)}(X) \rightarrow T_x^{(k)}(X) \otimes T_x^{(k)}(X);$$

we call it the coproduct attached to $T_x^{(k)}(X)$. Equipped with this coproduct, $T_x^{(k)}(X)$ is a coassociative and cocommutative coalgebra (A, III, p. 144–145); it admits as its counit the linear form which assigns to a point distribution its constant term (13.2.1). If $k' \leq k$, the inclusion of $T_x^{(k')}(X)$ into $T_x^{(k)}(X)$ is a morphism of coalgebras.

If $c = (U, \varphi, E)$ is a chart of X centered at x , the isomorphism

$$\theta_c : T^{(k)}(E) \rightarrow T_x^{(k)}(X)$$

is an isomorphism of coalgebras, $T^{(k)}(E)$ being equipped with the coproduct induced by that of $\text{TS}(E)$, cf. A, IV, § 5, no. 7.

If $\varphi : X \rightarrow Y$ is a morphism of manifolds of class C^r , the map from $T_x^{(k)}(X)$ to $T_{\varphi(x)}^{(k)}(Y)$ induced by φ_* is a morphism of coalgebras.

13.5.2. Let F_1, F_2 and F be separated polynormed spaces, and let $(u_1, u_2) \mapsto u_1 \cdot u_2$ be a continuous bilinear map from $F_1 \times F_2$ to F . Let $t \in T_x^{(r)}(X)$ and let $c(t) = \sum_j u_j \otimes v_j$, where $u_j, v_j \in T_x^{(r)}(X)$, be its image under the coproduct. Let $f_i : X \rightarrow F_i$ ($i = 1, 2$) be maps of class C^r . One has

$$\langle f_1, f_2, t \rangle = \sum_j \langle f_1, u_j \rangle \cdot \langle f_2, v_j \rangle,$$

which we write, by abuse of notation:

$$\langle f_1, f_2, t \rangle = \langle f_1 \otimes f_2, c(t) \rangle.$$

13.5.3. Let $t \in \mathbf{T}_x^{(k)}(\mathbf{X})$, where $k \leq r$.

- a) For $c(t) = t \otimes t$ it is necessary and sufficient that t be equal to 0 or to ε_x .
- b) For $c(t) = t \otimes \varepsilon_x + \varepsilon_x \otimes t$ it is necessary and sufficient that t be a tangent vector, i.e., that one has $t \in \mathbf{T}_x^{(1)+}(X)$.

13.6 Distributions with finite support

13.6.1. Let X be a manifold of class C^r , and let $k \leq r$. We denote by $\mathcal{T}^{(k)}(X)$ the direct sum of the spaces $\mathbf{T}_x^{(k)}(X)$, for $x \in X$. An element of $\mathcal{T}^{(k)}(X)$ is called a distribution with finite support on X of order $\leq k$. If $f : X \rightarrow \mathbf{F}$ is a function of class C^r with values in a separated polynormed space $\mathbf{F}$ and if $t = \sum_{x \in X} t_x$, where $t_x \in \mathbf{T}_x^{(k)}(X)$ for all $x \in X$, is a distribution with finite support; we set

$$\langle f, t \rangle = \sum_{x \in X} \langle f, t_x \rangle.$$

Similarly, we set $c(t) = \sum_{x \in X} c(t_x)$, which endows $\mathcal{T}^{(k)}(X)$ with the structure of a coassociative, cocommutative coalgebra with a counit $t \mapsto \langle 1, t \rangle$. The inclusions $\mathcal{T}^{(k')}(X) \rightarrow \mathcal{T}^{(k)}(X)$, where $k' \leq k$, are morphisms of coalgebras.

If $\varphi : X \rightarrow Y$ is a morphism of manifolds of class C^r , the maps $\mathbf{T}_x^{(k)}(\varphi) : \mathbf{T}_x^{(k)}(X) \rightarrow \mathbf{T}_{\varphi(x)}^{(k)}(Y)$ relative to the various points x of X define a linear map φ_* of $\mathcal{T}^{(k)}(X)$ to $\mathcal{T}^{(k)}(Y)$, which is a morphism of coalgebras.

If X_1 and X_2 are two manifolds of class C^r , the maps α_k^{-1} (cf. 13.4.5) define a linear map

$$\mathcal{T}^{(k)}(X_1 \times X_2) \rightarrow \mathcal{T}^{(k)}(X_1) \otimes \mathcal{T}^{(k)}(X_2)$$

which is injective; it is a morphism of coalgebras. If $k = \infty$ or ω , it is an isomorphism of coalgebras.

13.6.2. Let X be a manifold of class C^r and let V be a finite-dimensional K -vector space. An element of $\mathcal{T}^{(r)}(X) \otimes V$ is called a distribution with finite support on X with values in V . When $K = \mathbb{R}$, $V = \mathbb{C}$, such a distribution is also called a complex distribution with finite support on X . The definitions and results of the preceding numbers extend immediately, by linearity, to distributions with values in V .

13.7 Weakening of structure

We assume $K = \mathbb{R}$. Let $r' \in \mathbb{N}_K$ be such that $r' \leq r$. Let X be a manifold of class C^r , and let X' be the manifold of class $C^{r'}$ obtained by weakening of structure (5.13.1). Let $x \in X$, and let $t \in T_x^{(k)}(X)$, with $k \leq r'$. Choose a chart $c = (U, \varphi, E)$ of X centered at x , and let $\tilde{t}$ be the element of $TS^{(k)}(E)$ such that $\theta_c(\tilde{t}) = t$. Since c is a chart of X' centered at x , the element $t' = \theta_c(\tilde{t})$ of $T_x^{(k)}(X')$ is defined; it does not depend on the choice of c . The map $t \mapsto t'$ is a bijection of $T_x^{(k)}(X)$ onto $T_x^{(k)}(X')$ by which these two spaces are identified. The identifications thus obtained are compatible with the operations $\langle f, t \rangle$, $\varphi_*(t)$, $t_1 \otimes t_2$, $c(t)$, ... of the preceding numbers.

14 Differential operators

In this paragraph, X denotes a locally finite-dimensional manifold of class C^r , where $r \in N_K$; let E and F be two finite-rank vector bundles of class C^r with base X .

All manifolds and vector bundles considered are assumed to be locally of finite dimension.

14.1 Differential operators

14.1.1. Let k be an integer such that $0 \leq k \leq r$, and let $\mathbf{P}^k(E)$ be the bundle of jets of sections of order k from E (12.6.1). Set

$$\mathbf{D}^k(E, F) = \mathcal{L}(\mathbf{P}^k(E); F).$$

It is a vector bundle with base X and class C^{r-k} . A section of this bundle over an open set U of X is identified with a map

$$\mathbf{D} : P^k(E)|U \rightarrow F|U$$

which commutes with the projection onto U and is linear on each fiber; such a section is called a differential operator on U , of type $E \rightarrow F$, and of order $\leq k$. Let $h \in N_K \cup \{0\}$, with $0 \leq h \leq r - k$. We denote by $\mathcal{D}_U^{k,h}(E, F)$ the set of differential operators on U , of type $E \rightarrow F$ and of order $\leq k$, which are of class C^h (as sections of $D^k(E, F)$, or as morphisms of $P^k(E)|U$ into $F|U$, it amounts to the same thing). If $M = D^k(E, F)$, one has $\mathcal{D}_U^{k,h}(E, F) = \mathcal{S}_M^h(U)$; it is a module over $\mathcal{C}^h(U)$, cf. 7.4.1.

If $0 \leq k' \leq k$, the morphism $r^{k,k'} : P^k(E) \rightarrow P^{k'}(E)$ (cf. 12.6.6) defines an injection from $D^{k'}(E, F)$ to $D^k(E, F)$; every differential operator of order $\leq k'$ is thus identified with a differential operator of order $\leq k$. A differential operator of order $\leq k$ which is not of order $\leq k - 1$ is sometimes said to be of order k . If $h \in N_{\mathbb{K}} \cup \{0\}$ and $h \leq r - k$, and if U is an open subset of X , we have:

$$0 \subset \mathcal{D}_U^{0,h}(E, F) \subset \mathcal{D}_U^{1,h}(E, F) \subset \dots \subset \mathcal{D}_U^{k,h}(E, F).$$

If $r = \infty$ or ω , one denotes by $\mathcal{D}_U^{\infty,h}(E, F)$ the union of the $\mathcal{D}_U^{k,h}(E, F)$ for $k \geq 0$; an element of this union is called a differential operator on U of bounded order (of type $E \rightarrow F$, of class C^h) or sometimes simply a differential operator on U .

14.1.2 (“Operators of order zero”). We have $P^0(E) = E$ (12.5.1) and $D^0(E, F) = \mathcal{L}(E; F)$. If U is open in X , and if $h \in N_{\mathbb{K}} \cup \{0\}$ and $h \leq r$, an element of $\mathcal{D}_U^{0,h}(E, F)$ is a morphism of class C^h from $E|U$ to $F|U$.

14.1.3 (“Weakening of structure”). Suppose $K = \mathbb{R}$, and let $r' \in N_R$ with $r' \leq r$. Let X' , E' , and F' be the manifold and the vector bundles of class $C^{r'}$ obtained by weakening the structure from X , E , and F respectively. Let k be an integer such that $0 \leq k \leq r'$. Then $P^k(E')$ is a bundle of class $C^{r'-k}$ deduced from $P^k(E)$ by weakening the structure, and one has an analogous result for $D^k(E', F')$. In particular, if U is open in X , and if $0 \leq h \leq r - k'$, one has

$$\mathcal{D}_U^{k,h}(E', F') = \mathcal{D}_U^{k,h}(E, F) .$$

14.1.4. Let D be a differential operator on X , of type $E \rightarrow F$, of order $\leq k$ (where $0 \leq k \leq r$). Let U be an open subset of X , and let s be a section of class $\mathbf{C}^m$ of E over U , with $m \in N_K \cup \{0\}$ and $k \leq m \leq r$. Then $j^k(s)$ is a section of class $\mathbf{C}^{m-k}$ of $P^k(E)$ over U (12.6.1); its image under D is denoted $D_U(s)$, or $D(s)$. It is a section of F over U . Suppose D is of class $\mathbf{C}^h$, where $h = m - k$. Then $D_U(s)$ is of class $\mathbf{C}^h$ and the maps

$$D_U: \mathcal{S}_E^m(U) \rightarrow \mathcal{S}_F^h(U)$$

thus obtained enjoy the following properties:

- (1) D_U is K -linear.
- (2) For every $s \in \mathcal{S}_E^m(U)$ and every open set V of U , one has $D_V(s|_V) = D_U(s)|_V$.
- (3) For every $s \in \mathcal{S}_E^m(U)$ and every $x \in U$ such that $j_x^k(s) = 0$, one has $D_U(s)(x) = 0$.

(3') For every coordinate system $\xi = (\xi^1, \dots, \xi^n)$ in U and any frame $s = (s_1, \dots, s_d)$ of E over U (7.4.4), there exist sections $n_{i,\alpha}$ ($1 \leq i \leq d, \alpha \in \mathbf{N}^n, |\alpha| \leq k$) of F on U , of class $\mathbf{C}^h$ and such that, for every family $(f_i)_{1 \leq i \leq d}$ of elements of $\mathcal{C}^m(U)$, one has

$$D_U \left(\sum_{1 \leq i \leq d} f_i \cdot s_i \right) = \sum_{1 \leq i \leq d, |\alpha| \leq k} \Delta_\xi^\alpha(f_i) \cdot n_{i,\alpha},$$

- (4) For any function $f \in \mathcal{C}^m(U)$ and any mapping θ of $\mathcal{S}_U^m(E)$ to $\mathcal{S}_U^h(F)$, denote by $\text{ad}(f)\theta$ the map

$$s \mapsto f \cdot \theta(s) - \theta(f \cdot s)$$

of $\mathcal{S}_U^m(E)$ to $\mathcal{S}_U^h(F)$. Then, for every function $f \in \mathcal{C}^m(U)$, there exists a differential operator L on U , of type $E \rightarrow F$ of order $\leq k - 1$ and of class $\mathbf{C}^h$ such that

$$(\text{ad}(f)D_U)(s) = L_U(s) \quad \text{for all } s \in \mathcal{S}_E^m(U).$$

- (5) One has

$$\text{ad}(f_0) \dots \text{ad}(f_k) D_U = 0 \quad \text{for all } f_0, \dots, f_k \text{ in } \mathcal{C}^m(U).$$

If we set $I = \{0, \dots, k\}$ and $f_H = \prod_{i \in H} f_i$ for every subset H of I , the preceding relation is equivalent to:

$$\sum_{H \in I} (-1)^{\text{Card}(H)} f_H \cdot D_U(f_{I-H} \cdot s) = 0$$

for all $f_0, \dots, f_k$ in $\mathcal{C}^m(U)$ and s in $\mathcal{S}_E^m(U)$.

14.1.5 (“Characterization of differential operators”). Let k, m and $h = m - k$ be as in 14.1.4. For every open set U of X , let $\mathbf{D}_U$ be a map from $\mathcal{S}_E^m(U)$ to $\mathcal{S}_F^h(U)$. Suppose that conditions 1, 2, 3 (resp. 1, 2, 3’) of 14.1.4 are satisfied for every open subset U of X . Then there exists a unique element D of $\mathcal{D}_X^{k,h}(E, F)$ such that the $\mathbf{D}_U$ are the corresponding maps. ⁽¹⁾ In what follows, we identify D with the family of the $\mathbf{D}_U$.

When $m = \infty$ or ω , one has $h = m$ and we may replace conditions (3) and (3’) by one of the conditions (4) and (5).

14.1.6 (“Scalar operators”). Suppose that E and F are equal to the trivial bundle K_X . A differential operator of type $E \rightarrow F$ is then called a scalar differential operator or simply a differential operator; if it is of order $\leq k$, it is a section of the bundle $\mathcal{L}(P^k(X); K_X) = P^k(X)^*$, dual of the bundle $P^k(X)$; using the isomorphism

$$i^{-1}: P^k(X)^* \rightarrow T^{(k)}(X) \quad (\text{cf. 13.2.5}),$$

we see that a scalar differential operator of order $\leq k$ is identified with a section of the vector bundle $T^{(k)}(X)$, i.e. with a field of point distributions of order $\leq k$.

Take in particular for X an open set of K^n , where n is an integer ≥ 0 . The fields of point distributions

$$\Delta^\alpha: x \mapsto \Delta_x^\alpha \quad (\text{cf. 13.2.6})$$

are scalar differential operators of class C^ω . For every $h \in \mathbf{N}_K$, the Δ^α for $(|\alpha| \leq k)$ form

a basis of the $\mathcal{C}^h(\mathbf{X})$ -module $\mathcal{D}_X^{k,h}(\mathbf{K}_X, \mathbf{K}_X)$.

14.1.7 (“Complex differential operators on a real manifold”). Suppose $K = \mathbb{R}$ and that the vector bundles E and F are equipped with complex structures (8.8.1); let k be an integer such that $0 \leq k \leq r$. The bundle $P^k(E)$ is then equipped with a complex structure (12.6.3); the bundle $\mathcal{L}_C(P^k(E); F)$ (7.8.5) of complex linear maps from $P^k(E)$ to F is a vector sub-bundle of $\mathcal{L}(P^k(E); F) = D^k(E, F)$; we denote it $D_C^k(E, F)$. A section of $D_C^k(E, F)$ is called a complex differential operator, of type $E \rightarrow F$ and of order $\leq k$. If $h \in N_K$ and $h \leq r - k$, and if U is an open set of X , we denote similarly by $\mathcal{D}_U^{k,h}(E, F)_C$ the space of sections of class C^h of $D_C^k(E, F)$ on U ; it is a C -vector space. The other definitions and results of this paragraph extend analogously to complex differential operators; we leave their formulation to the reader.

14.1.8 (“Composition”). Let G be a vector bundle of class C^r with base X . Let k' and k'' be positive integers with sum $k \leq r$, and let D' (resp. D'') be a differential operator on X , of type $E \rightarrow F$ (resp. of type $F \rightarrow G$), of order $\leq k'$ (resp. $\leq k''$). Suppose $D' : P^{k'}(E) \rightarrow F$ is of class $C^{h'}$, with $h' \in N_K \cup \{0\}$ and $k'' \leq h' \leq r - k'$; the map

$$\mathbf{D}'_* = \mathbf{P}^{k''}(\mathbf{D}') : \mathbf{P}^{k''}(P^{k'}(E)) \rightarrow \mathbf{P}^{k''}(F)$$

This map is of class $C^{h'-k''}$ (12.6.3). Let β be the canonical homomorphism of $\mathbf{P}^k(E)$ to $\mathbf{P}^{k''}(P^{k'}(E))$, cf. 12.6.5. By composing the maps

$$P^k(E) \xrightarrow{\beta} P^{k''}(P^{k'}(E)) \xrightarrow{\mathbf{D}'_*} P^{k''}(F) \xrightarrow{D''} G,$$

we obtain a differential operator of type $E \rightarrow G$ and of order $\leq k$. This operator is called the composite of D'' and D' ; it is denoted by $D'' \circ D'$.

¹ It suffices, moreover, that condition (3') be satisfied for a family (ξ_λ) of coordinate systems and a family of frames (s_λ) whose domains cover X .

If U is an open subset of X , and if $s \in \mathcal{S}_E^k(U)$, one has

$$(\mathbf{D}'' \circ \mathbf{D}')(s) = \mathbf{D}''(\mathbf{D}'(s))$$

(where both sides are sections of G over U); this justifies the terminology and notation

adopted.

Suppose now $\mathbf{D}''$ is of class $\mathbf{C}^{h''}$, with $h'' \in \mathbb{N}_{\mathbb{K}} \cup \{0\}$ and $h'' \leq r - k''$. Then $\mathbf{D}'' \circ \mathbf{D}'$ is of class $\mathbf{C}^h$, with $h = \inf(h' - k'', h'')$.

Suppose we have $E = F = G$, and that $k' \leq h''$, in which case $D' \circ D''$ is defined. Then $D' \circ D'' - D'' \circ D'$ is a differential operator of type $E \rightarrow E$ and of order $\leq k - 1$; we denote it $[D', D'']$.

14.1.9 ("Associativity"). With the notations and hypotheses as in 14.1.8, let H be a vector bundle of class $\mathbf{C}^r$ and base X , and let $\mathbf{D}''$ be a differential operator on X , of type $\mathbf{G} \rightarrow \mathbf{H}$ and of order $\leq k^m$, with $k' + k'' + k'' \leq r$. We assume that $\mathbf{D}'$ is of class $\mathbf{C}^{h'}$, with $h' \in \mathbb{N}_{\mathbb{K}} \cup \{0\}$ and $k'' + k'' \leq h' \leq r - k'$, and that $\mathbf{D}''$ is of class $\mathbf{C}^{h''}$, with $h'' \in \mathbb{N}_{\mathbb{K}} \cup \{0\}$ and $k'' \leq h'' \leq r - k''$. Then the composites

$$\mathbf{D}''' \circ (\mathbf{D}'' \circ \mathbf{D}') \quad \text{and} \quad (\mathbf{D}''' \circ \mathbf{D}'') \circ \mathbf{D}'$$

are defined and are equal.

14.1.10. Examples

- a) Let U be an open subset of X , let $D \in \mathcal{D}_U^{k,h}(E, F)$ with $h \in \mathbb{N}_{\mathbb{K}} \cup \{0\}$ and $h \leq r - k$, and let $f \in \mathcal{C}^{h+k}(U)$ (resp. $f \in \mathcal{C}^h(U)$). Multiplication by f in E (resp. F) is a differential operator on U of order ≤ 0 , and of type $E \rightarrow E$ (resp. $F \rightarrow F$), cf. 14.1.2. The composites $D \circ f$ and $f \circ D$ are defined and belong to $\mathcal{D}_U^{k,h}(E, F)$; we set

$$\text{ad}(f)D = f \circ D - D \circ f \in \mathcal{D}_U^{k-1,h}(E, F),$$

cf. 14.1.4, (4). One thus endows $\mathcal{D}_U^{k,h}(E, F)$ with the structure of a $\mathcal{C}^{h+k}(U)$ -module on the right (resp. of a $\mathcal{C}^h(U)$ -module on the left). The left module structure coincides with that of 14.1.1.

- b) Suppose $r = \infty$ or $r = \omega$. The differential operators of type $E \rightarrow E$ and of class $\mathbf{C}^r$ form an associative $\mathbb{K}$ -algebra with identity element.
- c) Take for X an open subset of K^n . The scalar differential operators Δ^α (14.1.6) satisfy the

formulas

$$\Delta^\alpha \circ \Delta^\beta = ((\alpha, \beta)) \Delta^{\alpha+\beta}.$$

They commute with each other. Denote by D_i the operator Δ^{ε_i} ("i-th partial derivative"). If K has characteristic 0, we have

$$\Delta^\alpha = \frac{1}{\alpha!} D_1^{\alpha_1} \dots D_n^{\alpha_n}.$$

If K has characteristic $p \neq 0$, one has $\mathbf{D}_i^p = 0$ for all i .

14.1.11 ("Multidifferential operators"). Let $E_1, \dots, E_n$ be vector bundles of class C^r with base X , and let $k, k_1, \dots, k_n$ be integers belonging to $[0, r]$. Set

$$L(k_1, \dots, k_n) = \mathcal{L}(P^{k_1}(E_1) \otimes \dots \otimes P^{k_n}(E_n); F).$$

If $k_i \leq k$ for all i , the projections $r^k, k_i : P^k(E_i) \rightarrow P^{k_i}(E_i)$ make it possible to identify the bundle $L(k_1, \dots, k_n)$ with a subbundle of the bundle $L(k) = L(k, \dots, k)$. There exists a smallest subbundle $M(k)$ of $L(k)$ containing all the subbundles $L(k_1, \dots, k_n)$, for $k_1 + \dots + k_n = k$. A section H of $M(k)$ is called an n -differential operator (or multidifferential operator) of type $(E_1, \dots, E_n) \rightarrow F$ and of order $\leq k$. If s_i ($1 \leq i \leq n$) is a section of class C^m ($m \in \mathbb{N}_{\mathbb{K}}, m \geq k$) of E_i on an open set U of X , one defines $H_U(s_1, \dots, s_n) = H(s_1, \dots, s_n)$ as the image under H of the section $j^k(s_1) \otimes \dots \otimes j^k(s_n)$ of the bundle

$$P^k(E_1) \otimes \dots \otimes P^k(E_n).$$

It is a section of $F|U$. If H is of class C^h , this section is of class C^p with $p = \inf(m - k, h)$.

When $E_1, \dots, E_n$ and F are equal to K_x , one says that H is a scalar n -differential operator (or multidifferential operator).

Example. — Let J be a finite set, X an open subset of K^J . Let H be a scalar n -differential operator of order $\leq k$ on X . There exists a unique family of scalar functions

$$c(\alpha(1), \dots, \alpha(n))_{(\alpha(1), \dots, \alpha(n)) \in \mathbb{N}^J}, \quad \sum_{i=1}^n |\alpha(i)| \leq k$$

on X such that

$$H_X(f_1, \dots, f_n) = \sum_{\alpha(1), \dots, \alpha(n)} c(\alpha(1), \dots, \alpha(n)) \Delta^{\alpha(1)}(f_1) \dots \Delta^{\alpha(n)}(f_n)$$

for every family $(f_1, \dots, f_n)$ of functions of class $\mathbf{C}^k$ on X . For H to be of class $\mathbf{C}^h$, it is necessary and sufficient that the functions $c(\alpha(1), \dots, \alpha(n))$ be of class $\mathbf{C}^h$.

14.2 Symbols

14.2.1. Let k be a positive integer $\leq r$. Consider the exact sequence

$$0 \rightarrow P_k(T(X); E) \xrightarrow{\iota} P^k(E) \xrightarrow{j} P^{k-1}(E) \rightarrow 0$$

of no. 12.6.8, where we have set $j = r^{k, k-1}$. Applying the vector functor $M \rightarrow L(M; F)$ to this sequence, we obtain the exact sequence

$$0 \rightarrow \mathcal{L}(P^{k-1}(E); F) \xrightarrow{f'} \mathcal{L}(P^k(E); F) \xrightarrow{\sigma_k} \mathcal{L}(P_k(T(X); E); F) \rightarrow 0$$

which can be written:

by setting

$$0 \rightarrow D^{k-1}(E, F) \xrightarrow{j'} D^k(E, F) \xrightarrow{\sigma_k} S^k(E, F) \rightarrow 0$$

$$S^k(E, F) = \mathcal{L}(P_k(T(X); E); F).$$

The homomorphism $j' = \mathcal{L}(j; \text{Id}_F)$ is the canonical inclusion of $\mathbf{D}^{k-1}(E, F)$ into $\mathbf{D}^k(E, F)$; the homomorphism σ_k is by definition equal to $\mathcal{L}(\iota; \text{Id}_F)$. If D is a differential operator of type $E \rightarrow F$ and of order $\leq k$, its image under σ_k is a section $\sigma_k(\mathbf{D})$ of $S^k(E, F)$ which we call the k -symbol (or simply the symbol) of D ; we have $\sigma_k(\mathbf{D}) = 0$ if and only if D is of order $\leq k-1$.

If D is of class $\mathbf{C}^h$ so is its symbol.

14.2.2 (“Identifications of the symbol bundle”). One may write the bundle $S^k(E, F) = \mathcal{L}(P_k(T(X); E); F)$ in various ways. First, if $x \in X$, an element of $P_k(T_x(X); E_x)$ is identified (13.1.2) with an element of $\mathcal{L}(TS^k(T_x(X)); E_x)$. We thus obtain identifications of vector bundles

$$P_k(T(X); E) = \mathcal{L}(TS^k(T(X)); E) = (TS^k(T(X)))^* \otimes E$$

and, by applying the vector functor $\mathbf{M} \mapsto \mathcal{L}(\mathbf{M}; \mathbf{F}) = \mathbf{M}^* \otimes \mathbf{F}$ we obtain:

$$S^k(E, F) = TS^k(T(X)) \otimes E^* \otimes F = TS^k(T(X)) \otimes \mathcal{L}(E; F).$$

One can further transform the preceding expression. If $\mathbf{M}$ and $\mathbf{N}$ are two vector bundles, denote by $\text{Sym}^k(\mathbf{M}; \mathbf{N})$ the subbundle of $\mathcal{L}(\mathbf{M}, \dots, \mathbf{M}; \mathbf{N})$ consisting of symmetric k -linear maps from $\mathbf{M}^k$ to $\mathbf{N}$. There is a canonical isomorphism

$$\text{Sym}^k(\mathbf{M}; \mathbf{N}) \rightarrow \text{Sym}^k(\mathbf{M}; \mathbf{K}_X) \otimes \mathbf{N};$$

On the other hand, the duality between $\otimes^k \mathbf{M}$ and $\otimes^k \mathbf{M}^*$ identifies $\text{Sym}^k(\mathbf{M}; \mathbf{K}_X)$ with $TS^k(\mathbf{M}^*)$. One thus obtains an identification

$$\text{Sym}^k(\mathbf{M}; \mathbf{N}) = TS^k(\mathbf{M}^*) \otimes \mathbf{N}.$$

Applying this formula to $\mathbf{M} = T(X)^*$ and $\mathbf{N} = \mathcal{L}(E; F)$, we have:

$$S^k(E, F) = TS^k(T(X)) \otimes \mathcal{L}(E; F) = \text{Sym}^k(T(X)^*; \mathcal{L}(E; F)).$$

Suppose K has characteristic 0 or $> k$. If u is a symmetric k -linear map, denote $\tilde{u}$ the polynomial map

$$x \mapsto \frac{1}{k!} u(x, \dots, x)$$

This is the corresponding polynomial map. The map $u \mapsto \tilde{u}$ defines an isomorphism

$$S^k(E, F) = \text{Sym}^k(T(X)^*, \mathcal{L}(E; F)) \rightarrow P_k(T(X)^*; \mathcal{L}(E; F)).$$

Thus, every element σ of $S^k(E, F)_x$, with $x \in X$, can be identified with a homogeneous polynomial map $\tilde{\sigma}$ of degree k from $T_x(X)^*$ to $\mathcal{L}(E_x; F_x)$.

14.2.3 (“Computation of the symbol of a differential operator”). Let $x \in X$ and let D be a differential operator of type $E \rightarrow F$ of order $\leq k$ defined in an open neighborhood of x and of class C^{r-k} . We will now make explicit the value $\sigma_k(D)(x)$ of the symbol of D at x . Consider it as an element of $\text{Sym}^k(T_x(X)^*; \mathcal{L}(E_x; F_x))$ (cf. 14.2.2). Let $\omega_1, \dots, \omega_k$ be covectors at x , and choose functions $f_1, \dots, f_k$ of class C^r in an open neighborhood U of x such that $d_x f_i = \omega_i$ for $i = 1, \dots, k$. Set

$$\mathbf{D}' = (-1)^k \text{ad}(f_k) \dots \text{ad}(f_1) \mathbf{D} \quad (\text{cf. 14.1.10, } a).$$

The operator D is of order ≤ 0 ; it is a section of $\mathcal{L}(E; F)|_U$. The value of $\mathbf{D}'$ at x is an element $\lambda(\omega_1, \dots, \omega_k)$ of $\mathcal{L}(E_x; F_x)$ which depends only on $(\omega_1, \dots, \omega_k)$. The map λ thus defined is symmetric. It is equal to the symbol $\sigma_k(\mathbf{D})(x)$ of D at x .

Suppose $K = \mathbb{R}$ or $\mathbb{C}$, and let us make explicit $\sigma_k(\mathbf{D})(x)$ considered as a homogeneous polynomial map of degree k from $T_x(X)^*$ to $\mathcal{L}(E_x; F_x)$ (cf. 14.2.2). Let $\omega \in T_x(X)^*$ and $v \in E_x$; choose a function f of class C^r on an open neighborhood U of x such that $d_x f = \omega$, and a section s of class C^r of E over U such that $s(x) = v$. There exists a family $(\varphi_0, \varphi_1, \dots, \varphi_k)$ of sections of F over U , and only one, such that

$$e^{-tf} D(e^{tf} s) = \sum_{j=0}^k t^j \varphi_j \quad \text{for all } t \in K.$$

Then $\varphi_k(x)$ does not depend on the choice of f and s such that $d_x f = \omega$, $v(x) = s$; the map $v \mapsto \varphi_k(x)$ is an element $\lambda(\omega)$ of $\mathcal{L}(E_x; F_x)$, and λ is a homogeneous polynomial map of degree k from $T_x(X)^*$ to $\mathcal{L}(E_x; F_x)$. This map is equal to the symbol $\sigma_k(D)(x)$ of D at x .

14.2.4 (“The scalar case”). Take $\mathbf{E} = \mathbf{F} = \mathbf{K}_x$, in which case one identifies $\mathbf{D}^k(\mathbf{E}, \mathbf{F})$ with the bundle $\mathbf{T}^{(k)}(\mathbf{X})$ of point distributions of order $\leq k$, cf. 14.1.6. One has

$$S^k(E, F) = TS^k(T(X))$$

and the symbol

$$\sigma_k: T^{(k)}(X) \rightarrow TS^k(T(X))$$

is none other than the composite

$$T^{(k)}(X) \rightarrow T^{(k)}(X)/T^{(k-1)}(X) \xrightarrow{i_k} TS^k(T(X)),$$

where i_k is the isomorphism defined in 13.3.2.

Take for example X an open subset of K^n , and let $\{e_1, \dots, e_n\}$ be the canonical basis of K^n (identified with the tangent spaces $T_x(X)$). If $\alpha \in N^n$ is such that $|\alpha| \leq k$, the symbol of the operator Δ^α is given by the formulas (cf. 13.3.3):

$$\begin{aligned} \sigma_k(\Delta^\alpha)(x) &= 0 & \text{if } |\alpha| < k \\ \sigma_k(\Delta^\alpha)(x) &= \gamma_{\alpha_1}(e_1) \dots \gamma_{\alpha_n}(e_n) & \text{if } |\alpha| = k, \end{aligned}$$

the product of the $\gamma_{\alpha_i}(e_i)$ being the symmetric product, cf. 13.2.6 and 13.3.3.

14.2.5 ("Symbol of a composite"). The notations and hypotheses are those of 14.1.8. The composition of linear maps defines a pairing

$$\mathcal{L}(F; G) \times \mathcal{L}(E; F) \rightarrow \mathcal{L}(E; G).$$

On the other hand, the operation of symmetric product defines a pairing

$$TS^{k''}(T(X)) \times TS^{k'}(T(X)) \rightarrow TS^k(T(X)).$$

Since one has

$$\left. \begin{aligned} S^{k''}(F, G) &= TS^{k''}(T(X)) \otimes \mathcal{L}(F; G) \\ S^{k'}(E, F) &= TS^{k'}(T(X)) \otimes \mathcal{L}(E; F) \\ S^k(E, G) &= TS^k(T(X)) \otimes \mathcal{L}(E; G) \end{aligned} \right\} \quad \text{cf. 14.2.2,}$$

one deduces from this, by tensor product, a pairing

$$S^{k''}(F, G) \times S^{k'}(E, F) \rightarrow S^k(E, G). \quad (*)$$

One then has the formula

$$\sigma_k(\mathbf{D}'' \circ \mathbf{D}') = \sigma_{k''}(\mathbf{D}'') \cdot \sigma_{k'}(\mathbf{D}'),$$

where the product appearing in the right-hand term is defined by the pairing (*) above.

Now suppose K is of characteristic zero, and let $x \in X$. Denote by σ (resp. σ' , σ'') the k -symbol (resp. the k' -symbol, the k'' -symbol) of $D'' \circ D'$ (resp. D' , D'') at x . For every $\omega \in \mathbf{T}_x(\mathbf{X})^*$, one then has (with the notations of the end of 14.2.2):

$$\tilde{\sigma}(\omega) \in \mathcal{L}(E_x; G_x), \quad \tilde{\sigma}'(\omega) \in \mathcal{L}(E_x; F_x), \quad \tilde{\sigma}''(\omega) \in \mathcal{L}(F_x; G_x)$$

and

$$\tilde{\sigma}(\omega) = \tilde{\sigma}''(\omega) \circ \tilde{\sigma}'(\omega).$$

14.3 Transposition

In this number, one assumes that X is purely of dimension n .

14.3.1. Let $\Omega = \det(\mathbf{T}(\mathbf{X})^*)$, cf. 7.9.9; this is a vector bundle of rank 1 at each point, with base X , and of class $\mathbf{C}^{r-1}$. One has

$$\Omega = \bigwedge^n \mathbf{T}(\mathbf{X})^* = \text{Alt}^n(\mathbf{T}(\mathbf{X}); K_X).$$

Let M be a vector bundle with base X . One sets

$$\tilde{\mathbf{M}} = \mathcal{L}(\mathbf{M}; \Omega) = \mathbf{M}^* \otimes \Omega. \quad (1)$$

If $\mathbf{M}$ is of class C^h , with $h \in N_K \cup \{0\}$ and $h \leq r - 1$, then so is $\tilde{\mathbf{M}}$. The canonical map from $\mathbf{M}$ to $\tilde{\tilde{\mathbf{M}}} = \mathcal{L}(\mathcal{L}(\mathbf{M}; \Omega); \Omega)$ is an isomorphism; it is used to identify $\mathbf{M}$ with $\tilde{\tilde{\mathbf{M}}}$.

14.3.2 (“Definition of the transpose”). Let k be a positive integer $\leq r - 1$, and let $\mathbf{D}$ be a differential operator on X , of type $E \rightarrow F$, of order $\leq k$, and of class C^h , with $h \in N_K \cup \{0\}$ and $k \leq h \leq r - k$. There then exists a differential operator ${}^t\mathbf{D}$ on X , of type $\tilde{F} \rightarrow \tilde{E}$ and of order $\leq k$, having the following property:

Let $\xi = (\xi^1, \dots, \xi^n)$ be a coordinate system in an open set U of X and let $s = (s_i)_{1 \leq i \leq e}$ (resp. $t = (t_j)_{1 \leq j \leq f}$) be a frame of E (resp. of F) on U . Let (s_i^*) (resp. (t_j^*)) be the frame of E^* (resp. of F^*) such that $\langle s_i, s_j^* \rangle = \delta_{ij}$ (resp. $\langle t_i, t_j^* \rangle = \delta_{ij}$), and let $(\tilde{s}_i)$ (resp. $(\tilde{t}_j)$) be the frame of $\tilde{E}$ (resp. $\tilde{F}$) on U obtained by forming the tensor product of (s_i^*) (resp. of (t_j^*)) with the frame $\omega = d\xi^1 \wedge \dots \wedge d\xi^n$ of Ω . Let c_α^{ij} ($1 \leq i \leq e$, $1 \leq j \leq f$, $|\alpha| \leq k$) be the functions of class C^h on U such that one has

$$\mathbf{D} \left(\sum_i f_i s_i \right) = \sum_{i,j,\alpha} c_\alpha^{ij} \Delta_\xi^\alpha (f_i) \cdot t_j$$

for any functions f_i in $\mathcal{C}^{h+k}(U)$, cf. 14.1.4, (3'). One then has

$${}^t\mathbf{D} \left(\sum_j g_j \tilde{t}_j \right) = \sum_{i,j,\alpha} (-1)^{|\alpha|} \Delta_\xi^\alpha (c_\alpha^{ij} g_j) \tilde{s}_i$$

for any functions g_j in $\mathcal{C}^{h+k}(U)$.

The preceding property (which must be verified for any ξ , s and t) determines ${}^t\mathbf{D}$ uniquely.

(1) The operator ${}^t\mathbf{D}$ is called the transpose of $\mathbf{D}$; it is of class C^{h-k} . If $h \geq 2k$, the transpose of ${}^t\mathbf{D}$ is defined and equal to $\mathbf{D}$ (taking into account the identifications of $\tilde{E}$ and $\tilde{F}$ with E and F respectively).

The map $\mathbf{D} \mapsto {}^t\mathbf{D}$ is K -linear. If $k = 0$, i.e., if $\mathbf{D}$ is a morphism of E to F , ${}^t\mathbf{D}$ is the morphism $\mathcal{L}(\mathbf{D}; \text{Id}_\Omega)$ of $\tilde{F}$ to $\tilde{E}$.

The symbol of tD is deduced from the symbol of D by means of the isomorphism

$$\mathrm{TS}^k(\mathbf{T}(\mathbf{X})) \otimes \mathcal{L}(\mathbf{E}; \mathbf{F}) \rightarrow \mathrm{TS}^k(\mathbf{T}(\mathbf{X})) \otimes \mathcal{L}(\tilde{\mathbf{F}}; \tilde{\mathbf{E}})$$

which is the tensor product of the automorphism $(-1)^k$ of $\mathrm{TS}^k(\mathbf{T}(\mathbf{X}))$ and the isomorphism $u \mapsto \mathcal{L}(u; \mathrm{Id}_\Omega)$ of $\mathcal{L}(\mathbf{E}; \mathbf{F})$ onto $\mathcal{L}(\tilde{\mathbf{F}}; \tilde{\mathbf{E}})$.

14.3.3 (“Transpose of a composite”). The notations and hypotheses being those of 14.1.8, suppose that D' is of class $C^{h'}$ and D'' of class $C^{h''}$, with $h', h'' \in \mathbf{N}_\mathbf{K} \cup \{0\}$, $h' \geq k' + 2k''$ and $h'' \geq k'' + 2k'$.

Then the transposes of $\cdot$, $\cdot$, and $\circ$ are defined, as is the composite of ${}^t\mathbf{D}'$ and ${}^t\mathbf{D}''$, and we have

$${}^t(\mathbf{D}'' \circ \mathbf{D}') = {}^t\mathbf{D}' \circ {}^t\mathbf{D}''.$$

In particular, the hypotheses and notations being those of 14.3.2, and f being a function of class $C^{h'}$ on X with $2k \leq h'$, one has ${}^t(\mathbf{D} \circ f) = f \circ {}^t\mathbf{D}$.

Example. — Take for X an open subset of K^n ; denote by $\xi^1, \dots, \xi^n$ the coordinate functions on X and identify Ω with K_x by means of the frame $\omega = d\xi^1 \wedge \dots \wedge d\xi^n$. If $E = F = K_x$, one has

$$\tilde{\mathbf{E}} = \tilde{\mathbf{F}} = \mathcal{L}(K_x; \Omega) = \mathcal{L}(K_x; K_x) = K_x,$$

and the operation $\mathbf{D} \mapsto {}^t\mathbf{D}$ transforms scalar differential operators into scalar differential operators. One has, for example

$${}^t(\Delta^\alpha) = (-1)^{|\alpha|} \Delta^\alpha \quad \text{for all } \alpha \in \mathbf{N}^n.$$

If r is infinite, transposition is an anti-automorphism of the algebra $\mathcal{D}_\mathbf{x}^{\infty, r}(K_\mathbf{x}, K_\mathbf{x})$.

14.3.4. Let k be a positive integer such that K has characteristic 0 or $> k$. Let Ω_1 be the vector bundle $\bigwedge^{n-1} T(X)^* = \mathrm{Alt}^{n-1}(T(X); K_X)$. The notations and hypotheses being those of 14.3.2, let moreover D' be a differential operator on X , of type $\tilde{F} \rightarrow \tilde{E}$ and of order $\leq k$. We call any multidifferential operator G (cf. 14.1.11) of type $(E, \tilde{F}) \rightarrow \Omega_1$, of order $\leq k-1$ ⁽²⁾ and of class C^h (with $h \geq 1$), such that one has the identity

$$\langle D(u), v \rangle - \langle u, D'(v) \rangle = d(G(u, v)) \quad (1)$$

for every open set U of X and all elements $u \in \mathcal{S}_E^k(U)$, $v \in \mathcal{S}_F^k(U)$. Let us specify that, in this formula, d denotes exterior differentiation (8.3.5) and $\langle D(u), v \rangle$ (resp.

$\langle u, D'(v) \rangle$) denote the sections of Ω over U obtained from the pair $(D(u), v)$ (resp. the pair $(u, D'(v))$) by the canonical pairing of $F \times \tilde{F}$ (resp. of $E \times \tilde{E}$) into Ω . The existence of G implies $D' = {}^t D$; one also says that G is a Green operator for D .

14.3.5. Let D be a differential operator on X , of type $E \rightarrow F$, of order $\leq k$, of class $\mathbf{C}^h$ with $h \in N_K \cup \{0\}$ and $k \leq h \leq r - k$. There exists a Green operator of class $\mathbf{C}^{h-k+1}$ for D in each of the following cases:

- a) $\mathbf{K} = \mathbf{R}$, $r = \infty$ and X is paracompact.
- b) $\mathbf{K}$ is of characteristic 0 or $> k$, X is isomorphic to an open subset of a space K^n , and the bundles E and F are isomorphic to trivial bundles.
- c) The operator D is of order ≤ 1 (cf. 14.3.7).

14.3.6. We keep the hypotheses and notations of 14.3.3. If G' (resp. G'') is a Green operator for D' (resp. D''), there exists a Green operator H for $D'' \circ D'$ and only one such that one has

$$\mathbf{H}(u, v) = \mathbf{G}''(D'(u), v) + \mathbf{G}'(u, {}^t D''(v))$$

for every open set U of X and all elements $u \in \mathcal{S}_E^k(U)$ and $v \in \mathcal{S}_G^k(U)$.

14.3.7. We have $\mathbf{S}^1(E, F) = \mathbf{T}(X) \otimes E^* \otimes F$ (cf. 14.2.2). On the other hand, the right interior product $(\xi, \omega) \mapsto i(\xi)\omega$ (cf. A, III, p. 158) defines an isomorphism of $\mathbf{T}(X) \otimes \Omega$ onto Ω_1 ; by tensor product with the dual Ω^* of Ω , one thus obtains an isomorphism of $\mathbf{T}(X)$ onto $\Omega^* \otimes \Omega_1$, whence identifications of vector bundles

$$\begin{aligned} \mathbf{S}^1(E, F) &= \Omega^* \otimes \Omega_1 \otimes E^* \otimes F = (E \otimes F^* \otimes \Omega)^* \otimes \Omega_1 \\ &= (E \otimes \tilde{F})^* \otimes \Omega_1 = \mathcal{L}(E \otimes \tilde{F}; \Omega_1). \end{aligned}$$

That said, let D be a differential operator on X , of type $E \rightarrow F$, of order ≤ 1 , of class $\mathbf{C}^h$

with $h \in N_K \cup \{0\}$ and $h \leq r - 1$. There exists a unique Green operator for D ; it is of order 0; it is a morphism of class C^h from $E \otimes \tilde{F}$ to Ω_1 , and it is deduced from the symbol of D by the preceding identification.

14.3.8. Suppose that $K = \mathbb{R}$, that X is separated and oriented (10.2.4), and let A be a closed piece (11.1.2) of X . Endow A with the orientation induced by that of X , and ∂A with the corresponding orientation (11.2.1). Let k be an integer such that $2k \leq r$, D a differential operator on X , of type $E \rightarrow F$, of order $\leq k$ and of class C^h . Let G be a Green operator for D . Let U be an open set of X , $u \in \mathcal{S}_E^k(U)$, $v \in \mathcal{S}_F^k(U)$ and suppose that the supports of u and of v meet A in a compact set. One then has

$$\int_A \langle D(u), v \rangle - \int_A \langle u, {}^t D(v) \rangle = \int_{\partial A} G(u, v)$$

("Green's formula"). In particular, if $\partial A = \emptyset$, one has

$$\int_A \langle D(u), v \rangle = \int_A \langle u, {}^t D(v) \rangle.$$

14.4 Examples

In this number, we assume X is pure of finite dimension n . We also assume it is endowed with a manifold structure of class C^{r+1} compatible with the given structure of class C^r .

14.4.1 ("Infinitesimal transformations"). Let τ be a finite-dimensional vector functor for isomorphisms, of class C^r ; we assume that, for every finite-dimensional vector space V , $\tau(V)$ is of finite dimension. We denote by E_τ the vector bundle $\tau(T(X))$; it is of class C^r .

Let ξ be a vector field of class C^r on X . There exists a differential operator D of type $E_\tau \rightarrow E_\tau$ and of order ≤ 1 such that

$$D_U(s) = \theta_{\xi} . s$$

for every open set U of X and every $s \in \mathcal{S}_{E_\tau}^r(U)$, cf. 8.4.3. Such an operator D is unique; we denote it by $(\theta_{\xi})_\tau$ or simply θ_{ξ} . Its symbol is the section $\xi \otimes \text{Id}_{E_\tau}$ of the vector bundle

$$S^1(E_\tau, E_\tau) = T(X) \otimes \mathcal{L}(E_\tau; E_\tau).$$

If V is a finite-dimensional K -vector space, set

$$\tilde{\tau}(V) = \mathcal{L}(\tau(V); \text{Alt}^n(V; K)) = \tau(V)^* \otimes \bigwedge^n V^*,$$

and if $u: V_1 \rightarrow V_2$ is an isomorphism, define $\tilde{\tau}(u): \tilde{\tau}(V_1) \rightarrow \tilde{\tau}(V_2)$ by transport of structure. One thus obtains a vector functor in finite dimension $\tilde{\tau}$. The bundle $E_{\tilde{\tau}} = \tilde{\tau}(T(X))$ is identified in an obvious manner with the bundle $(E_\tau)^\sim$; the transpose of $(\theta_\xi)_\tau$ is $-(\theta_\xi)_{\tilde{\tau}}$ and formula (1) of no. 14.3.4 takes the form

$$\langle \theta_\xi . u, v \rangle + \langle u, \theta_\xi . v \rangle = d(i(\xi)(u.v)).$$

14.4.2 (“Exterior differential”). For every integer $p \geq 0$ one sets

$$\Omega^p = \text{Alt}^p(T(X); K_X).$$

There exists a differential operator D of type $\Omega^p \rightarrow \Omega^{p+1}$ and of order ≤ 1 such that

$$D_U(\omega) = d\omega$$

for every open set U of X and every $\omega \in \mathcal{S}_{\Omega^p}^r(U)$. Such an operator D is unique; it is denoted d (or d_p if one wishes to specify the integer p).

Its symbol is the element of $S^1(\Omega^p, \Omega^{p+1})$ obtained in the following manner: one first has

$$\begin{aligned} S^1(\Omega^p, \Omega^{p+1}) &= T(X) \otimes \mathcal{L}(\Omega^p; \Omega^{p+1}) \\ &= \mathcal{L}(\Omega^1; \mathcal{L}(\Omega^p; \Omega^{p+1})) = \mathcal{L}(\Omega^1, \Omega^p; \Omega^{p+1}), \end{aligned}$$

where $\mathcal{L}(\Omega^1, \Omega^p; \Omega^{p+1})$ denotes the bundle of bilinear maps from $\Omega^1 \times \Omega^p$ to Ω^{p+1} . Now the exterior product $(\alpha, \beta) \mapsto \alpha \wedge \beta$ defines a canonical section of this latter bundle; this section is the symbol of d_p .

To determine the transpose of d_p , one first notes that the exterior product defines a pair-

ing $\Omega^p \times \Omega^{n-p} \rightarrow \Omega^n = \Omega$ which allows one to identify $(\Omega^p)^\sim$ with Ω^{n-p} . One identifies in the same manner $(\Omega^{p+1})^\sim$ with Ω^{n-p-1} . With these conventions, the transpose of d_p is $(-1)^{p+1} d_{n-p-1}$ and the corresponding Green operator, of type $(\Omega^p, \Omega^{n-p-1}) \rightarrow \Omega_1 = \Omega^{n-1}$, is simply the exterior product (on each fiber).

14.4.3 (“Laplacian”). One takes $K = \mathbb{R}$, $r = \infty$, $X = \mathbb{R}^n$, $E = F = K_X$; one denotes ξ^i ($1 \leq i \leq n$) the coordinate functions on X ; one sets $D_i = \partial/\partial \xi^i$, and $L = \sum_{i=1}^n D_i^2$. The operator L is a scalar differential operator of order ≤ 2 . If one identifies Ω with K_X by means of the frame $\omega = d\xi^1 \wedge \cdots \wedge d\xi^n$, one has $\tilde{E} = \tilde{F} = K_X$ (cf. 14.3.3, example) and ${}^t L = L$.

For $1 \leq i \leq n$, set $\omega_i = (-1)^{i-1} d\xi^1 \wedge \cdots \wedge d\xi^{i-1} \wedge d\xi^{i+1} \wedge \cdots \wedge d\xi^n$. The ω_i form a frame of $\Omega_1 = \Omega^{n-1}$. If $f \in \mathcal{C}^1(X)$, denote by $\text{grad}(f)$ the section $\sum_{i=1}^n D_i(f) \omega_i$ of Ω_1 . There exists a Green operator G for L such that one has

$$\mathbf{G}(u, v) = v \cdot \text{grad}(u) - u \cdot \text{grad}(v) \text{ for every open set } U \text{ of } X \text{ and } u, v \in \mathcal{C}^1(U).$$

In particular, let A be a compact piece of $\mathbb{R}^n$. Orient $\mathbb{R}^n$, endow A with the orientation induced by that of $\mathbb{R}^n$ and ∂A with the corresponding orientation (11.2.1). One has

$$\int_A (\mathbf{L}u) \cdot v \omega - \int_A u \cdot (\mathbf{L}v) \omega = \int_{\partial A} (v \cdot \text{grad}(u) - u \cdot \text{grad}(v))$$

for u, v in $\mathcal{C}^1(X)$.

15 Manifolds of maps ¹

In this section, one assumes $K = \mathbb{R}$ or $\mathbb{C}$. The letter r denotes an integer ≥ 1 and the letter s an element of N_K such that $s \geq r$; we have $s = \omega$ if $K = \mathbb{C}$.

15.1 Functions of class $C^{r,s}$

In this no., E denotes a real Banach space, F_1 and F_2 Banach spaces over K , V an open subset of $E \times F_1$ and f a map from V to F_2 . One endows E , F_1 and F_2 with norms compatible with their Banach space structures.

15.1.1 (“Differentiable case”). One assumes $K = \mathbb{R}$ and $s \neq \omega$. We say that f is of class $C_R^{r,s}$ (or simply $C^{r,s}$) if it possesses iterated partial derivatives $D_E^p D_{F_1}^q f$ (1.7.2) continuous for

every pair of positive integers (p, q) such that $p \leq r$ and $p + q \leq s$. It is equivalent to say that the iterated partial derivatives $D_{F_1}^q f$ exist and are of class C^r for every integer q such that $0 \leq q \leq s - r$. If $s = r$, this is equivalent to saying that f is of class C^r ; if $s \geq r + 1$, this is equivalent to saying that f is of class C^r and $D_{F_1} f$ of class $C^{r, s-1}$.

If f is of class $C^{r, s}$ and if $p \leq r, p + q + 1 \leq s$, the function $D_E^p D_{F_1}^q f$ has a partial derivative with respect to F_1 given by:

$$\mathbf{D}_{F_1}(\mathbf{D}_E^p \mathbf{D}_{F_1}^q f) = \mathbf{D}_E^p \mathbf{D}_{F_1}^{q+1} f. \quad (1)$$

15.1.2 (“Analytic case”). We assume $s = \omega$. For every integer $k \geq 0$, denote by $\mathbf{P}_k(\mathbf{F}_1; \mathbf{F}_2)$ the space of continuous homogeneous polynomials of degree k on $\mathbf{F}_1$ with values in $\mathbf{F}_2$, equipped with the norm defined in the Appendix of the first fascicle, p. 88, A.2.

One says that f is of class $\mathbf{C}_K^{r, \omega}$ (or simply $C^{r, \omega}$ if there is no ambiguity about K) if, for every point $(x_0, y_0) \in \mathbf{V}$, there exist a neighborhood U of x_0 in E , real numbers δ and η , and functions $h_k: U \rightarrow \mathbf{P}_k(\mathbf{F}_1; \mathbf{F}_2)$ ($k \in \mathbb{N}$), of class C^r (for the differentiable-manifold structure underlying the structure of the K -manifold $\mathbf{P}_k(\mathbf{F}_1; \mathbf{F}_2)$), such that:

- (i) $U \times (y_0 + \mathbf{B}(\mathbf{R})) \subset \mathbf{V}$, where $\mathbf{B}(\mathbf{R})$ is the open ball of radius $\mathbf{R}$ in $\mathbf{F}_1$;
- (ii) if $x \in U$ and $p \leq r$, one has $\sum_{k=0}^{\infty} \|D^p h_k(x)\| R^k \leq M$;
- (iii) if $x \in U$ and $t \in \mathbf{B}(\mathbb{R})$, one has $\sum_{k=0}^{\infty} h_k(x)(t) = f(x, y_0 + t)$.

These properties imply:

- (iv) For every $t \in \mathbf{B}(\mathbf{R})$, the function $x \mapsto f(x, y_0 + t)$ is of class C^r in U , and one has

$$\mathbf{D}_E^p f(x, y_0 + t) = \sum_{k=0}^{\infty} \mathbf{D}_E^p h_k(x)(t) \quad \text{if } p \leq r \text{ and } x \in U. \quad (2)$$

One does not change the definition of functions of class $C^{r, \omega}$ if one replaces condition (ii) above by one of the following two conditions:

(ii') The h_k define a morphism of class C^r from U into the real Banach space underlying the K -Banach space $\mathcal{H}_{\mathbb{R}}(\mathbf{F}_1; \mathbf{F}_2)$ (3.1.1 and 3.1.2).

(ii'') The h_k define a morphism of class C' from U into the real Banach space underlying the K -Banach space $\mathcal{H}_R(F_1; F_2)$ (3.1.5).

15.1.3. Suppose $K = \mathbb{R}$. For f to be of class $C_R^{r,\omega}$, it is necessary and sufficient that f be of class $C^{r,\infty}$ (15.1.1) and that, for every $(x_0, y_0) \in V$, there exists a neighborhood V_0 of (x_0, y_0) in V and positive real numbers A and M such that

$$\frac{1}{q!} \|D_E^p D_{F_1}^q f(x, y)\| \leq A.M^q \quad (3)$$

for all $p \leq r$, every $q \geq 0$ and every $(x, y) \in V_0$.

15.1.4. Suppose $K = \mathbb{C}$. For f to be of class $C_C^{r,\omega}$, it is necessary and sufficient that the following two conditions be satisfied:

- a) f is of class $C_R^{r,\omega}$ for the real structures underlying F_1 and F_2 .
- b) For every $(x, y) \in V$, $D_{F_1} f(x, y)$ is a $\mathbb{C}$ -linear map from F_1 to F_2 .

When F_1 is of finite dimension, one may replace a) by:

a') f is of class C^r .

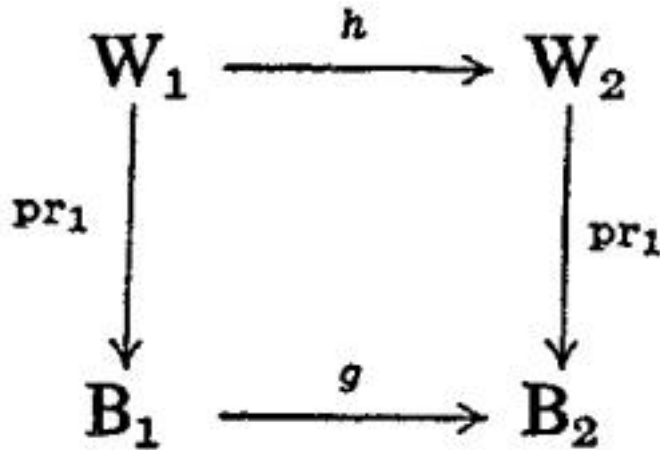
15.1.5. The set of maps of class $C^{r,s}$ from V into F_2 does not depend on the choice of norms on E, F_1, F_2 .

15.1.6. Let B be a differentiable manifold of class C^r , let W be an open subset of $B \times F_1$, and let ρ be a map from W into F_2 . Let $c = (U, \varphi, E_c)$ be a chart of the manifold B . Denote by W_c the open subset of $E_c \times F_1$ that is the image under $\varphi \times Id_{F_1}$ of $W \cap (U \times F_1)$ and let ρ_c be the map from W_c to F_2 such that

$$\rho_c(\varphi(x), y) = \rho(x, y) \quad \text{if} \quad (x, y) \in W \cap (U \times F_1).$$

The map $\rho: W \rightarrow F_2$ is said to be of class $C^{r,s}$ if, for every chart c of the manifold B , the map $\rho_c: W_c \rightarrow F_2$ is of class $C^{r,s}$ in the sense of 15.1.1 and 15.1.2; it suffices that this condition be satisfied for a family of charts whose domains cover B . Such a map ρ is of class C^r (for the real structures underlying $B \times F_1$ and F_2); its restriction to the set $W \cap (\{b\} \times F_1)$ is a K -morphism of class C^s for every $b \in B$.

15.1.7. Let $\mathbf{B}_1$ and $\mathbf{B}_2$ be differentiable manifolds of class C^r , and let $g : \mathbf{B}_1 \rightarrow \mathbf{B}_2$ be a



morphism of class C^r . Let $\mathbf{W}_1$ (resp. $\mathbf{W}_2$) be an open subset of $\mathbf{B}_1 \times \mathbf{F}_1$ (resp. of $\mathbf{B}_2 \times \mathbf{F}_2$) and let $h : \mathbf{W}_1 \rightarrow \mathbf{W}_2$ be a map. We say that h is a g -morphism of class $C^{r,s}$ if $\text{pr}_1 \circ h = g \circ \text{pr}_1$ and if $\text{pr}_2 \circ h$ is a map of class $C^{r,s}$ (15.1.6) from $\mathbf{W}_1$ to $\mathbf{F}_2$.

Similarly, let $\mathbf{B}_3$ be a manifold of class C^r , let $\mathbf{W}_3$ be an open subset of $\mathbf{B}_3 \times \mathbf{F}_3$, where $\mathbf{F}_3$ is a Banach K-space, and let g' be a morphism of class C^r from $\mathbf{B}_2$ to $\mathbf{B}_3$. If h (resp.

h') is a g -morphism (resp. g' -morphism) of class $C^{r,s}$ of $\mathbf{W}_1$ to $\mathbf{W}_2$ (resp. of $\mathbf{W}_2$ to $\mathbf{W}_3$), the composite $h' \circ h$ is a $(g' \circ g)$ -morphism of class $C^{r,s}$ of $\mathbf{W}_1$ to $\mathbf{W}_3$.

15.2 Manifold of class $C^{r,s}$ over a manifold of class C^r

Throughout this section, we denote by $\mathbf{B}$ a differentiable manifold of class C^r .

15.2.1. Let $\mathbf{X}$ be a set equipped with a map $p : \mathbf{X} \rightarrow \mathbf{B}$. We call a chart of $\mathbf{X}$ over $\mathbf{B}$ a triple $\theta = (\mathbf{V}, \psi, \mathbf{F})$, where $\mathbf{V}$ is a subset of $\mathbf{X}$, $\mathbf{F}$ is a Banach space over $\mathbf{K}$, and ψ is a bijection from $\mathbf{V}$ onto an open subset of $\mathbf{B} \times \mathbf{F}$ such that $\text{pr}_1 \circ \psi = p|_{\mathbf{V}}$.

Let $(\theta_i)_{i \in I} = ((V_i, \psi_i, F_i))_{i \in I}$ be a family of charts of $\mathbf{X}$ over $\mathbf{B}$. We say that the θ_i are $C^{r,s}$ -compatible if, for every pair of elements i, j of I , the following two conditions are satisfied:

- a) $\psi_i(V_i \cap V_j)$ is open in $\mathbf{B} \times F_i$ and $\psi_j(V_i \cap V_j)$ is open in $\mathbf{B} \times F_j$;
- b) the map $\psi_i \circ \psi_j^{-1}$ from $\psi_j(V_i \cap V_j)$ to $\psi_i(V_i \cap V_j)$ is an $\text{Id}_{\mathbf{B}}$ -morphism of class $C^{r,s}$ (15.1.7).

A family $((V_i, \psi_i, F_i))_{i \in I}$ of charts of $\mathbf{X}$ over $\mathbf{B}$ that are $C^{r,s}$ -compatible and whose domains

V_i cover X is called a $C^{r,s}$ -atlas of X (over B). Two $C^{r,s}$ -atlases are said to be equivalent if their union is again a $C^{r,s}$ -atlas; the relation of equivalence between $C^{r,s}$ -atlases is an equivalence relation. An equivalence class of $C^{r,s}$ -atlases is called a manifold structure of class $C^{r,s}$ over B on the set X ; when one wishes to specify K , one writes $C_{\mathbb{K}}^{r,s}$ instead of $C^{r,s}$. If X is a manifold of class $C^{r,s}$ over B , we say that $p: X \rightarrow B$ is the projection of X ; if θ is a chart of X over B , we say that θ is a chart of the $C^{r,s}$ -manifold X if θ belongs to an atlas of the equivalence class defining the structure of X .

Let X be a manifold of class $C^{r,s}$ over B , with projection p . There exists on X

a differentiable-manifold structure of class C^r , and only one, such that, for every chart (V, ψ, F) of the $C^{r,s}$ -manifold X , V is open in X and ψ is a C^r -isomorphism of the open submanifold V of X onto the open submanifold $\psi(V)$ of $B \times F$. If one equips X with this structure, the map $p: X \rightarrow B$ is a submersion of class C^r .

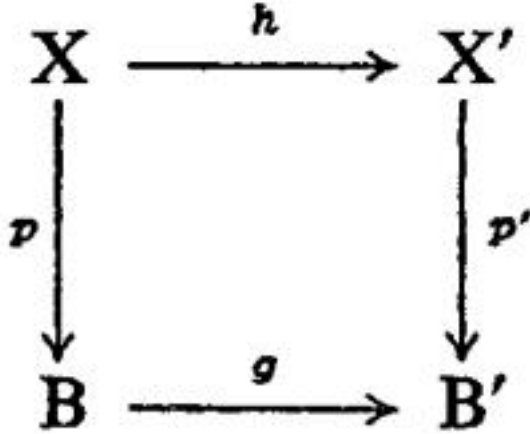
Let $b \in B$ and let $X_b = p^{-1}(b)$. There exists a K -manifold structure of class C^s on X_b , and only one, such that, for every chart $\theta = (V, \psi, F)$ of the $C^{r,s}$ -manifold X , the triple $\theta_b = (V \cap X_b, \text{pr}_2 \circ \psi|_{(V \cap X_b)}, F)$ is a chart of the manifold X_b . The differentiable-manifold structure of class C^r underlying this structure is induced by the differentiable-manifold structure of class C^r on X defined above. 15.2.3. Examples

- (i) When B is reduced to a point, the notion of a manifold of class $C_{\mathbb{K}}^{r,s}$ over B is equivalent to that of a K -manifold of class C^s .
- (ii) Let Z be a K -manifold of class C^s ; take $X = B \times Z$ and $p = \text{pr}_1$. There exists on X a structure of manifold of class $C^{r,s}$ over B , and only one such structure, such that, for every chart (U, ψ, L) of Z , $(B \times U, \text{Id}_B \times \psi, L)$ is a chart of the $C^{r,s}$ -manifold X .
- (iii) Let X be a manifold of class $C^{r,s}$ over B , and let Y be an open subset of X . The structure of manifold of class $C^{r,s}$ of Y induced by that of X is defined as in the case of manifolds (cf. 5.2.3).
- (iv) Let M be a vector bundle over K with base B and of class C^r (7.3.4), and let π be its projection onto B . There exists on M a structure of manifold of class $C_{\mathbb{K}}^{r,\omega}$ over B , and only one, such that, if (U, ψ, F) is a K -vector chart of M (8.8.1), the triple $(\pi^{-1}(U), \psi, F)$

is a chart of the $C_K^{r,\omega}$ -manifold M .

15.2.4. Let X be a manifold of class $C^{r,s}$ over B , p its projection, L a K -Banach space, and f a mapping of X into L . We say that f is of class $C^{r,s}$ if, for every chart (V, ψ, F) of the $C^{r,s}$ -manifold X , the mapping $(f|V) \circ \psi^{-1}$ of $\psi(V)$ into L is of class $C^{r,s}$ (15.1.6). Such a mapping is of class C^r and its restriction to X_b for $b \in B$, is of class C^s .

15.2.5. Let $g : B \rightarrow B'$ be a morphism of differentiable manifolds of class C^r and let



X (resp. X') be a manifold of class $C^{r,s}$ over B (resp. B'), and let h be a mapping from X into X' . We say that h is a g -morphism of class $C^{r,s}$ if the following three conditions are satisfied:

- a) $p' \circ h = g \circ p$;
- b) h is continuous;
- c) for every chart (V', ψ', F') of the $C^{r,s}$ -manifold X' , the map $pr_2 \circ \psi' \circ h$ of $V = h^{-1}(V')$ to F' is of

class $C^{r,s}$ (in the sense of 15.2.4) when one endows the open set V with the structure induced by that of X (15.2.3, (iii)).

When $B = B'$ and $g = \text{Id}_B$, we also say that h is a B -morphism of class $C^{r,s}$.

For a B -morphism of class $C^{r,s}$ to be a B -isomorphism of class $C^{r,s}$, it suffices that it be an isomorphism of class C^1 .

Let $g' : B' \rightarrow B''$ be a morphism of manifolds of class C^r and let X'' be a manifold of class $C^{r,s}$ over B'' . If $h : X \rightarrow X'$ is a g -morphism of class $C^{r,s}$ and $h' : X' \rightarrow X''$ is a g' -morphism of class $C^{r,s}$, then $h' \circ h$ is a $(g' \circ g)$ -morphism of class $C^{r,s}$.

15.2.6. Let X be a manifold of class $\mathbf{C}_R^{r,s}$ over B and p its projection; we assume $s \neq \omega$, X is separated, and p is proper (TG, I, § 10, no. 1). Then, for every $b \in \mathbf{B}$ there exists an open neighborhood U of b and a $\mathbf{C}^{r,s}$ -isomorphism from $p^{-1}(U)$ onto $U \times \mathbf{X}_b$, where $\mathbf{X}_b = p^{-1}(b)$.

15.2.7. Let X be a manifold of class $\mathbf{C}_K^{r,s}$ over B , p its projection, and $\sigma : B \rightarrow X$ be a section of class $\mathbf{C}^r$. We call a tubular neighborhood of class $\mathbf{C}_K^{r,s}$ of σ a triple (M, N, φ) where M is a K -vector bundle, with base B and of class $\mathbf{C}^r$, N is an open neighborhood of $\sigma(\mathbf{B})$ in X , and φ is a B -isomorphism of class $\mathbf{C}_K^{r,s}$ from N onto an open subset of M (cf. 15.2.3 (iv)), mapping the section σ to the zero section of M .

Suppose that $s \geq r + 1$ and that the manifold B is paracompact and admits partitions of unity of class $\mathbf{C}^r$ (5.3.6); there then exist tubular neighborhoods of class $\mathbf{C}_K^{r,s}$ of σ .

15.3 Varieties of sections and varieties of mappings

15.3.1 ("Banach space structure on a space of sections"). Let B be a compact differentiable manifold of class $\mathbf{C}^r$; let M be a vector bundle over K (7.3.4) with base B and of class $\mathbf{C}^r$; denote by M_r the differentiable manifold of class $\mathbf{C}^r$ underlying M , and let $\mathcal{S}_M^\tau(B)$ be the K -vector space of sections of class $\mathbf{C}^r$ of M (7.4.1). The topology on $\mathcal{S}_M^\tau(B)$ induced by the topology of $\mathbf{C}^r$ -uniform convergence on $\mathcal{C}^\tau(B; M_r)$ (12.3.10) makes $\mathcal{S}_M^\tau(B)$ a K -Banach space.

Let $(U_i)_{i \in \mathbb{I}}$ be a finite open covering of B and, for every $i \in I$, let $c_i = (U_i, \varphi_i, E_i)$ be a chart of B and $t_i = (U_i, \psi_i, F_i)$ be a K -vector chart of M (8.8.1). Let each of the E_i (resp. F_i) be equipped with a norm compatible with its structure of Banach space over $\mathbb{R}$ (resp. K), and let $(V_i)_{i \in \mathbb{I}}$ be a closed covering of B , such that $V_i \subset U_i$ for all $i \in I$. For $f \in \mathcal{S}_M^\tau(B)$ and $i \in I$, denote by f_i the map from $\varphi_i(U_i)$ to F_i obtained by composition:

$$\varphi_i(U_i) \xrightarrow{\varphi_i^{-1}} U_i \xrightarrow{f} M \Big| U_i \xrightarrow{\psi_i} U_i \times F_i \xrightarrow{\text{pr}_2} F_i.$$

Set

$$\|f\| = \sup_{i \in I, p \leq r, x \in V_i} \|D^p f_i(\varphi_i(x))\|.$$

The map $f \mapsto \|f\|$ is a norm on $\mathcal{S}_M^\tau(B)$, which gives $\mathcal{S}_M^\tau(B)$ the structure of a Banach space

over K .

If N is an open subset of the space M , the set $\mathcal{S}^\tau(\mathbf{B}; N)$ of sections $f \in \mathcal{S}_M^\tau(\mathbf{B})$ such that $f(\mathbf{B}) \subset N$ is open in $\mathcal{S}_M^\tau(\mathbf{B})$.

15.3.2. Let B be a compact differentiable manifold of class C^r , and let X be a manifold of class $C_K^{r,s}$ ($s \geq r + 1$) over B . If N is an open subset of X , denote by $\mathcal{S}^r(B; N)$ the set of sections of class C^r of X over B whose image is contained in N . There exists on $\mathcal{S}^r(B; X)$ a K -manifold structure of class C^{s-r} , and only one, such that, for every section $\sigma \in \mathcal{S}^r(B; X)$ and every tubular neighborhood (M, N, φ) of class $C_K^{r,s}$ of σ (15.2.7), the triple $(\mathcal{S}^r(B; N), \varphi^*, \mathcal{S}_M^r(B))$, where φ^* is the map $f \mapsto \varphi \circ f$, is a chart of the K -manifold $\mathcal{S}^r(B; X)$.

In what follows, we equip $\mathcal{S}^r(\mathbf{B}; X)$ of this structure.

15.3.3. Let X be a compact differentiable manifold of class C^r , and let Y be a K -manifold of class C^s ($s \geq r + 1$). Let $\mathcal{C}^r(X; Y)$ be the set of maps of class C^r from X into Y (in other words, taking values in the manifold of class C^r underlying Y , cf. 5.13.1 and 5.14.2). We equip $\mathcal{C}^r(X; Y)$ with the structure of a K -variety of class C^{s-r} obtained by identifying it with $\mathcal{S}^r(X; X \times Y)$, cf. 15.2.3, (ii) and 15.3.2. It is called the variety of mappings of class C^r of X into Y . Its topology is that of C^r -uniform convergence (12.3.10).

Suppose Y is separated; then so is $\mathcal{C}^r(X; Y)$. Moreover, the set of immersions (resp. embeddings, submersions, surjective submersions, étale morphisms, isomorphisms) of class C^r from X into Y is an open subset of $\mathcal{C}^r(X; Y)$.

Let X' be a compact differentiable manifold of class C^r , $\varphi : X' \rightarrow X$ a morphism of differentiable manifolds of class C^r and $\psi : Y \rightarrow Y'$ a morphism of K -varieties of class C^s . The map $f \mapsto \psi \circ f \circ \varphi$ is a morphism of class C^{s-r} of $\mathcal{C}^r(X; Y)$ to $\mathcal{C}^r(X'; Y')$.

15.3.4 (“Weakening of structure”). Keep the hypotheses and notations of 15.3.3. Moreover, let $s' \in \mathbf{N}_R$ satisfy $r < s' \leq s$ and let $\mathbf{Y}_{s'}$ be the real manifold of class $C^{s'}$ underlying $\mathbf{Y}$ (5.13 and 5.14.2). The structure of real manifold of class $C^{s'-r}$ on $\mathcal{C}^r(\mathbf{X}; \mathbf{Y}_{s'})$ underlies the structure of K -manifold of class C^{s-r} on $\mathcal{C}^r(\mathbf{X}; \mathbf{Y})$.

15.3.5. Let X and X' be compact differentiable manifolds of class C^r and $C^{r'}$, respectively, with $r' < r$; let Y be a K -variety of class C^s , with $s \geq r + 1$. Let $t = \inf(r - r', s - r)$. The map $(f, g) \mapsto g \circ f$ is a morphism of class C^t from $\mathcal{C}^r(X'; X) \times \mathcal{C}^r(X; Y)$ to $\mathcal{C}^{r'}(X'; Y)$.

15.3.6 (“Tangent space to $\mathcal{C}^r(X; Y)$ ”). Given the hypotheses of 15.3.3, let $f \in \mathcal{C}^r(X; Y)$ and let ξ be a tangent vector at f to the manifold $\mathcal{C}^r(X; Y)$. If $x \in X$, denote by ε_x the map $g \mapsto g(x)$ of $\mathcal{C}^r(X; Y)$ into Y ; it is a morphism of class $\mathbf{C}^{s-r}$. The image of ξ under $T_f(\varepsilon_x)$ is an element ξ_x of $T_{f(x)}(Y)$; the map $x \mapsto \xi_x$ is a lifting of class $\mathbf{C}^r$ of f in $T(Y)$ (cf. 8.6.1) and it can be identified with a section of class $\mathbf{C}^r$ of the vector bundle $f^*T(Y)$ of base X . In this way, one obtains an isomorphism of the tangent space $T_f(\mathcal{C}^r(X; Y))$ onto the Banach space $\mathcal{S}_{f^*T(Y)}^r(X)$.

15.3.7 (“Interpretation of $\mathcal{C}^p(X; \mathcal{C}^q(Y; Z))$ ”). Let X and Y be compact differentiable manifolds of class $\mathbf{C}^p$ and $\mathbf{C}^q$ respectively ($1 \leq p \leq +\infty, 1 \leq q < +\infty$), and let Z be a K -variety of class $\mathbf{C}^s$, with $s \geq p + q$. Let $f: X \times Y \rightarrow Z$ be a continuous map;

for every $x \in X$, denote by f_x the map $y \mapsto f(x, y)$ from Y to Z . The following two properties are equivalent:

- a) For any charts $c = (U, \varphi, E)$ of X , $d = (V, \psi, F)$ of Y and $e = (W, \theta, G)$ of Z such that $f(U \times V) \subset W$, the partial derivatives $D_E^{p'} D_F^{q'} f'$ of the expression f' of f in the charts $c \times d$ and e (5.3.2) exist and are continuous for $p' \leq p$ and $q' \leq q$.
- b) For every $x \in X$, the map f_x is of class $\mathbf{C}^q$ and the map $x \mapsto f_x$ is a morphism of class $\mathbf{C}^p$ from X into the variety $\mathcal{C}^q(Y; Z)$.

Z

15.3.8. Let Y be a compact K -variety of class $\mathbf{C}^s$, with $s \geq r + 1$, and let $\text{Diff}^r(Y)$ be the group of automorphisms of class $\mathbf{C}^r$ of the real manifold Y_r of class $\mathbf{C}^r$ underlying Y . The set $\text{Diff}^r(Y)$ is open in $\mathcal{C}^r(Y; Y) = \mathcal{C}^r(Y_r; Y)$; we endow it with the structure of a K -variety of class $\mathbf{C}^{s-r}$ induced by that of $\mathcal{C}^r(Y; Y)$. The composition law $(f, g) \mapsto g \circ f$ makes $\text{Diff}^r(Y)$ a topological group. On the other hand, the manifold structure of class $\mathbf{C}^{s-r}$ is not in general compatible with the group structure of $\text{Diff}^r(Y)$. For $f \in \text{Diff}^r(Y)$, the map $g \mapsto g \circ f$ of $\text{Diff}^r(Y)$ into itself is of class $\mathbf{C}^{s-r}$, but the map $g \mapsto f \circ g$ is not in general of class $\mathbf{C}^1$. In particular, $\text{Diff}^r(Y)$ is not in general a Lie group.⁵

⁵Suppose that $s = \infty$. Let $\text{Diff}^\infty(Y)$ be the group of automorphisms of class $\mathbf{C}^\infty$ of Y ; equip it with the coarsest

topology making the canonical injections $\text{Diff}^\infty(Y) \rightarrow \text{Diff}^r(Y)$, for every integer $r \geq 1$, continuous. On the other hand, let $\text{Diff}^0(Y)$ be the group of homeomorphisms of Y , endowed with the topology of uniform convergence (TG, X, § 3, no. 5, Proposition 11). If $\dim Y \geq 1$, it is impossible to find a Lie group G and continuous homomorphisms from $\text{Diff}^\infty(Y)$ to G and from G to $\text{Diff}^0(Y)$ whose composite is the canonical injection (cf. LIE, III, § 4, Exercise 7).

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